Panasonic

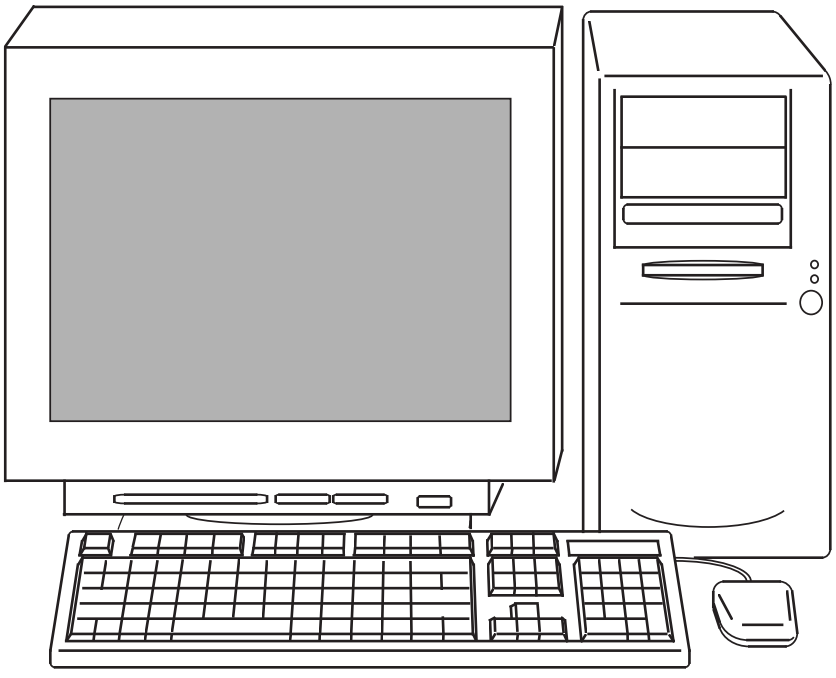
Operating Instructions

Material Verification

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”
- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A445E19 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

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Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

This chapter explains the Overview of PanaCIM-EE / Material Verification.

1.1 Overview

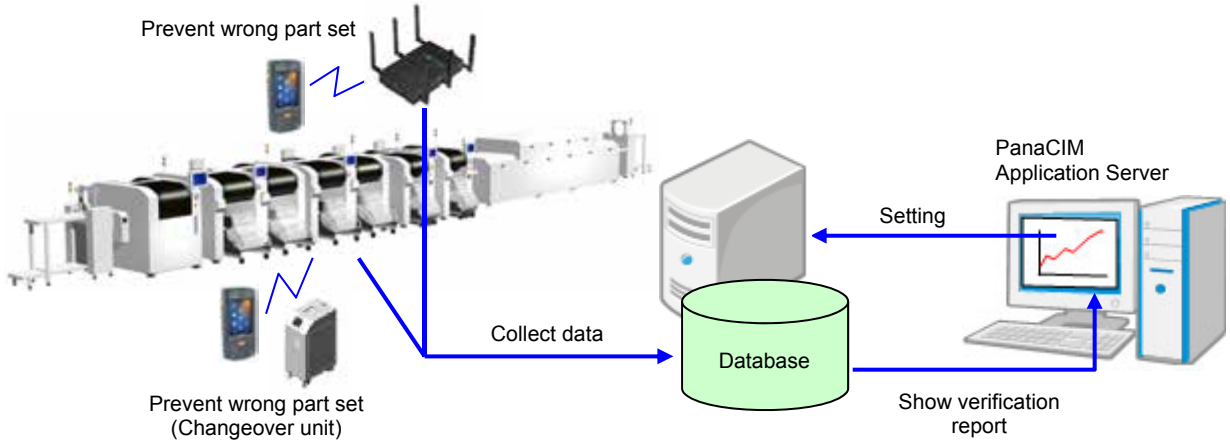
PanaCIM-EE (PanaCIM Enterprise Edition: PanaCIM) provides PanaCIM/Material Verification system to prevent wrong parts set by verifying parts while setting them at setup, changeover, splicing and part exhaust.

Parts are verified by comparing part information read with a barcode reader and the information in recipe. If unnecessary parts for production were scanned, warning appears. If unnecessary parts are already set, machine is controlled not to start production. Printer material (solder/mask) can also be verified same as tray parts.
Material verification work history (parts scan history) can be displayed on report. Scanned part information (parts number, vendor name) can be traced linking with PanaCIM/Traceability.

The following information can be obtained by scanning barcode.

- Machine slot/position
- Part set time
- Operator ID
- Vendor
- Lot
- Quantity
- User defined data

Function configuration is shown below.

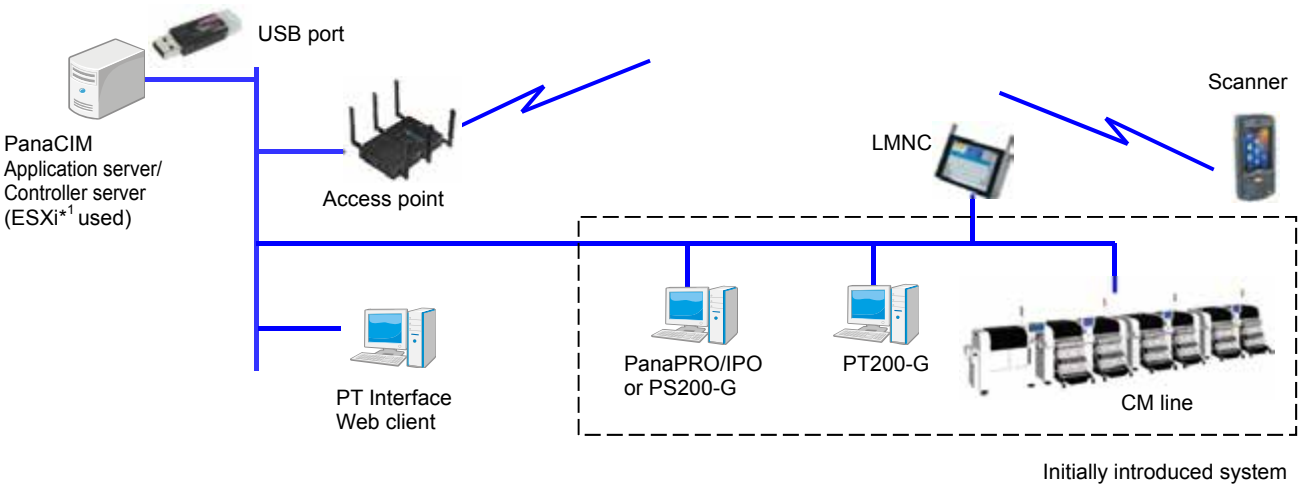


1.2 System Configuration

This section shows the schematics of systems with PanaCIM for small-scale and large-scale production.

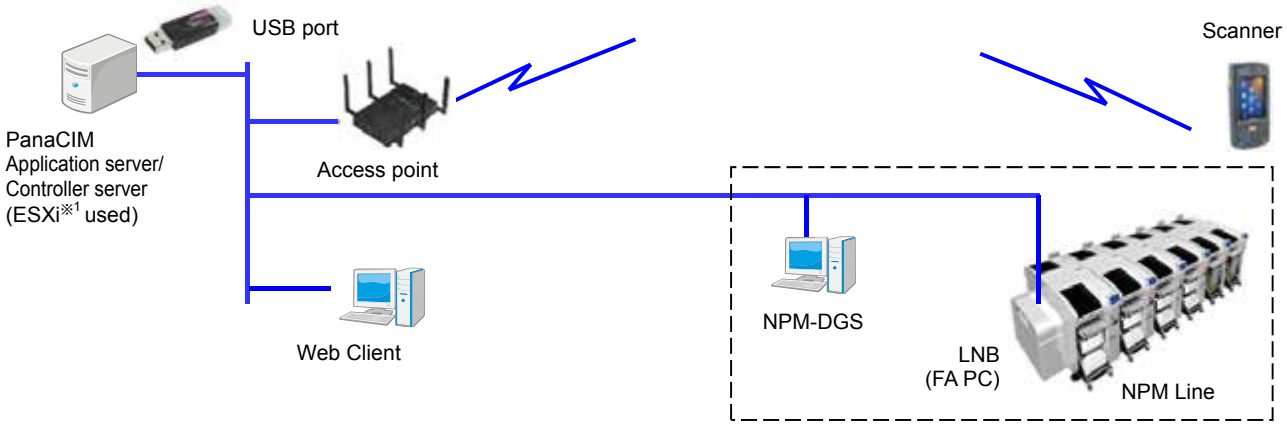
■ System with PanaCIM (for small-scale production)

CM Line



Initially introduced system

NPM Line *2

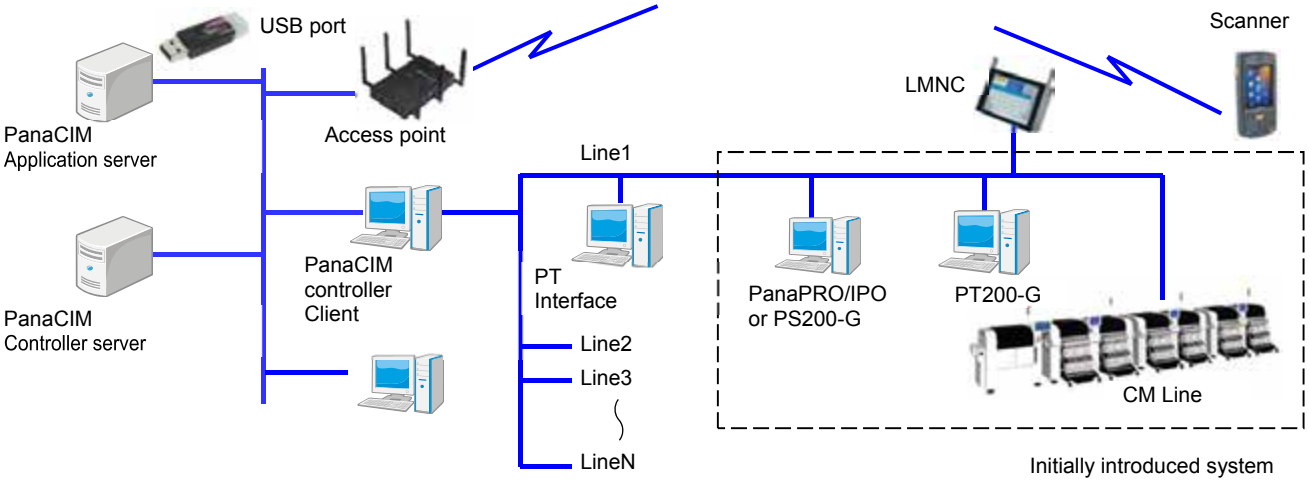


Initially introduced system

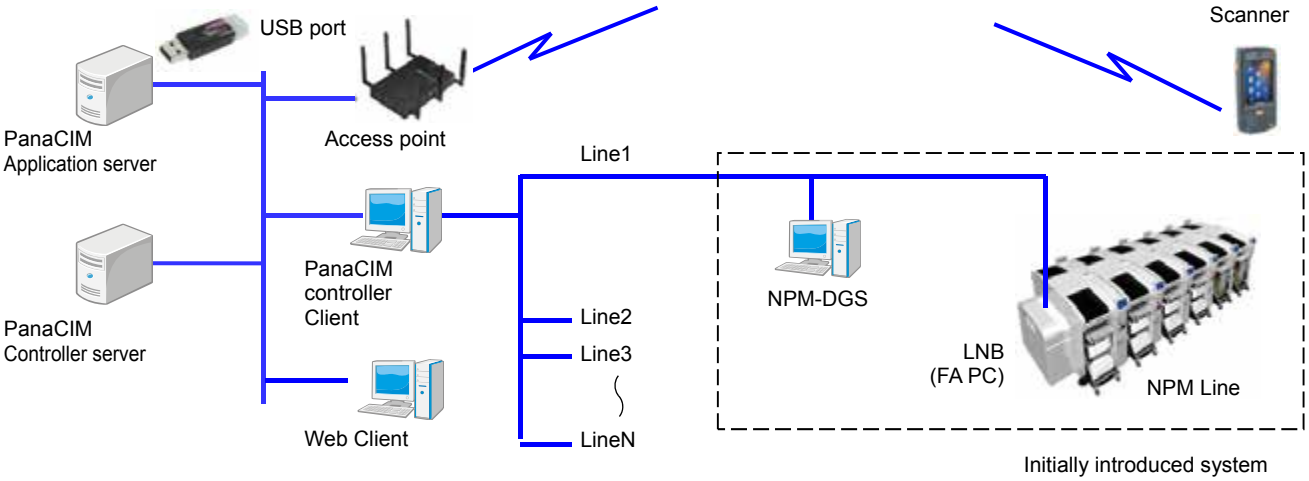
*1. ESXi: A virtualization tool that allows multiple applications to run on a single server.
*2. NPM line can also include the models such as AM100, SPD, SPG, SPV and SP70 as NPM-related machines.

■ System with PanaCIM (for large-scale production)

CM line



NPM line *1



*1. NPM line can also include the models such as AM100, SPD, SPG, SPV and SP70 as NPM-related machines.

1.3 Functional Configuration

■ Material verification functions

Main function	Detailed functions		Specifications
	Large category	Small category	
Prevent wrong material use	Part replace		Material verification possible when replacing parts
	Production start control		Stop production when parts are set wrong by production start control
	Substitute parts		Use substitute parts instead of defined original parts
			Both original and substitute parts are recorded for trace
	Invalid parts		Invalid parts can be set. If scanned parts are invalid, they cannot be set.
	Tray parts		Material verification possible at part replacement
	Printer material		Verify solder/mask.
	Common	Scanner	Scanner reads 1D or 2D barcode
		Barcode format	The following barcode format supported
			1D Barcode: Code 39/Code 128/Code 93/Code 2of5 Interleaved/UPC/EAN/CODEBAR/MSI
			2D Barcode: Data Matrix/QR Code/PDF417
Data Collection	Data Type		Part number, Lot, Vendor, Quantity, User defined item
	Max. Machines to Collect Data		100 stages (per PanaCIM controller server)
	Max. Lines to Collect Data		Equivalent number of line(s) (per PanaCIM controller server)
	When to Collect Data		Every time when scanning parts
Data Storage	Backup	Auto Backup	Backs up data automatically at the specified intervals.
		Store Outside	Stores the backed up data outside (on other media).
	Restore	Restore	Data can be restored from the backup data.
Charts and Reports	Report	Save	Parameters necessary for generating reports (attribute, range) are saved as report configuration file
		Open	Call saved report configuration file to display reports
Settings	Report setting		Parameters such as interval and graph kind from reports to be displayed.
	Barcode setting		Part information barcode setting can be changed

1.4 The Process of Material Verification

1.4.1 Material Verification for Intelligent Feeders

The CM/DT/NPM series machines (including AM100) use the intelligent feeders to supply reel parts. The reel parts are verified by comparing the feeder memory with the part information of the product program. This operation is enabled via the machine or offline unit. The information will be held unless the feeder memory is cleared even after the program is changed.

When splicing other reel into an existing reel, new part information must be written to the splice memory of the feeder by the operator. If the joint is detected by the machine during production, the splice memory will be used as the current memory.

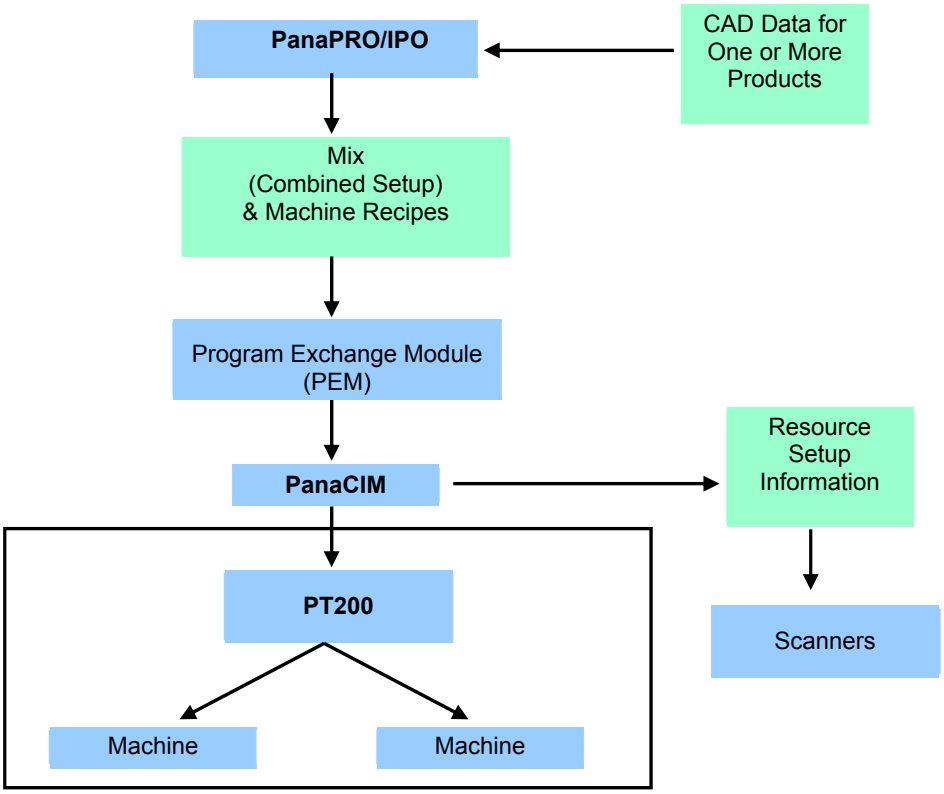
Before starting production after setup, the machine checks whether the part numbers of the feeders agree with those in the product program. In addition, the part numbers are checked also when components are replenished or joint is detected during production. If the part information does not agree, production is stopped until correct part information is confirmed.

A product program can be imported into the PanaCIM from the PanaPRO system via the PEM application or from the PT200 via the product import application (launched from “Product Setup” application).

1.4.2 Material Verification Data Setup

To import a program into the PanaCIM, use the program exchange module (PEM) to download one from the PanaPRO or the product import feature of the PanaCIM product setup to upload one from the PT200.

■ Data Flow Using PanaPRO



Recipe and setup data flow using PanaPRO/IPO

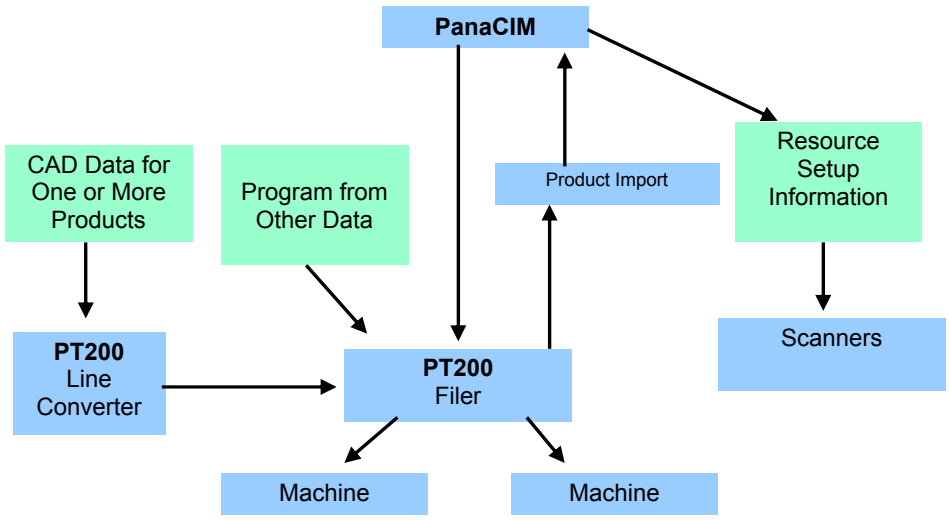
- Product programs can be created by importing CAD data for the product using PanaPRO/IPO.
- PanaPRO/IPO combines one or more product programs to make setup in consideration of common arrangement, etc. A file containing multiple product programs and combined setups is referred to as 'Mix'.
- Product programs or mix created on the PanaPRO/IPO are downloaded to the PanaCIM system via the Program Exchange Module (PEM).
- New information such as model IDs or PT200 folder names can be added to the product programs and mix downloaded to the PanaCIM system via the PanaCIM product setup.
- Programs are downloaded to the PT200 via the program exchange application.
- Product programs are downloaded from the PT200 to placement machines as necessary.
- PanaCIM system sends the material setup information required for verification to the scanner once the product program is downloaded to the machine.

■ Data Flow Using Program Import

Programs may be created on the PT200 in the following cases:

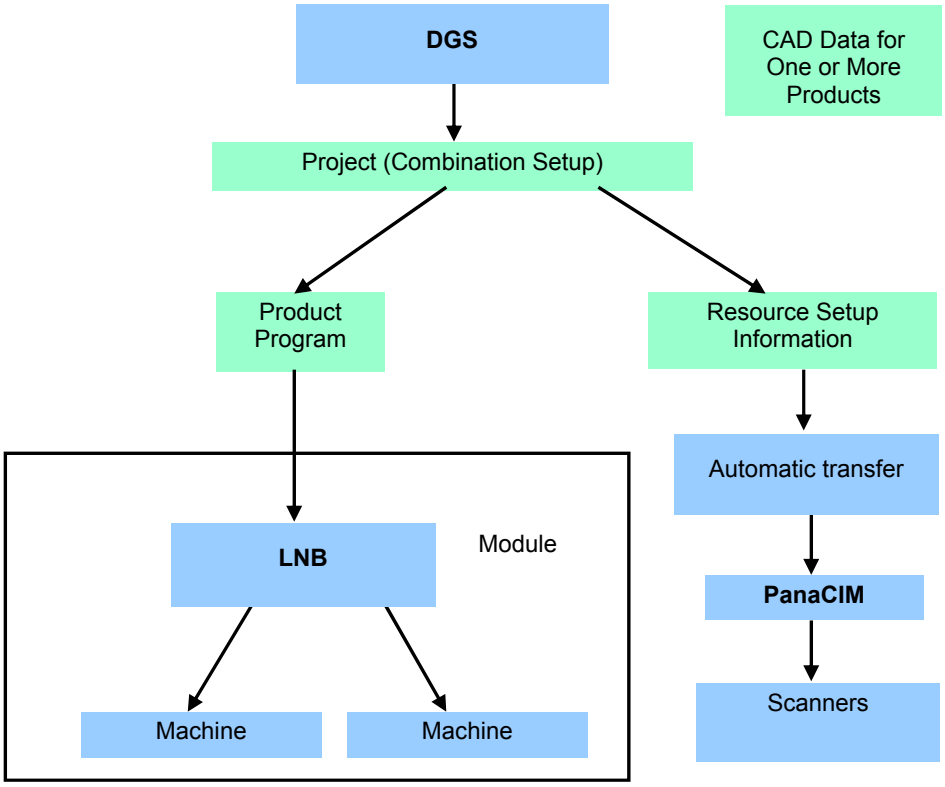
- Product program has been created or modified via the PT200 filer.
- Product program has been imported from the PT200 Line Converter into the PT200 Filer.
- Product program has been created from the program other than the PanaPRO/IPO.
- Product program was transferred to the PT200 in another line.

In such a case, the Product Import function of the PanaCIM product setup is used to upload programs to PanaCIM. This Product Import function is available only for the product programs. (Not available for the mix programs.)



Recipe and setup data flow using PT200

■ Data Flow using DGS



Recipe and setup data flow using DGS

- Using the DGS, product programs can be created by importing the product CAD data.
- The DGS combines one or more product programs to make setup in consideration of common arrangement, etc. A file containing multiple product programs and combined setups is referred to as ‘Mix’
- Product program or mix created on the DGS is automatically downloaded to the PanaCIM system when the data is committed on the DGS.
- Product programs are downloaded from the LNB to placement machines as necessary.
- PanaCIM system sends the material setup information required for verification to the scanner once the product program is downloaded to the machine.

2. SETTINGS

This chapter explains the various setting items about Material Verification.

The following PanaCIM applications perform a setup about Material Verification.

- Configure Licenses
- System Config: Configure the factory whole managed by PanaCIM
- Route Editor: Configure Route (Line)
- Inventory Manager: Configure Invalid Components

2.1 Overview

Before performing Material Verification operations, the Material Verification Module must be configured for use. This section guides you through the Material Verification configuration process that corresponds to the CM/DT/NPM series machines[*1].

*1: NPM series also include the models AM100, SPD, SPG, SPV and SP70. The same applies to the NPM series machines described in the subsequent sections.

Refer to the below table for a brief overview of what Applications will be used to fulfill various configuration settings.

Application	Overview
System Configuration	The System Configuration Application is used to configure the site-dependent aspects of the PanaCIM system. Use the System Configuration Application to configure the following settings for the Material Verification Module: • Consumables for Screen Printers • Filter Components (Define Groups) • Assign Part Numbers to a Group • Configure Barcode Expressions • Configure Mobile Scanners • Operator Access to Mobile Scanners • Other Configurations ○ Invalid/Restricted Components ○ Verify Tray Component Forced Setup during Changeover ○ Scan Standard Barcodes • Configure Intelligent Feeder Programming Mode for Substitute Parts
Route Editor	Route Editor Settings are specific to a particular production line. Changes are made for Material Verification specific to the Machine, to the Module (modular machines only) and to the Scanner. The settings can be observed while the object is running, but the object *must be disabled or stopped* in order to make any changes. **Note:** It is strongly recommended to only make changes in Route Editor when the Production Line is *not* assembling panels. Use the Route Editor Application to configure the following settings for the Material Verification Module: • Configure Licenses • Edit Hosts • Configure Route Settings • Configure Machine Interface Settings • Configure Periodical Checks • Configure Offline Units

Product Setup	The Product Setup Application allows you to take a product that has been setup and optimized in PanaPRO and customize it for the PanaCIM route that it is being built on. Use the Product Setup Application to configure the following settings for the Material Verification Module: • Configure the Setup • Configure Component Filtering • Edit the Feeder Setup • Edit the Substitute Parts
Inventory Manager	Use the Inventory Manager Application to update and edit Invalid components.
Product Transfer	Use the Product Transfer Application to transfer products to or from selected Zones in the Route.
Machine Settings	To use the Material Verification Module with the modular machines, change the following settings on the machine. • Serviceman Setup (CM/DT) • Machine Option Setup (CM/DT) • LNB Machine Setting (NPM*1) • NPM Machine Setting (NPM *1)

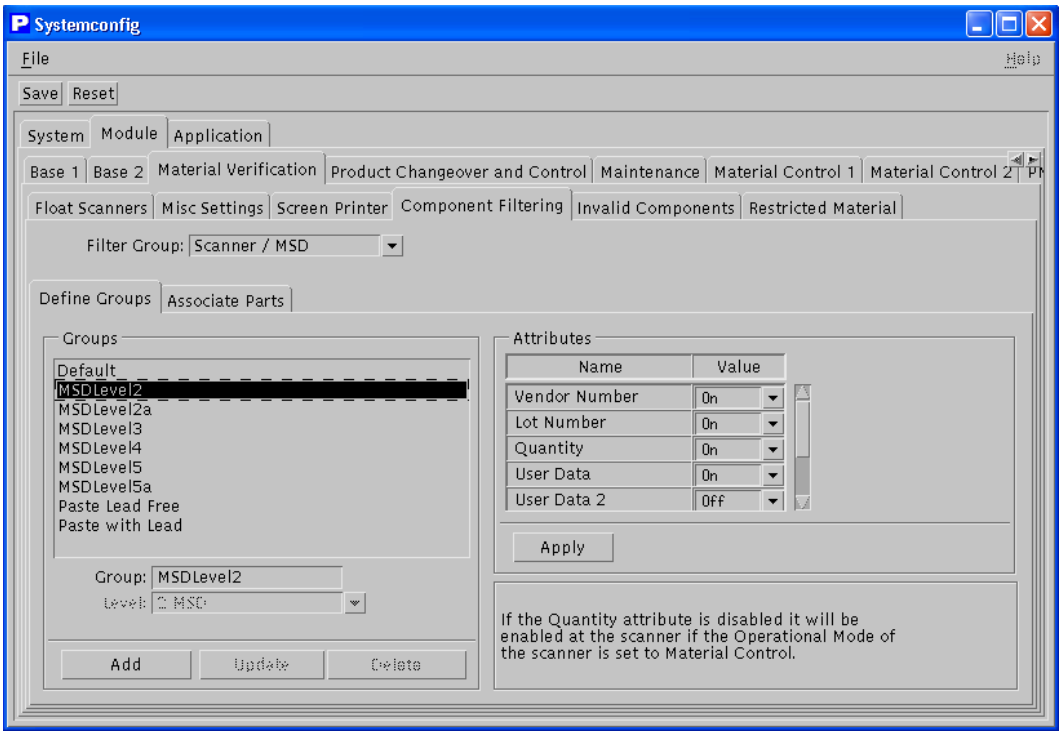
2.2 Settings

2.2.1 Setting of CIMC

■ System Configuration

● Component Filtering

PanaCIM provides the ability to establish groups for components and assign different information required (Lot, Vendor, Quantity, etc.) when the components of a particular group are scanned.



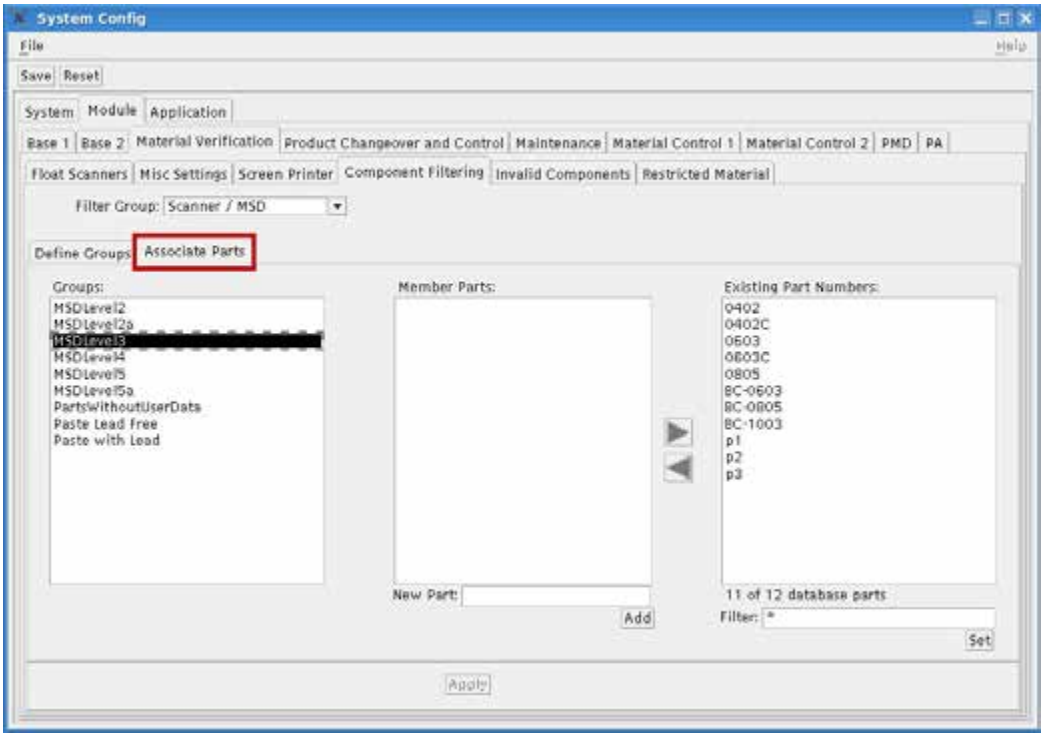
Initially, PanaCIM includes groups for “Default” and each of the typical “Moisture Sensitive Device (MSD)” values. You may add additional groups as needed. As a further step, you may assign each component part number to the appropriate group. Any part number not assigned to a particular group is assumed to be in the “Default” group.
In case of the screen printers (SPV/SPG/SPD/SP70), two groups ‘Paste Lead Free’ and ‘Paste With Lead’ will be added. These respectively correspond to lead-free solder and leaded solder, and each of them includes a unique attribute ‘Exposure Time (Hrs)’ that cannot be used by other groups. ‘Exposure Time’ shows the expiration time of the solder group, i.e., the maximum period of time the solder may be exposed to moisture after it has been printed by the screen printer (SPV/SPG/SPD/SP70). After the lapse of this time, the solder concerned cannot be used for screen printing. Exposure time varies from group to group.

* Part number to group association can also be made for specific products in the Product Setup Application.

Step	Procedure
1	Open **System Configuration** from the PanaCIM Main Menu.
2	Click the **Module** tab.
3	Click the **Material Verification** tab.
4	Click the **Component Filtering** tab.
5	Click the **Define Groups** tab and select **Scanner/MSD** from the **Filter Group** drop-down list.
6	Enter the **Group** name and click **Add**. To define the solder group, select **Paste** from the **Level** drop-down list.
7	In the Attributes section, select **On** from the Value drop-down for those values required when scanning components of the indicated group. • Vendor Number • Lot Number • Quantity • User Data
8	Click **Apply**.
9	Click **Save**.

Assign Part Numbers to a Group

Once you have created the desired group and set the required information to be scanned, you must assign what part numbers are in the group. The group association made in this menu can be overwritten for a particular product by setting the group association in the **Product Setup** Application. Anytime a new product is downloaded into PanaCIM, new components are added to the Existing Part Number List.

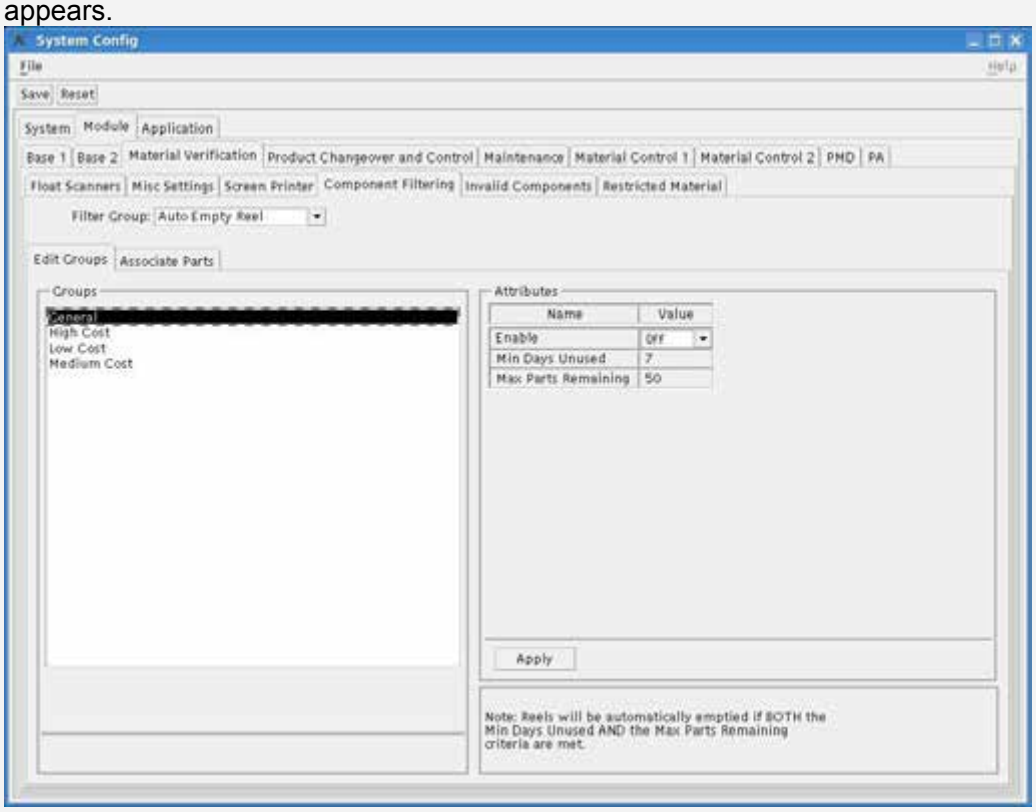


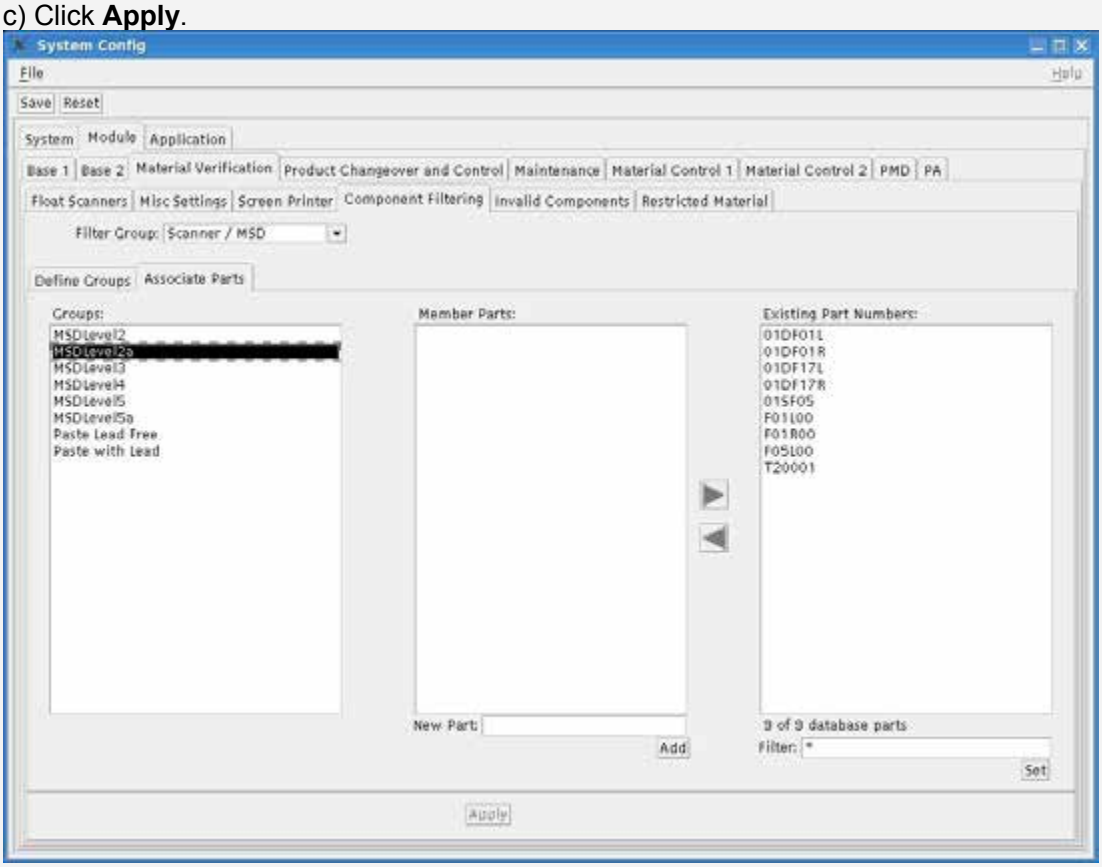
Refer to the action table below to configure the System Configuration Application to assign part numbers to a group.

Step	Procedure
1	Open the **System Configuration** from the PanaCIM Main Menu.
2	Click the **Module** tab.
3	Click the **Material Verification** tab.
4	Click the **Component Filtering** tab.
5	Click the **Associate Parts** tab. Check that **Scanner/MSD** is selected from the **Filter Group** drop-down list.
6	Select a **Group** from the **Groups** list.
7	Select one or more part numbers from the **Existing Part Numbers** list.
8	Use the arrow keys to associate the selected part numbers with the selected group.
9	Click **Apply**, then click **Save**.

- **Auto part exhaust reel option setting**

  Part attributes can be set depending on which reel automatically have part exhaust. Attributes are ‘Minimum Days Unused’ and ‘Maximum Parts Remaining’.
  If ‘Minimum Days Unused’ allows a large value such as 9999, the reel detects part exhaust after the specified number of days regardless of the value set here.
  ‘Minimum Days Unused’ allows a value between 2~99999 and ‘Maximum Parts Remaining’ between 1~99999. If number of parts on reel is within ‘Minimum Days Unused’ and ‘Maximum Parts Remaining’ range, the reel detects part exhaust automatically. If the number is out of the range, the value will be automatically changed to a value within the range when saving the group. (Example: If 1 or less is set for ‘Maximum Parts Remaining’, change it to 1 and saved.)
  In this way, any parts can be managed by ‘Automatic Empty Reel’ and other parameters. See the table below to set ‘Automatic Empty Reel’.

Step	Procedure
1	Go to System Configuration dialog **Module** - **Material Management** - **Part Filtering** tab.
2	Select **Auto Empty Reel** from **Filter Group** drop-down list. The following screen appears.  As seen above, four groups **General**, **High Cost**, **Low Cost** and **Medium Cost** are created. If parts on reel belong to any of these groups, part exhaust is detected automatically. If parts do not belong to other three groups than ‘General’, they belong to **General**. **General** group’s attribute **Enable** is set to **Off** as default. **General** group’s **Enable** is set to **Off**, parts that do not belong to the group do not detect parts exhaust at default. Other three groups do not have valid attributes and automatically detect part exhaust at default. **Note:** You cannot create another new group than those four above.
3	Enter values to **Minimum Days Unused** and **Maximum Parts Remaining** for groups **General**, **High Cost**, **Low Cost** and **Medium Cost** and click **Apply**.

Step	Procedure
4	Click **Related parts** tab and link parts with groups **General**, **High Cost**, **Low Cost** and **Medium Cost**. a) Click one or more parts from **Existing part number** list. b) Click  and make sure the selected parts appear on **Parts** list of the selected group. c) Click **Apply**. 
5	Or enter part number at **New parts** and click **Add**. If entered parts are not in the list, they appear at **Parts** list. If they are already in the list, a dialog appears to inform that. If the parts are in **Existing part number** list, the parts name will be deleted from the list.
6	Click **Save** to save the settings.

Configure Barcode Expressions

Please refer to the Functional Specifications that came with the PanaCIM-EE Base documentation for detailed instruction on how to configure barcode expressions.

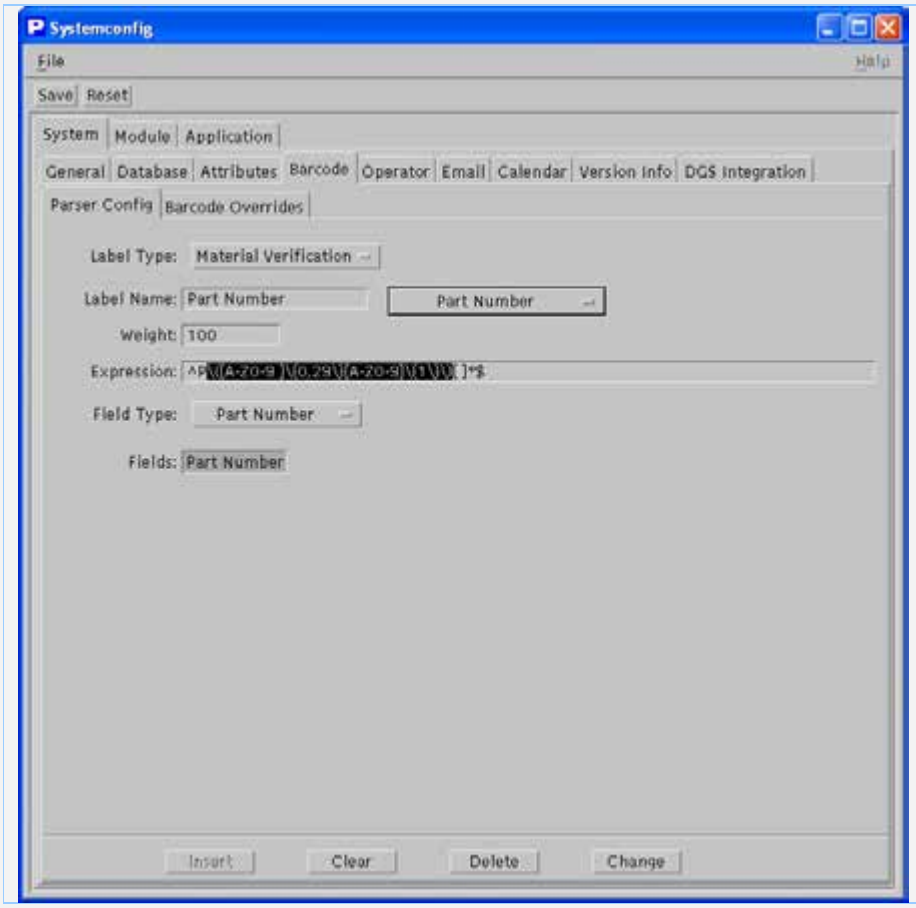
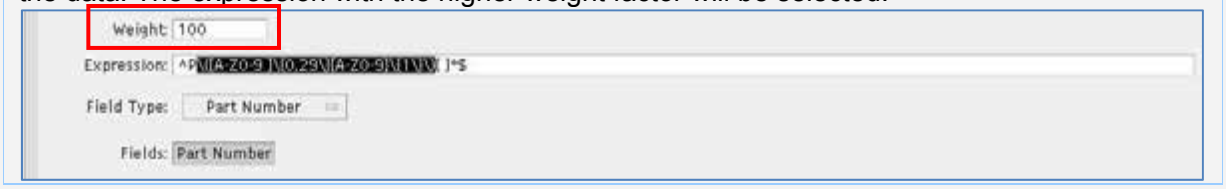
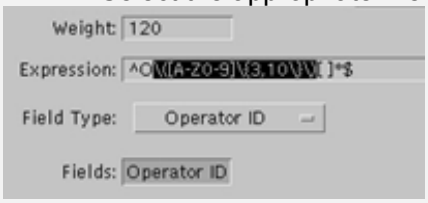
You are required to configure how Material Verification information is formatted for scanning into PanaCIM. This is done using a formatting method called Regular expressions. Oftentimes this determines how barcodes are formatted, but in most cases you can configure the Route Editor Application to allow for manual entry.

In some cases, the barcode expression can contain more than one field type.

Field	Description
Part Number	Component part Number associated with a material supply.
Vendor	The name of the supplier of a component.
Lot	The number assigned by the manufacturer for a batch of components. Multiple reels can have the same Lot number.
Quantity	The number of physical parts in the supply unit.
User Data	A user configured value that is scanned for tracking.
Operator ID	The number associated with employees that are used to track who performed an operation.
Cart ID	Number assigned to identify the feeder cart
MCID	Unique identifier associated with part number

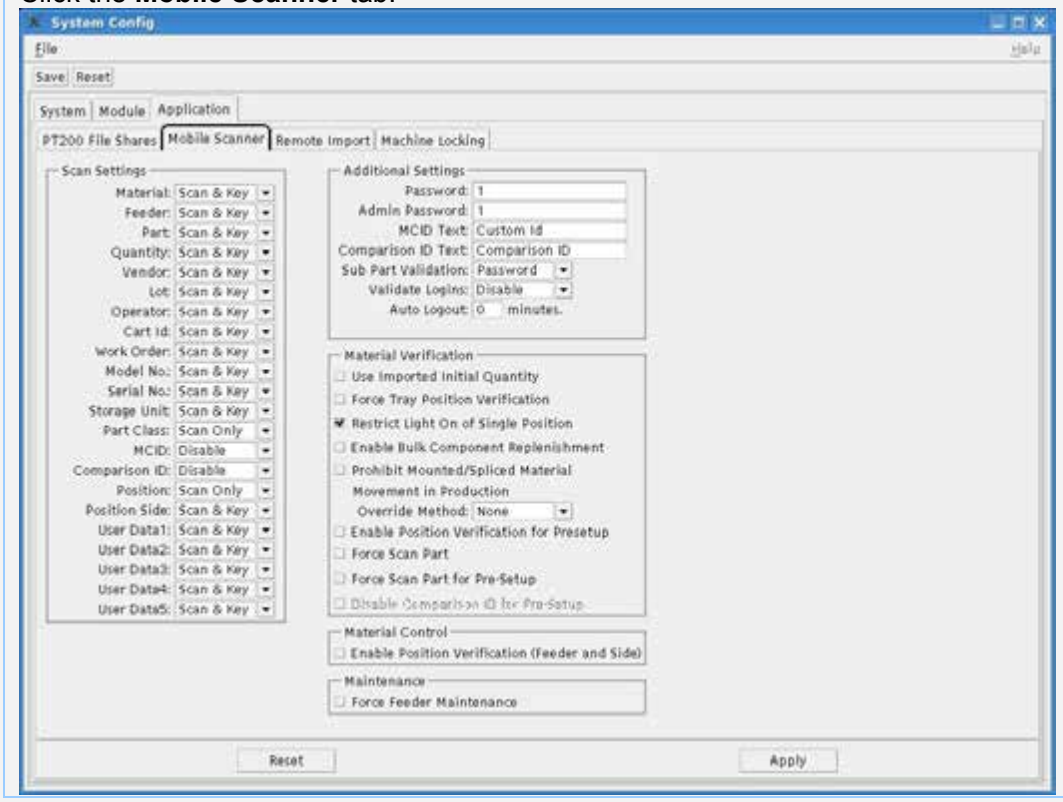
Refer to the action table below to configure barcode expressions using the System Configuration Application. More than one expression can be entered if desired.

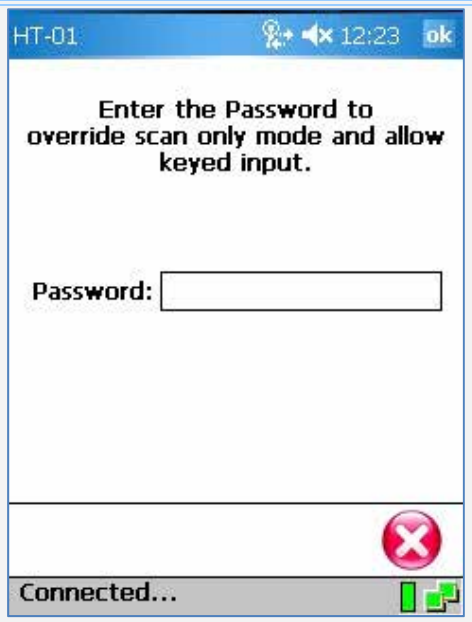
Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Select the **Barcode** tab from the **System** tab.

Step	Procedure
	
3	Select **Material Verification** as the Label Type.
4	Enter a Label name of your choice. This name can be selected in the future to review or modify the expression.
5	Enter the **Weight** factor. The Weight factor is used when more than one expression will apply to the data. The expression with the higher weight factor will be selected. 
6	Enter the **Expression**. (Refer to the Functional Specification document for Regular expression descriptions)
7	• Select the appropriate **Field Type**.  Repeat the process if there are more than one data term in the expression.
8	Click **Change** when configuration changes have been completed.
9	Click **Save**.

Mobile Scanner Settings

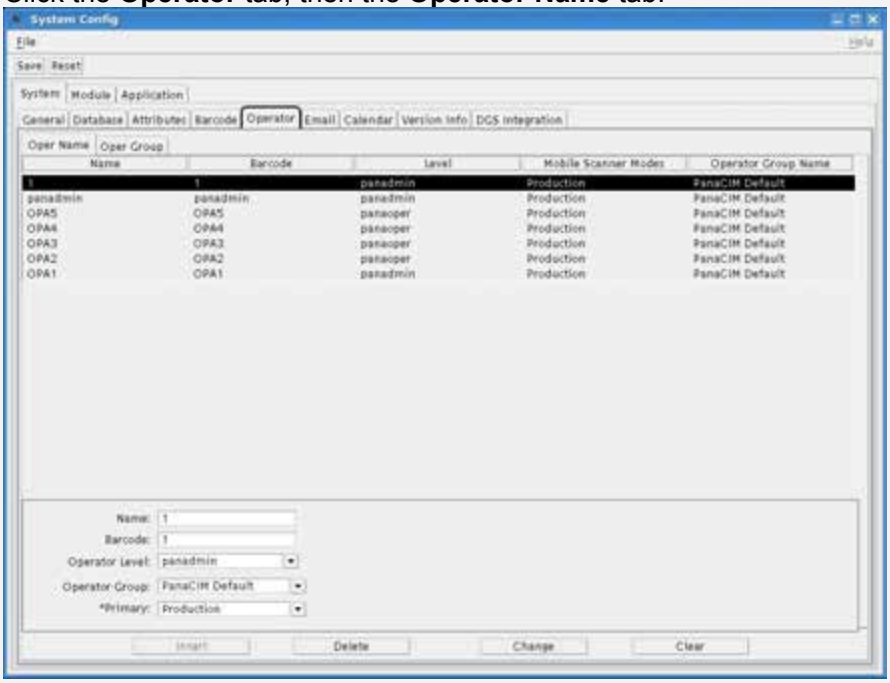
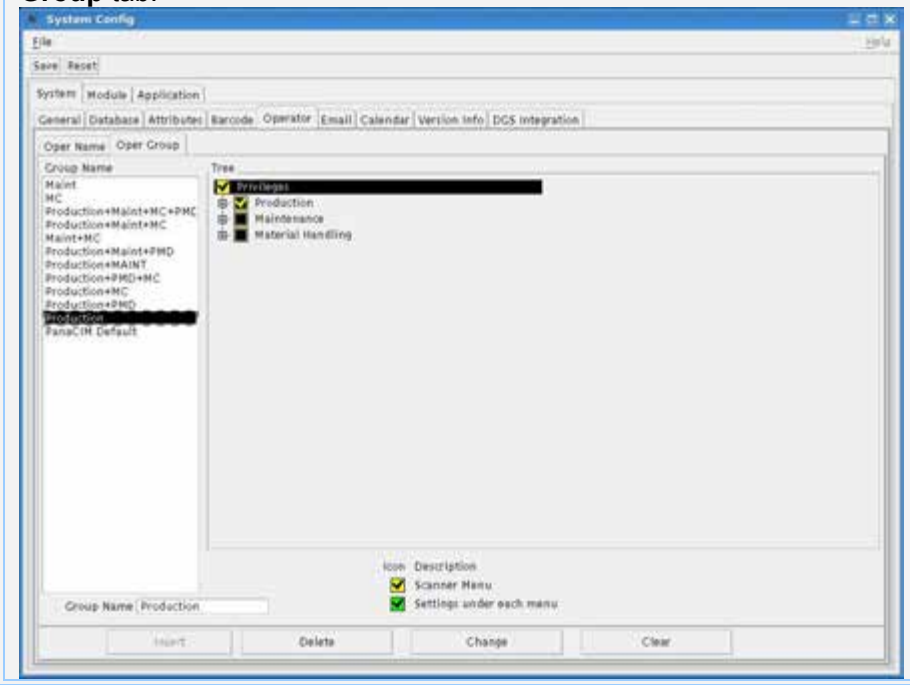
The mobile scanner is one of the standard features provided with the PanaCIM. It enables a single scanner to be used in the several lines. (CM/DT/NPM series machine only)
Refer to the action table below to configure the mobile scanner in the System Configuration Application.

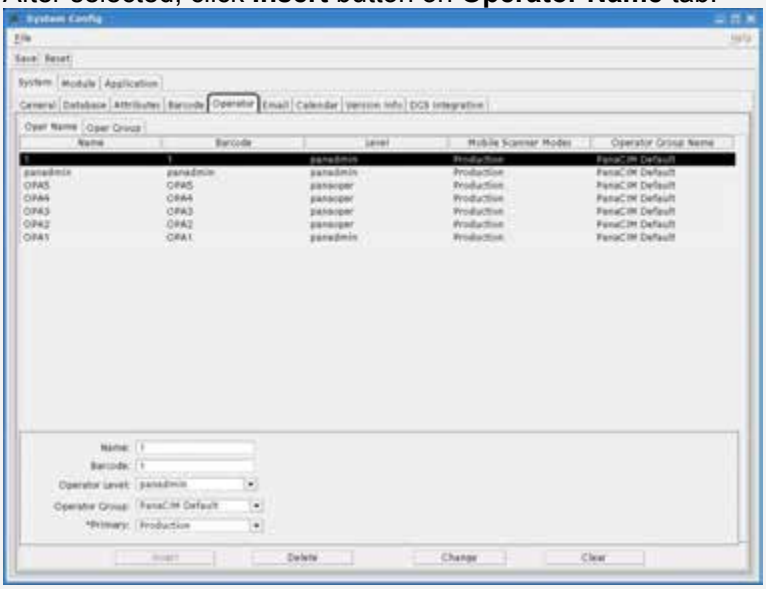
Step	Procedure
1	Select the **System Configure** icon from the main menu.
2	Click the **Application** tab.
3	Click the **Mobile Scanner** tab. 
4	Configure the **Scan Settings** by choosing the options in the drop-down lists. • **Scan Only**: Only scanning is enabled. • **Key Only**: Only key input is enabled. • **Scan & Key**: Scanning or key input is enabled. • **Disable**: Disabled (Some items do not have this option. Select this option when the item is not used.) **Note:** When the **Scan Only** option has already been set, but key input is required, the operators having the panadmin privileges may enter the password to override the scan only mode on the scanner screen, to allow key input.

	
5	**Options** field includes: • **Password:** Password for removing any restrictions on the scanner features to the operators with panadmin privileges • **Administrator password:** Required to exit the PanaCIM scanner application. This keeps the operators from changing the OS of the scanners or network settings. • **Verify ID string:** When this option is enabled, the scanner prompts you to input available reel list ID before continuing component scanning. It is also possible to define a text in this field when the reel list ID is displayed. (Default text is **Comparison ID:** After the material ID is validated, the barcode expression needs to be defined on the System Configuration screen. The text of the verification ID except for the leading identifier must perfectly match the substring of the part number. Normally, verification ID entry is made before component scanning; however, you can make a verification ID change by scanning a verification ID during component scanning. • **Verify Operator:** Choosing **Enable** verifies the operator ID when logging in the scanner. You cannot log in with unregistered ID. Choosing **Disable** allows you to log in the scanner even with an unregistered ID, thus registering the ID as a new operator (with panaoper privilege). • **Feeder Jam:** When this option is enabled, feeder jam can be reset and verification can be made on the mobile scanner. Each time this setting is changed, it will be sent to the scanner in a second and a feeder jam icon will be displayed. • **Auto Logout:** If the scanner is not used for the time specified here, it automatically logs off. To disable this option, set to **0**.
6	**Material Verification** field includes: • **Imported Initial Quantity:** Check this box to import the initial quantity of the primary or substitute components without scanning the field information. The initial quantity will be obtained from the information saved in the parts library. **Note:** Once the scanner is checked, it does not prompt you to input the quantity during component scanning or presetup. • **Verify Tray Forcedly:** Check this box to scan the barcode at the tray position to see if the component and position are correct. **Note:** Make sure that the **Position** field within the scan setting area has already been configured before using this option.
7	Click the **Apply** button, then click the **Save** button.

Access to mobile scanner

See the table below to set access to mobile scanner on system configuration screen.

Step	Procedure
1	Open system configuration application from PanaCIM main menu.
2	Click the **System** tab.
3	Click the **Operator** tab, then the **Operator Name** tab. 
4	Enter **Name**, **Barcode** (Operator ID) and select **Operator Level** from drop-down list.
5	Set necessary items and select operation mode from **Primary**. (**Production**, **Maintenance**, **Material**)
6	To use multiple operation modes, select them from **Operator Group** drop-down list. Only privileged operators can do this operation.
7	Operator group and privileges of each group can be seen from **Operator Group** tab. 

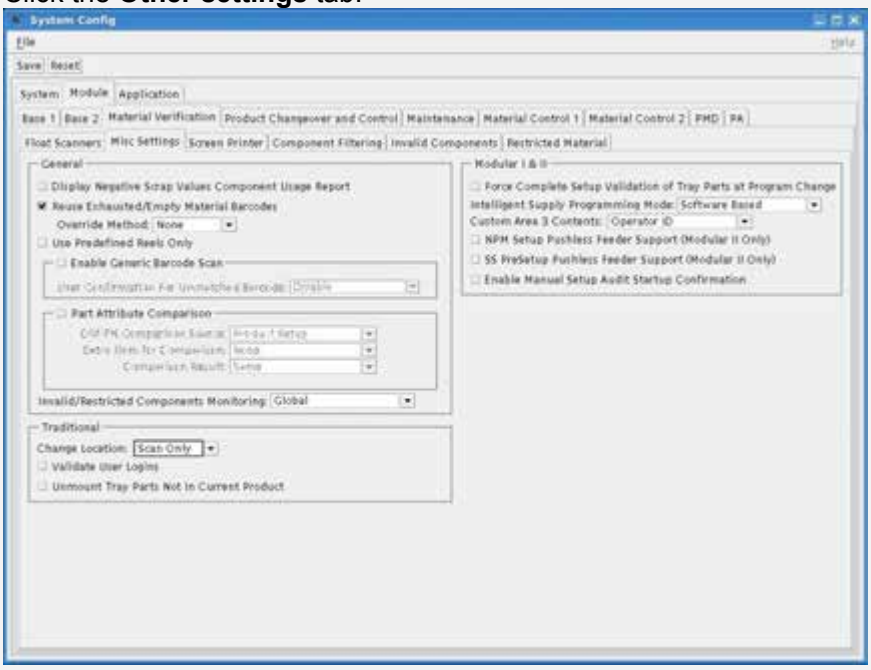
Step	Procedure
	Right pane tree shows items available for privileges **Production**, **Maintenance** and **Material Handling**.
8	After selected, click **Insert** button on **Operator Name** tab. 
9	Repeat this for each operator. Click **Save** to save the settings.

Material verification and other settings

This section explains other material verification settings done from system configuration application.

- Set **Standard barcode scan** option to **Valid** to allow Vendor, Lot, User data barcode input that do not match with the barcode definition.
- Set **Tray parts force setup verification at changeover** to **Valid** to force scanning parts again before starting production.
- Invalid/limited parts monitoring control level setting

* These settings are not mandatory.

Step	Procedure
1	Select **System Configuration** icon from main menu.
2	Select the **Application** tab.
3	Select the **Material Verification** tab.
4	Click the **Other settings** tab. 
5	**Standard barcode scan**: Select the checkbox to allow inputting Vendor, Lot and User data barcode that do not match the definition. **Barcode mismatch check**: Select Disable/Enable from the drop-down list.
6	**Compare new/old reel:** Request to scan already set (old) reel before scanning a new to make sure operator set reel to the right position. • **Old parts:** Select recipe setup information or current setup information to compare with old parts • **Additional comparison item:** Set whether to compare Vendor or Lot or User data in addition to parts number (optional). • **Result:** Set whether Same or Different is allowed as the result when additional items are to be compared.

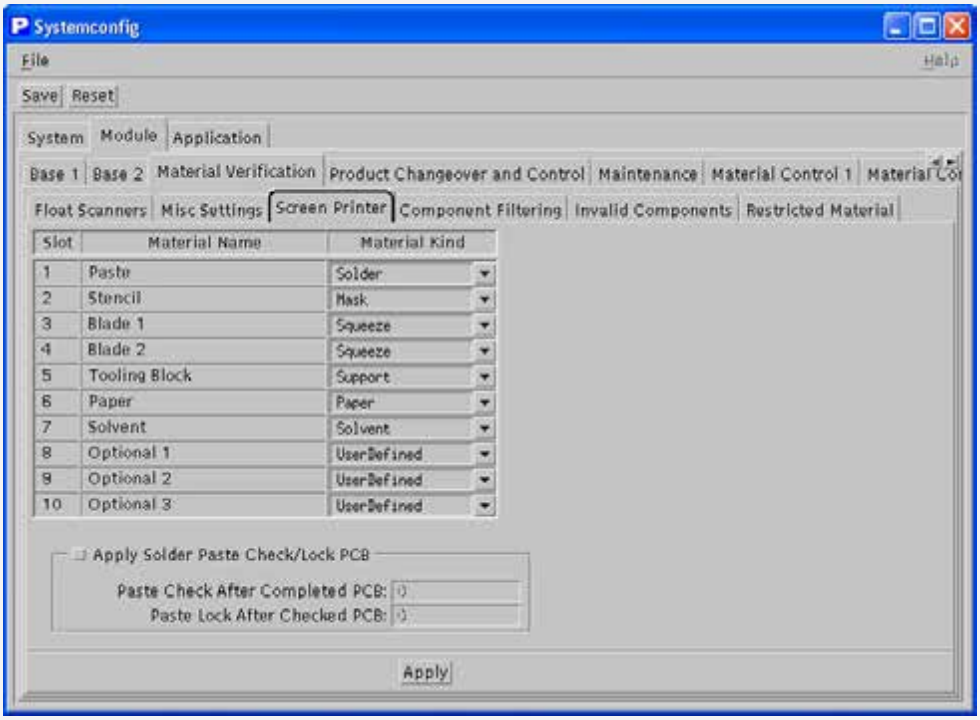
7	**Tray parts force setup verification at changeover**: Select the checkbox to enable force tray setup validation to request part rescan any time before starting production. If the box is not checked, tray parts used in both current and previous model are held when removing tray parts that are not used by the current model. This is the same as **Unmount Tray Parts Not In Current Product** as in old machines.
8	**Invalid/Restricted parts monitoring control level**: Select **Global** or **Route Based** • **Global:** Monitor invalid/restricted parts in all modules • **Route Based:** Set invalid/restricted part monitoring per route. **Note:** If you select Route Based, you need to enable 'Invalid/Limited part monitor' from Route Editor for target route. See Route Editor configuration section for more details.
9	Click **Apply** - **Save**.

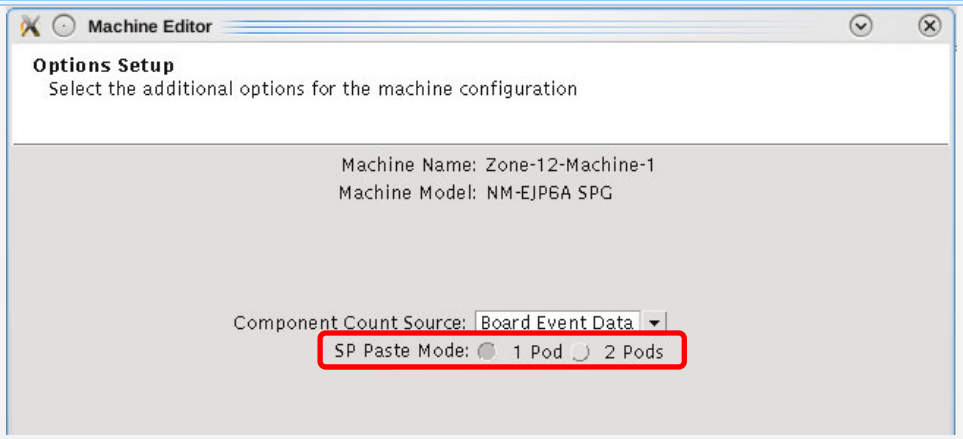
Configure Screen Printer Consumables

Material Verification can be used to record significant information for consumables used in the screen printing process.

* This configuration is optional and only applicable to the screen printers SPV, SPG, SPD, SP70, SP60 and SP80.

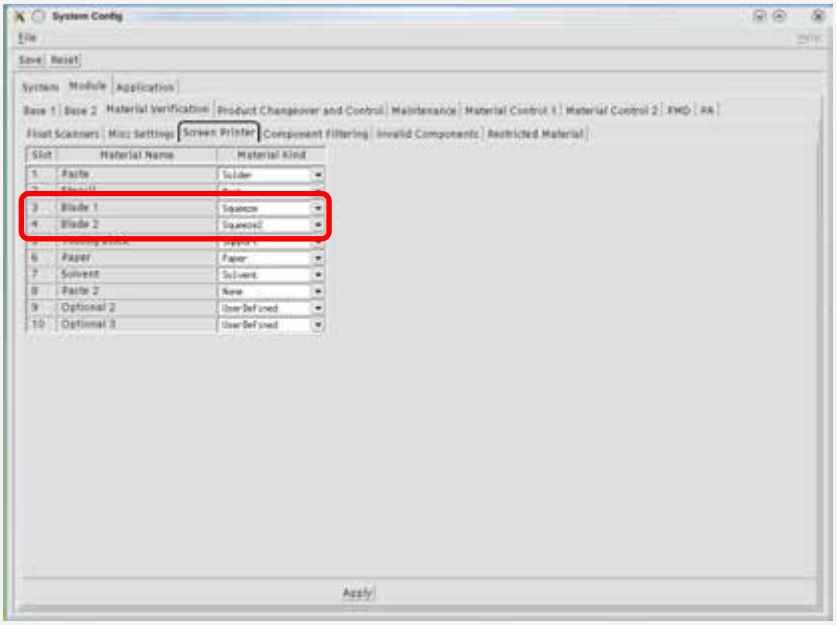
Refer to the action table below to configure screen printer consumables using the System Configuration Application.

Step	Procedure
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Click the **Module** tab.
3	Click the **Material Verification** tab.
4	Click the **Screen Printer** tab. You can set a Material Kind for each of the indicated consumable materials by selecting an option from underneath the **Material Kind** column. You may also change the **Material Name** as desired. **Note:** Anytime a material in the list is added, the operator can manually initiate verification of a configured item.  <SPD/SPG/SPV with dual solder pod> ***** The SPD does not support the dual solder pod. If two solder pods (dual solder pod) are used, select “2 Pods” in SP Paste Mode of Options Setup from Route Editer->Machine Editor.



<SPD/SPV with dual squeeze (both the front and rear sides)>

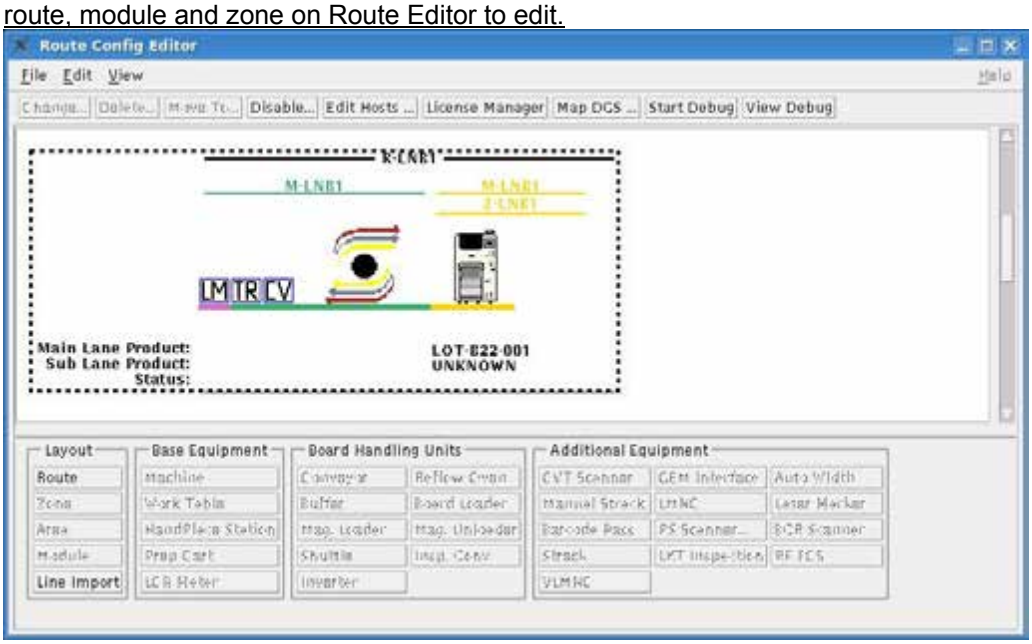
To use the dual squeeze, set as follows.
・For Material Name [Blade 1], select [Squeeze] as its Material Kind
・For Material Name [Blade 2], select [Squeeze] as its Material Kind
*To set up only one squeeze, select [None] as the Material Kind of [Blade 2]. Squeeze2 will be deselected and no longer displayed on the scanner setup screen.



5	Check **Apply Solder Paste Check/Lock PCB**.
6	Click **Save**.

■ Route Editor

Route Editor is to be set per line. Change Material Verification per Machine or Module or Scanner. Machine, Module and Scanner settings can be viewed during operation but disable route, module and zone on Route Editor to edit.

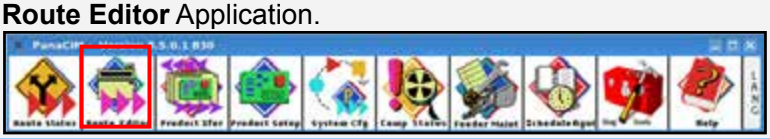
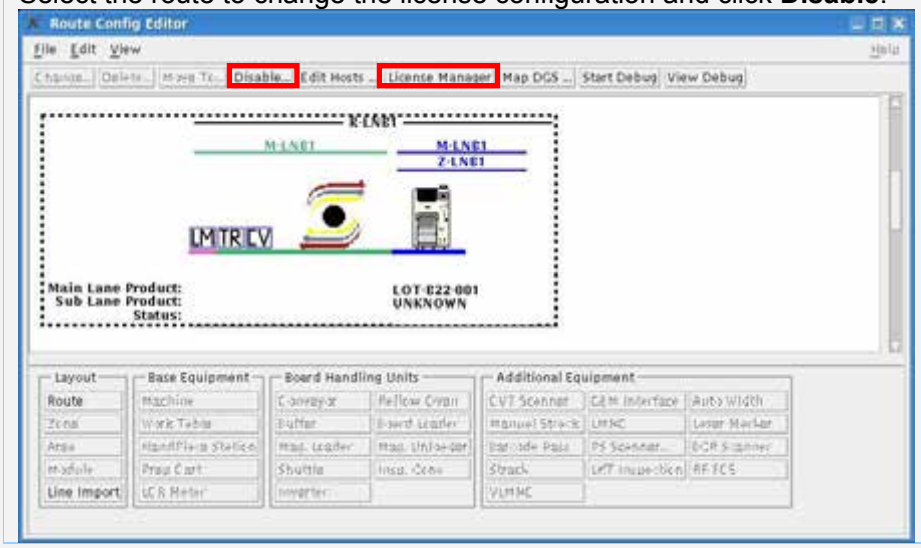
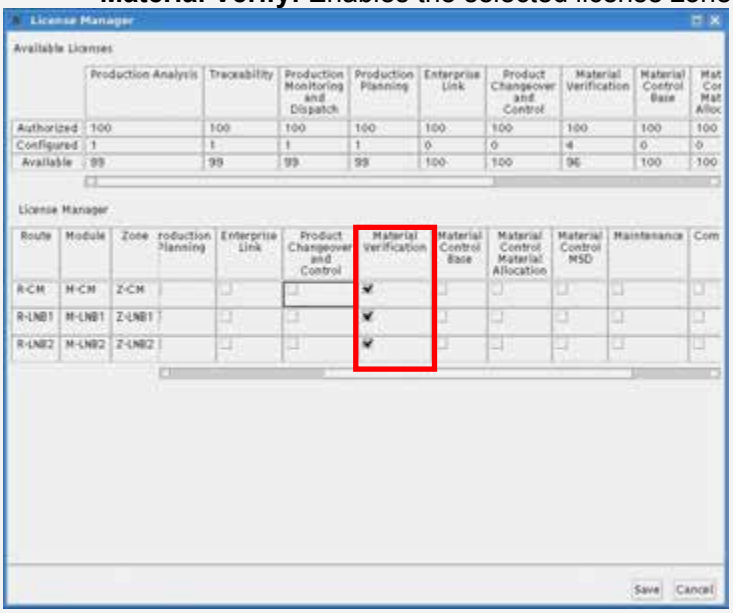


☞ See the PanaCIM-EE Base documentation for more details on Route Editor.

- Clicking ‘CIMC stop’ button stops all PanaCIM controller server services and interfaces. Press ‘CIMC start’ button to resume.
- Selecting a route and clicking ‘Disable’ button stops route and interface. After changed settings, select the route and press ‘Enable’ button.
- Selecting a module and click ‘Disable’ button stops interface with the module. After disabled module, change scanner settings.
- Selecting a zone and clicking ‘Disable’ button to stop interface with the zone. After disabled the zone, change machine settings.

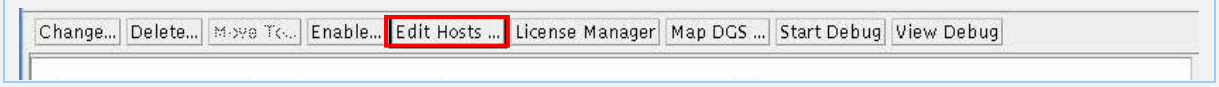
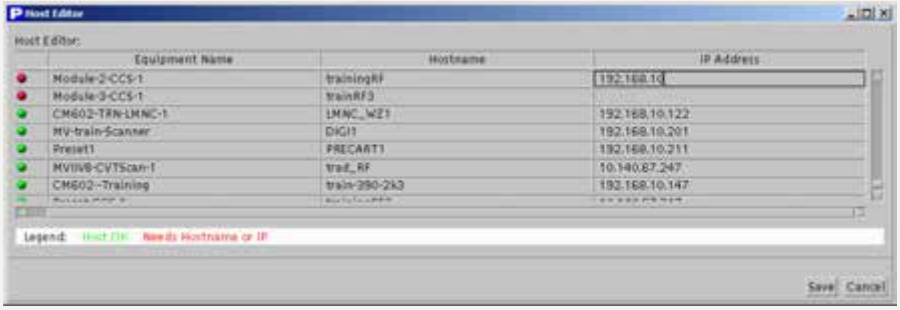
Configuring Licenses

Before using the PanaCIM/material Verification, the license needs to be enabled via the Route Editor of the PanaCIM controller server.

Step	Procedure
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**. 
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.
4	Under **Available licenses**, select the module for modules and zones. To use the PanaCIM/Material Verification, place a check mark in the following checkbox: • **Material Verify:** Enables the selected license zone by zone.  **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box.
5	After configuring the license, click **Save** and exit the License Manager

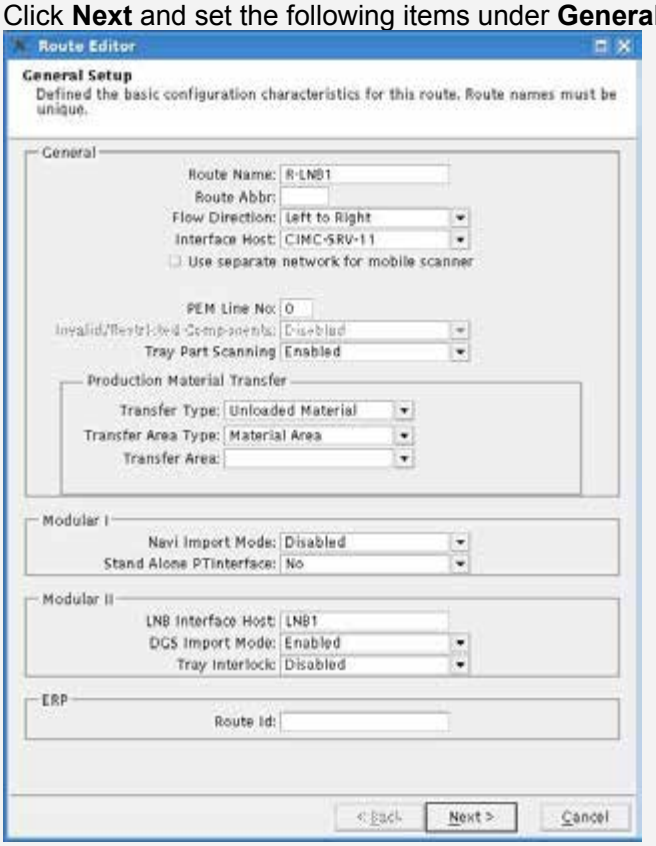
Edit Hosts Function

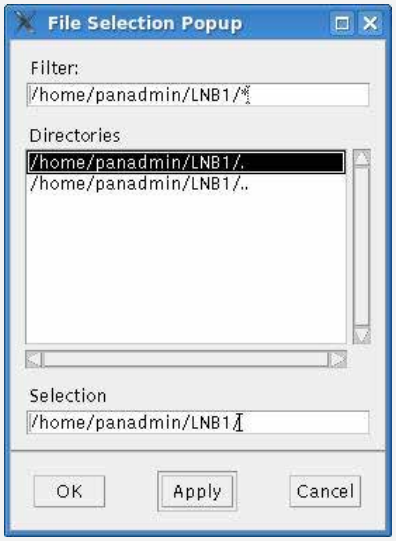
Whenever a device or machine interface is added to PanaCIM, the /etc/hosts file must be updated to map to the correct IP address to the Host Name of the device or machine. This feature included in the Route Editor enables the hosts files to be checked or updated.

Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu. 
2	Click the **Edit Host** button to invoke the Host Editor. 
3	The Equipment Name refers to all machines and devices that have already been referenced in the Route Editor. A **red** light in front of a machine or device indicates that the IP address has not been set. Enter the correct **IP Address** in the edit field for the machine or device.  **Note:** If a machine or device in the Route Editor does not appear, it may be because it was saved without a Hostname.
4	Click **Save**.
5	The following dialog will prompt you to perform the following: • Enter the 'root' password. *The text will not appear as you type but will after pressing **Enter***. • Close the dialog. • Select **Edit Hosts** and verify the Equipment now has a **green** light before it. 

Configure Route Settings

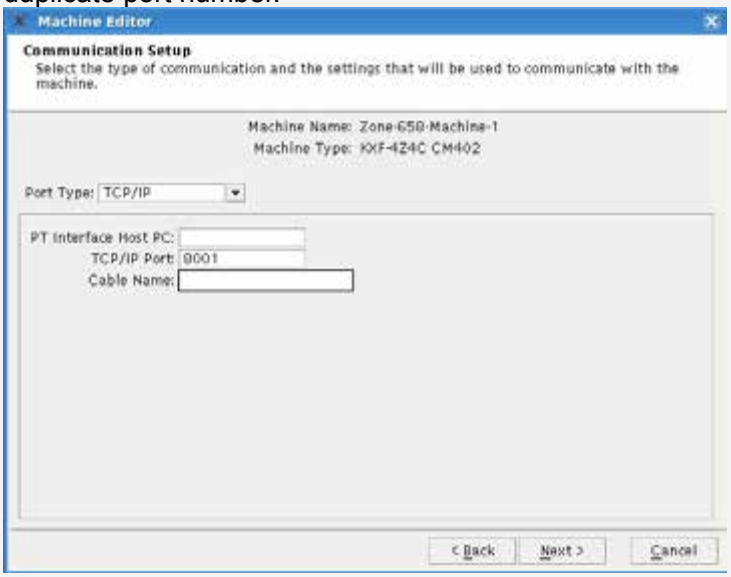
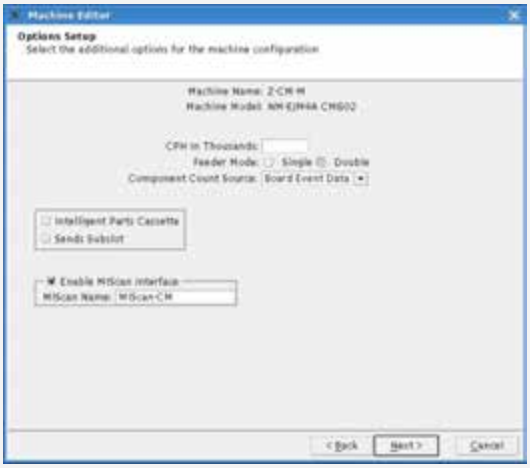
Refer to the table below to ensure the following route settings have been made in the Route Editor Application. (It assumes that the line configuration has already been created.)

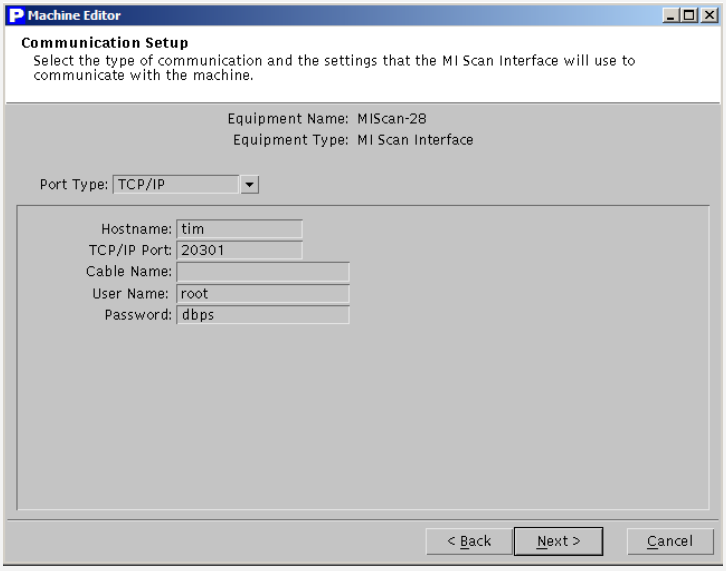
Step	Procedure
1	Open the **Route Editor** from the PanaCIM main menu.
2	Click **Next** and set the following items under **General**.  **General** (Common to CM/DT/NPM series machines): Set **Route Name**, **Route Abbr**, **Flow Direction**, **Interface Host** and **PEM Line No**. Refer to the Base documentation for details. • **Invalid/Restricted Components:** ○ Select **Enabled** from the drop-down list if **Route based** is set for **Invalid/Restricted Components Monitoring** in System Configuration. ○ Select **Disabled** from the drop-down list if **Global based** is set for **Invalid/Restricted Components Monitoring** in System Configuration. **CM/DT:** • **Navi Import Mode:** Select **Enabled** to get the setup information automatically when the product program is downloaded from the PT200. • **Stand Alone PT Interface:** Select **Enabled**. **NPM:** • **LNB Interface Host:** Enter the Hostname of the LNB unit. • **DGS Import Mode:** Select **Enabled**.

3	Click **Next** to proceed to the Substitute Part Import Management dialog. Set the following items.  Select one of the following options, dependent upon whether you are managing import at Product Creation, or Product Activation: • Select **Product Creation**. If the source of your substitute parts is **PanaPRO/IPO** or **Panasonic DGS Cerberus file or Panasonic DGS Dat File**, select the radio button next to the appropriate selection. • Select **Product Activation** and then specify either: ○ **<MixName>.CSV File Parse:** Select this option to import Mixname .csv files. Click [...] to locate the path and select a file. The **File Selection Popup** allows you to filter the directories by first entering searchable text into the **Filter** field. Select the file from the **Directories** list. Review the selection made by viewing the Selection field, then click **Apply**, then **OK**.  ○ **NAVI File Parse (CM/DT series machines only)**. Select this option to import substitute parts from a Navi file. **Note:** If the Route contains any machine(s) other than a CM/DT series machine, the Navi parse mode will be disabled.
4	Click **Next** to proceed to the dialog page, then click **Save** to save the setting.

Machine Interface Settings (CM/DT Series Only)

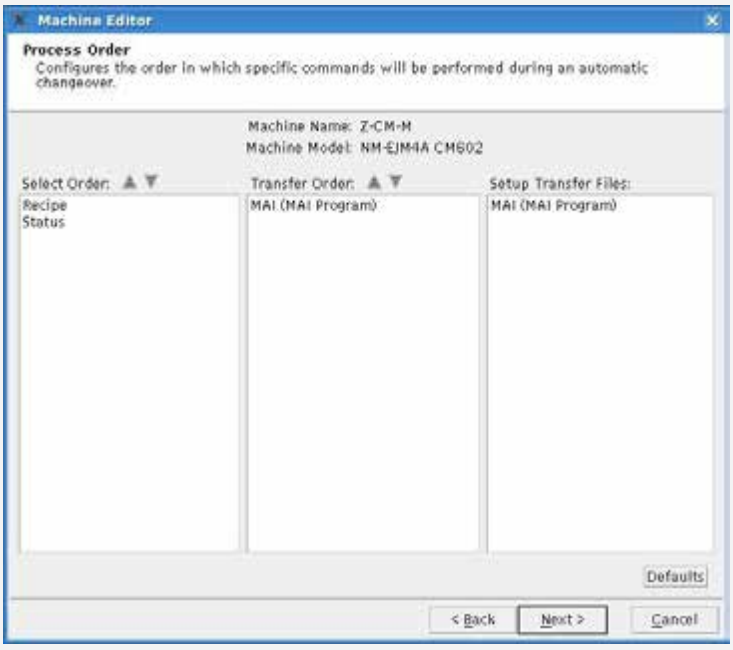
Refer to the table below to set the machine interface required for PanaCIM/Material Verification. (It assumes that the line configuration has already been created.)

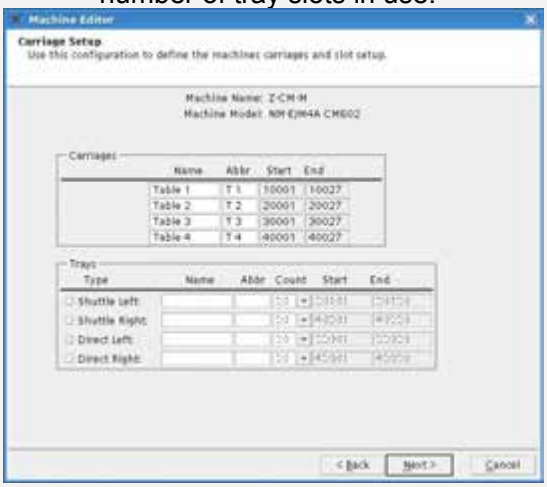
Step	Procedure
1	On the Route Editor screen, double-click the **Machine** to open the **Machine Editor**.
2	Click **Next** and set the following items on the **Communication Setup** screen. • **PT Interface Host PC:** Set the host name of the PT interface. • **TCP/IP Port:** Set the port number. **Note:** TCP/IP Port accepts any odd number between 8001 and 8099. Avoid duplicate port number.  Click **Next**.
3	Set the following items on the **Options Setup** screen. • **Feeder Mode:** Select **Double**. • **Intelligent Parts Cassette/Sends Subslot:** Uncheck these boxes. • **Enable MIScan Interface:** Check this box. • **MIScan Name:** To be populated automatically. **MIScan Name** shows the communication terminal unit Digi-One (optional kit for CM/DT series machines) incorporated in the PanaCIM and CM/DT machines. The unit is used to communicate with the intelligent feeders mounted on via the machine.  Click **Next**.

4	The following must be configured on the *second* Communications Setup screen: • **TCP/IP** Port Type. • The Hostname value for the **DigiOne Terminal Server** matches the setting in the /etc/hosts file (see Edit Hosts section above). You should be able to ping DigiOne in the machine by the Hostname. • TCP/IP Port is **20301** for a DigiOne brand Terminal Server. • The username is **root** and password is **dbps** for the DigiOne brand terminal server.  Click **Next**.
5	The **Supporting Commands** page appears. To verify the tray materials, set the following item. (To verity the reel parts only, this step is not necessary.) • Check **Tray Release** under **Supported Commands**.  Click **Next**.

6	Click **Next** until the **Supporting Data** page appears. To verify the tray materials, set the following item to enable tray locking. (To verity the reel parts only, this step is not necessary.) • Check **Part Setup Status** under **Set Supported**.



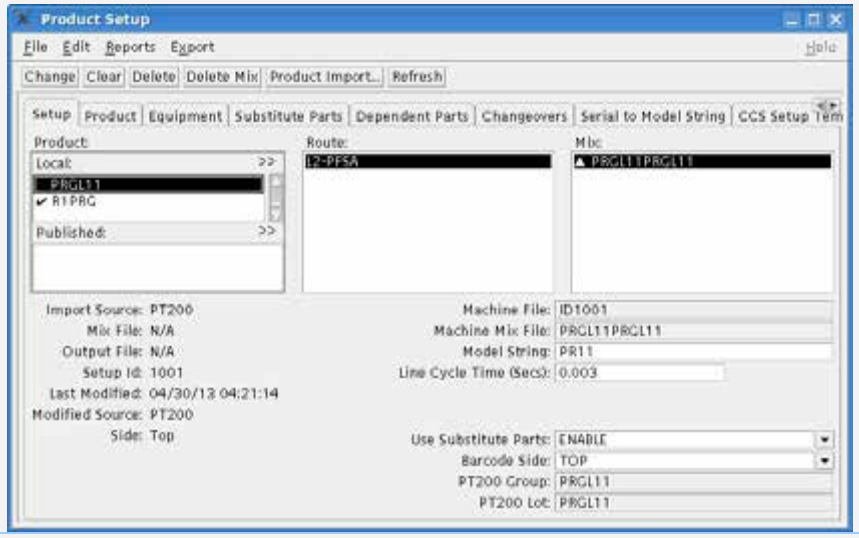


7	Set the following items on the **Cart Setup** page. • When the machine is provided with the tray feeders, check the checkboxes of the tray feeders concerned. • Change **Name** and **Symbol** (abbreviation) of the cart if necessary. • Make sure that the quantity of the tray reflects the number of available tray slots. • Make sure that the **Start** and **End** values reflect the actual tray feeder and number of tray slots in use.  Click **Next** until the last page appears.
8	Click **Save**.

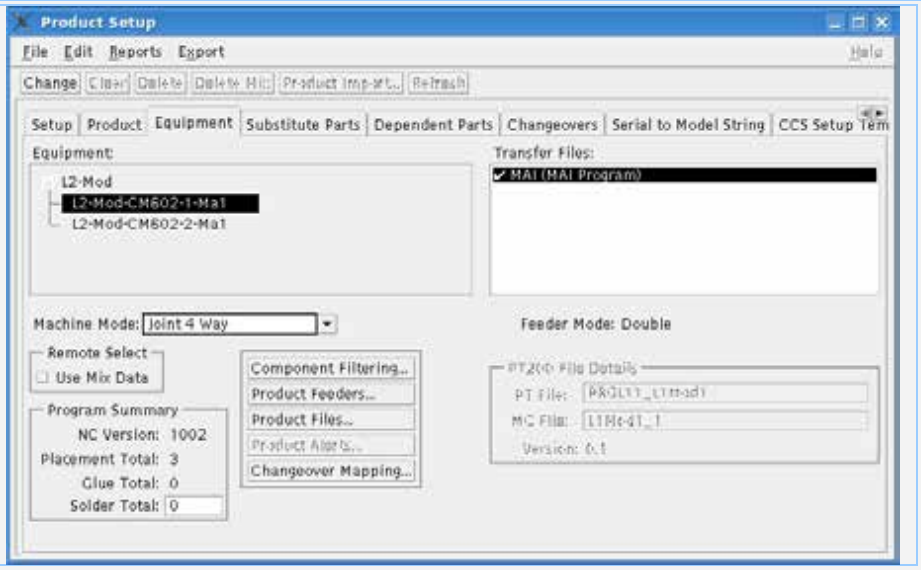
■ Product Setup

● Configure Setup in Product Setup

After a product has been successfully downloaded from PanaPRO or DGS, the product must be set up in the Product Setup Application. Product Import can also be used to create a Product from an existing Modular file on the PT200.

Step	Procedure
1	Open the **Product Setup** Application from the PanaCIM Main Menu.
2	Perform the following on the Product Setup screen: • Click the **Setup** tab. • Select the **Product** from the **Product** list. • Select the **Route**. The product may be setup to run on different production Lines. • Select the **Mix** (or project). The same product may be in different mixes (combined setups). • Set the **Model String**. • Enter the **PT200 Group** and **PT200 Lot**. (May be populated automatically if imported from the PT200) For CM/DT series machines only • Select **Enable** from the Use Substitute Parts drop-down in order to allow a substitute part listed in the Substitute Part list to be accepted in place of the original programmed part number. • Select the **Barcode Side**. 
3	Perform the following steps on the Product Setup screen: • Click the **Equipment** tab. • Select the **machine name** from the **Equipment** area. • For CM/DT series machines: Select **MAI (MAI Program)** from the **Transfer Files** list. For NPM series machines (including AM100): Select **CRM (Cerberus Program)** from the **Transfer Files** list. • Click the **Change** button after all changes have been made to the product before exiting Product Setup.

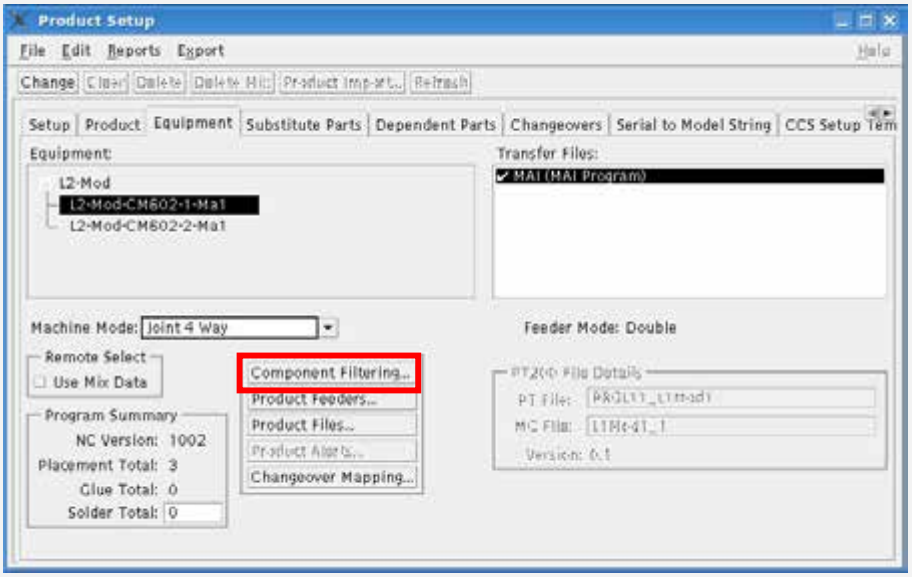
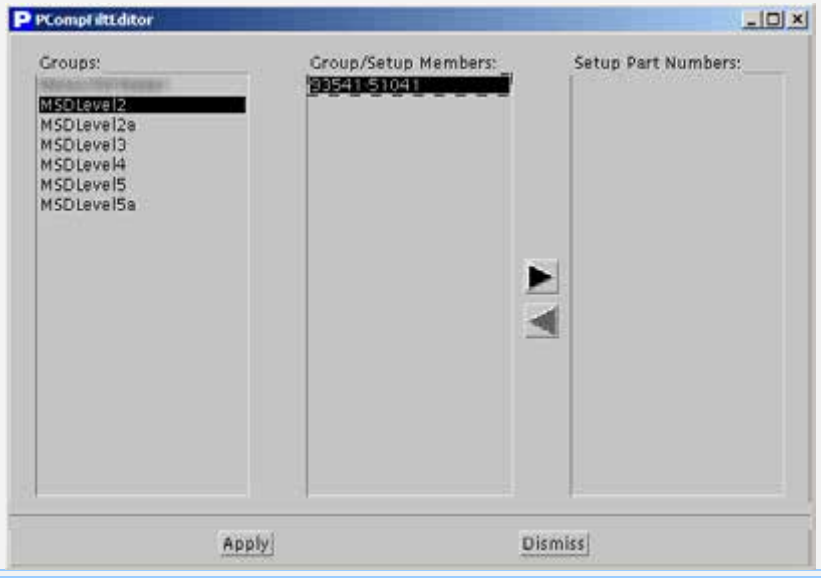
2.2 Settings


Product Setup
File Edit Reports Export
Change
Setup Product Equipment Substitute Parts Dependent Parts Changeovers Serial to Model String
Equipment:
L2-Mod
L2-Mod-CM602-1-Ma1
L2-Mod-CM602-2-Ma1
Transfer Files:
MAI (MAI Program)
Machine Mode: Joint 4 Way
Feeder Mode: Double
Remote Select
Use Mix Data
Program Summary
NC Version: 1002
Placement Total: 3
Glue Total: 0
Solder Total: 0
Component Filtering...
Product Feeders...
Product Files...
Changeover Mapping...

Component Filtering

You can assign part numbers from the product to specific groups, which will override settings made in System Configuration. These specific groups determine what information is required when the component is scanned.

Refer to the action table below to configure the setup using the Product Setup Application.

Step	Procedure
1	Open the **Product Setup** Application from the PanaCIM Main Menu. • Click the **Equipment** tab. • Select the applicable machine and click the **Component Filtering** button. 
2	Perform the following next steps on the Component Filtering Editor screen: • Choose a **Group** from the list. • Select a **Setup Part Number**. • Press the arrow key to associate a part number with the indicated group. • Click **Apply**. 

Edit the Feeder Setup

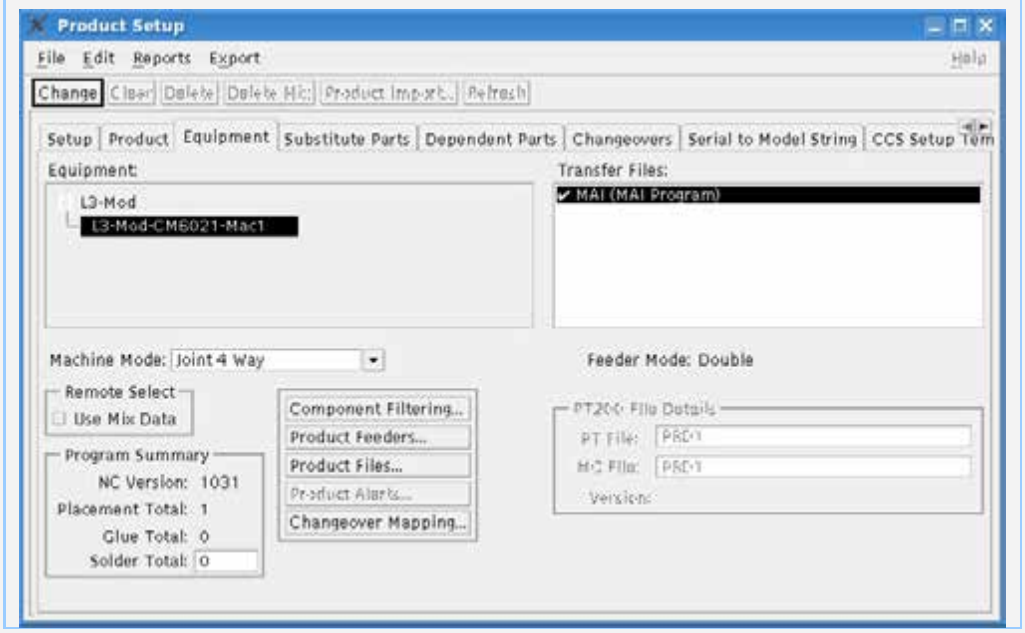
Changes made in the Product Setup Application's Feeder Setup screen allow you to change the setup to match changes that have already been made in the actual machine program.

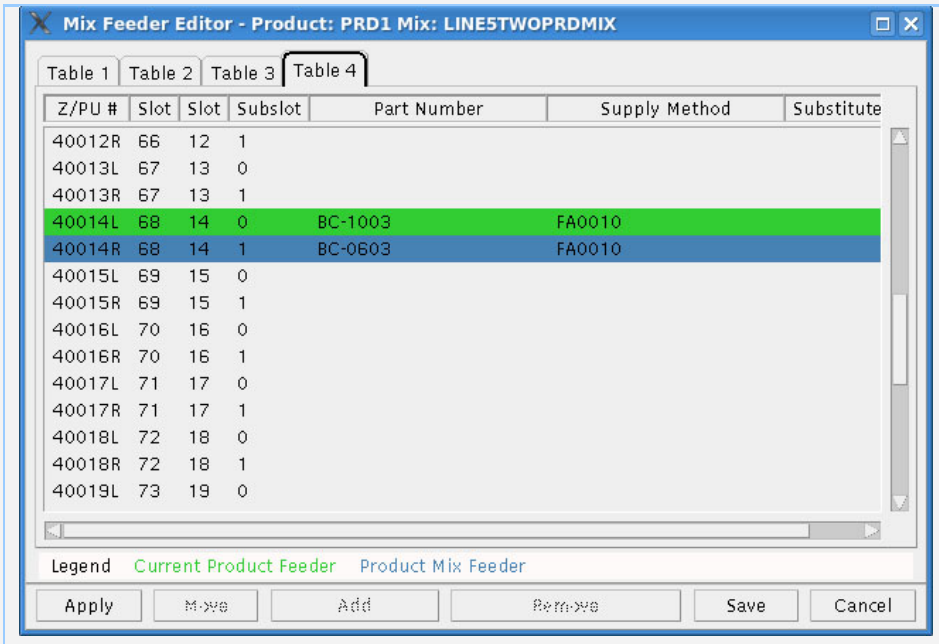
Mix Feeders

Mix feeders are the set of component part numbers and their location on a carriage of feeders to build a specific mix of products.

The Mix Feeder Editor enables you to modify this mix by adding, moving and removing feeders from the Mix and Product. The Mix Feeder Editor is organized by carriages. For each carriage that is configured on the placement machine, there is a tab with the feeder range specified by their Z # and Slot/ Subslot numbered locations.

Refer to the action table below to edit the feeder setup using the Product Setup Application.

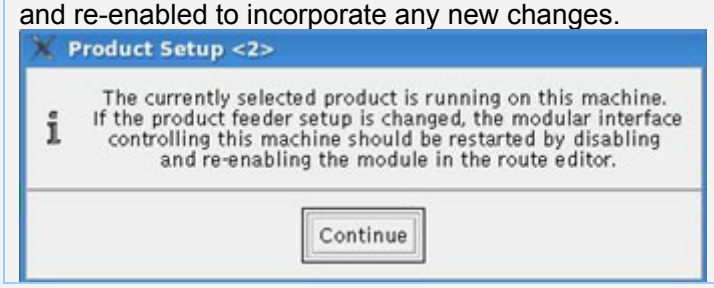
Step	Procedure
1	Open the **Product Setup** Application and click the **Equipment** tab.
2	Click **Product Feeders**. 
3	The **Mix Feeder Editor** appears. Choose your desired action following the appropriate steps on the following page. All feeders that are a part of the current product are identified in green. All feeders that are a part of the Mix but not a part of the current product are identified in blue.



Note: If you do not click **Apply**, additional actions to this feeder such as moving, adding to the mix, etc., will not be allowed. Clicking **Save** will update any new feeders, but will close the Mix Feeder Editor.

4

After selecting your action the following message appears, the selected program is currently loaded on the machine, and the Module must be disabled and re-enabled to incorporate any new changes.



5

Add a New Product Feeder:

1. Select the **tab** corresponding to the desired carriage.
2. Click the Part Number column of the desired empty feeder location, and enter the **Part Number**.
3. Click in the Supply Method column and enter the **Supply Method** for the component.
4. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.

6

Modify a Mix Feeder:

1. Select the **tab** corresponding to the desired carriage.
2. Click the Part Number column of the desired empty feeder location, and enter the new **Part Number**.
3. Click in the Supply Method column and enter the new **Supply Method** for the component.
4. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.

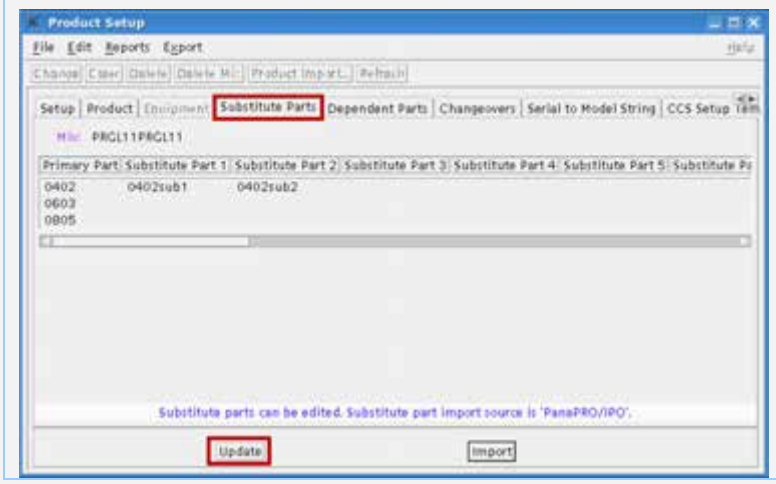
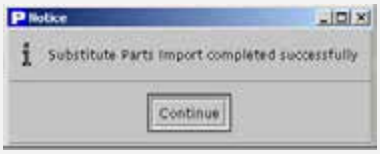
7	**Move a Mix Feeder**: 1. Select the **tab** corresponding to the desired carriage. 2. Select the **feeder** to be moved. 3. Click **Move**. A notice appears requesting that the destination feeder location be selected. 4. Select the **tab** corresponding to the desired carriage. 5. Select the desired **destination** feeder location. 6. Click **Continue** to perform the move, or **Cancel** to abort. 7. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.
8	**Add a Mix Feeder to the Product**: 1. Select the **tab** corresponding to the desired carriage. 2. Select the **mix feeder**. 3. Click **Add to Product**. 4. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.
9	**Remove a Product Feeder**: A product feeder can be removed from the current product. A product feeder may also be included in other parts of the current mix. Removing a feeder from the product will not remove it from the entire mix. If the feeder is part of other products in the mix, when the feeder is removed from the product, the feeder color will change to indicate that it is still a part of the mix. 1. Select the **tab** corresponding to the desired carriage. 2. Select the **product feeder**. 3. Click **Remove From Product**. 4. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.
10	**Remove a Mix Feeder**: A mix feeder can be removed from the current mix. Removing a feeder from the mix will completely remove the feeder from the entire mix. 1. Select the **tab** corresponding to the desired carriage. 2. Select the **mix feeder**. 3. Click **Remove From Mix**. 4. Click **Apply**. The Mix Feeder Editor updates its visual mix feeder representation.
11	Click **Save** to permanently commit the edits to the database, or **Cancel** to disregard all editing performed in the Editor.

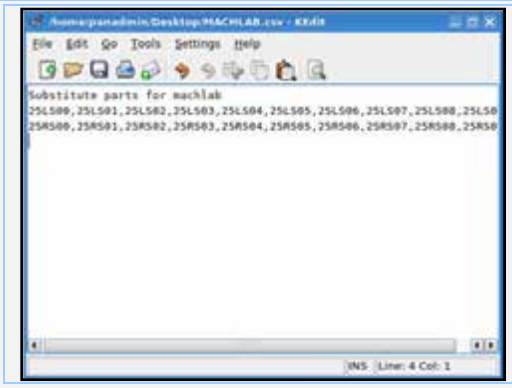
Edit Substitute Parts

If a particular part becomes unavailable, a replacement part may be substituted if the Software has been configured to permit this type of action. Whether or not data can be modified on this page depends on the Automatic Import setting in the Route Editor application.

Automatic Import Setting in Route Editor	Import Source	Edit Substitute Parts via Product Setup Allowed?
Product Creation	PanaPRO/IPO	Yes
Product Creation	Panasonic DGS	Yes
Product Activation	<mix_name>.CSV File	Yes
Product Activation	NAVI file	No

Follow the steps below to register the substitute parts in the Substitute Parts page on the Product Setup screen.

Method	Procedure
Manually Enter the Substitute Part Number	Substitute parts can be manually entered into the Product Setup. 1. Double-click the blank text field to the right of the part number. 2. Enter the part number to be used as the substitute part. 3. Click **Update** when complete. 
Import Substitute Part from a .CSV File in Product Setup	Substitute parts can be imported into the Product Setup from a .CSV file. 1. Click **Import**. 2. A dialog asking you to specify the source file appears. Set the path to the CSV file and click **OK**. 3. A confirmation dialog appears. Click **Continue**. The new substitute parts should appear and replace any previous substitute parts.  **Example of CSV file format** • The first row contains a comment or explanatory note. This row will not be imported. • On the second row, enter the part number registered in the mix/product, followed by the substitute part numbers delimited by commas.



***** The Substitute Parts tab will be read only if substitute parts import source is either MAI file or Navi file. This tab is only available for IPO and .CSV file import.

■ Inventory Manager

● Configure Invalid or Restricted Components

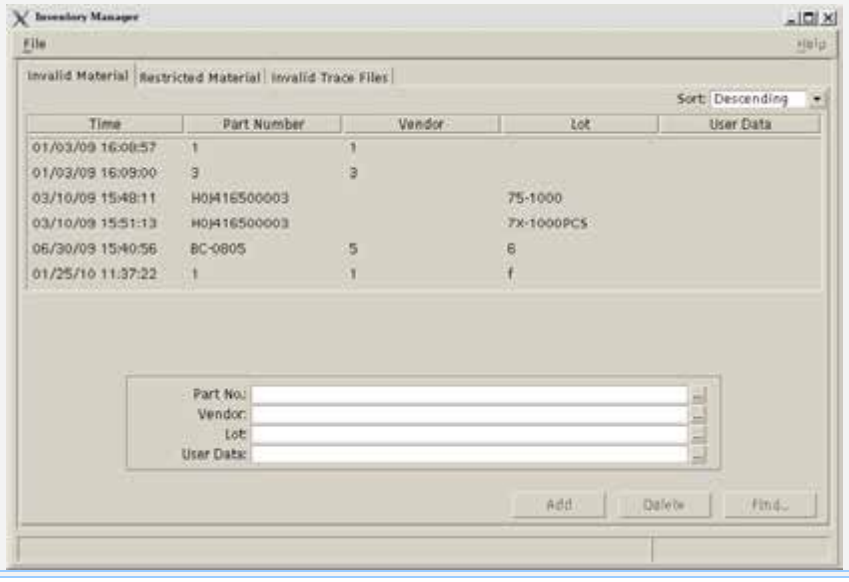
The Inventory Manager Application can be used to add and edit Invalid or Restricted components. Differences between Invalid and Restricted components include:

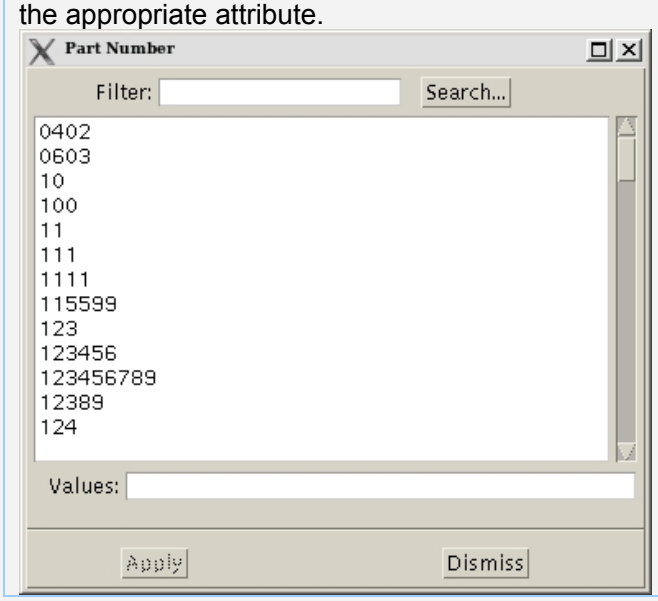
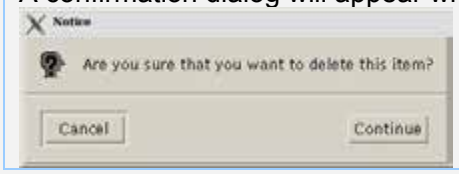
- **Invalid Components**. This is a component that is prohibited from use in any setup. They are prohibited for use and considered defective.
- **Restricted Components**. This is a component that is not defective but is prohibited for use on specific products. They are prohibited for use on a product basis.

When PanaCIM detects an Invalid or Restricted component at the machine, the machine will lock, stopping Production. Locking can occur at Production start, or when the configuration of a component changes during Production. The PanaCIM checks for any prohibited components and restricts the continuation of production.

After component information has been edited in Inventory Manager, a “Configure Change” message specific to Invalid/Restricted components will be sent to the PanaCIM database after changes are saved and the Inventory Manager is already open.

* Activating imports of Invalid/Restricted components via Inventory Manager will disable the importing of these components via ELink. (*please reference the ELink documentation for further guidance*). (Under development)

Step	Procedure
1	Open the **Inventory Manager** Application from the PanaCIM Main Menu.
2	Select: • The **Invalid Material** tab to edit the current the current list of components listed as invalid. • The **Restricted Material** tab to edit the current list of components that are restricted by product. **Note:** The Invalid Material and Restricted Material tabs display identical criteria. The following steps apply to either tab.
3	Use the **Sort** drop-down to determine how the list is sorted (Ascending or Descending). It will be set to Descending by default. 

4	Enter a **Part No** and then optionally enter **Vendor**, **Lot**, and/or **User Data** into the applicable text fields. Maximum character length for each field is 20. **Note:** Click ... to generate a search dialog that will allow you to search for the appropriate attribute. 
5	Click **Add** to add the entry into the list. The Add button will become enabled only when there is text entered into the **Part No.** field. **Note:** Click **Delete** to delete the selected entry from the list. The Delete button will become enabled only when there is an active selection in the list. A confirmation dialog will appear when selected. 

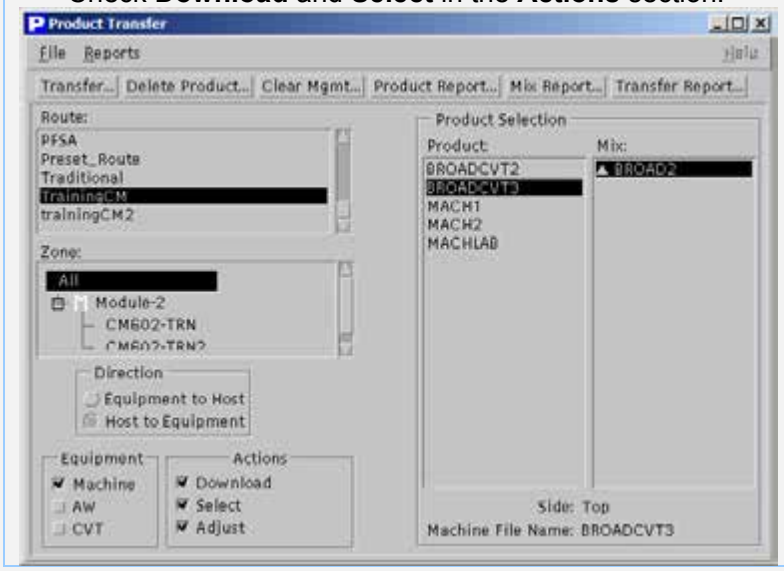
■ Product Transfer

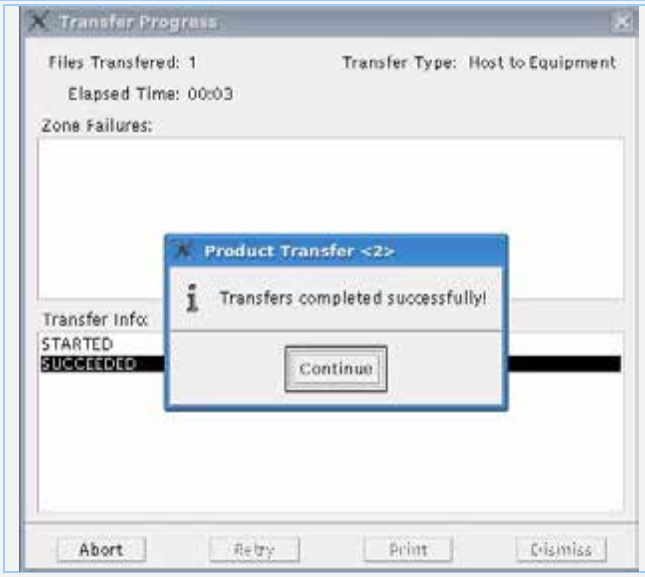
● Product Transfer Settings

After a successful product transfer, the product becomes available on the PT200 Line Controller. The PT200 operator must open the product file and perform a successful data check prior to downloading the file to the machine.

* Products must be properly set up using the Product Setup Application prior to transfer. Reference the Product Setup section of the Base document for guidance on how to set up products using the Product Setup Application.

Refer to the action table below to configure Product Transfer Application settings.

Step	Procedure
1	Open the **Product Transfer** Application from the PanaCIM Main Menu.
2	Perform the following on the Product Setup screen: • Select the **Route**. • Select the **Module** from the Zone section. • In the Direction section: ○ Select **Host to Equipment** to transfer the product *to* the PT200 Line Controller. ○ Select **Equipment to Host** to transfer the product *from* the PT200 Line Controller. In this case the machine program on the PT200 will overwrite the program that is stored on the PanaCIM system. The Setup will NOT be changed. The setup will require modification to match the uploaded file. • Check **Machine** in the **Equipment** section. • Check **Download** and **Select** in the **Actions** section. 
3	A message will generate notifying you that the file was successfully transferred to the PT200 and activated in the module.


Transfer Progress
Files Transfered: 1
Transfer Type: Host to Equipment
Elapsed Time: 00:03
Zone Failures:
Product Transfer <2>
Transfers completed successfully!
Continue
Transfer Info:
STARTED
SUCCEEDED
Abort
Retry
Print
Dismiss

2.2.2 Setting of Machine

■ CM/DT Series Machine Setting

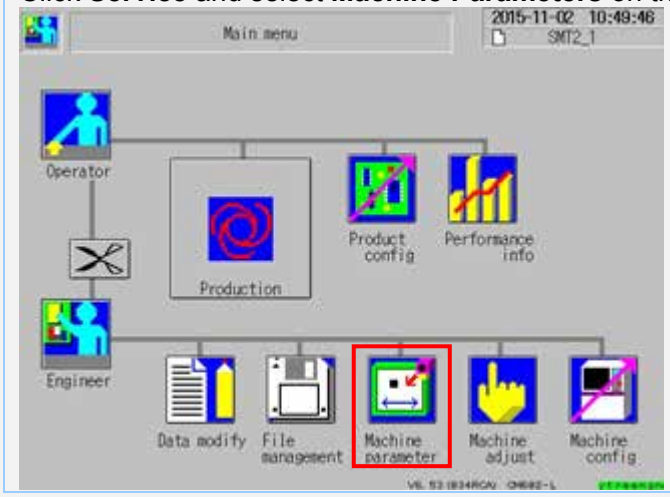
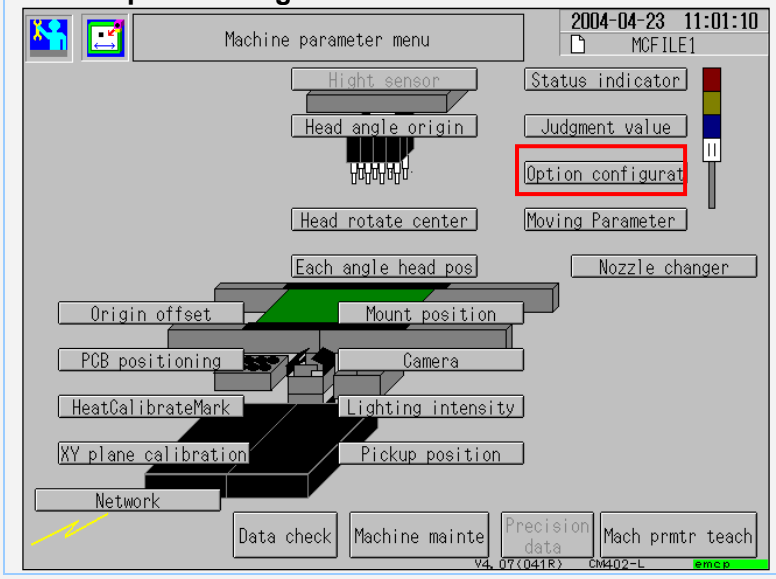
To use the Material Verification on the CM/DT series machines, the settings of the machine need to be changed as follows:

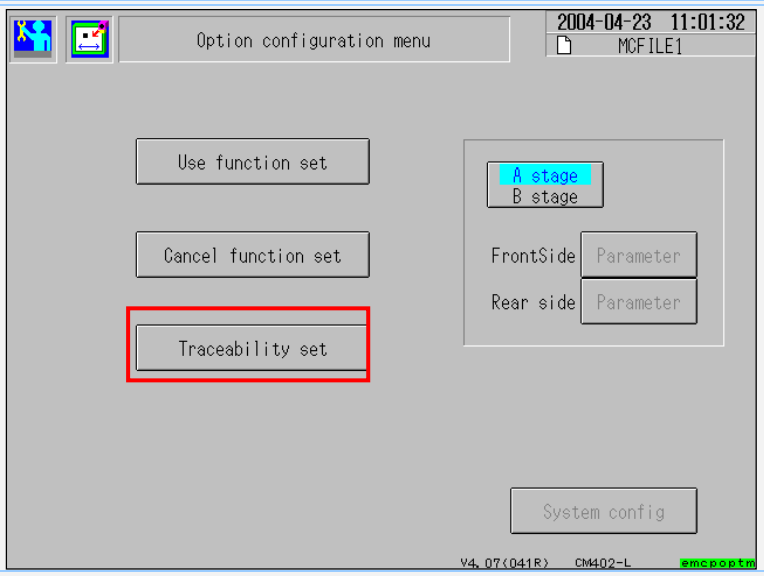
- **Panasonic Service Setup**

The following machine option setup is available through the settings dedicated to Panasonic service engineers.

- **Setting Machine Options**

The table below summarizes the procedure for setting the options for the CM/DT series machines.

Step	Procedure
1	Click **Service** and select **Machine Parameters** on the main menu. 
2	Select **Option Configuration**. 
3	Select **Traceability Set**.

	
4	Set the parameters according to the “CM/DT Machine Settings for Material Verification” table.
5	After setting the parameters, the main menu screen returns.

- **CM/DT Machine Settings for Material Verification**

No.	Option Name	Setting	Description
1	Splice Warning	Use	**When using Splicing.** Splice warning is given according to the preset feeder reel length. According to the length of a reel, feed pitch and the number of components are calculated.
2	Splicing Stop	Use	**When using Splicing.** The machine will perform a single or immediate stop when the length of tape on a reel is 2 meters or shorter, and splice memory of the feeder has not been programmed already.
3	Part Barcode check	Use	The scanner checks the part bar code(s).
4	Feeder Part Info Write	Use	When automatic operation begins, part information is loaded from feeders. When parts have run out, this information is updated. Whenever one board is complete, it is transferred.
5	Feeder Chip Remain Write (Select One)	Use	Machine will write the quantity that Material Verification sends.
		Unuse	Use this setting when not scanning the quantity of parts on the reel. The default PT200 quantity will be used instead.
6	Part Info Trace	Use	When automatic operation begins, part information is loaded from feeders. When parts have run out, this information is updated. Whenever one board is complete, it is uploaded to PT.
7	Clist Trace	Use	Each time the status of a feeder on the machine is changed, the part info is read from that feeder and uploaded to PT.

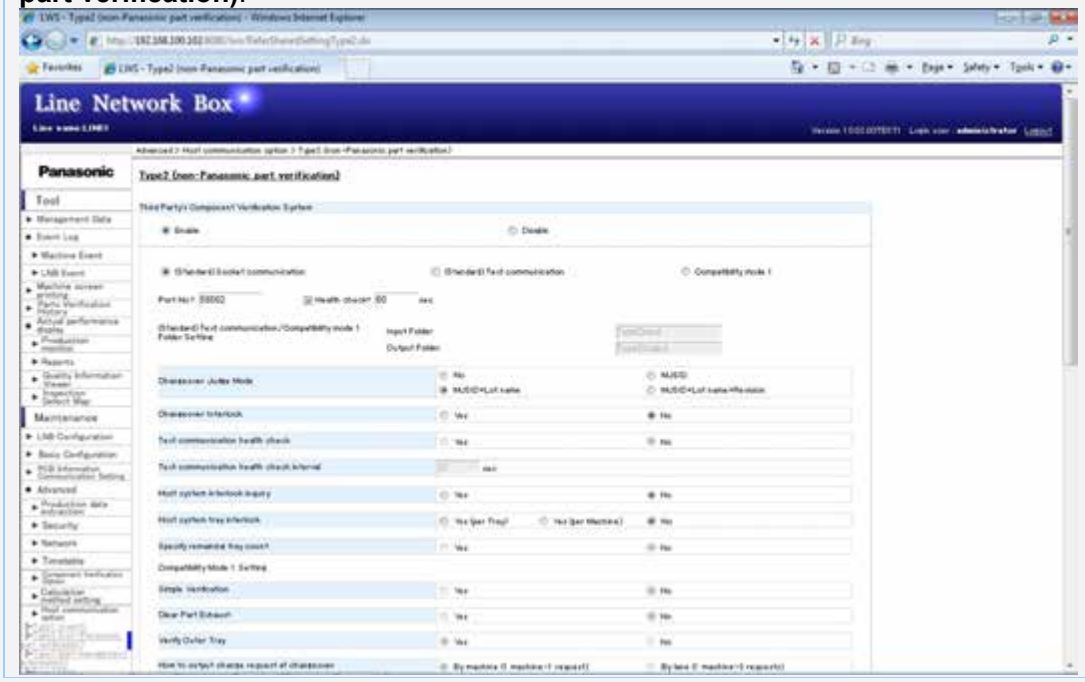
8	Feeder Info Gang Clear (Select One)	Use	The operator can clear all the feeder memories on the table at once.
		Unuse	The operator must clear each feeder memory individually.
10	Chip Verify After Stop (Select One)	Use	Operator will have to rescan part barcodes when feeders are pulled out and reset.
		Unuse	Machine does not care if the feeder is pulled out and reset.
11	Parts Data No Check Write	Use	**When using Splicing.** It will allow you to scan a tape splice reel and will overwrite the current data in the splice memory.
13	Plurality Scanner Concurrent Process	Use	Allows more than one scanner to be used on the machine at the same time.
18	Parts Name Check	Use	Part names are verified for accuracy. An error will result if the parts in the feeder differ from the setup.
28	Tray external collation	Use (in checking tray parts)	Required for Part Exhaust Verification.
30	Easy collation data	Unuse (in checking tray parts)	Differentiates splicing from part exhaustion at the request of the material verification from the host system. (Select **Use** to ignore.)
31	Operation Start Interlock	Use (in checking tray parts)	Required for Tray Locking.
—	C-List SID	Use	Select **Detail Config** button first. SID info is written in the C-List file.

■ NPM Series Machine Setting

In order to use the Material Verification on the NPM machines, changes are required at the machine (NPM) and LNB.

• LNB Settings

Refer to the action table below to configure LNB machine settings.

Step	Procedure
1	Log on to LNB and select **Host communication option** > **Type 2 (non-Panasonic part verification)**. 
2	Set the parameters according to the “NPM (LNB) material verification setting” table shown below.

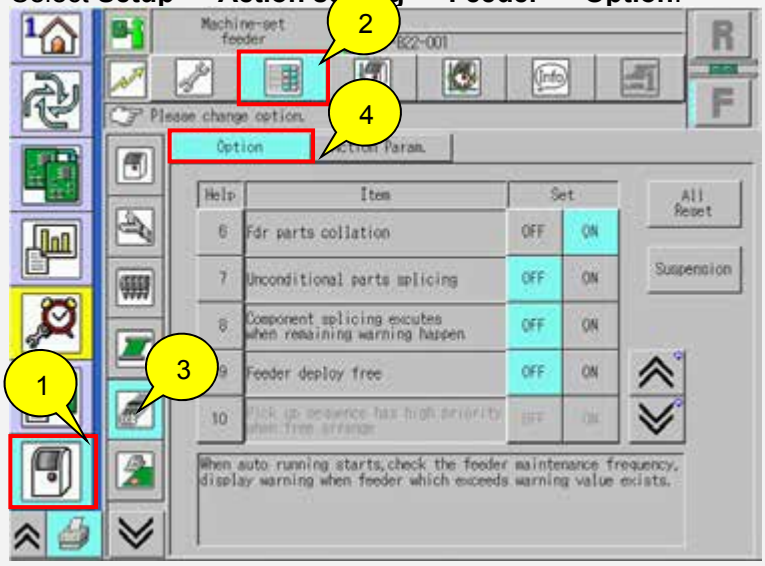
• NPM (LNB) Type 2 (non-Panasonic part verification) setting table

Setting item	Setting	Explanation
Non-Panasonic material verification	Valid	Check ‘Valid’ to enable PanaCIM Material Verification.
Communication method	Socket communication	Material information communication method between PanaCIM and LNB
Port number	58002	Set port number for communicating with PanaCIM via socket communication

• Machine setting

To use the Material Verification on the NPM series machines, follow the steps below.

The NPM option Feeder Free Arrangement is not supported. Set to 'OFF' when the option is available.

Step	Procedure
1	Select **Setup** → **Action setting** → **Feeder** → **Option**. 
2	Set the parameters according to the "NPM material verification setting" table shown below.

• NPM machine material verification setting table

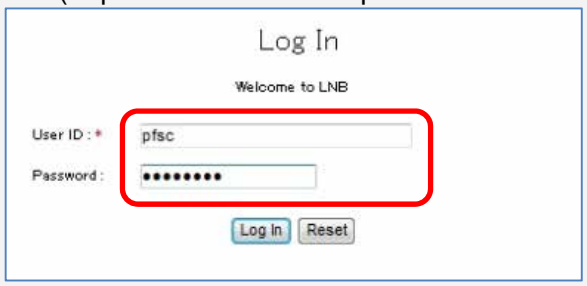
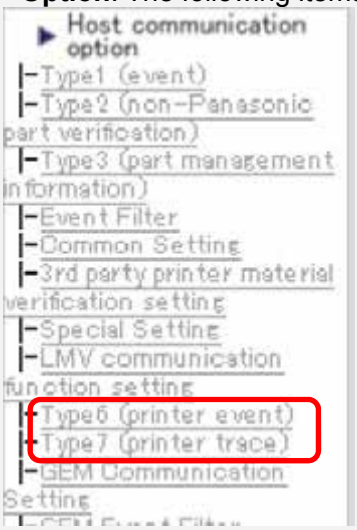
Setting item	Setting	Explanation
Part remainder number warning	ON	An alert message appears if the number of the remained components is less than this setting.
Fdr parts collation	ON	Carries out feeder parts verification by reading barcode on the part reel via the scanner.
Splicing length stop	ON	When the reel length reaches 1,200 mm or less (minimum splicing length), the machine comes to a single stop.
Fdr parts collation 2	ON	Disables the machine from stopping unless material verification is performed for the feeders removed in the single stop.
Part splicing at part remainder warning	ON	Enables splicing by the scanner only for the feeders with which a remainder warning has occurred.
Thin-type single feeder	ON	Set to ON when using thin-type single feeders.

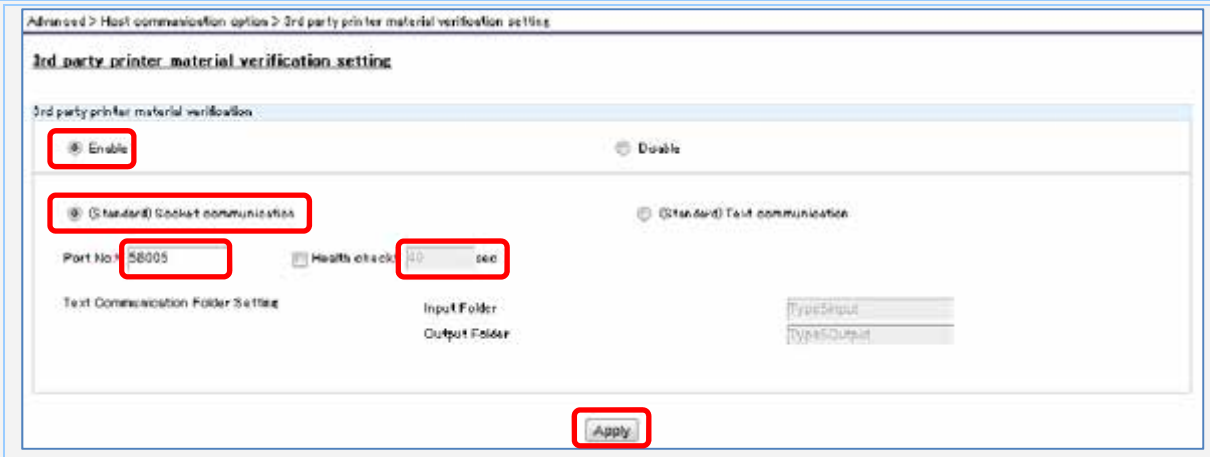
■ SP Series Machine Setting

To use the material verification on the SP series machines (SPV/SPG/SPD/SP70), change the settings of the machine and LNB.

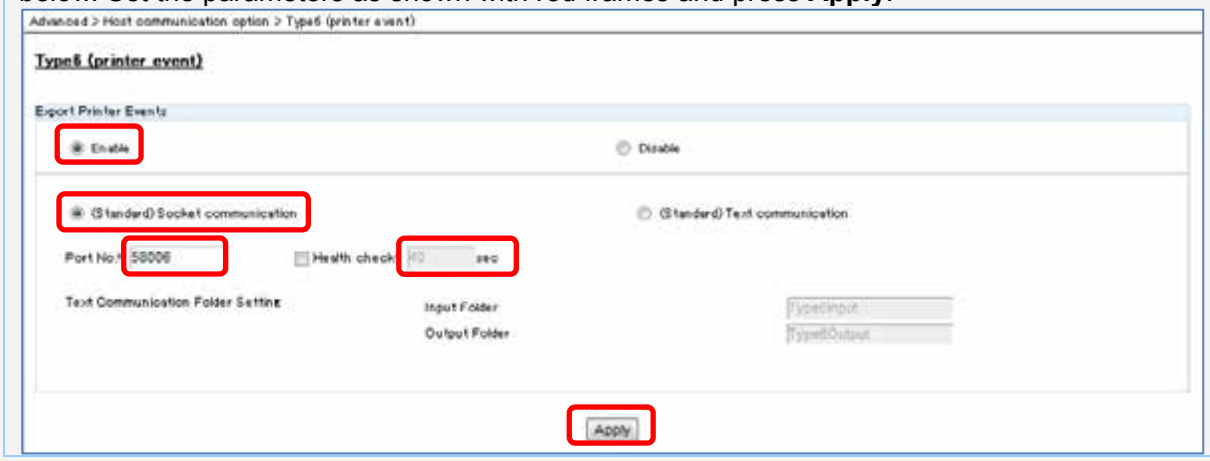
● LNB setting

To use the Material Verification on the SP series machines, perform the following steps on the LNB.

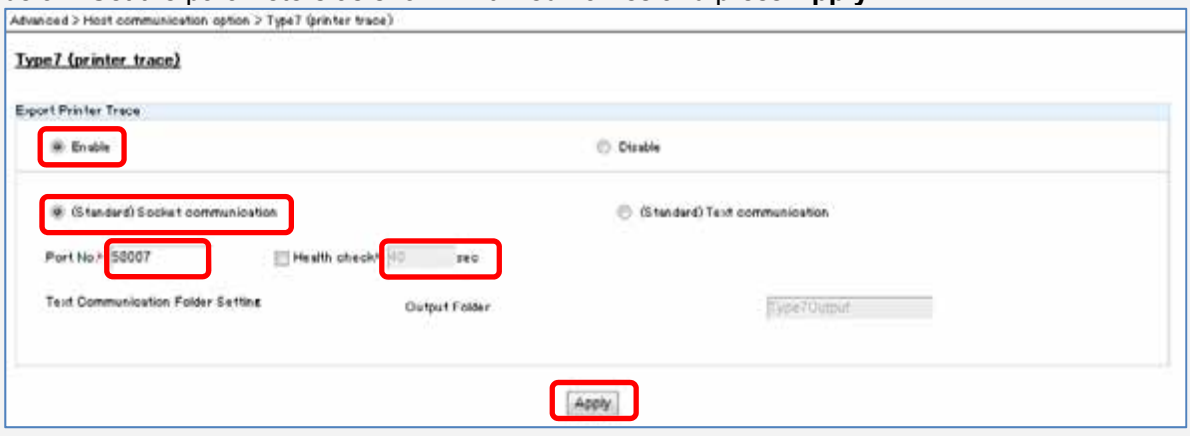
Step	Procedure
1	Log on the LNB with the following user information. **Login information** ・ User ID: **pfsc** ・ Password: **pfsc password** (Inquire of the service representative for the password.)  * After logged in as a pfsc user, select **Maintenance** - **Advanced** - **Host Communication Option**. The following items for screen printers are displayed. 
2	Select **Non-Panasonic printer material verification** to open up the screen below. Set the parameters as shown with red frames and press **Apply**.



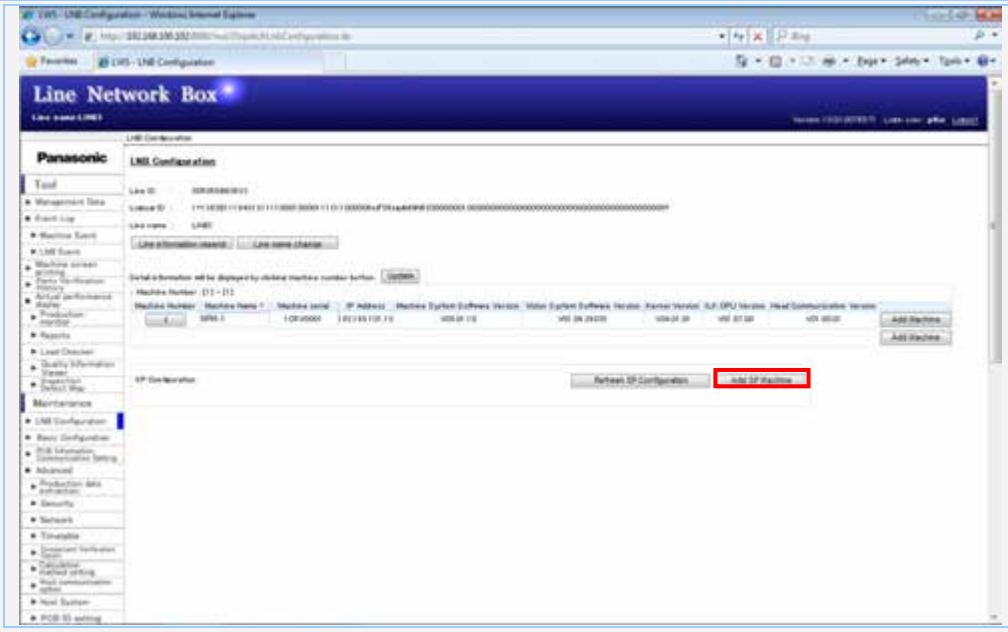
3	Select **Host Communication Option** - **Type 6 (Printer event)** to open up the screen shown below. Set the parameters as shown with red frames and press **Apply**.



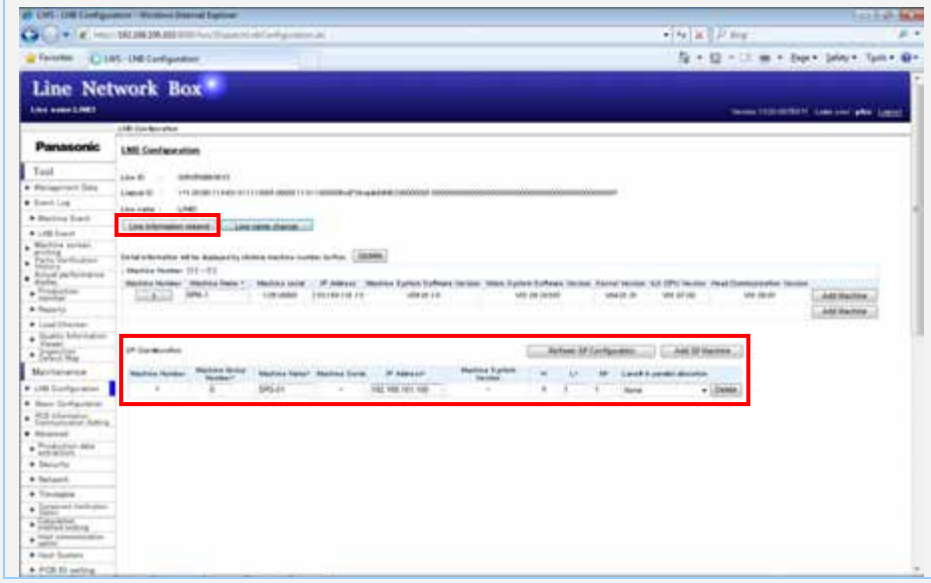
4	Select **Host Communication Option** - **Type 7 (Printer trace)** to open up the screen shown below. Set the parameters as shown with red frames and press **Apply**.



5	Select **Maintenance** - **LNB Configuration Information** and press **Add SP Machine**.



6	Enter the values set on Machine Setting - Network Setting screen as **SP Configuration** information. Press **Update SP configuration information** and **Resend line information** to register the line information in DGS.
7	Set the parameters according to the “SP configuration setting” table shown below.



- **Non-Panasonic printer material verification setting**

Setting item	Setting	Explanation
Non-Panasonic material verification	Valid	Check ‘Valid’ to enable PanaCIM Material Verification.
Communication method	Socket communication	Material information communication method between PanaCIM and LNB
Port number	58005	Set port number for communicating with PanaCIM via socket communication
Health check	Valid	Enable health check function to check connection status.
	40	Set interval in second for connection status check.

- **Type 6 (Printer event)**

Setting item	Setting	Explanation
Printer event information output function	Valid	Check ‘Valid’ to enable PanaCIM Material Verification.
Communication method	Socket communication	Material information communication method between PanaCIM and LNB
Port number	58006	Set port number for communicating with PanaCIM via socket communication
Health check	Valid	Enable health check function to check connection status.
	40	Set interval in second for connection status check.

- **Type 7 (Printer trace)**

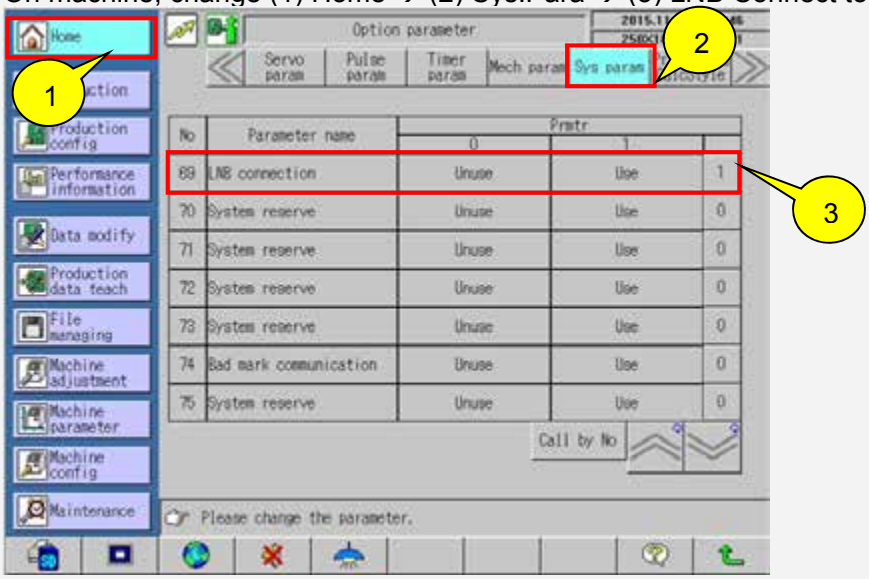
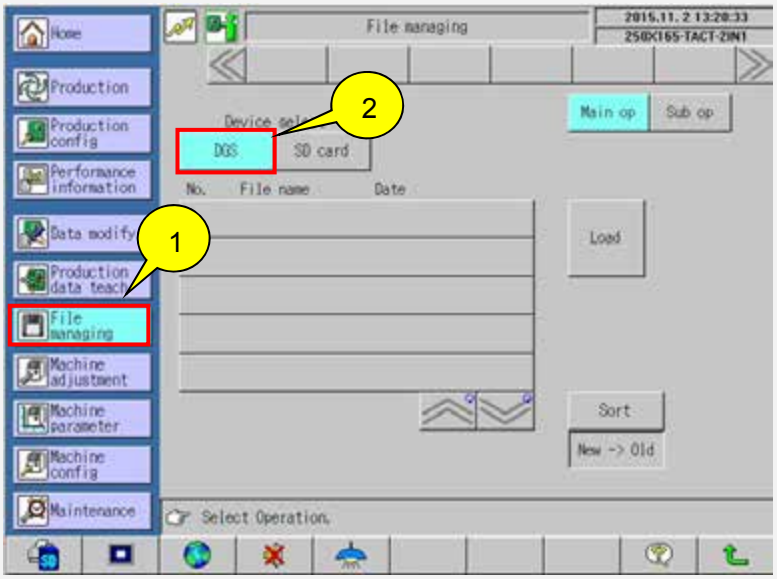
Setting item	Setting	Explanation
Printer trace information output function	Valid	Check ‘Valid’ to enable PanaCIM Material Verification.
Communication method	Socket communication	Material information communication method between PanaCIM and LNB
Port number	58007	Set port number for communicating with PanaCIM via socket communication
Health check	Valid	Enable health check function to check connection status.
	40	Set interval in second for connection status check.

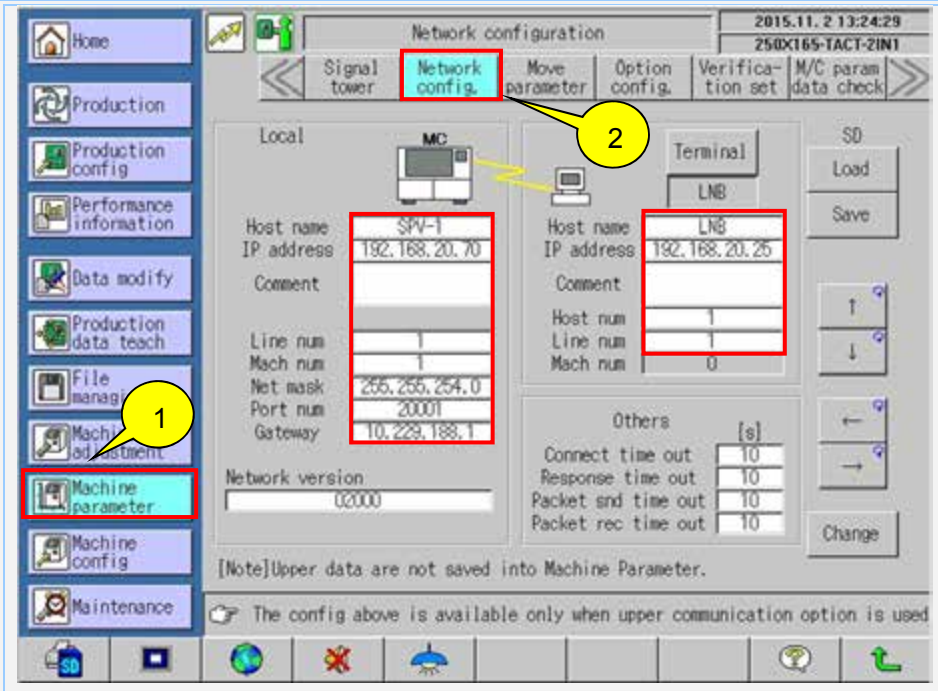
- **SP configuration setting**

Setting item	Setting	Explanation
Machine group number	0	Set local machine group number which was set at SP machine network setting
Machine name	SP series machine name	Set local host name which was set at SP machine network setting.
IP address	SP series machine IP address	Set IP address which was set at SP machine network setting.
L	Line number	Set local line number which was set at SP machine network setting.
M	Machine number	Set local machine number which was set at SP machine network setting.

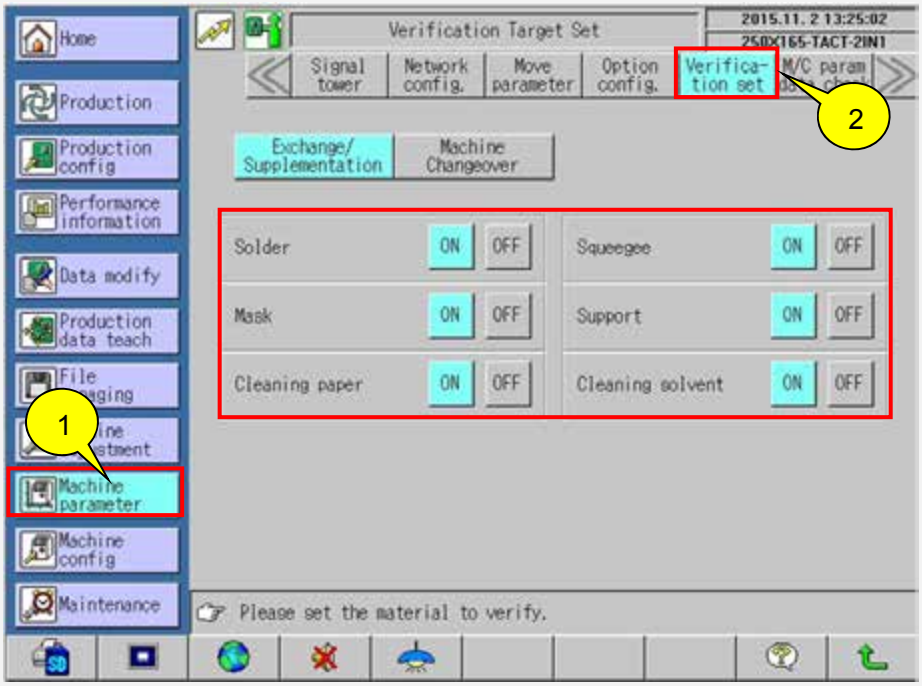
- **Machine setting**

To use material verification on SPD series machines, follow the steps below.

Step	Procedure
1	On machine, change (1) Home → (2) Sys.Para → (3) LNB Connect to [ON]. 
2	On machine, change (1) File Operation → (2) Select device to [DGS]. 
3	Select (1) Machine parameter → (2) Network setting.



4	Set the parameters according to “SPD series machine network setting” table.
5	On machine, select (1) Machine parameter → (2) Verification setting.



6	Set the parameters according to “SPD series machine verification target setting” table below.

● SPD series machine network setting

Option name	Setting	Explanation
Local		Shows set information on SPD series machine side.
Host name	SPD series machine's name	Set the name of SPD series machine registered in LNB.
IP address	SPD series machine's IP address	Set IP address on SPD series machine.
Line number	1	Fixed value
Machine number	1	Fixed value
Net Mask	Subnet Mask	Set subnet mask on SPD series machine side.
Port number	20001	
Gateway	Default Gateway	Set IP address of default gateway on SPD series machine side.
Remote		Shows the setting information on LNB side.
Host name	LNB name	Set name of LNB to connect.
IP address	LNB IP address	Set IP address of LNB to connect.
Host number	1	Fixed value
Line number	1	Fixed value

● SPD series machine target setting

Option name	Setting	Explanation
Solder	ON	Turn this on to verify solder.
Mask	ON	Turn this on to verify mask.
Cleaning paper	ON	Turn this on to verify cleaning paper.
Squeegee	ON	Turn this on to verify squeegee.
Support	ON	Turn this on to verify supports.
Cleaning solvent	ON	Turn this on to verify cleaning solvent.

3. FUNCTION - PREVENTION FROM AN INCORRECT SET - PARTS SETUP

This chapter explains the Part Setup.

3.1 Overview

3.1.1 Functional Overview

Mobile scanner now supports multiple defined operations from one menu. Setup and part supply operation in the current mobile scanner (Ver.8.2) are separated and caused confusions. New scanner application allows performing necessary operations without moving from Part Scan screen.

The following operations can be done from a Part Scan screen.

- Setup
- Replenish
- Recheck
- Bulk replenish

* See “4. FUNCTION - PREVENTION FROM AN INCORRECT SET - PARTS SUPPLY 4.3 Operation”for replenishing parts and recheck.

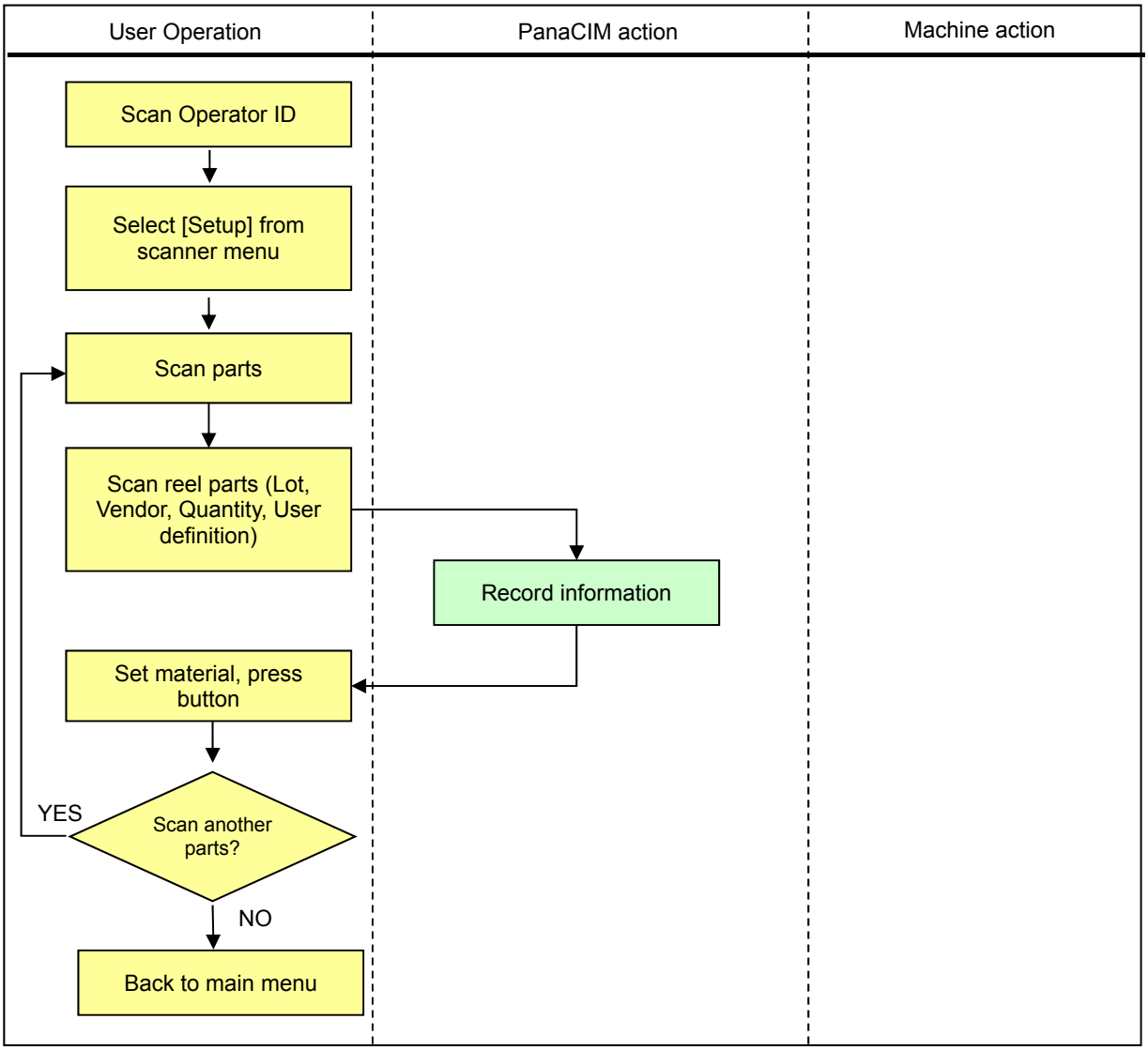


There are the following two modes available for the verification system: non-integrated and integrated versions (*1).

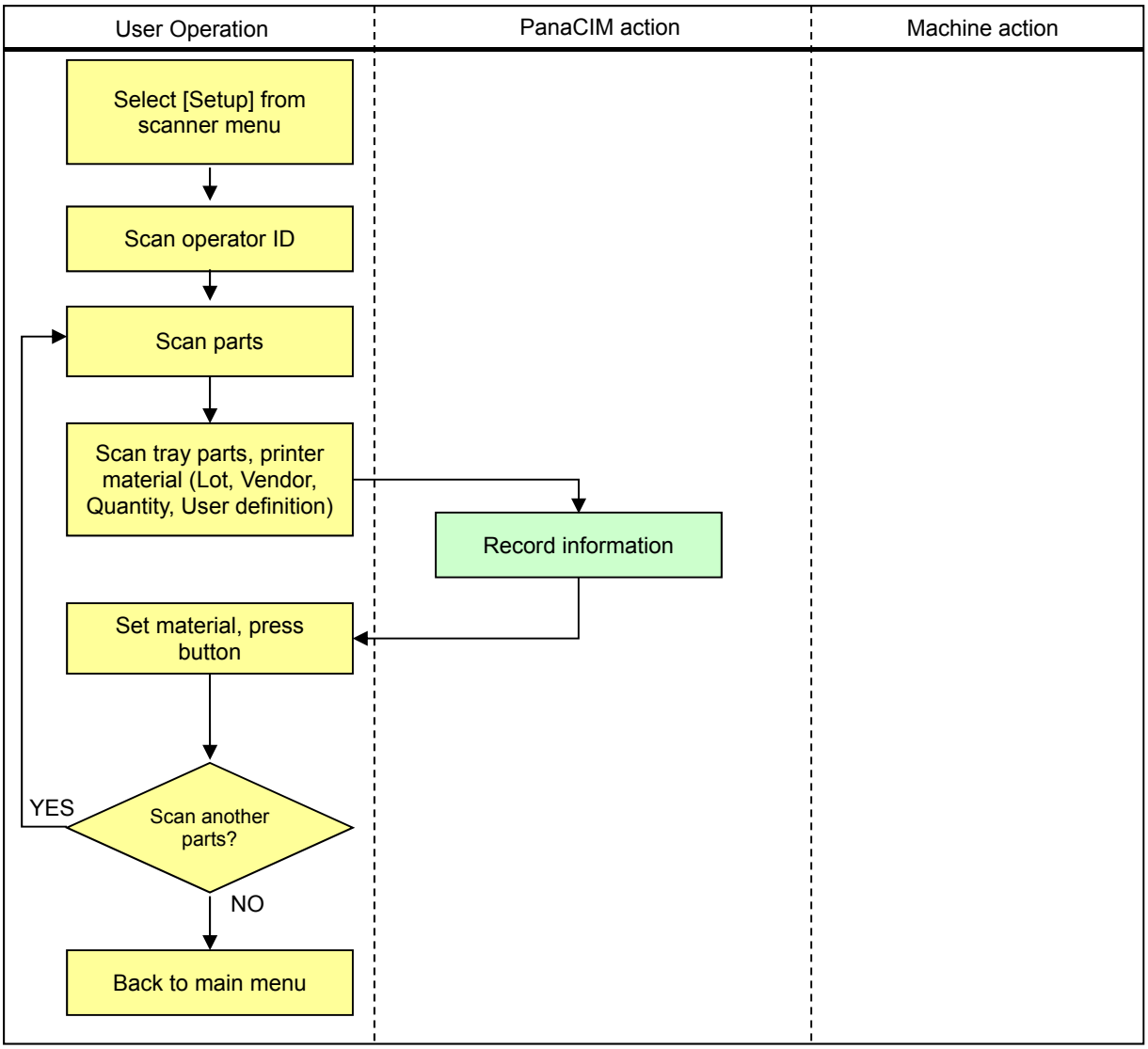
*1 The integrated version supports only the NPM system connected to LNB. Any line including NPM-X does not support the non-integrated version.

3.1.2 Material Verification Operation Flow

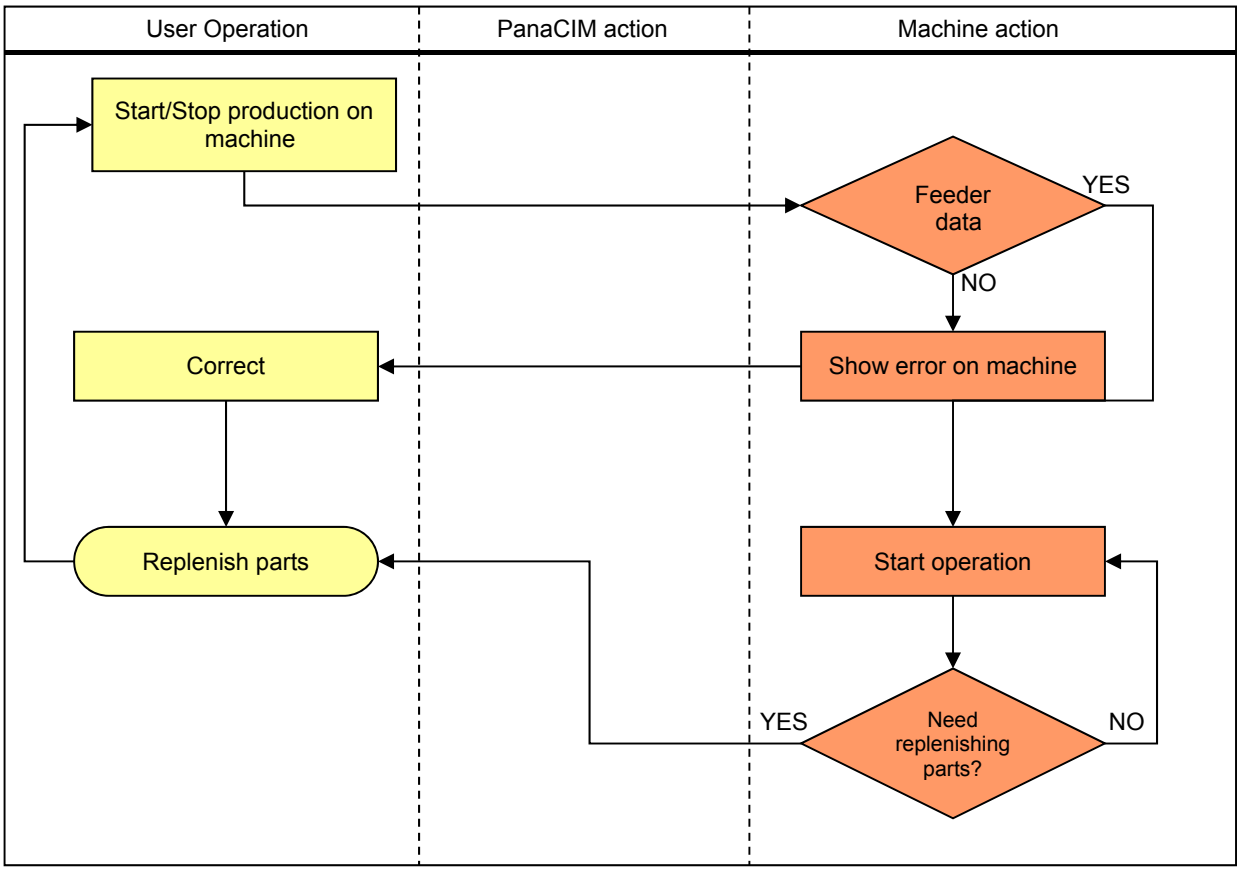
■ Replacing parts (reel parts) flow



■ Replacing parts (Tray parts/Printer material) flow



■ Production start control flow



3.2 Settings

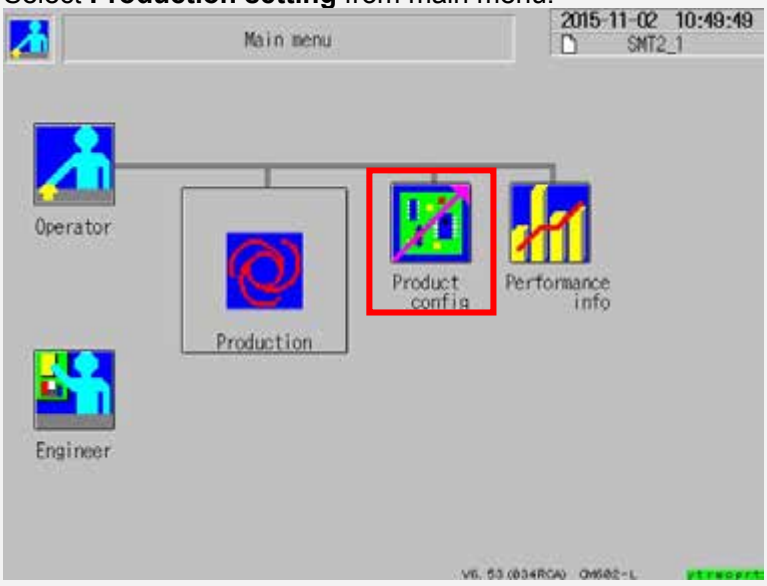
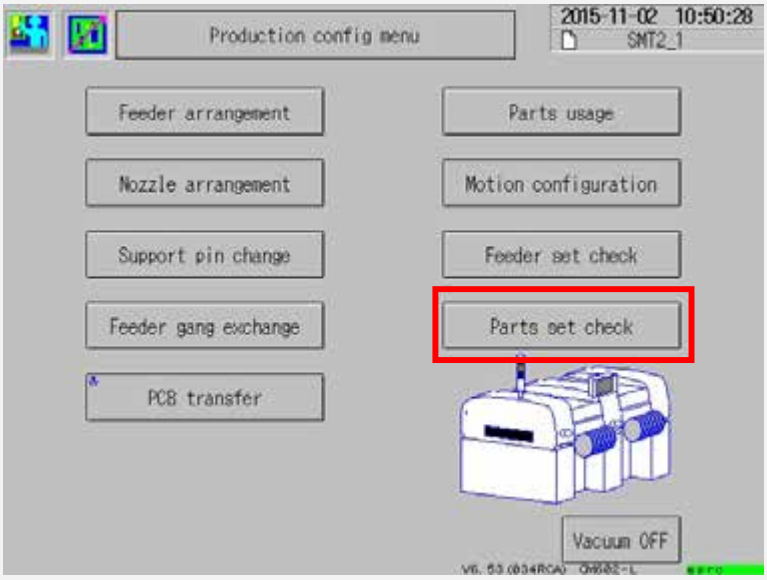
3.2.1 Feeder Parts

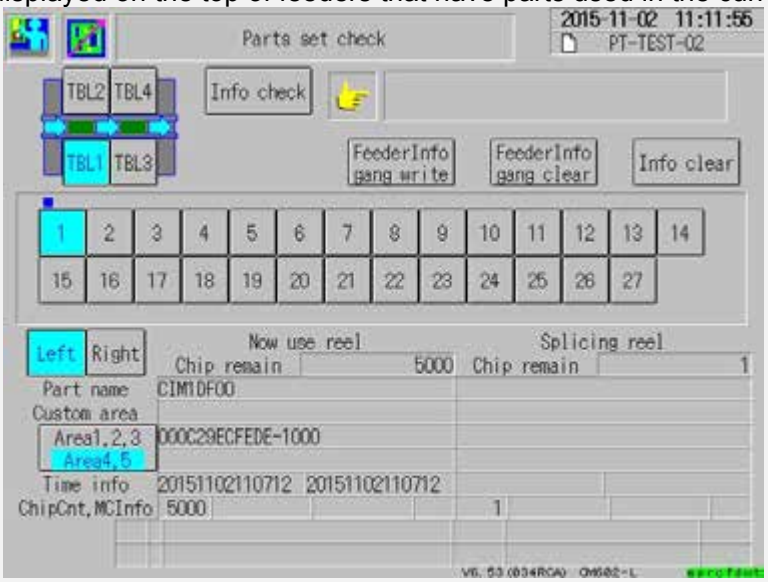
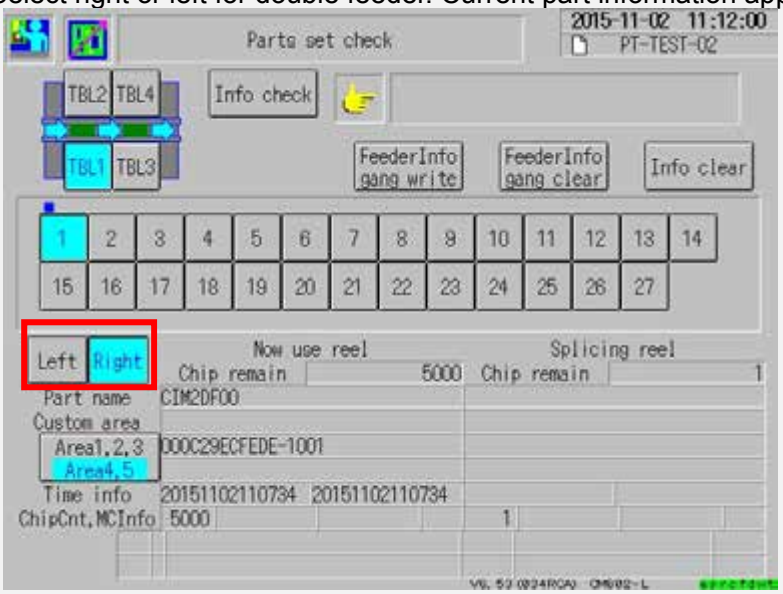
■ CM/DT Series Machine

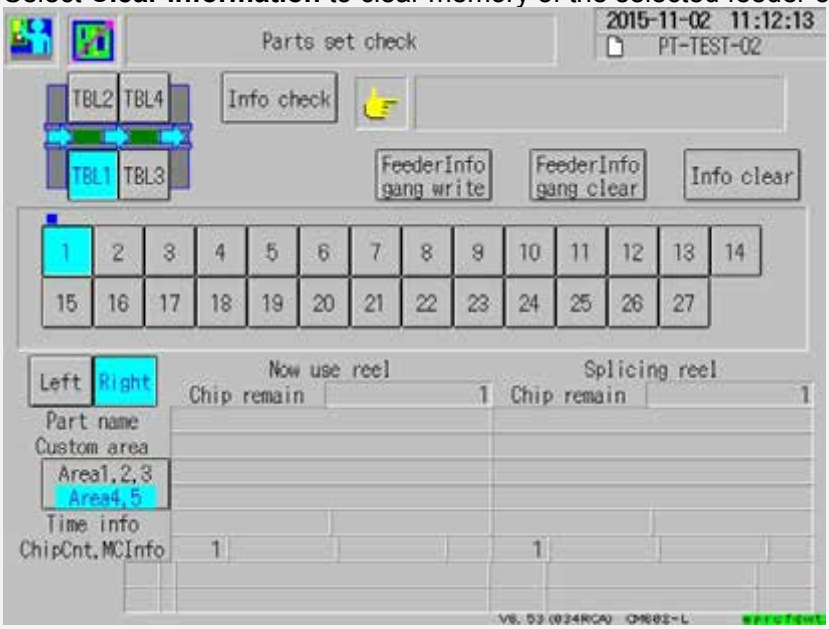
● Feeder Memory

Clear feeder information before writing part information on feeder memory. On <Parts set check> screen, select [Feeder information batch clear] and all feeder information on the table will be cleared. Use this button to verify entire table. If this function is set to off on machine, this button does not appear. To verify only some of feeders on table, you can clear memory on individual feeder.

The following flow is explained with CM/DT series machine CM602 as example.

Step	Procedure
1	Select **Production setting** from main menu. 
2	Click **Parts set check**. 

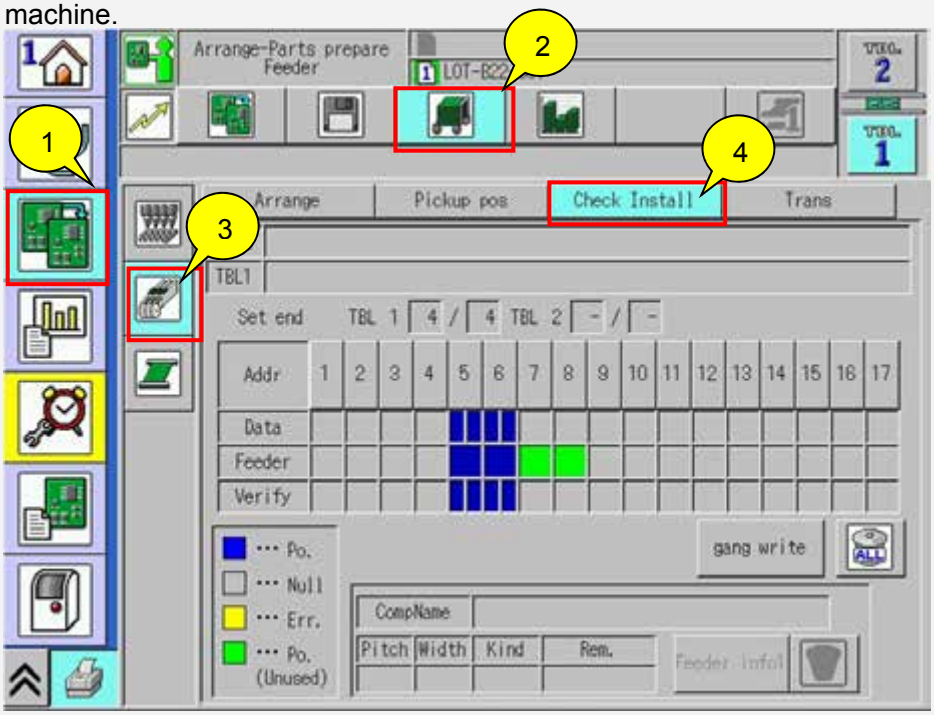
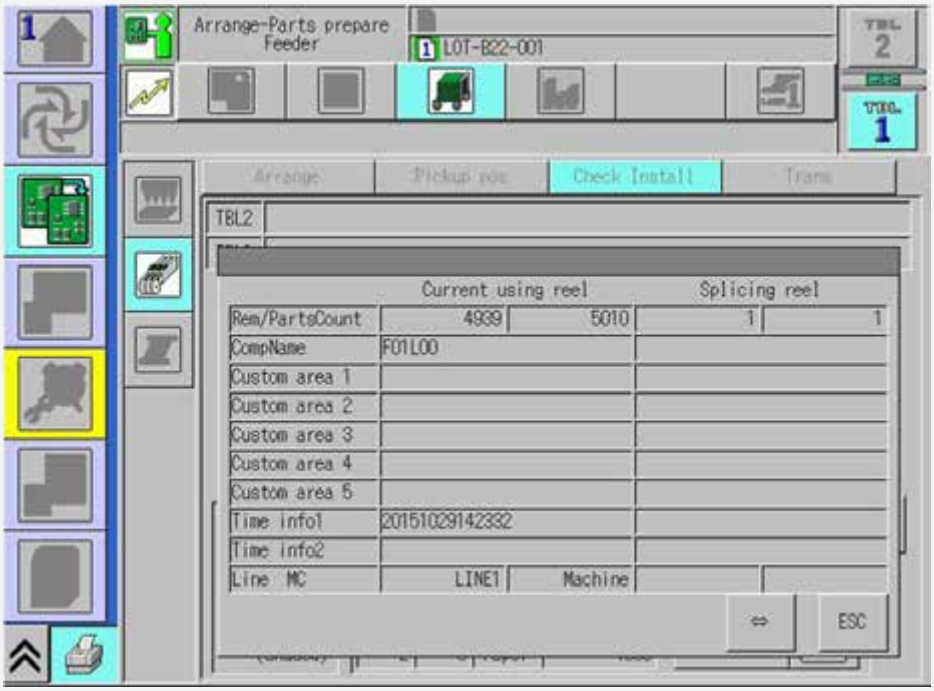
3	Parts set check menu appears on machine screen. Select a slot number. Blue square is displayed on the top of feeders that have parts used in the current production. 
4	Select right or left for double feeder. Current part information appears. 

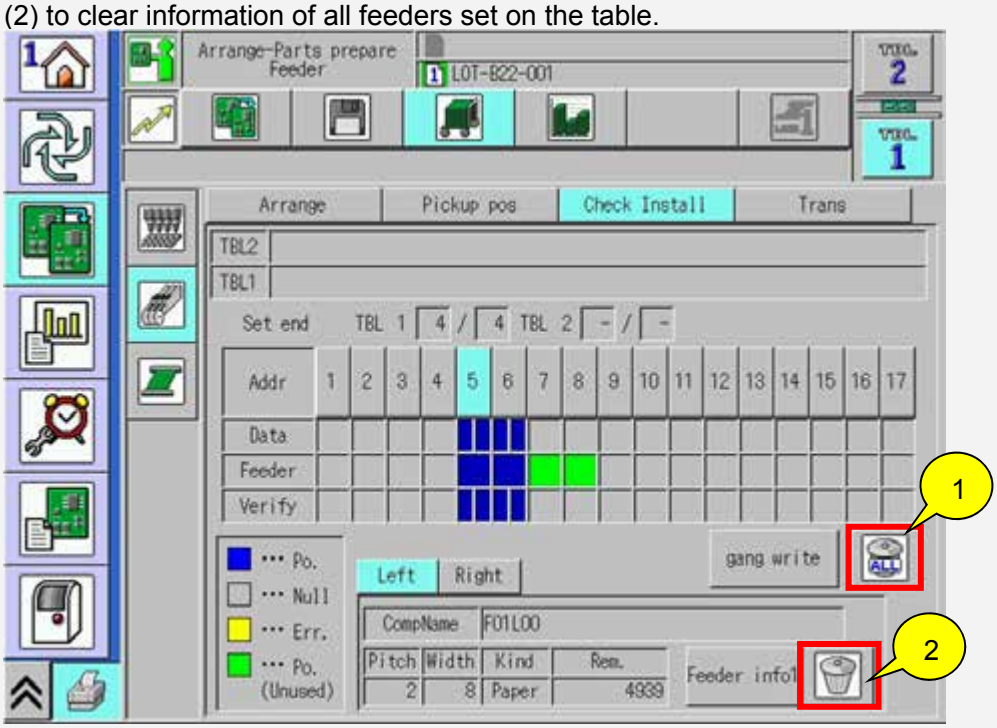
5	Select **Clear information** to clear memory of the selected feeder or right/left feeder. 

■ NPM Series Machine

● Feeder Memory

Clear feeder information before writing part information on feeder memory. All feeder information on the table will be cleared. To verify only some of feeders on table, you can clear memory on individual feeder.

Step	Procedure
1	Select (1) Changeover → (2) Material Preparation → (3) Feeder → (4) Set check on machine. 
2	Set check menu appears on machine screen. Select a slot number. Select right or left for double feeder/thin-type single feeder. Select **Feeder information1** to see feeder information. 

3	Press button (1) to clear memory of the selected feeder or right/left feeder. Press button (2) to clear information of all feeders set on the table. 

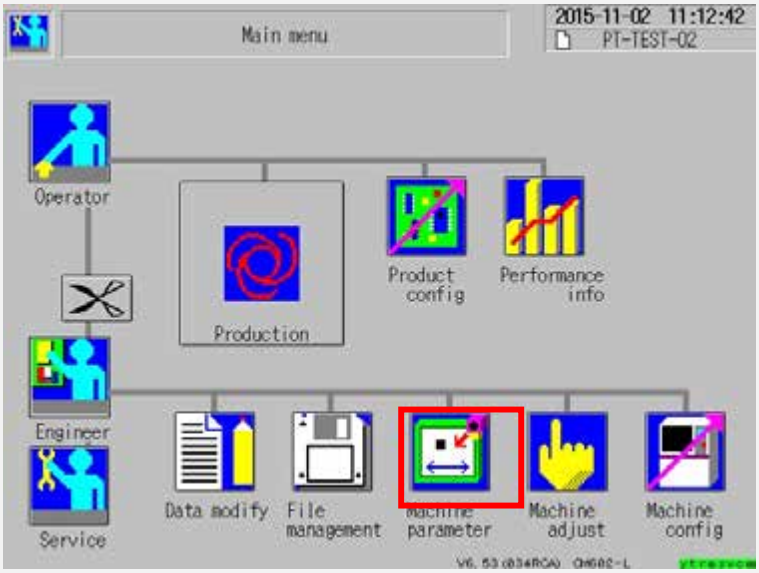
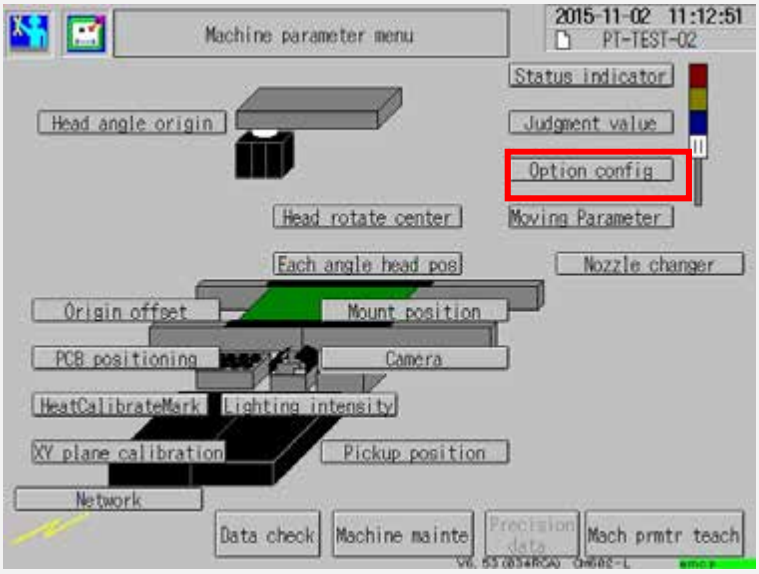
3.2.2 Tray Parts

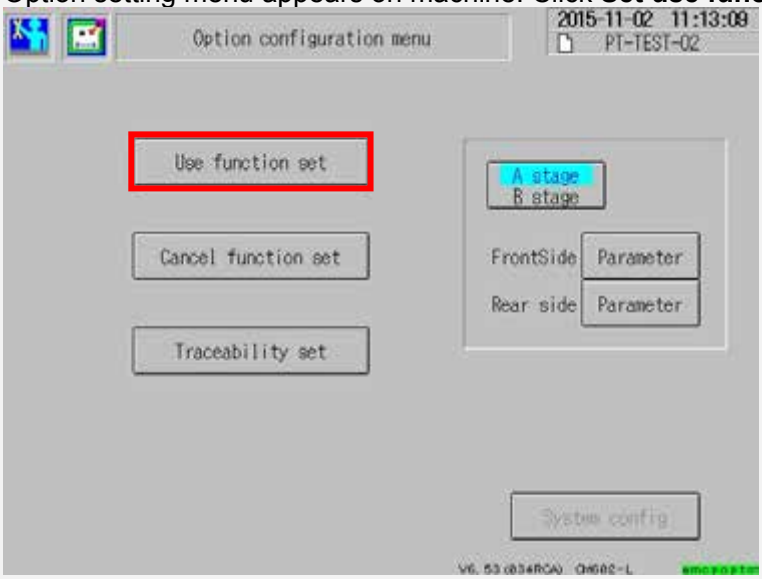
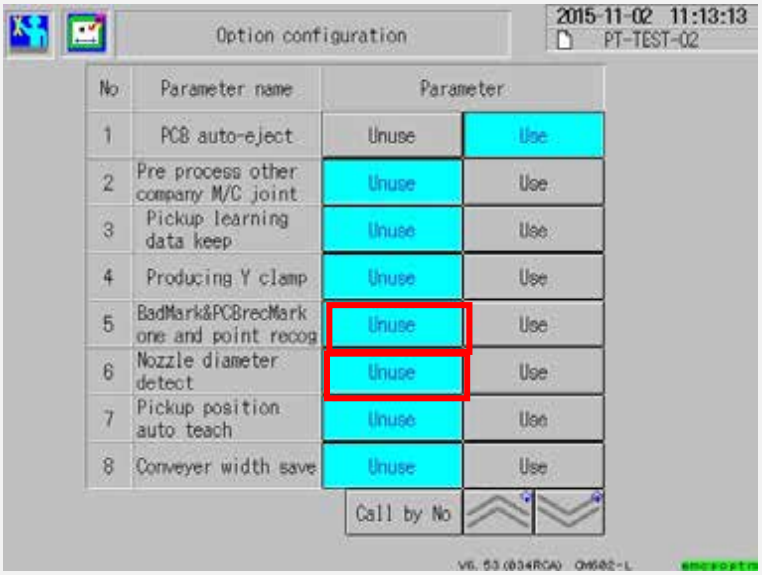
■ CM/DT Series

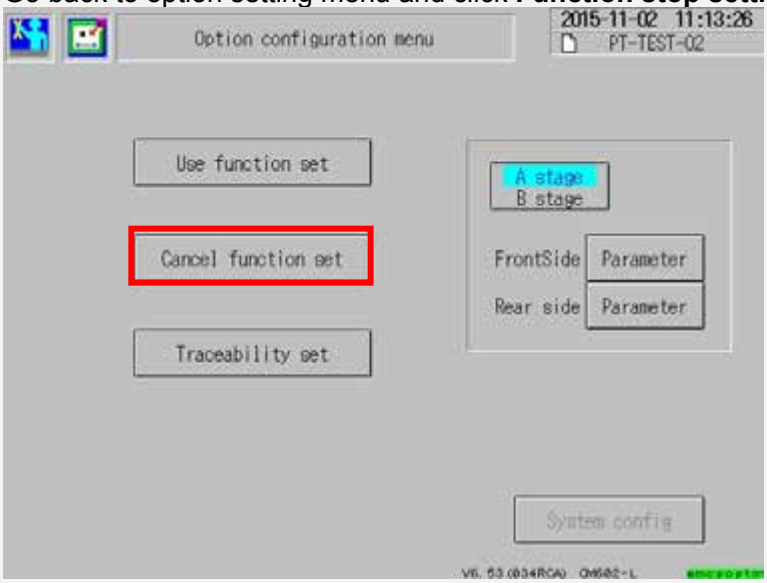
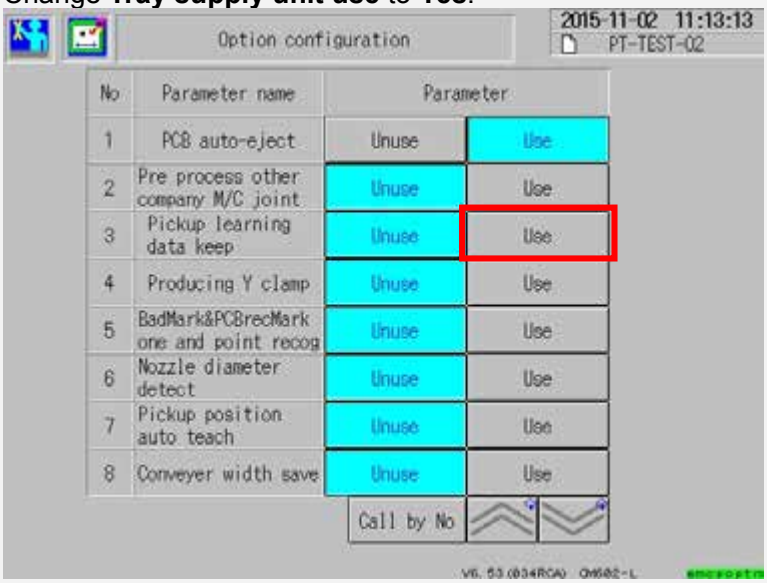
● Option

Change the following options to use tray parts.

The following flow is explained with CM/DT series machine CM602 as example.

Step	Procedure
1	Select **Machine parameter** from main menu. 
2	Machine parameter menu appears on machine screen. Click **Option Setting**. 

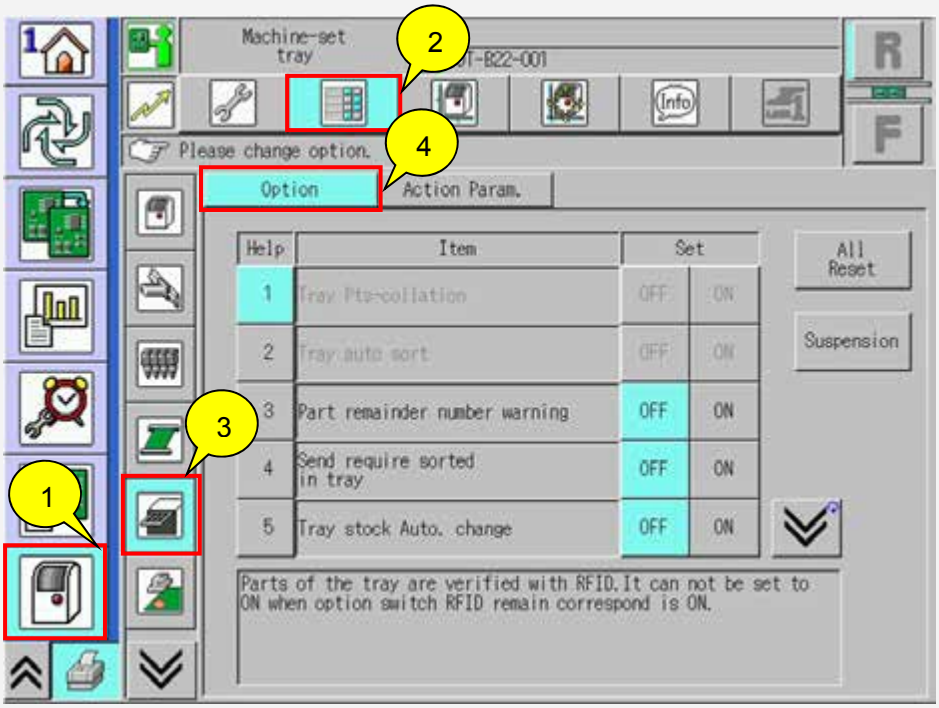
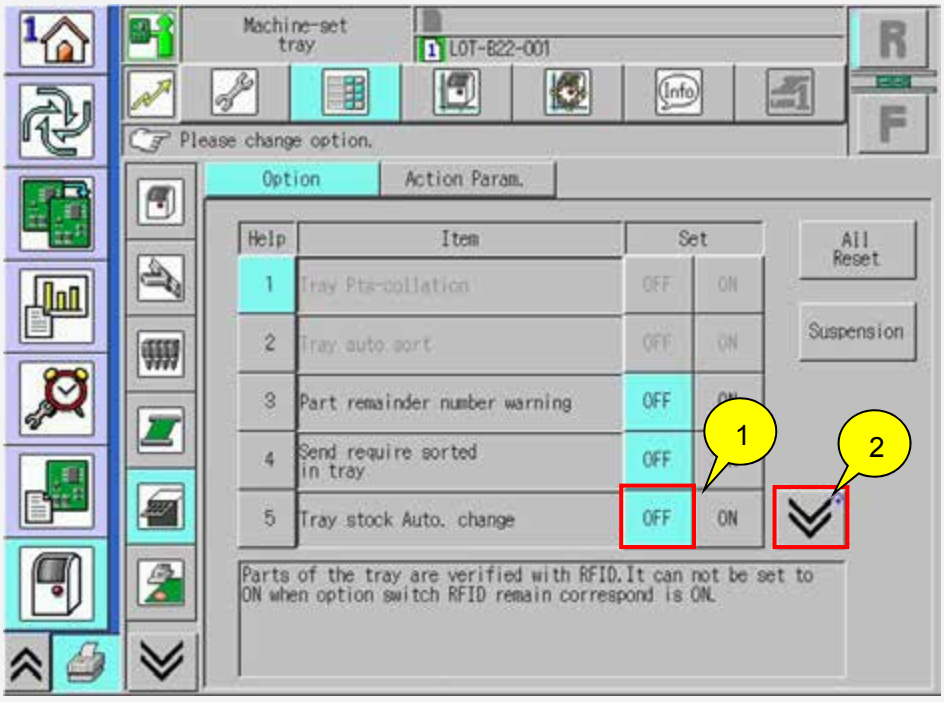
3	Option setting menu appears on machine. Click **Set use function**. 
4	Change **Tray stock auto exchange** and **Supply position drawer tray selection** to **No**. 

5	Go back to option setting menu and click **Function stop setting**. 
6	Change **Tray supply unit use** to **Yes**. 

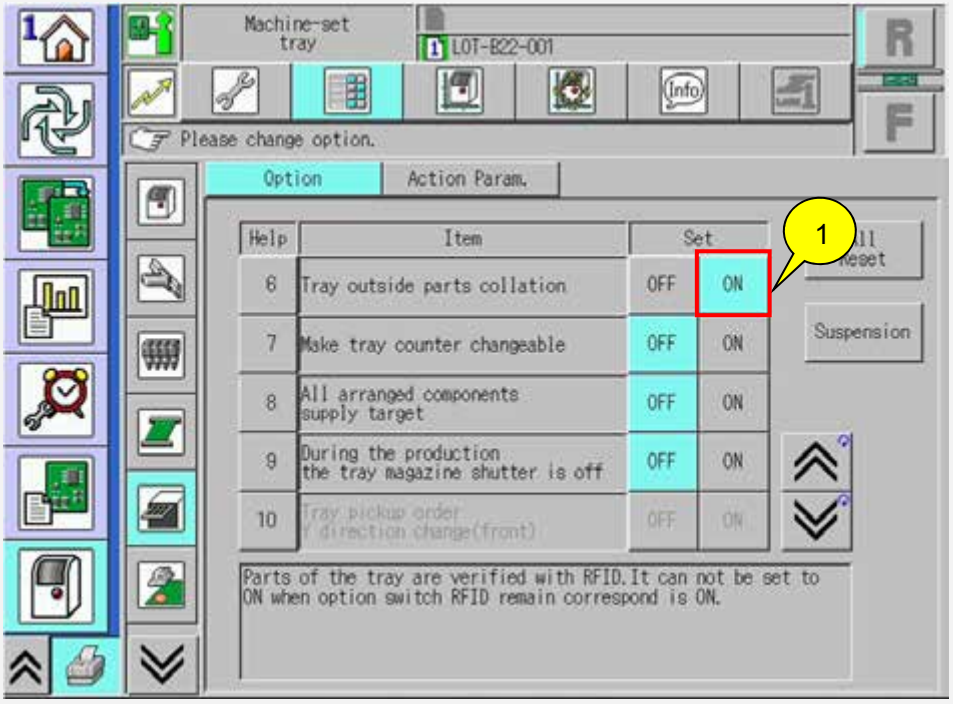
■ NPM Series

● Option

Change the following options to use tray parts.

Step	Procedure
1	Select (1) Machine → (2) Setting → (3) Tray → (4) Option on machine. 
2	Option menu appears on machine screen. Set **Tray stock auto change** to **OFF**.  (2) Select the next page.

3	Set **Tray supply unit use** to **ON**.



3.3 Operation

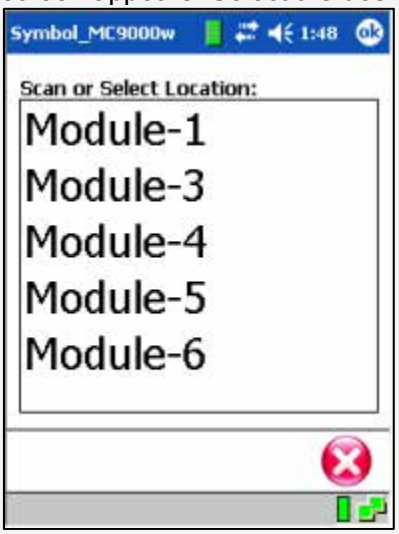
3.3.1 MV Mode

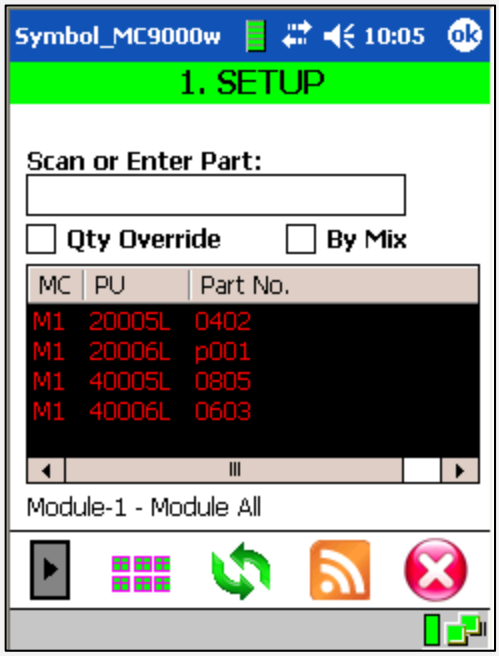
■ CM/DT Series and NPM Series

- **Part scan**

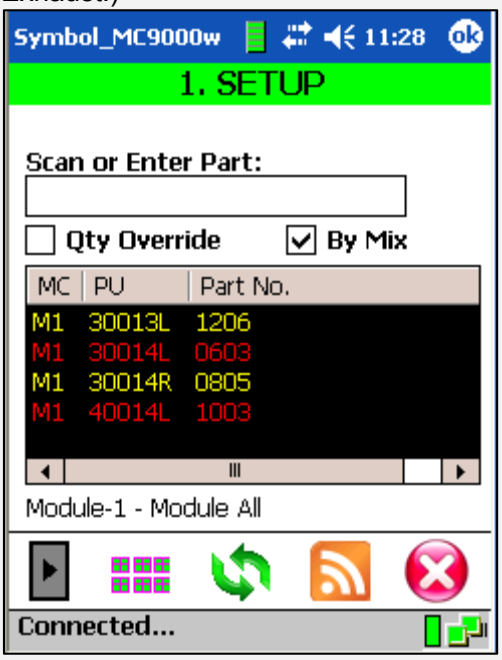
This section describes the procedure for setting up the components involving changeover.

Refer to the table below to set up the components.

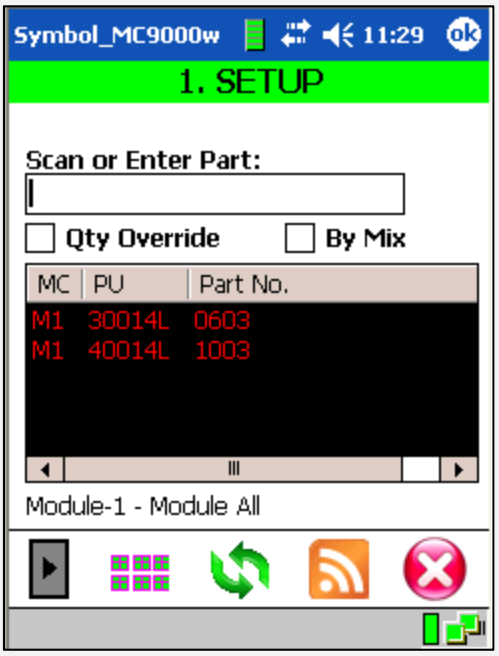
Step	Procedure
1	Clear the feeder memory at the Machine.
2	From the main menu screen, select **Part Scan** as shown below: 
3	After selecting **Part Scan** from main menu screen, **Scan or Select Location** screen appears. Select the desired location from the list. 

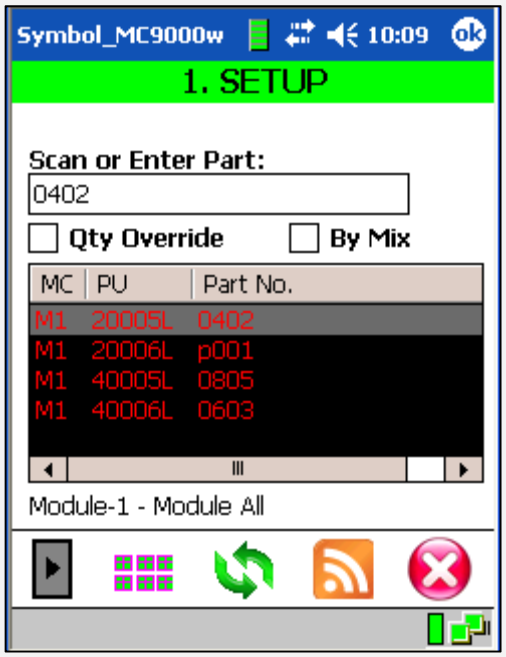
4	After selecting location, **Scan or Select Work Area** screen appears as shown below: 
5	Scan or select a work area and after doing so, following screen appears to scan or enter the part number.  Part numbers to be scanned or entered are listed below under the heading **Part No.** **Note:** In case of mix data sharing the part location information with multiple product programs for NPM series machines, setup list will contain and display list of setup locations for which setup is not completed from current and other products in mix, if any, i.e. the locations marked only in red and yellow color. Click the **By Mix** checkbox to view these all these feeder locations as shown below: For the CM series machines, components not to be used for the current product program will also appear in the same color. (The display remains unchanged even if the **By Mix** checkbox is checked.)

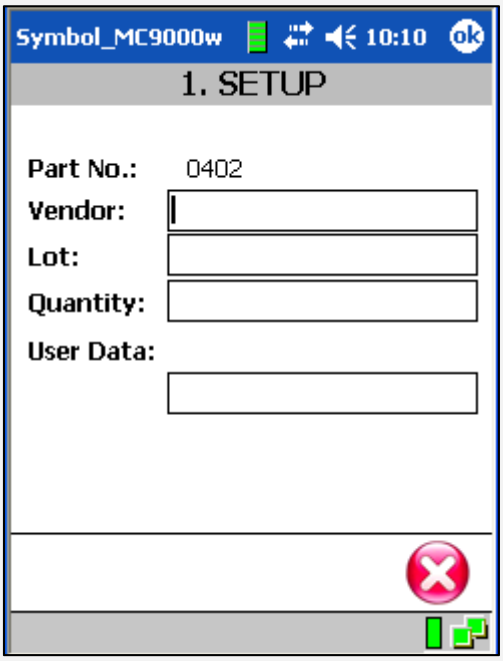
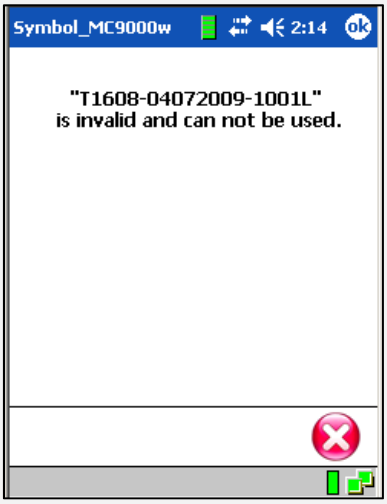
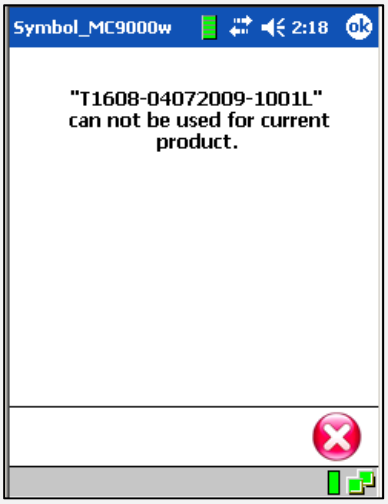
Note: If the menu other than '1. SETUP' is displayed, tap the highlighted menu at the top of the display until '1. SETUP' is shown. (Refer to 4.3.1 Parts Exhaust.)

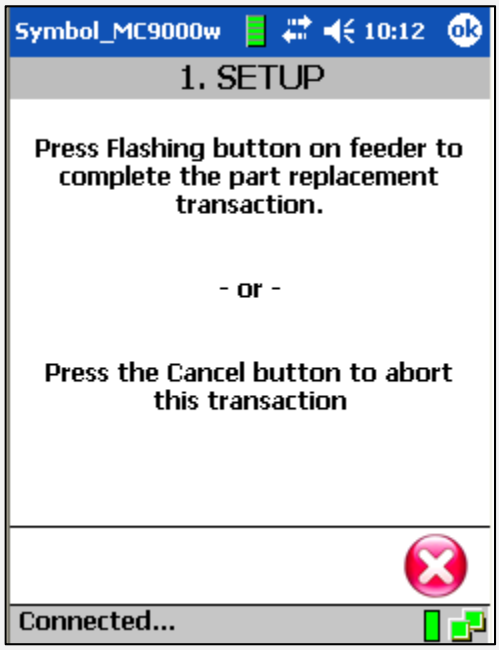
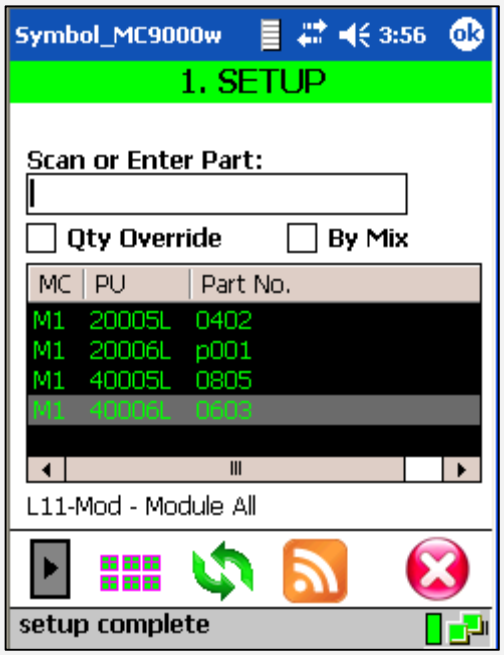


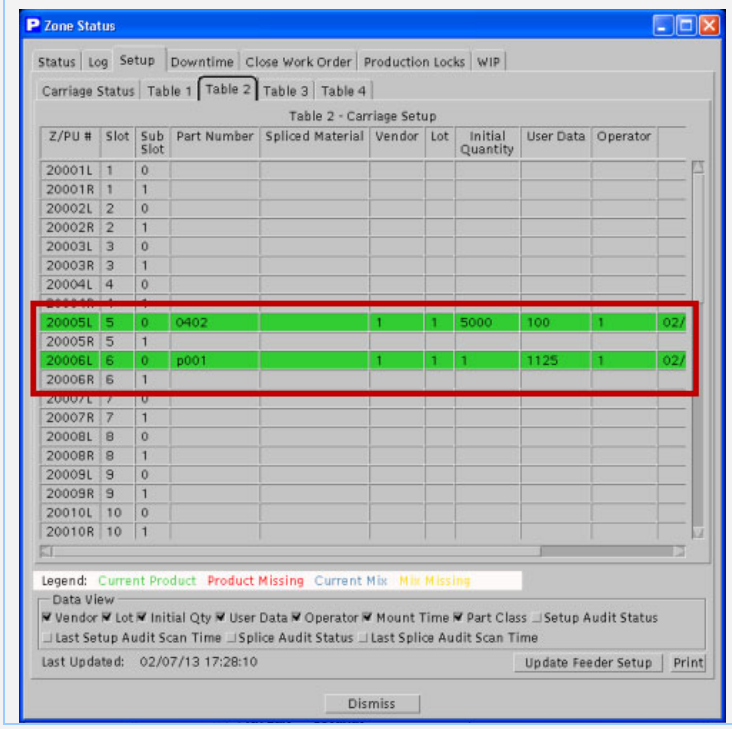
If **By Mix** checkbox is unchecked, then only feeder locations from current product are displayed for which setup is not completed. This is shown below:



6	Scan or enter the part number existing in the program downloaded to the scanner. In the screen shot shown below, part number is entered as 0402.  **Note:** If a part number scanned or entered does not match with the part number existing at the feeder location (as per the program configuration downloaded to the scanner), a message appears alerting the operator that scanned /entered part number is not existing in the product. a. Scan or enter the **Part Number**. b. **Quantity Override** is for barcodes that have a quantity value concatenated with the Part Number barcode in Material Verification mode. Check the **Qty Override** checkbox for the ability to key in a quantity value instead of taking the full original value when scanning partial reels.

7	After correct part number is scanned or entered, following screen appears to enter additional part attributes like vendor, lot no, quantity and user data. This is shown below:  **Note:** If you are using **Imported Initial Quantity** as configured in the System Configuration Application, the scanner will not prompt for Quantity.   **Note:** If the scanned part is detected as an Invalid or Restricted component, an error message will appear.

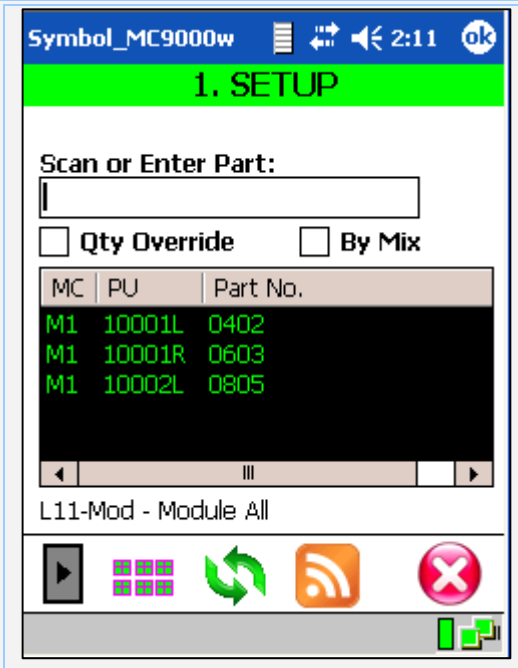
8	After entering the part attributes, following screen appears asking the operator to press the feeder button.  This operation writes the part information to the feeder memory.
9	**Repeat** this procedure for every component that is required by the program that does not already contain the proper component information. After setup is successfully completed for all the feeder locations (irrespective of whether these belong to current product or other products in a mix), the rows containing these feeder locations are removed from the setup screen. Also a message **"Setup Complete"** appears at the bottom with a short beep. This is shown below: 

10	**Verify the setup** is complete at the machine. Details can also be seen in the route status screen as shown below: 

- **Viewing an Active Setup**

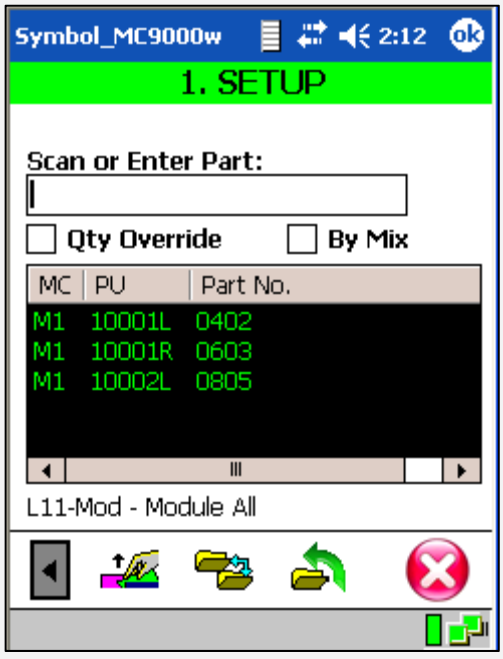
Take the steps below to display the name of the part scan in active.

Step	Procedure
1	Click  in the Production Main menu screen and navigate to the **Setup** Screen. Refer to the section Part Scan in 3.3.1 to navigate to the **Setup** Screen.
2	Click the  button at the bottom of **SETUP** Screen shown below:



3

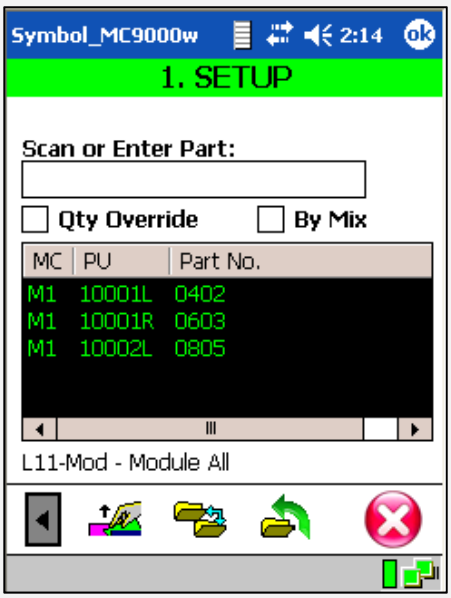
After clicking  button,  button appears at the bottom as shown below:

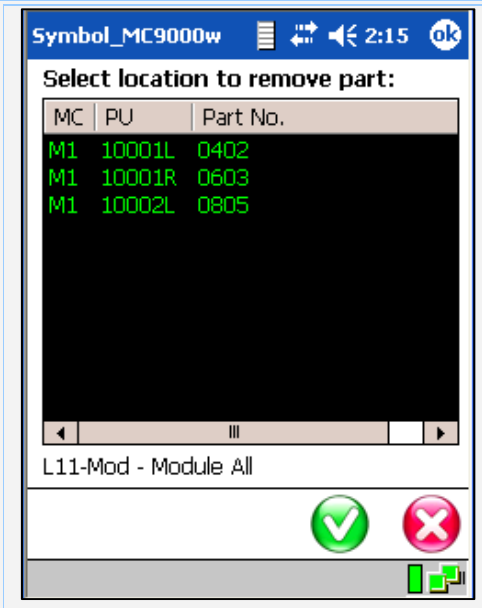


● Removing a Feeder from a Setup(Non-consolidated Edition Only)*1

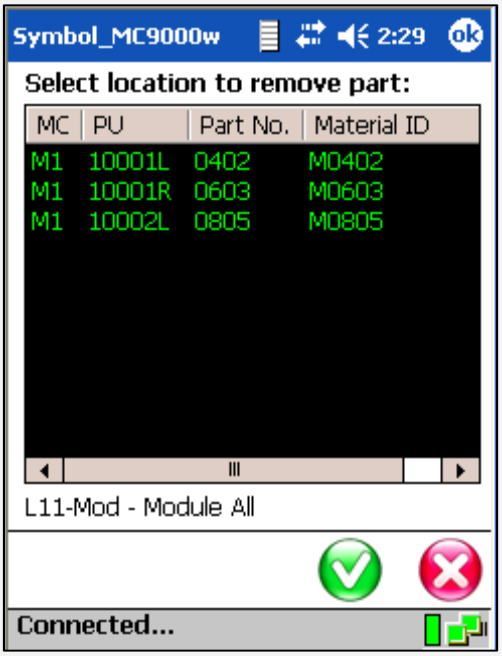
Refer to the action table below for the steps to remove feeders from an active setup. It is to be noted that only those parts can be Unmounted which belong to the currently running product.

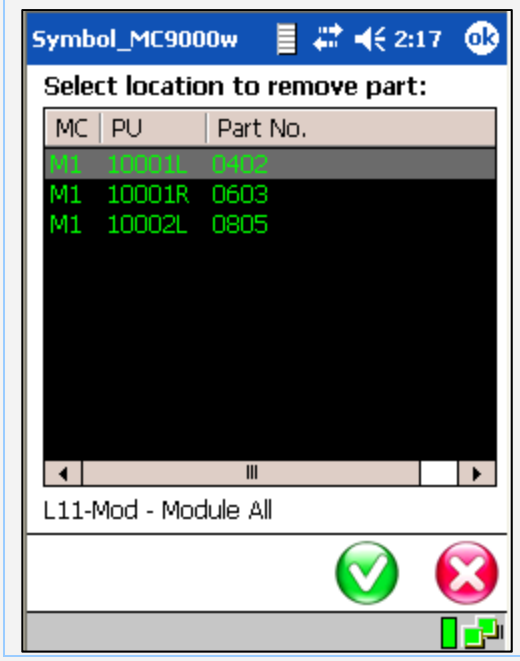
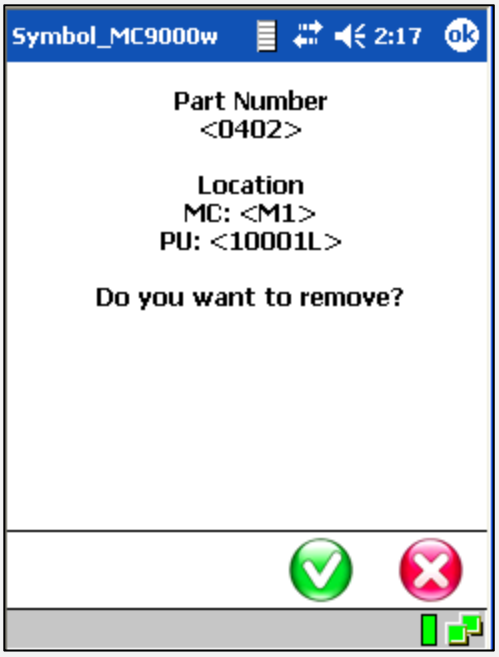
※1 Operation only for non-consolidated edition.
In the consolidated edition, because removal information of the machine is directly reflected, this operation is not necessary.
 button is not displayed.

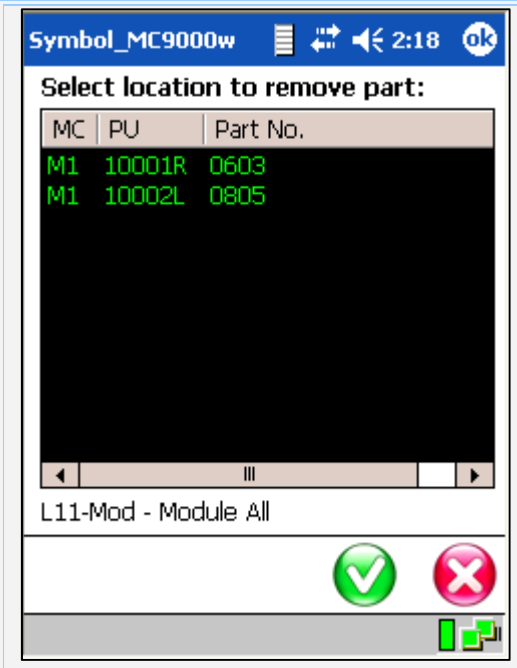
Step	Procedure
1	Click the  button at the bottom until  button is displayed at the bottom as shown below: 
2	Click the  at the bottom of the **SETUP** screen. A screen appears showing the feeder locations existing in the current product at which part is mounted. This is shown below:



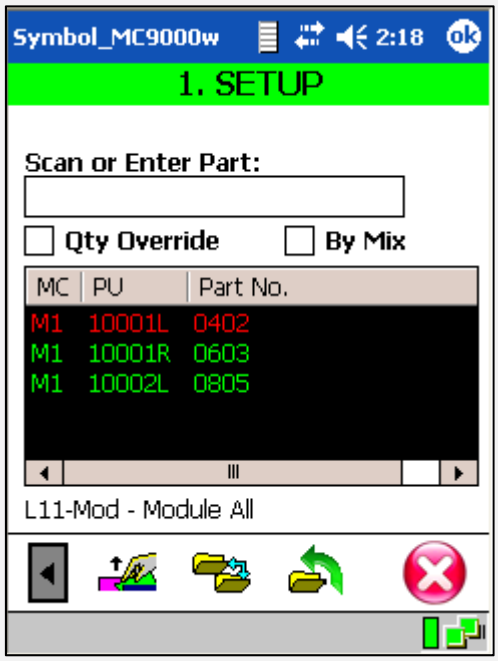
Note: In Material control mode, material ID is listed as an additional column as shown below:



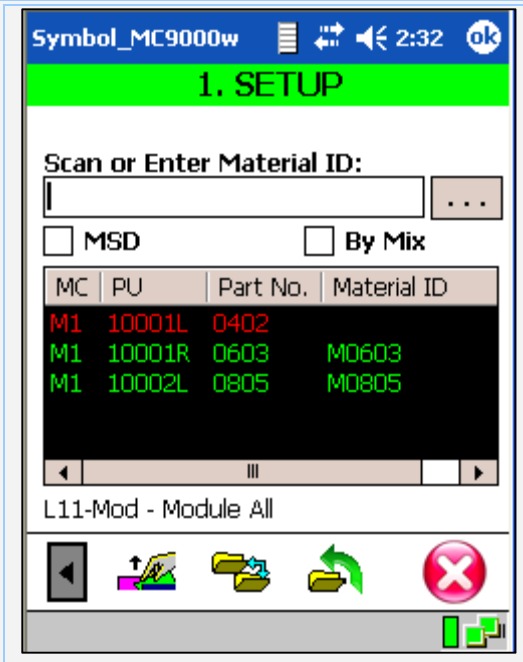
3	Select a location from which feeder / part needs to be removed as shown below: 
4	Click the  at the bottom to remove the feeder/part at the selected location. A message appears seeking your confirmation to remove part as shown below:  Click the  in the above shown message to proceed with the removing the feeder/part. Selected part is Unmounted as shown below:



Feeder/part that was removed from the setup is colored red in the main setup screen.

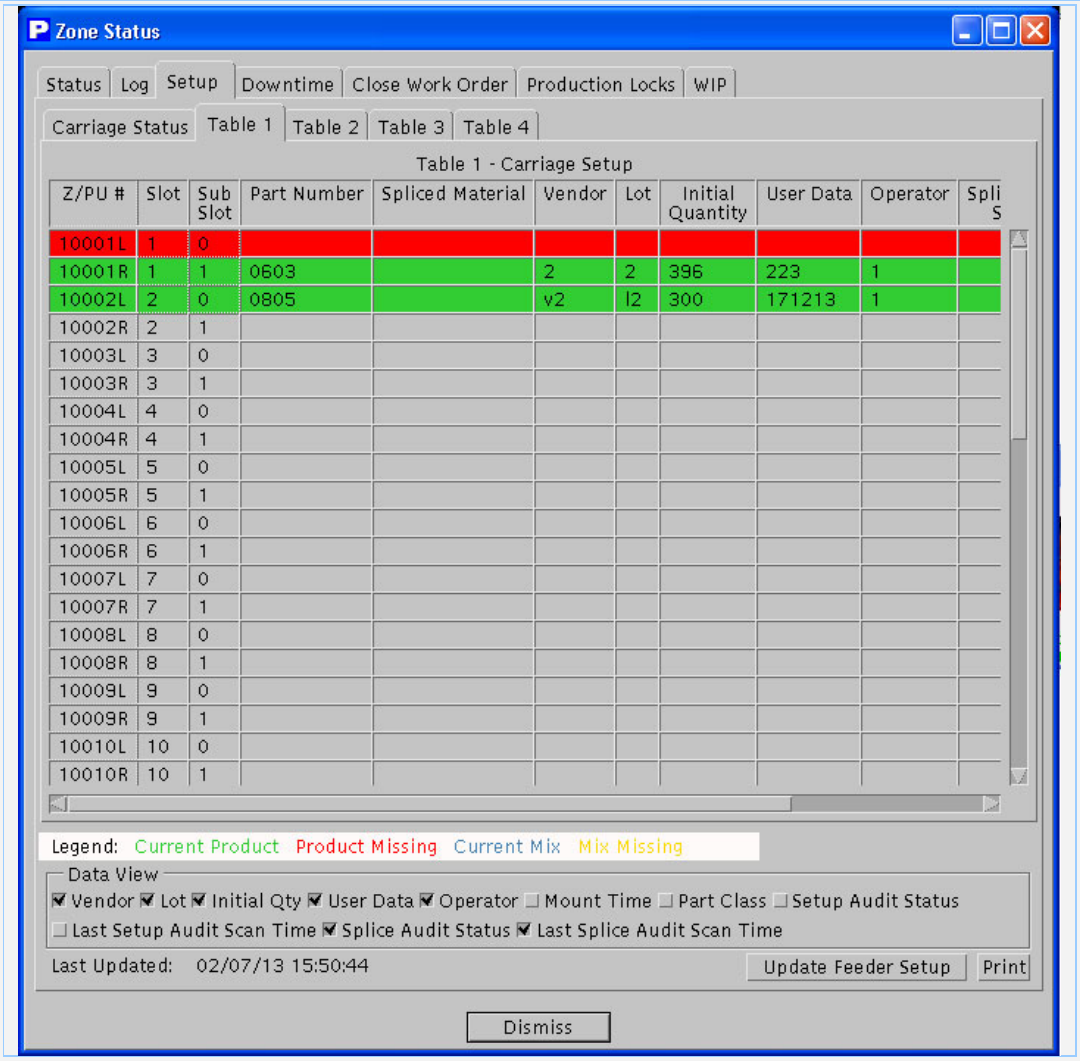


In Material control mode, screen appearance after feeder is removed from the setup is shown below:



Note: Feeder removal is allowed for one location at a time. There is no option to remove multiple locations in one go. The removed feeder location shall be considered as not mounted yet.

5	Route status reflects the current feeder setup as shown below. Feeder location from which a feeder was removed from the setup is highlighted with red color as shown below:



* Before performing step 3, operator has to physically remove a feeder from a feeder location.

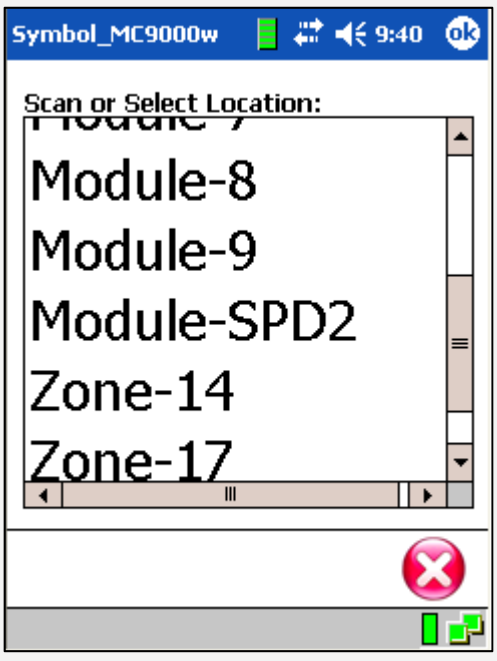
■ SP Series

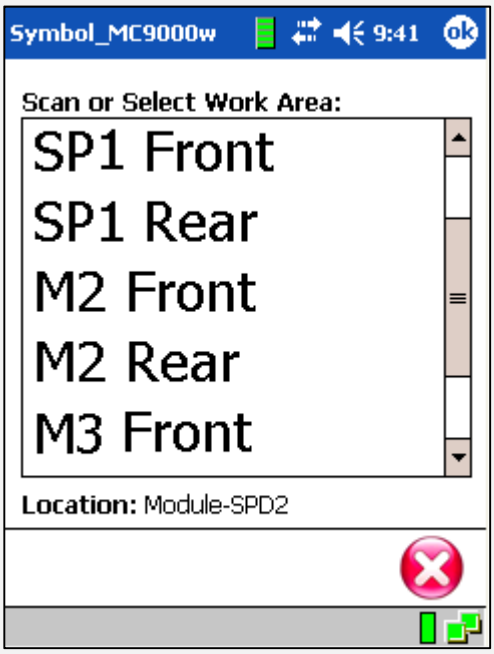
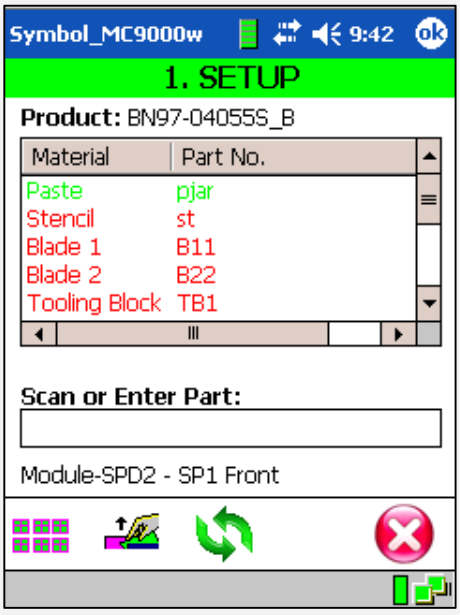
• Part Scan (SPV/SPG/SPD/SP70)

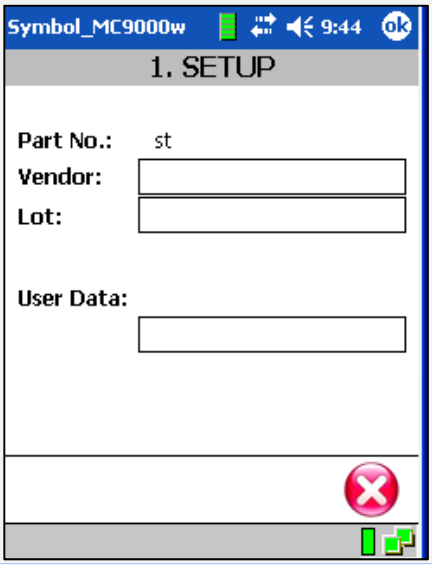
This section contains the procedure for scanning components used with the screen printers (SPV/SPG/SPD/SP70).

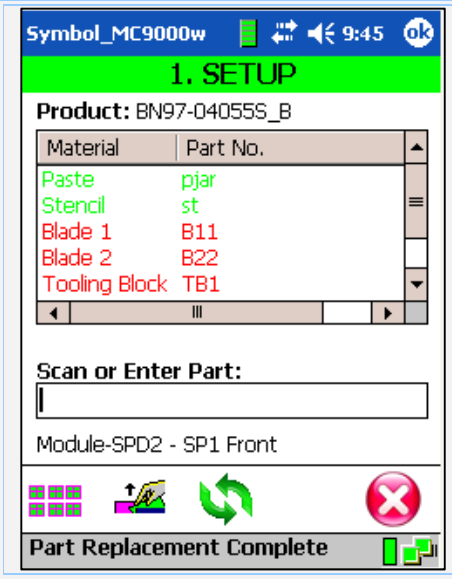
* The SP70 does not support the substitute parts, therefore cannot use them.

Use the steps below to scan components.

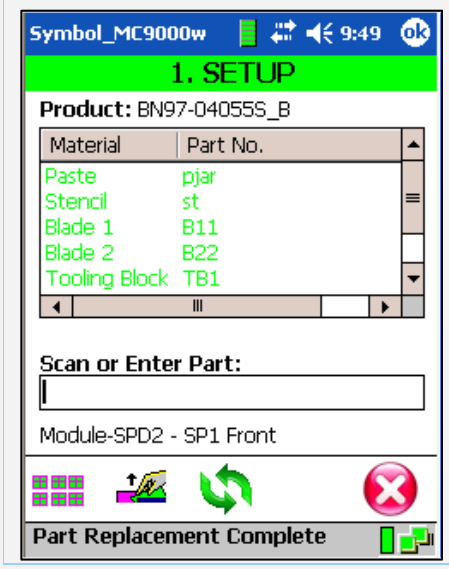
Step	Procedure
1	Launch **Production** Main menu screen from the mobile scanner as shown below: 
2	Click  . Following screen appears: 

3	Select the Work Area containing SPV/SPG/SPD/SP70 screen printer. Since there are three machines in the module and the Screen Printer is the first machine in the module, screen printer work area is named **SP1 Front** and **SP1 Rear** as shown below:  **Note:** Usually, Screen Printer is the first machine in the module. SP1 Rear is displayed only for SPD/SPV (Dual specifications, Single specifications with two machines in parallel).
4	Scan or Select the SPG/SPD/SP70 screen printer work area. Following screen appears:  As shown above, parts that are successfully scanned appear in green color while parts that are yet to be scanned appear in red color. <SPV/SPG with dual solder pod> *SPV does not yet support the dual solder pod, but we are working on support in the future.

	If two solder pods have been set, two solders are displayed as materials to be verified. ・As the first solder pod, [**Paste**] appears under Material ・As the second solder pod, [**Paste 2**] appears under Material.
5	Scan or enter part number used in screen printer. This is shown below: 
6	After scanning/entering part number, you are prompted to specify part attributes like **Vendor**, **Lot** and **User Data**. 
7	After a part in screen printer program is successfully scanned, a message **"Part Replacement Complete"** appears and parts appear in red will turn green.

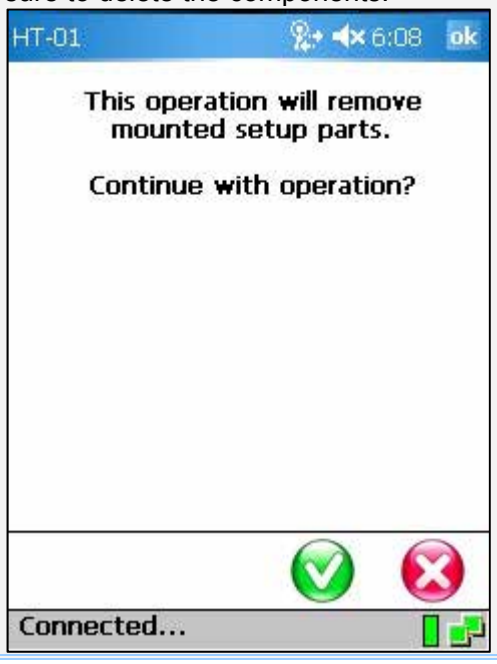


8 After all parts are successfully scanned, they appear in green color as shown below:



- **Removing parts from the screen printer (SPV/SPG/SPD/SP70)**

A new feature of the mobile scanner permits components to be removed from all positions by selecting the SP work area from the NPM module. All paste reels for the screen printer (SPV/SPG/SPD/SP70) need to be rescanned. Refer to the table below to remove components from the screen printer (SPV/SPG/SPD/SP70).

Step	Procedure
1	The following screen assumes that components have been loaded to the screen printer (SPV/SPG/SPD/SP70).  Click  to remove all components in the screen printer.
2	After  is clicked, a confirmation message appears asking you if you are sure to delete the components. 

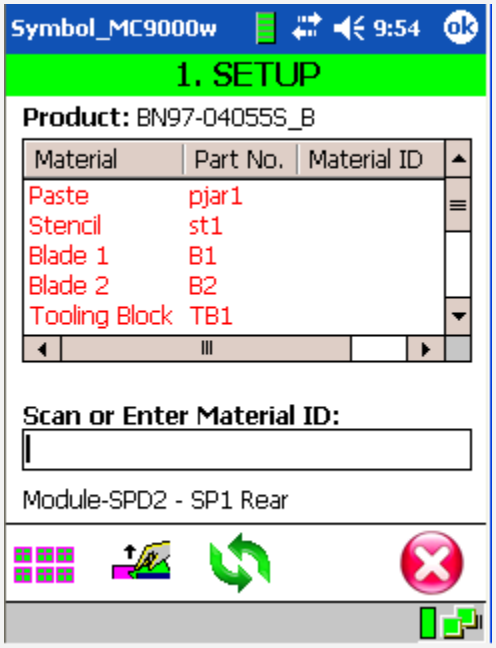
3	Click  on the screen in step 2. All components in the screen printer will be removed as shown below.  **Note:** Note that the components will not turn red but remains in green even after they have been removed.

3.3.2 MC Mode

■ SP Series

● Part Scan for SPG/SPD/SP70 Screen Printer

Step	Procedure
1	Activate the Screen Printer work area (in this case SP3 Rear) to display the part list.



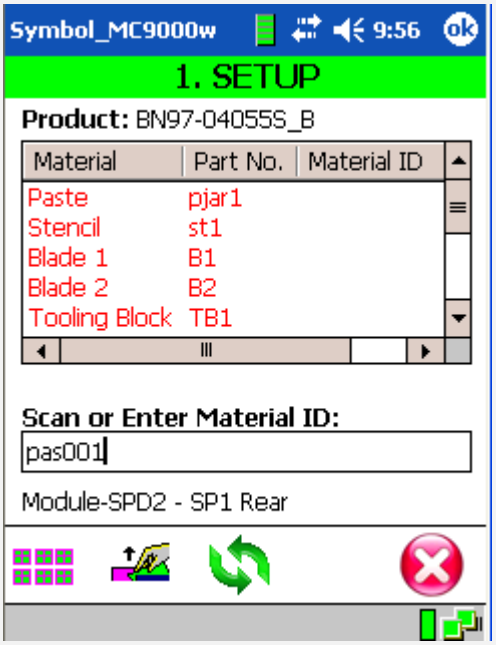
As shown above, parts appear in red color because they are yet to be scanned.

<SPV/SPG with dual solder pod>

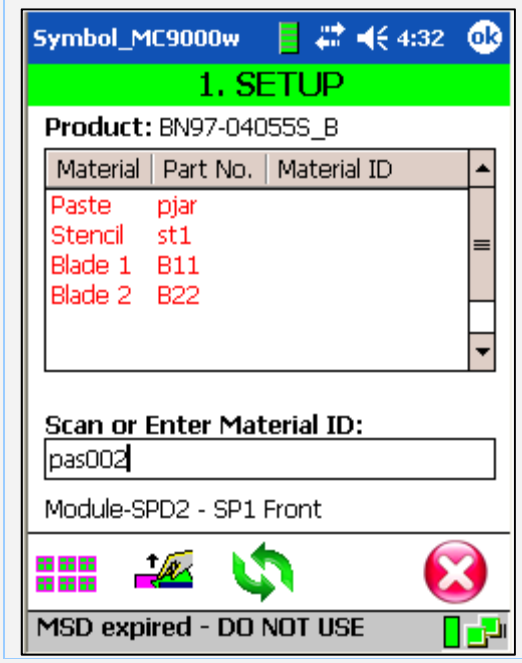
- As the first solder pod, [**Paste**] appears under Material
- As the second solder pod, [**Paste 2**] appears under Material.

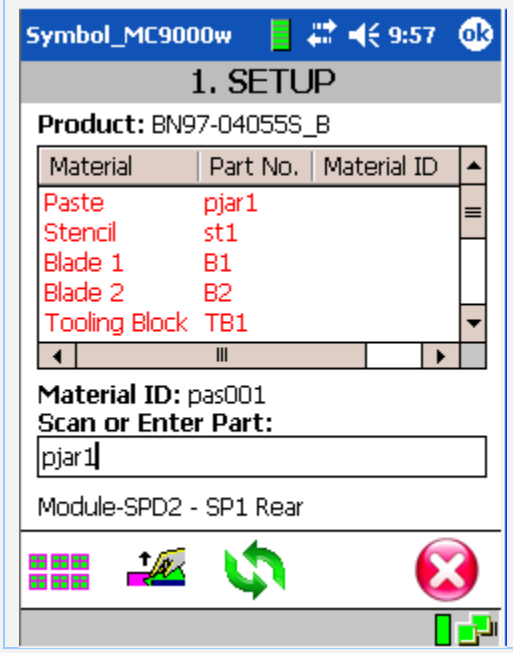
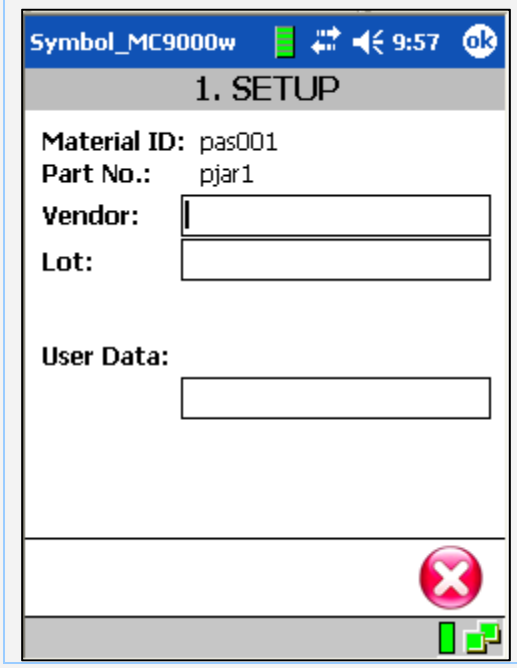


2	Scan/Enter Material ID as shown below:

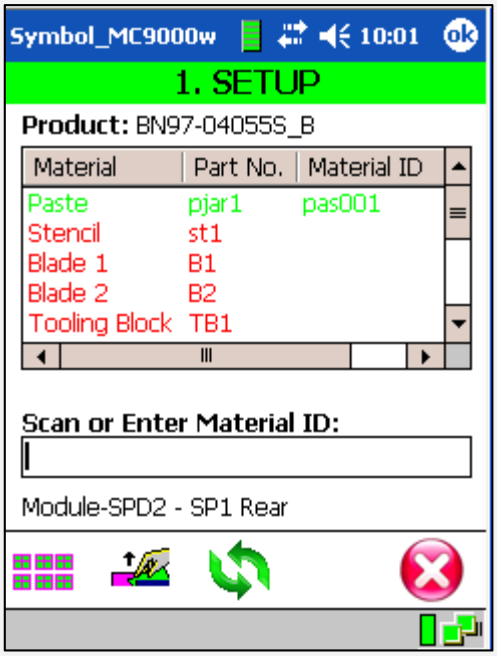


Note: If you try to mount a solder paste part whose floor time (time for which it can be exposed to moisture) has expired, it is forbidden from being mounted as shown below:

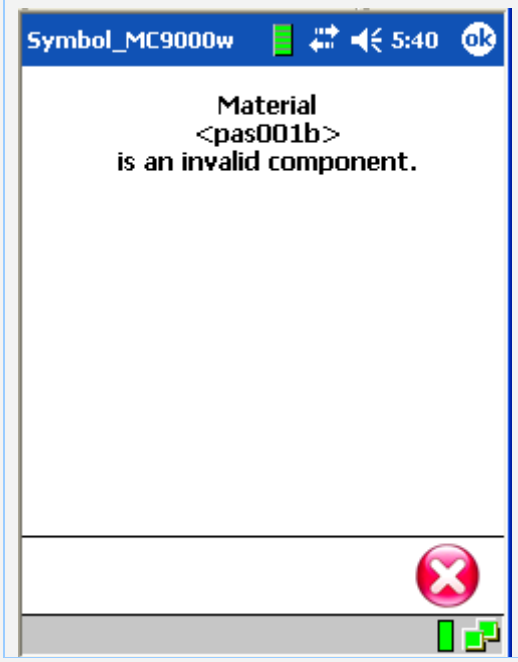


3	After successfully scanning the Material ID, you are prompted to enter part number as shown below: 
4	After entering part number, you are prompted to enter part attributes like Vendor, Lot and User Data. 

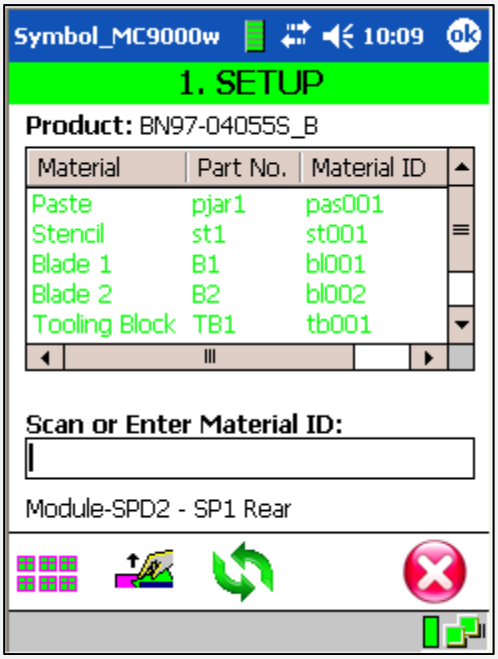
5 After a part is successfully scanned, it turns green as shown below:



Note: If solder paste part is defined as invalid in the **Invalid Components** tab of Inventory Manager application, following message appears after specifying part attributes in step 4 and part is not mounted:



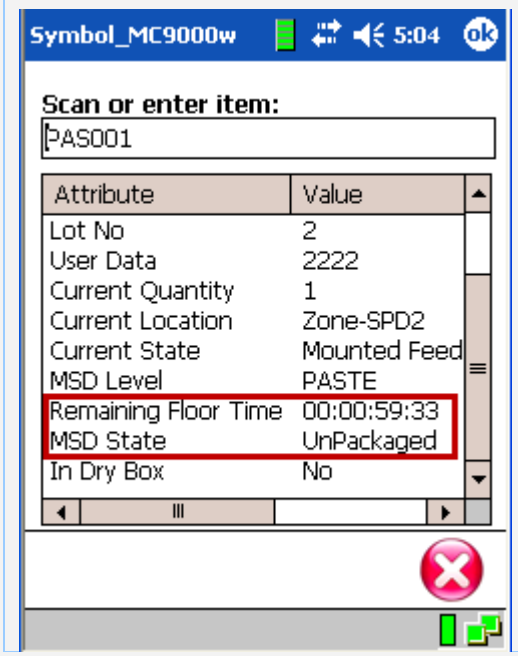
6	After all parts are successfully scanned, parts turn green as shown below:



Note: Immediately when solder paste part is mounted, floor time counter starts to decrease. **Remaining Floor time** can be seen by clicking 

-> 

as shown below:



■ Bulk Tray Replenishment

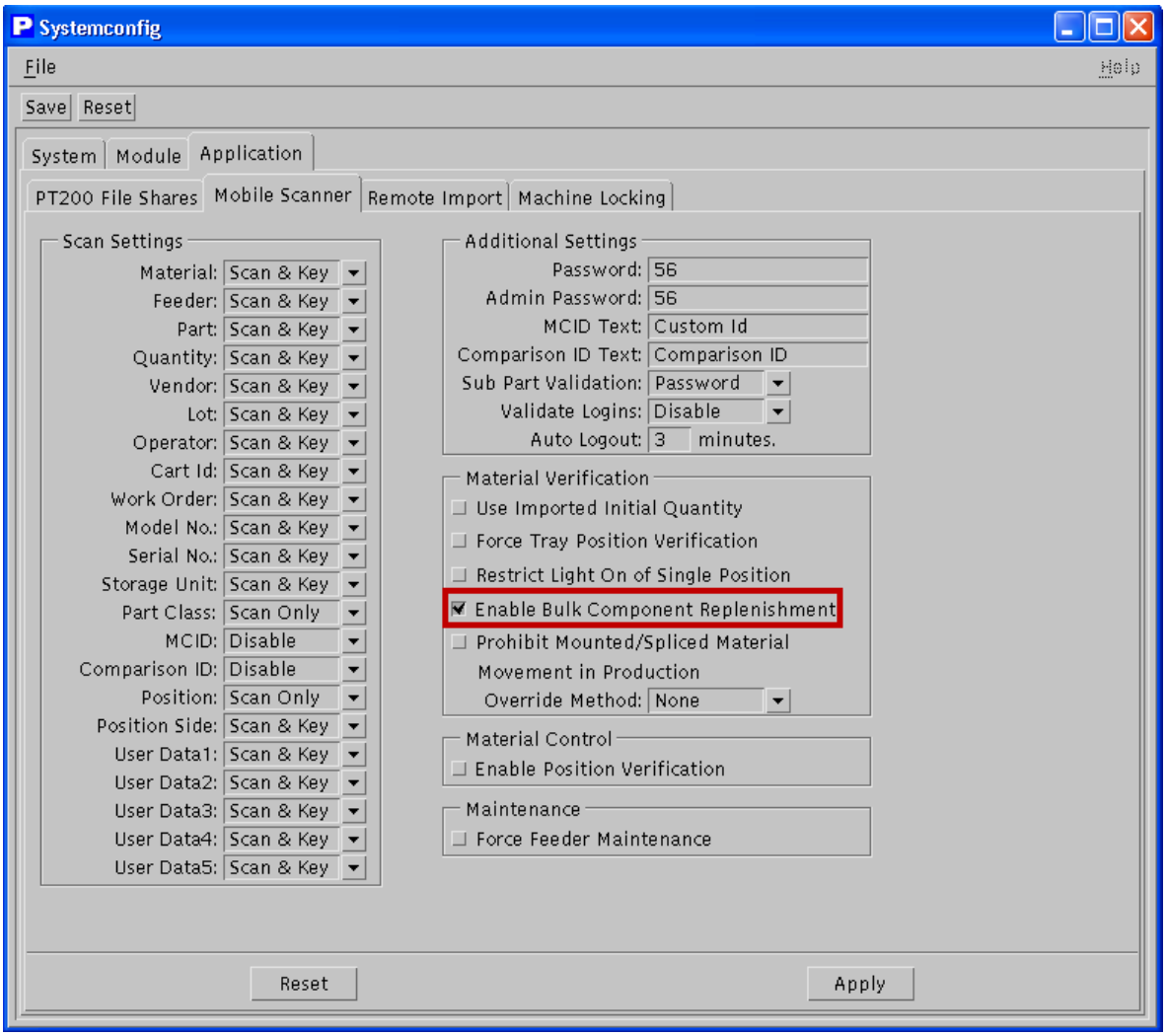
● Process flow

Refer to the action table below that lists the sequence of steps involved when tray part is loaded right up to its exhaust and its subsequent replenish.

Step	Description
1	During initial machine load, the operator will scan the bag followed by part number. The scanner will then request the tray size (components per tray). The mobile scanner will then present the slots expecting that component. Operator will then be able to confirm all locations with Yes/No type response.
2	As a tray exhausts, PanaCIM will be informed and the machine will attempt to move the tray to the tray exhaust slot if it is not already occupied. *Only when [Tray supply unit use] is set to ON on the machine side, the target tray will be moved to the tray exhaust slot. Exhausts will queue up as they happen and have not been replenished.
3	At the point, the operator decides to replenish the machine trays, he will choose the **Exhaust** function on the scanner, scan the bulk material ID and will be presented with a list of slots that are exhausted corresponding to that bulk material ID.
4	The operator will take the tray from the exhausted slot (including the tray exhaust slot) and place new tray components into the slot. Once he has replenished all tray components he will return to the scanner and select all of the slots that were replenished and commit the changes. PanaCIM will send the clear lock message to the machine to release all slot locks. Press the button to supply tray components to the machine. *This is available only when [Tray supply unit use] is set to ON on the machine side.
5	The operator will continue processing the exhaust using old bulk material IDs until the tray components with the old bulk material IDs are completely consumed. When using new bulk material IDs, the operator will choose the exhaust function on the scanner to scan the new bulk material IDs, enter the tray quantity and complete the rest of the slot replenishments.
6	Upon completion of the job, if the tray components remaining on the machine are to be removed from the machine and returned to a storage area, the remaining trays need to be rejoined to their original storage area. For control of materials, scan storage areas and their bulk material IDs, and place them in the storage areas.
7	In performing a setup again using the components remaining after production, use the bulk material IDs along with the scanner. When a setup is performed using remaining components, the screen to enter the remaining quantity of components on the first tray appears. So, enter the quantity of components remaining on the tray first and then select all of the slots where the components will be set up using a bulk material ID. *Those tray components in different positions will be set up in the quantity initially entered at the initial setup.

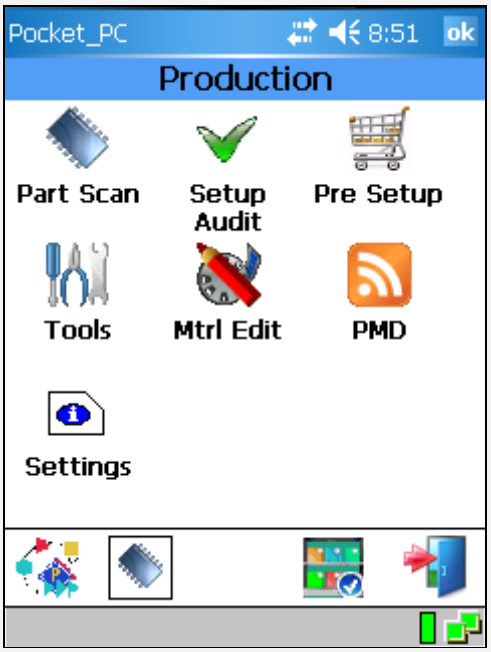
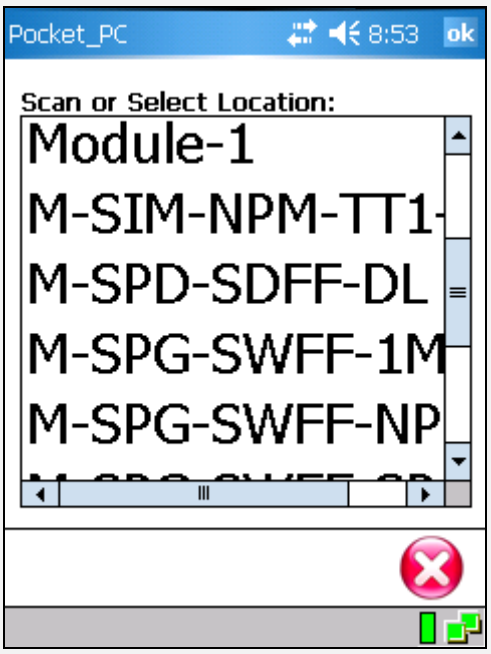
- **Enabling Bulk Component Replenishment**

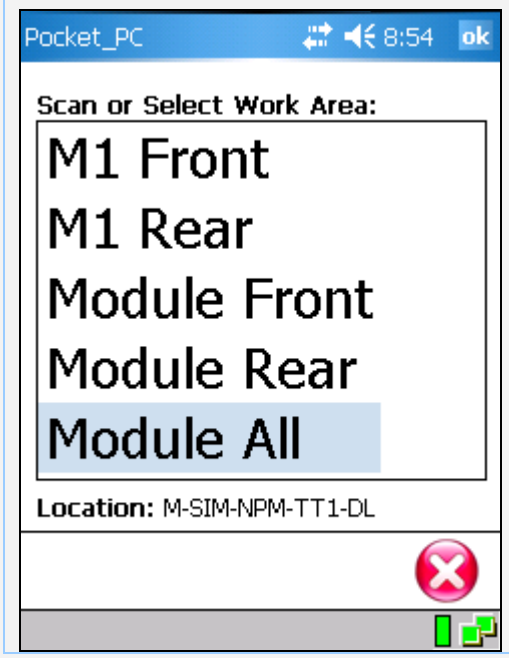
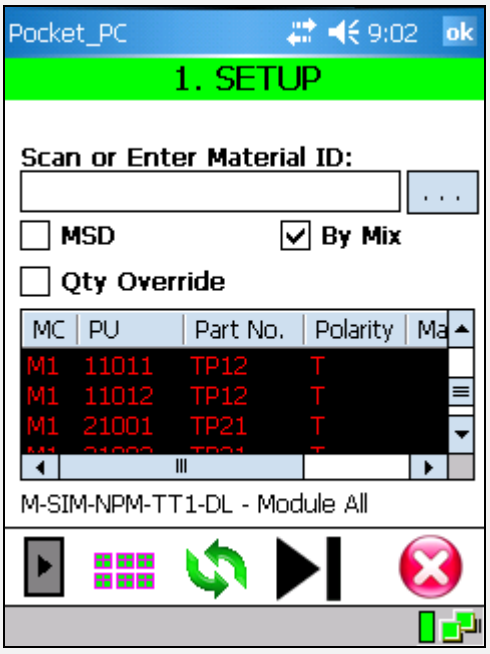
This feature is enabled by checking the **Enable Bulk Component Replenishment** checkbox underneath **Mobile Scanner** tab of the **System Config** dialog as shown below:

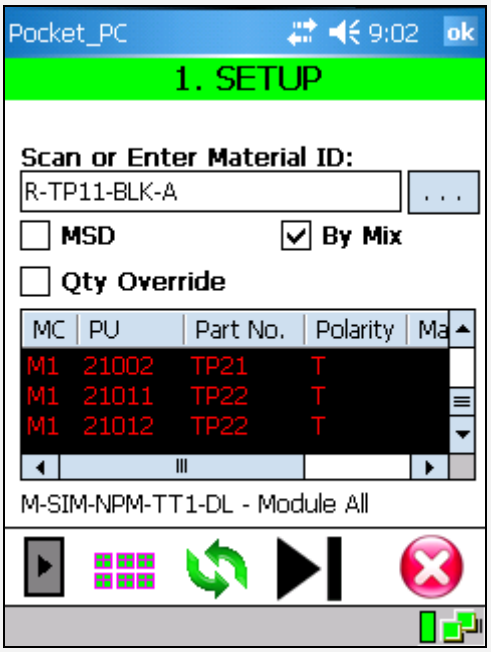
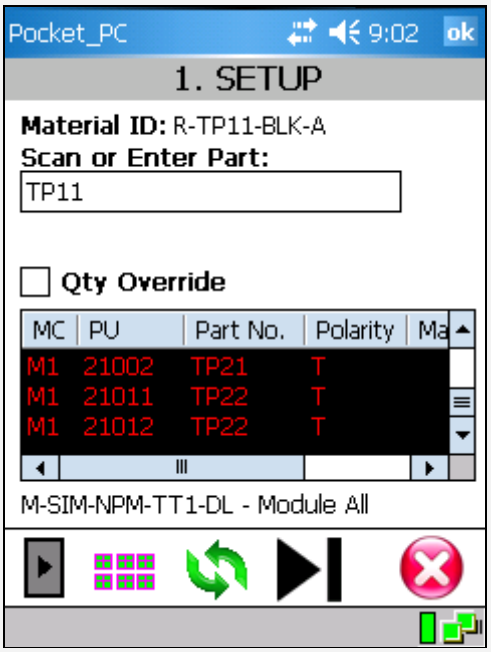


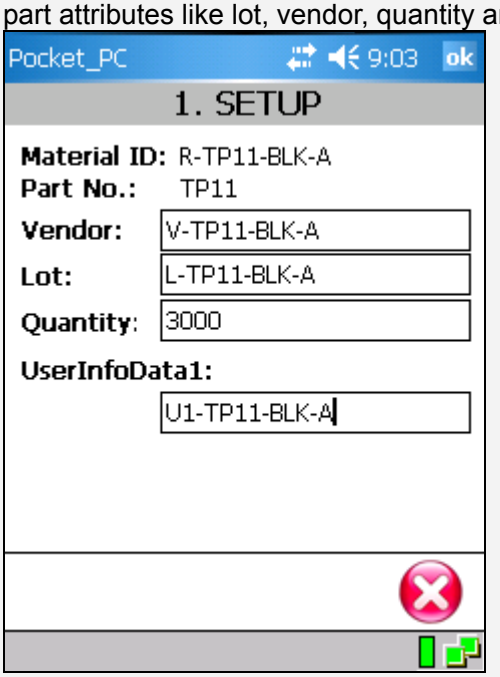
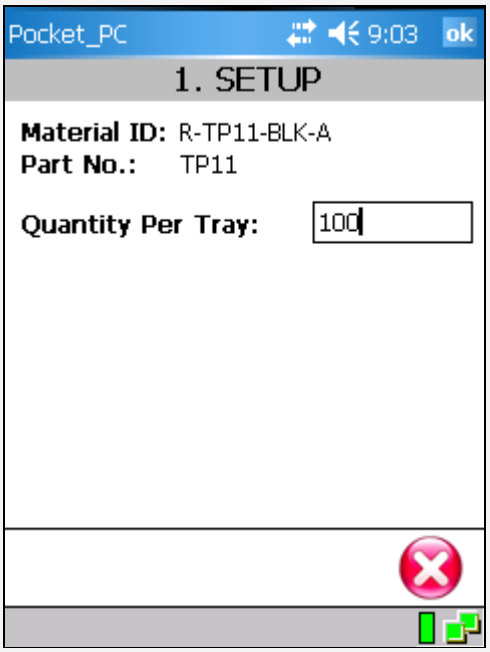
- **Bulk Tray Part Scan (Initial Quantity Use)**

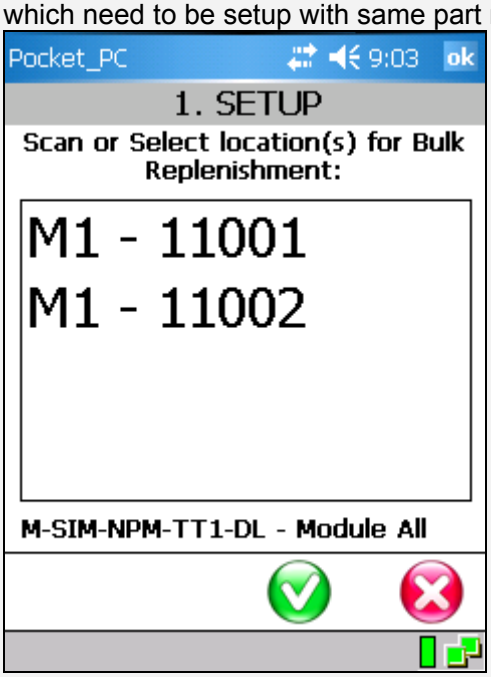
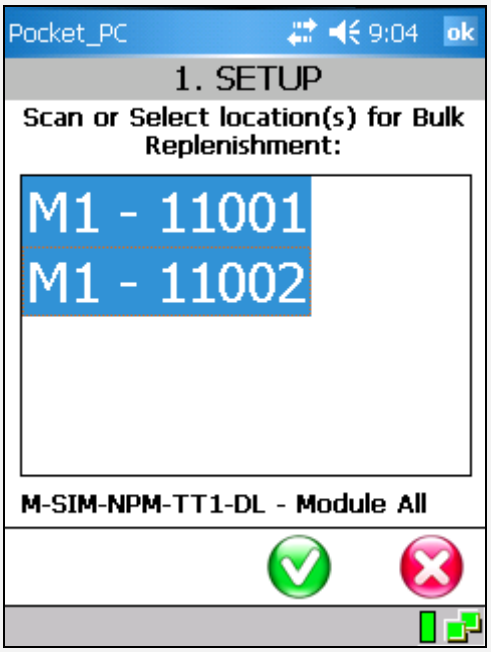
Refer to the action table below for the sequence of steps involved in Bulk Tray Part Scan. Before proceeding with the below mentioned steps, operator has to physically load the tray locations with components.

Step	Procedure
1	From the **Production** main menu, select **Part Scan** 
2	After Selecting **Part Scan**, following screen appears to select a module. 

3	After selecting module name, following screen appears to select a work area: 
4	After selecting work area, following screen appears in order to perform setup. 

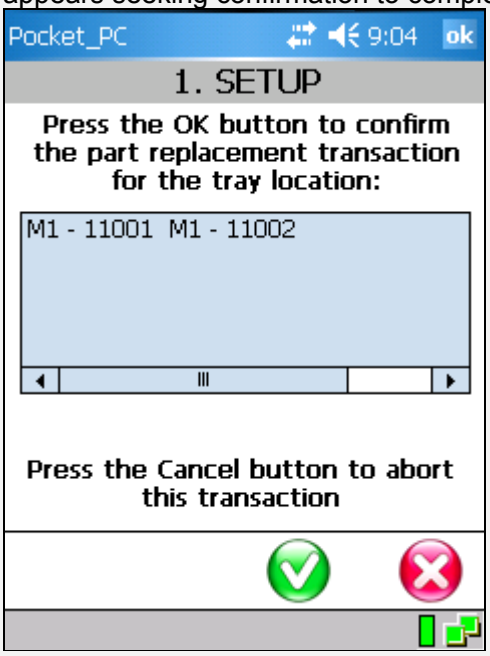
5	Scan or enter material ID of the bulk tray components. This is shown below.  This ID is the bulk material ID for the origin of a tray part and all placements of this tray part will ultimately be traced back to this bag ID /bulk material ID.
6	After scanning or entering bulk material ID, following screen appears to scan or enter tray part number: 

7	After scanning or entering tray part number, following screen appears to enter part attributes like lot, vendor, quantity and user data: 
8	After tray part attributes are entered, following screen appears to enter components per tray:  **Note:** ・This new input screen will be for tray parts only, and will be utilized only in MC mode to obtain the tray pallet component count size. ・This is for defining the component quantity per tray slot to be used. ・This will not be displayed when exhaustion of tray parts occurs in a slot and the slot is replenished. It is because the per-tray-slot quantity entered at the time of setup is maintained.

9	After entering quantity per tray, following screen appears listing tray locations which need to be setup with same part number: 
10	Select one or more tray locations loaded with same part for which setup needs to be performed and click  . When multiple tray locations are selected, it is highlighted as shown below:  **Note:** All the two locations are selected.

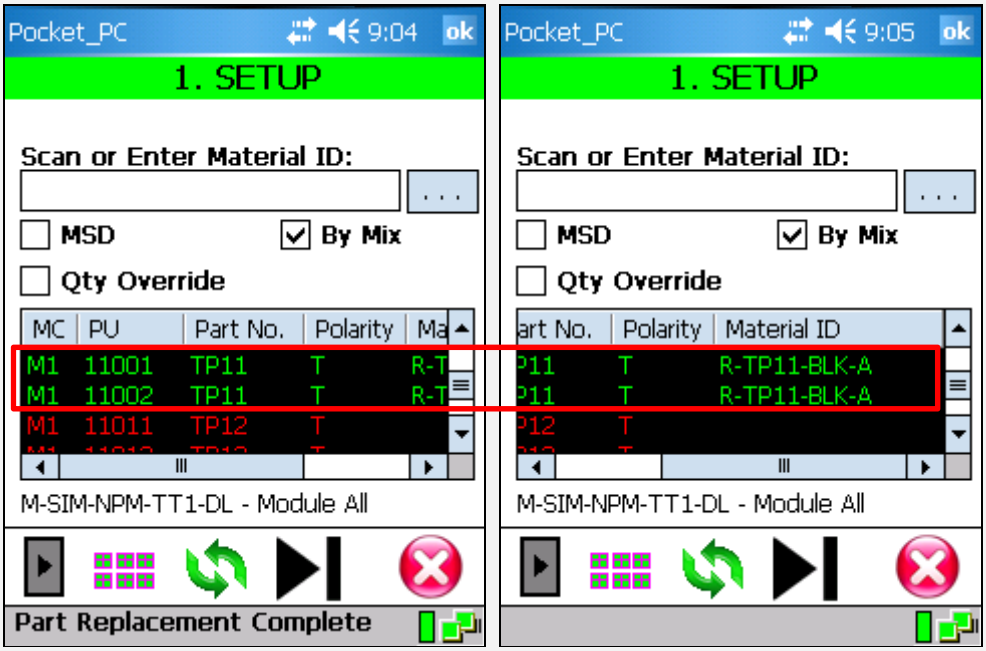
11

After selecting one or more tray locations and clicking  , following screen appears seeking confirmation to complete setup for selected locations:



12

Click  in the screen shown in step 11 to complete the setup. When setup is completed for selected tray locations, the parts in the list turn green as shown below:



* The above captured screens show the entire screen by scrolling horizontally.

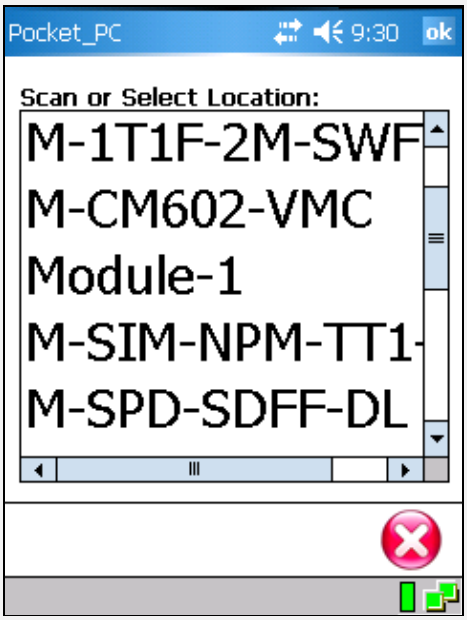
13	Repeat the steps mentioned above until setup is completed for all tray parts. When setup is completed for all tray parts, the screen changes as follows:

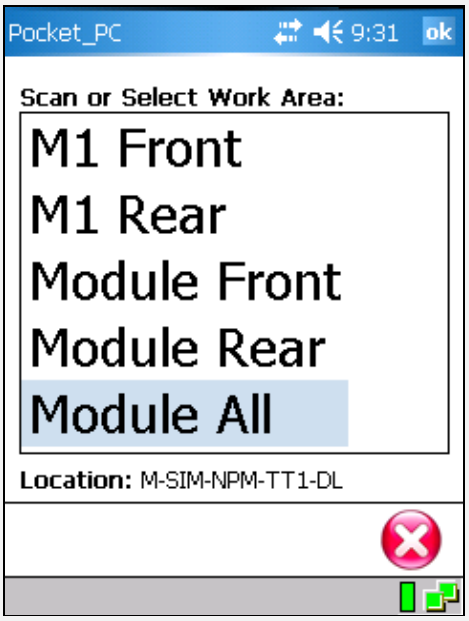
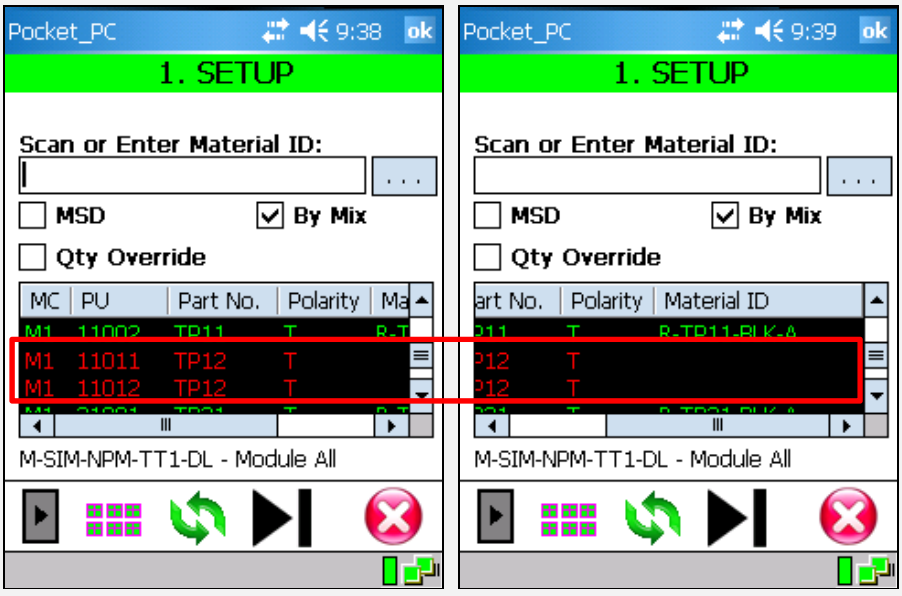


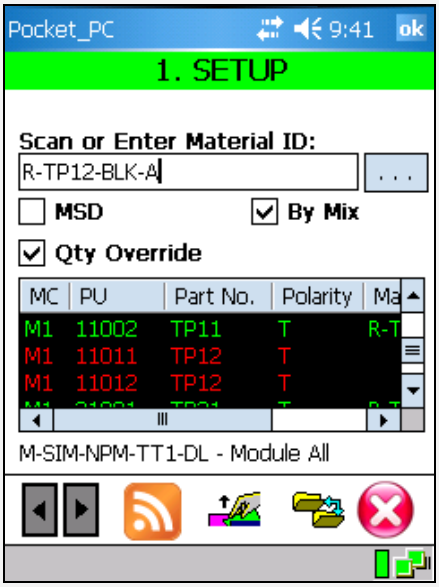
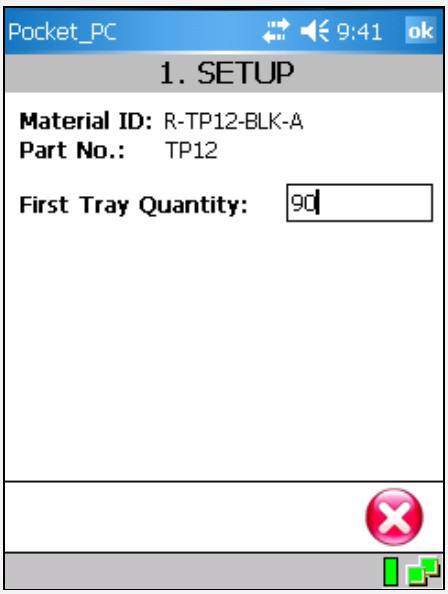
* The above captured screens show the entire screen by scrolling vertically and horizontally.

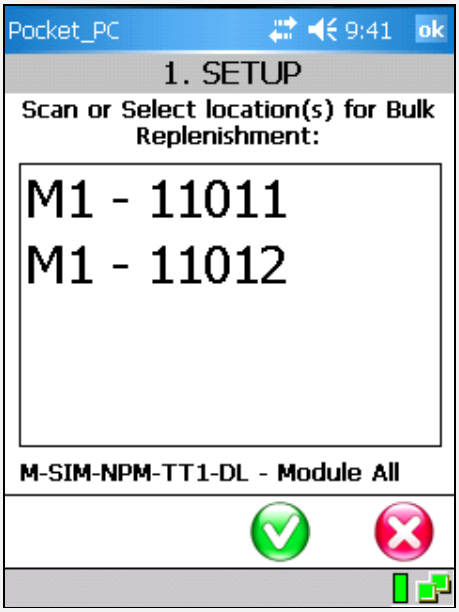
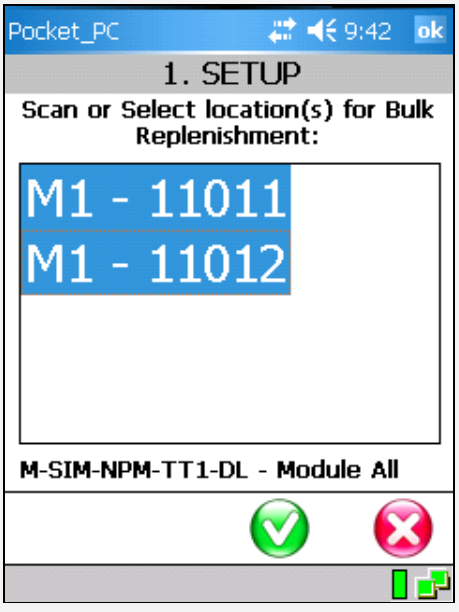
- **Bulk Tray Part Scan (Remaining Quantity Use)**

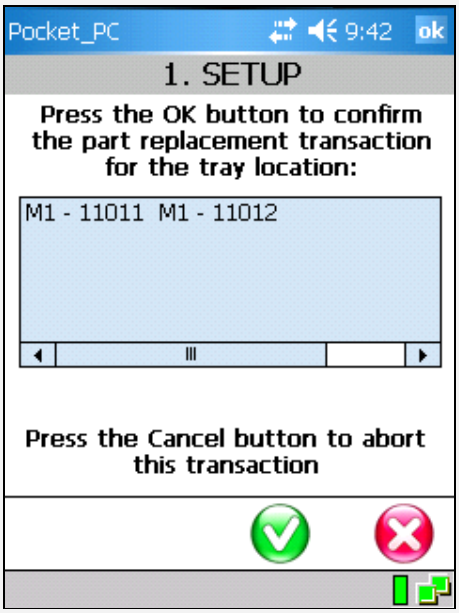
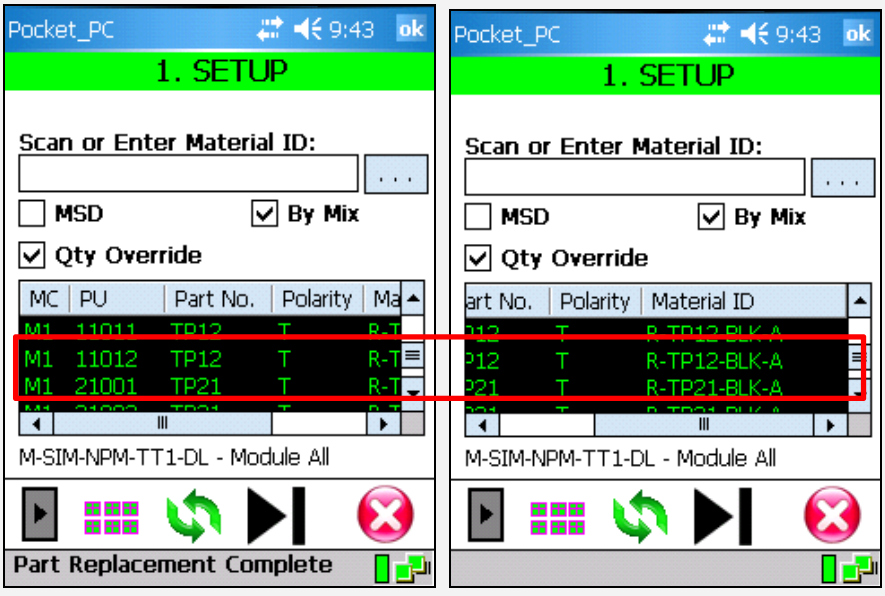
In reference to the table below, perform a setup task using the remaining quantity of bulk tray components on the machine. Additionally, change / adjust the X / Y value so that the remaining quantity set up on the [ ] screen before the start of production remains the same.

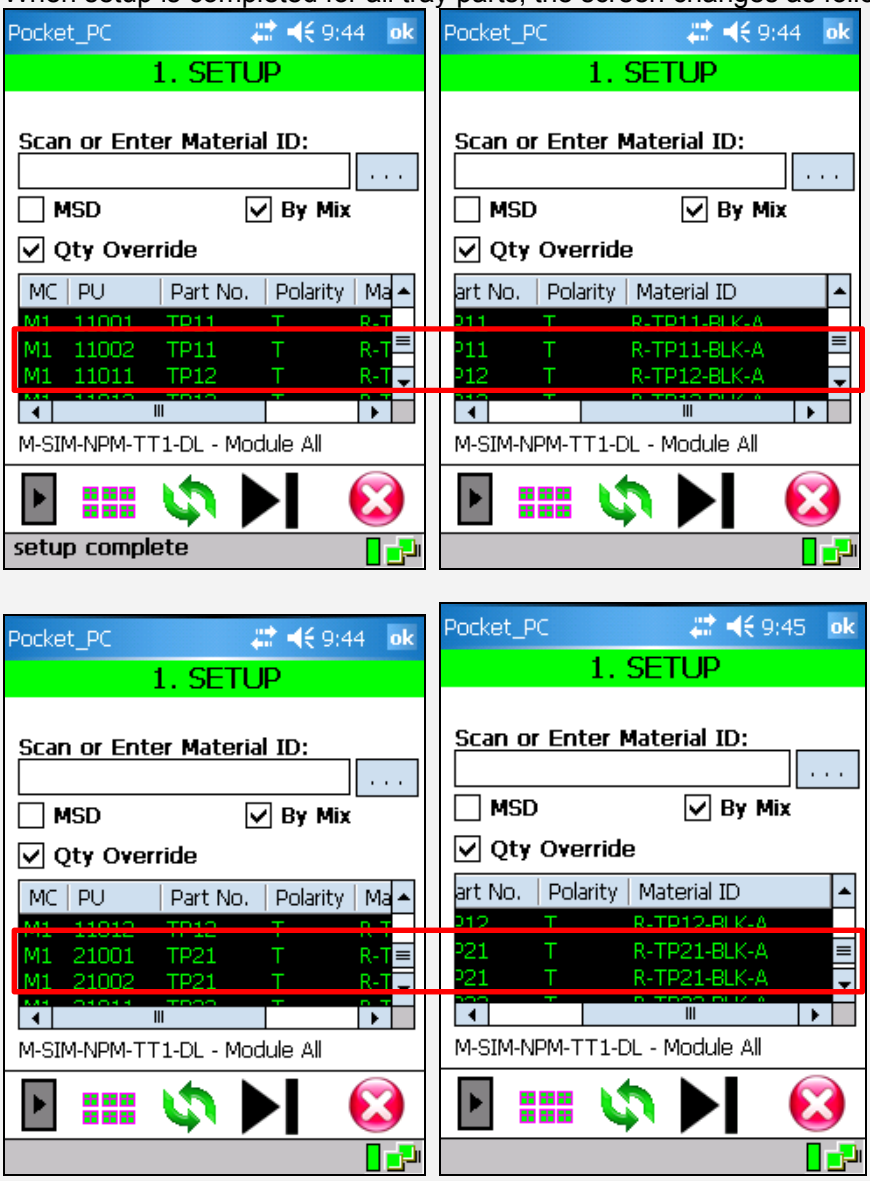
Step	Procedure
1	On the **Production** main menu, select Part Scan. 
2	Select a module name on the screen which is displayed after **Part Scan** is selected. 

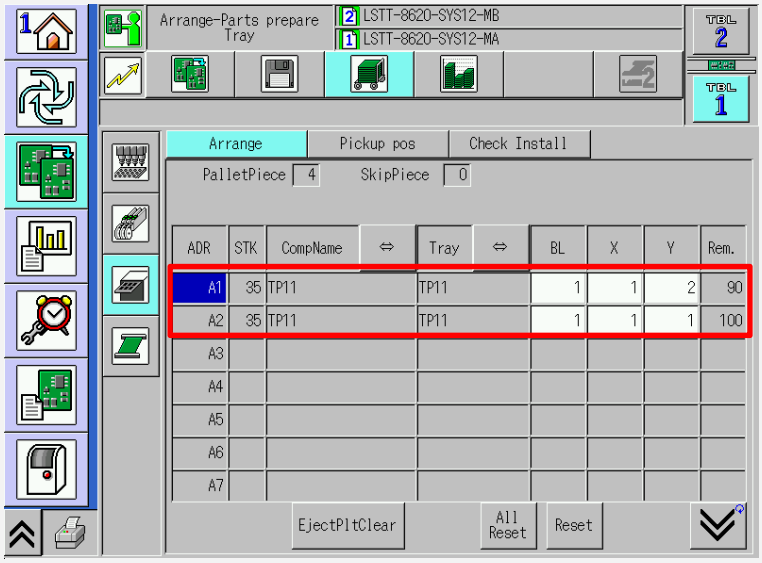
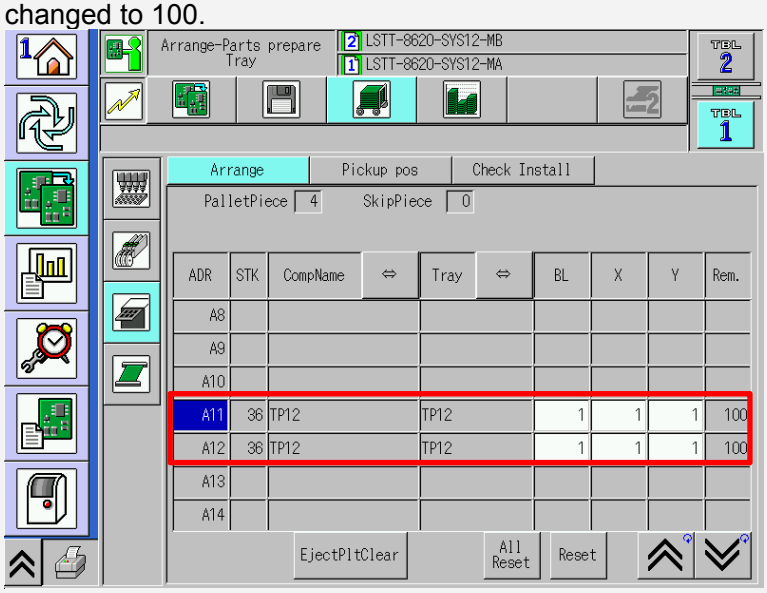
Step	Procedure
3	Select a work area on the screen which is displayed after the module name is selected. 
4	After the work area is selected the SETUP screen is displayed.  * The above captured screens show the entire screen by scrolling horizontally.

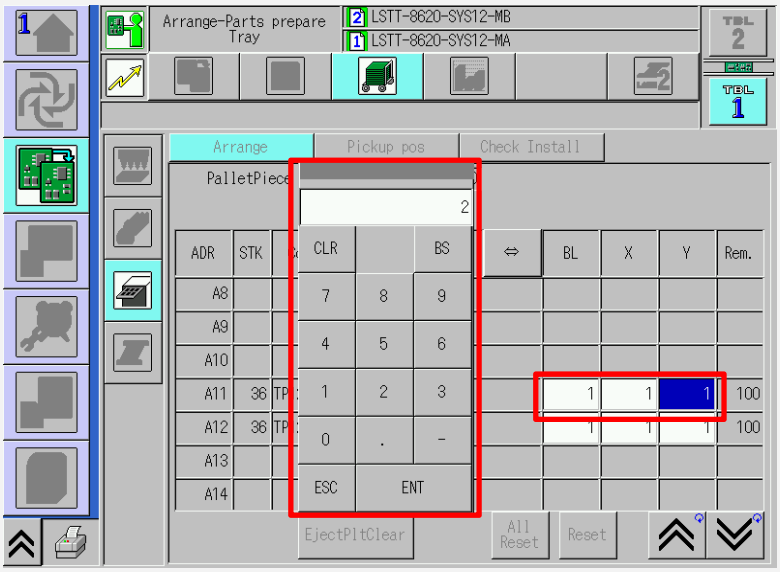
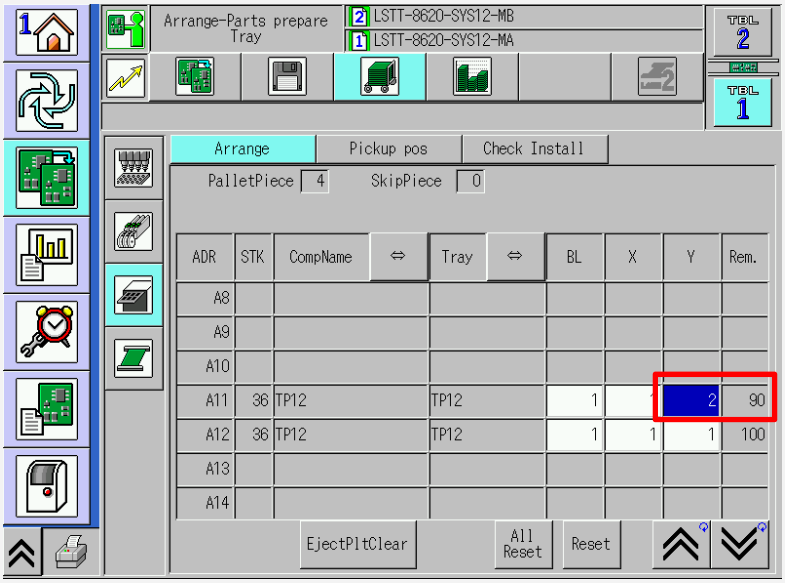
Step	Procedure
5	Mark the checkbox of **Qty Override** and scan or type the batch material ID of the bulk tray component. *Type the batch material ID used in production before product changeover.  The batch material ID is the batch management ID of the bulk tray component, therefore, all placement points on this bulk tray component can be traced by the batch material ID.
6	On the screen below, type the number of components actually remained on the pallet as first tray quantity set to the machine.  **NOTE)** ・This new input screen will be for tray parts only, and will be utilized only in MC mode to obtain the tray pallet component count size. ・This is only intended for tray components in first locations in tray slots. ・This is used to perform a setup using the quantity of components actually remaining on the tray. ・This will not be displayed when exhaustion of tray components occurs in a slot and the slot is replenished. It is because the per-tray-slot quantity entered at the time of setup is maintained.

Step	Procedure
7	After entering **First Tray Quantity**, the following screen appears listing tray locations which need to be setup with same part number. 
8	Select one or more tray locations loaded with same part for which setup needs to be performed and click  . When multiple tray locations are selected, it is highlighted as shown below:  **NOTE)** All the two locations are selected.

Step	Procedure
9	After selecting tray locations and clicking  , following screen appears seeking confirmation to complete setup for selected locations: 
10	Click  in the screen shown in step 9 to complete the setup. When setup is completed for selected tray locations, the parts in the list turn green as shown below:  *The above captured screens show the entire screen by scrolling horizontally.

Step	Procedure
11	Repeat the steps mentioned above until setup is completed for all tray parts. When setup is completed for all tray parts, the screen changes as follows:  *The above captured screens show the entire screen by scrolling vertically and horizontally.

Step	Procedure
12	Select [Arrange] - [Parts prepare] - [Tray] for the machine, and check the arrangement and remaining quantity of tray components after product changeover. **<u>＜For the same tray component arrangement＞</u>** In the following example, the remaining tray component (TP11) quantity is kept at 90.  **<u>＜For different tray component arrangement＞</u>** After a changeover (production data where tray components are arranged in different slot positions), the remaining tray component quantity is reset to the default value. *Note that it remains unchanged even if it is verified by using the scanner (Bulk Tray Part Scan (Remaining Quantity Use))! In the following example, the remaining tray component (TP12) quantity is changed to 100. 

Step	Procedure
13	Change the remaining quantity of tray components on the machine side so that it will be equal to the quantity (example: 90) of components actually remaining on the tray. *Make sure that it is changed to the same value as the remaining quantity entered during verification using the scanner (Bulk Tray Part Scan (Remaining Quantity Use). 1) Select which value (BL, X or Y) to change, and in the pop-up window, press **ENT** after entering a value. (As an example, enter [2] in [Y])  2) As the pop-up window disappears and the entered value takes the place, the remaining quantity is recalculated and then displayed. (As an example, [Y] changes to [2] and the remaining quantity is updated to [90]) 

Auto Bulk Component Replenishment

When a bulk tray location has exhausted, it may be replenished with another tray from the bulk package as long as there is a current quantity in the bulk material. This may be accomplished without user scanning intervention. The machine sends real time events when a component has been replenished on the elevator. If a scan is not performed and there is a remaining count in the bulk master material when this event is detected, the host performs an auto replenish of that tray location.

The auto replenish of the tray location presumes that there has not been any user scan operation where a new bulk part is not in use for this location. If the user decides to use a different bulk material for this location, then there will not be an auto replenish.

The requirement for an auto replenish is the following:
1. Tray location must have been previously setup using a bulk material.
2. Tray location must be in exhaust state.
3. The appropriate real time event is received for that tray during exhaust state.

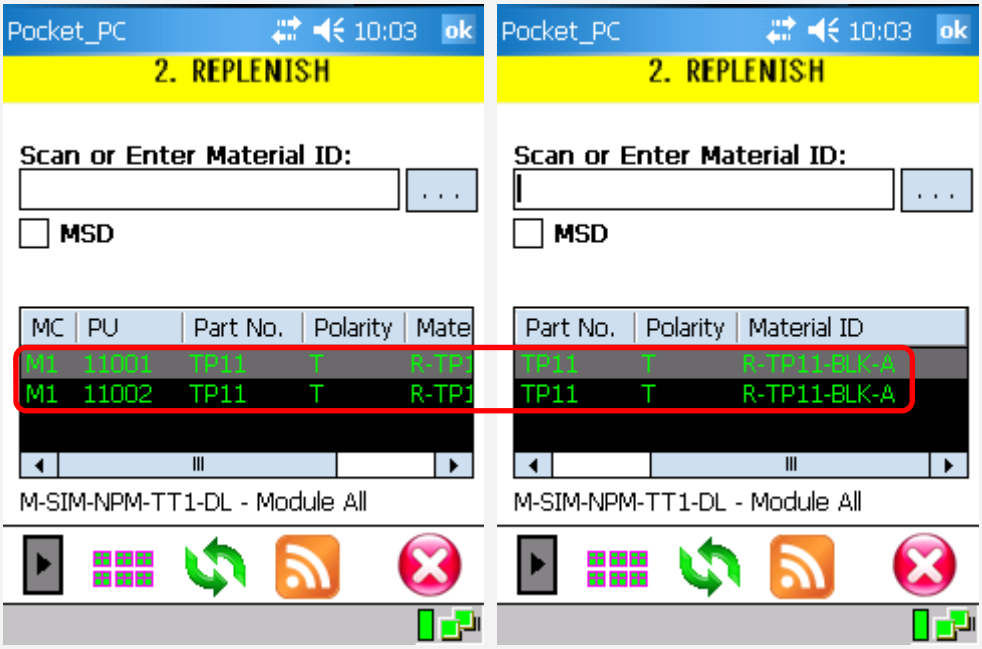
When the appropriate real time event has been received and the above conditions exist, then the PanaCIM will perform an auto replenish and clear out the exhaust status of that tray location. If tray locking is enabled, then normal processing to release the tray will be performed.

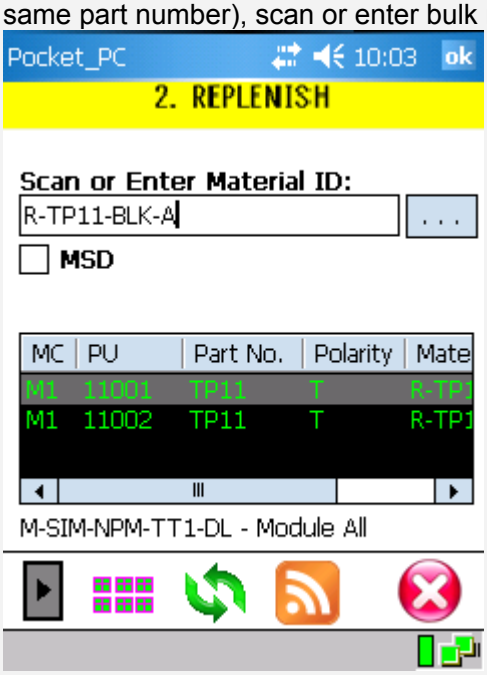
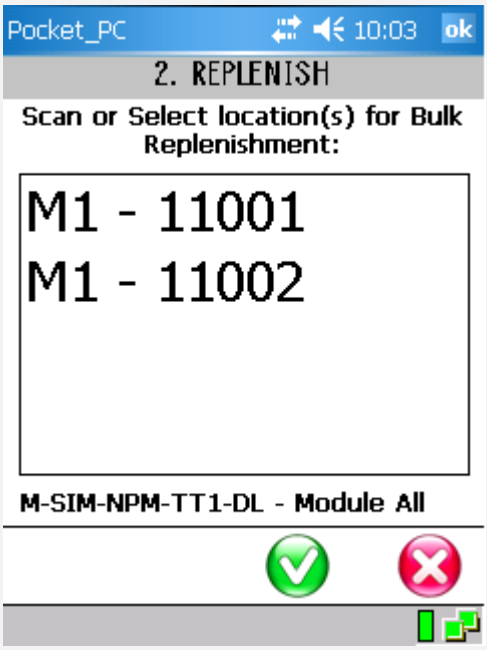
Following table lists the real time event for BM and CM/NPM series machines that trigger auto replenish.

	PanaCIM Event ID	PanaCIM Event Name	Machine Event
BM series	121	EE_KX_OCCURED_PARTS_EMPTY	MEC_OCCURED_PARTS_EMPTY
CM/NPM series	120	EE_KX_CANCEL_PARTS_EMPTY	Event Code = 03 Event Sub Code = 35000 Exhaust Code = 00

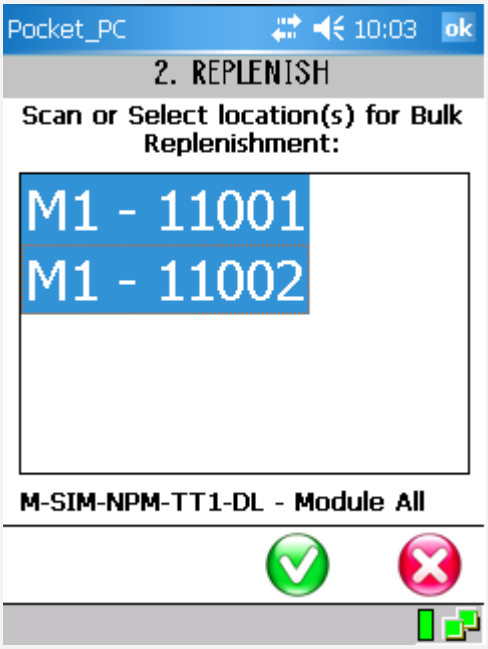
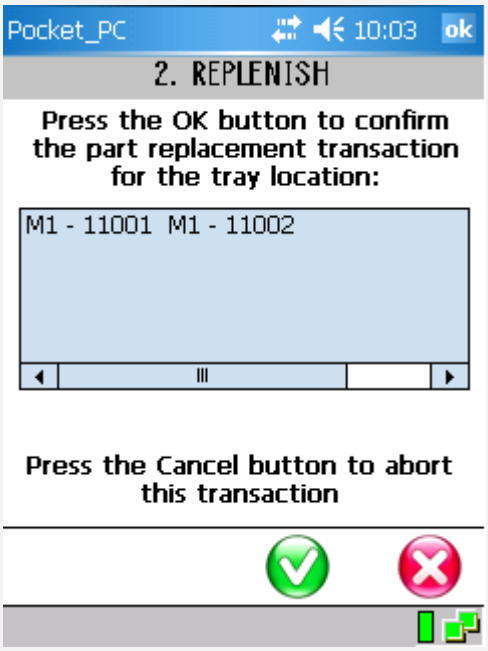
● Part exhaustion

Refer to the table below to replenish tray parts after the parts in the tray are exhausted.

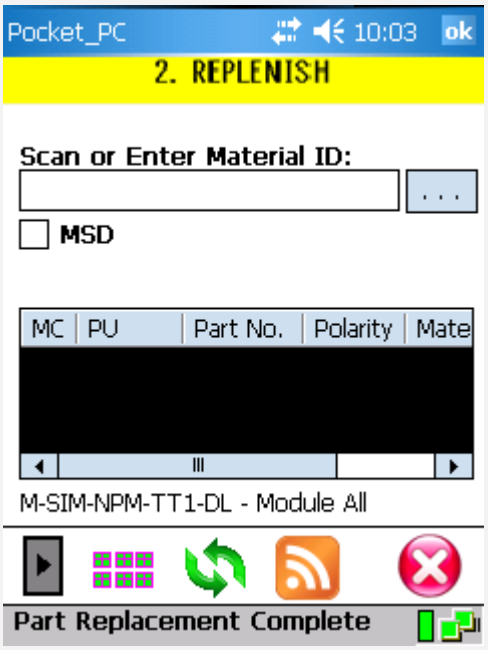
Step	Procedure
1	Select **Part Scan** from the **Production** main menu, followed by selection of module and work area. Name of the last performed transaction appears at the top. 
2	Change the transaction type to **Replenish** to replenish exhausted tray locations. When **Replenish** transaction type is activated, locations at which exhaust occurs are displayed  * The above captured screens show the entire screen by scrolling horizontally. As shown in the screen above, there are two tray locations at which exhausted part needs to be replenished. The Material ID column contains a bulk material ID (bag ID).

	The Material ID column is only displayed in the MC mode, and so is not displayed in the MV mode.
3	To replenish exhausted tray locations belonging to a group (which contains same part number), scan or enter bulk material ID as shown below:  **Note:** The important thing to note here is that operator selects a tray slot location before scanning used Bulk Material ID. TP11 is the bulk material ID identified by material ID R-TP11-BLK-A.
4	After scanning the used bulk material ID shown in step 3, following screen appears listing additional locations belonging to a group (containing same part number) which need to be replenished: 

+

5	Select one or more tray locations to replenish parts at these exhausted locations and click  . When multiple tray locations are selected, it is highlighted as shown below:  Each time a tray location is replenished, tray quantity size is deducted from the master bulk.
6	After selecting one or more locations from the screen shown in step 5 and clicking  , following screen appears in order to confirm replenishing part at the locations selected in step 5.  **Note:** If you scan or enter material ID corresponding to an individual tray location in step 3, you will not get the screens shown in step 4 or 5 but a screen will be displayed to complete part replacement transaction for a single location as shown below:

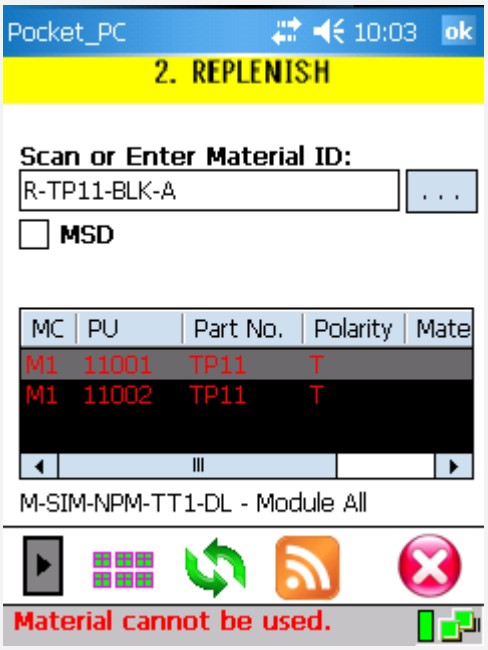
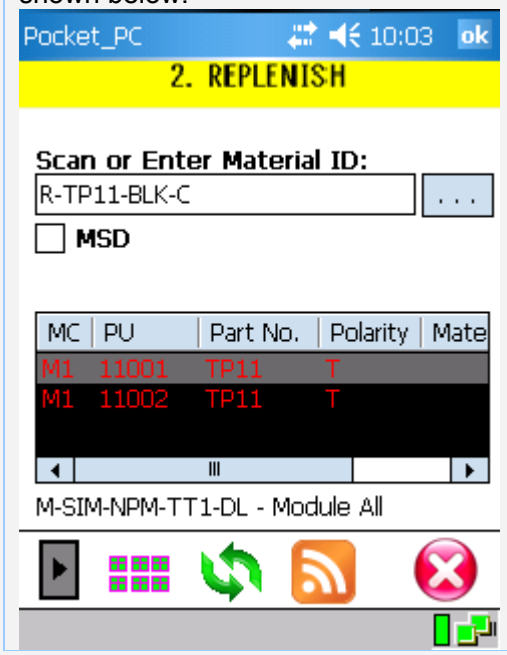

Pocket_PC
10:03
ok
2. REPLENISH
Scan or Enter Material ID:
R-TP11-BLK-A
. . .
MSD
MC
PU
Part No.
Polarity
Mate
M1
11002
TP11
T
R-TP1
M-SIM-NPM-TT1-DL - Module All
Pocket_PC
10:03
ok
2. REPLENISH
Press the OK button to confirm the part replacement transaction for the tray location:
M1 - 11002
- or -
Press the Cancel button to abort this transaction

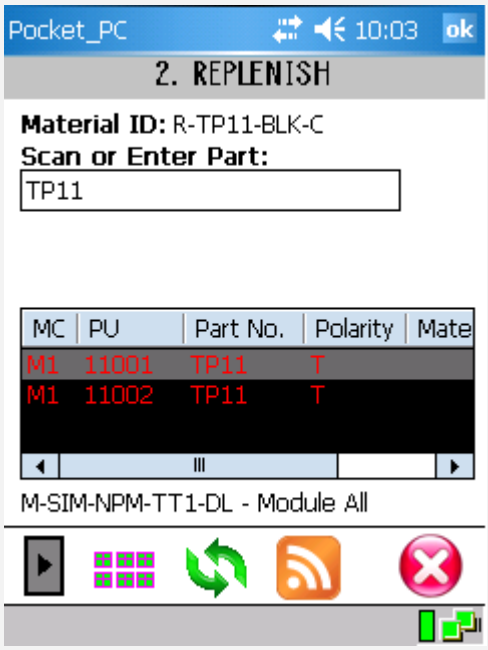
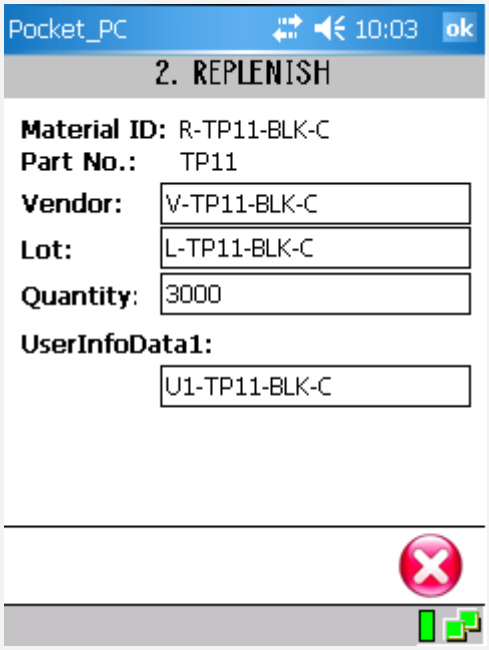
7	Click  to complete the part replenishment operation for multiple tray locations containing same part. Once all locations have been replenished, these are removed from the grid (underneath **MSD** checkbox) as shown in the following screen 

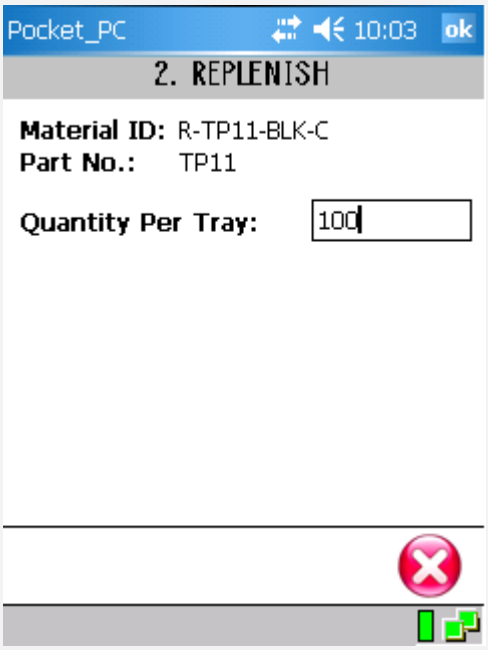
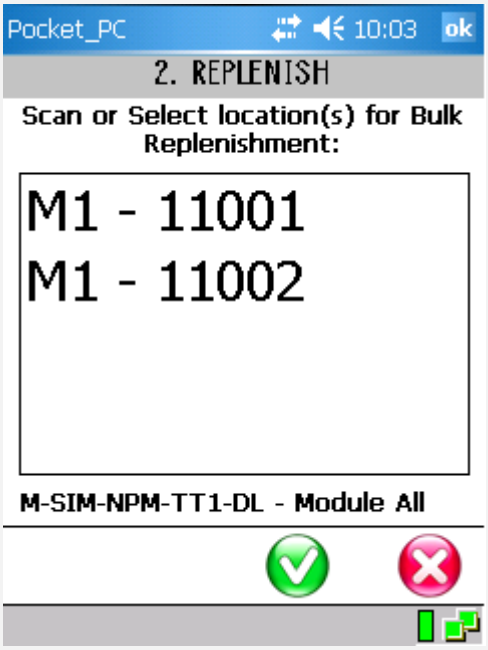
- **Tray Part Replenish When Bulk Material ID is consumed**

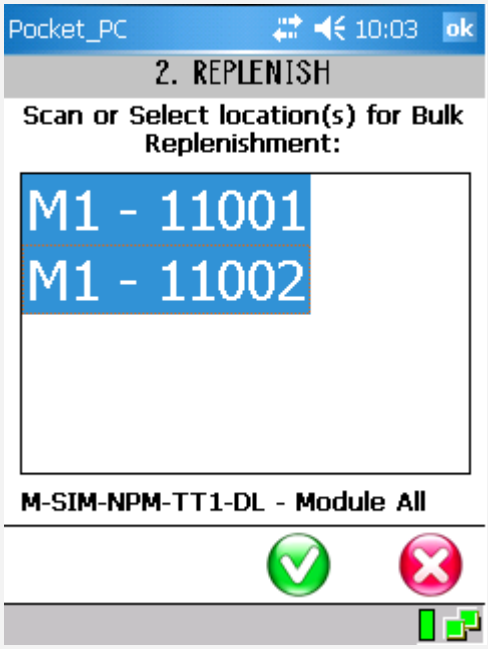
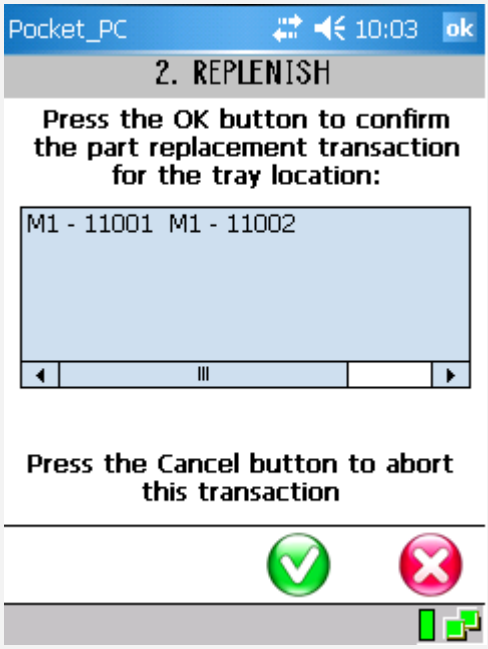
  Part exhaustion can be reset with the same bulk material ID until the old bag is completely consumed, but once the bag is consumed, the bulk material ID can no longer be used.
  In such a case, it is necessary to enter a new bulk material ID and tray quantity to reset part exhaustion.

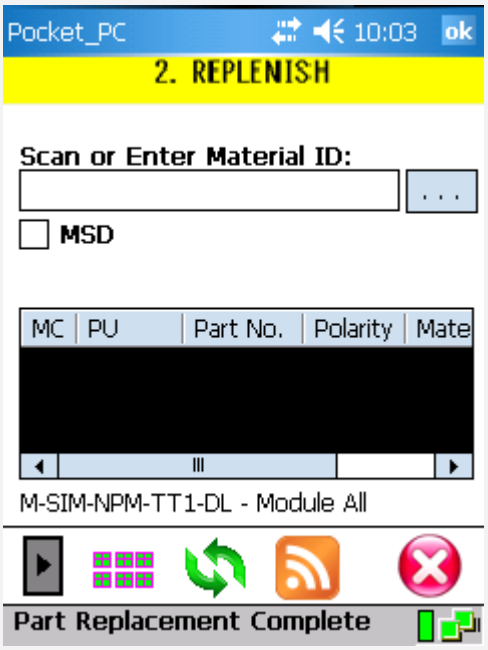
  Refer to the action table below that captures this transition.

Step	Procedure
1	Once old bag is completely consumed, you will get a message **"Empty material cannot be used"** as shown below: 
2	At this juncture, new bulk material ID needs to be scanned by the operator as shown below: 

3	After scanning the new bulk material ID, following screen appears to scan/enter part number. 
4	After scanning/entering part number, following screen appears to enter part attributes: 

5	After specifying part attributes, following screen appears to enter tray quantity----size (components per tray). 
6	After entering tray quantity, following screen appears showing locations for Bulk replenishment. These locations contain same part number. 

7	Select one or more tray locations to replenish parts at these exhausted locations and click  . When multiple tray locations are selected, it is highlighted as shown below: 
8	The following screen appears where you should confirm the selected location and click  to complete the replenishment of the selected tray components. 

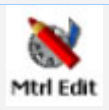
9	Once all locations have been replenished, these are removed from the part exhaust list. 

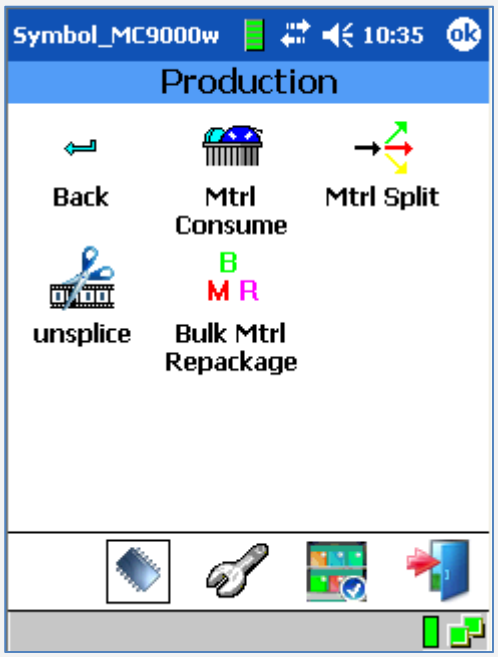
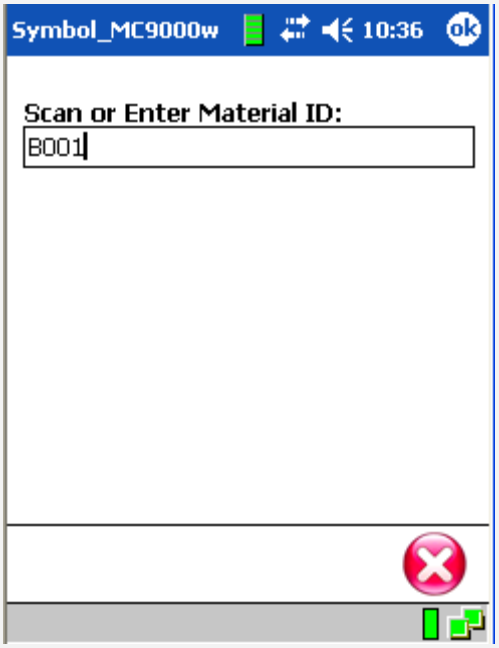
Bulk Material Repackaging

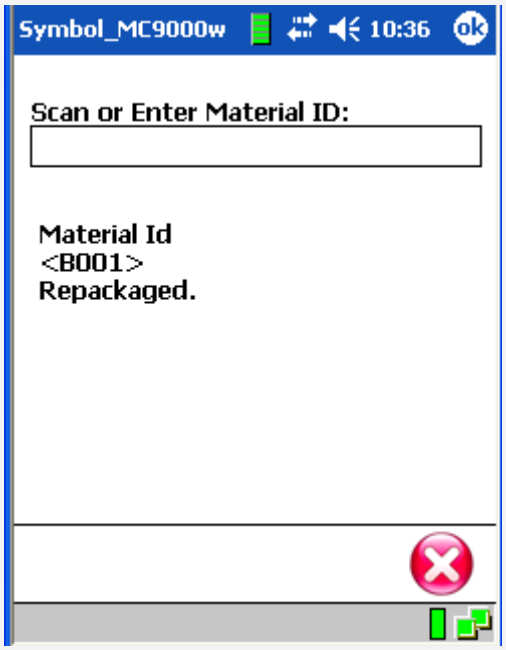
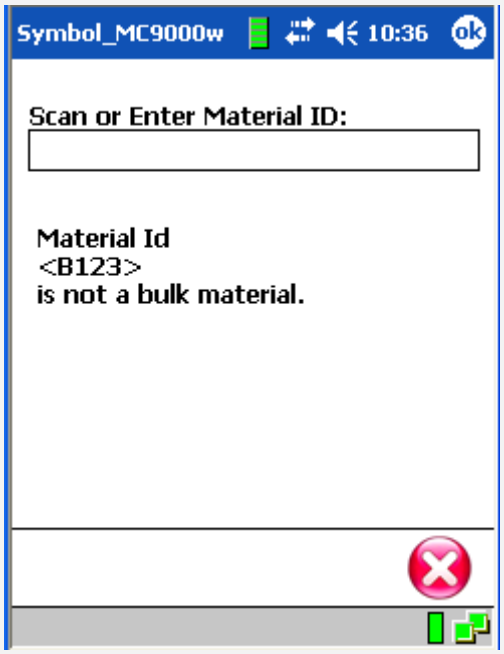
This will be a new function on the mobile scanner to enable the operator to repackage the bulk material. This new function will be available under the Production and Material modes on the scanner. It will also require Material Control license. Operator will enter the original bulk material barcode for the repackaging operation. Bulk material repackaging will add the parts remaining in the different trays to the same bulk material ID so that these parts can be used subsequently.

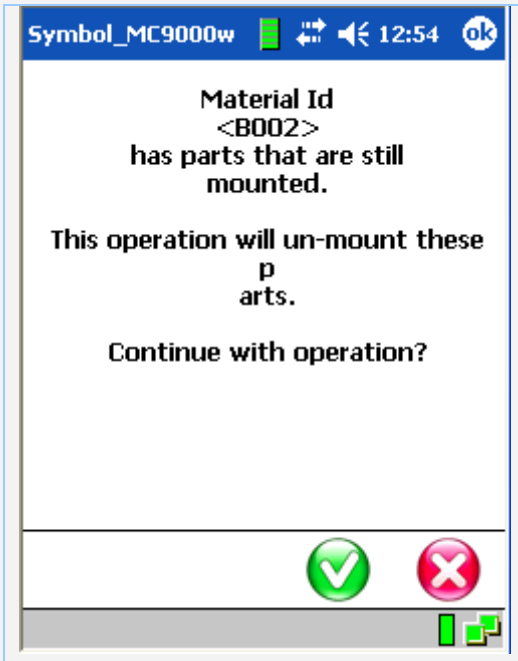
If the operation is performed under the Production menu, then the current location will not be updated. Whatever location the material is at is where it will stay. For Material mode, the location will change to the location selected when invoking this operation.

Refer to the following action below listing steps to perform bulk repackaging.

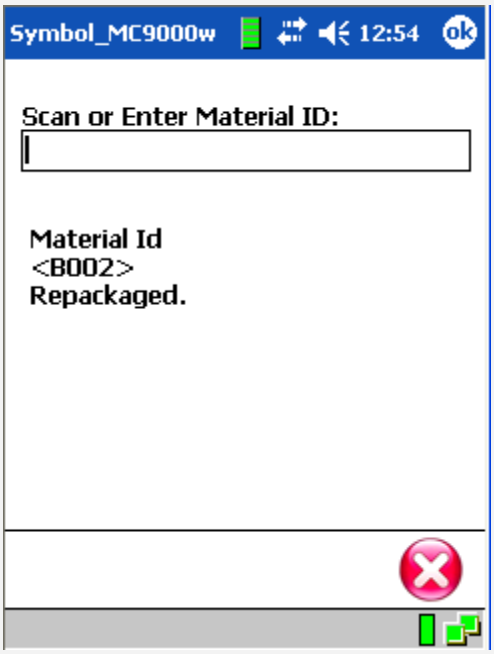
Step	Procedure
1	Select  from the **Production** or **Material** scanner operational modes. 

2	After Selecting **Mtrl Edit**, following screen appears: 
3	Click  . Following screen appears prompting to scan or enter bulk material ID: 

4	After scanning or entering bulk material ID, a message appears stating that bulk material is repackaged:  **Note:** a) If material ID entered is not a bulk material, following screen appears:  b) If bulk material ID scanned/entered is already mounted, you will get the following message:



Click  to complete repackaging the material ID.



- **Bulk Tray Replacement in MV operational mode**

For MV operations the bulk tray replacement is much simpler since there is no component count requirement. The operator will be able to specify a part number and simply replace tray components. All that the host needs to track is the component attributes of the part which is the Part No., Vendor, Lot and User Data. Since there is no uniqueness with the reel data in MV mode, there is no need to manage any linkage between the bulk material and what was put into the individual trays.

When bulk replenishment is performed the host will simply duplicate the reel and associate that reel with each replenished location. There is no need to track any component usage like Material Control.

3.3.3 Other Scan Procedure

■ Scan tray parts

Part information is held in PanaCIM as tray unit does not have a memory to hold tray part information.



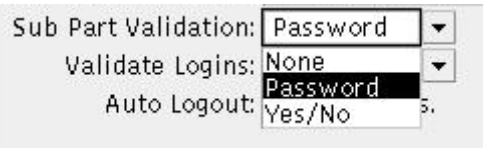
Parts are scanned in the same way as intelligent feeder, but its set check method is different. After scanned tray parts, you will need to check if parts are set on the correct tray.

Step	Procedure
1	Scan tray part number in the same was as reel part scanning.
2	If scanned parts are allocated on multiple tray slots, the following menu appears. Do the following operation. • Select tray position. • Select **Apply**. 
3	Make sure tray is set at the specified slot.

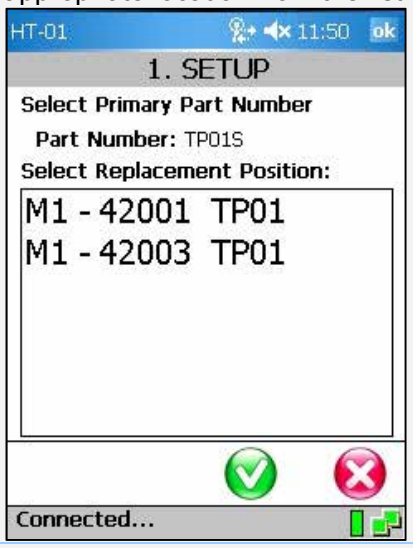
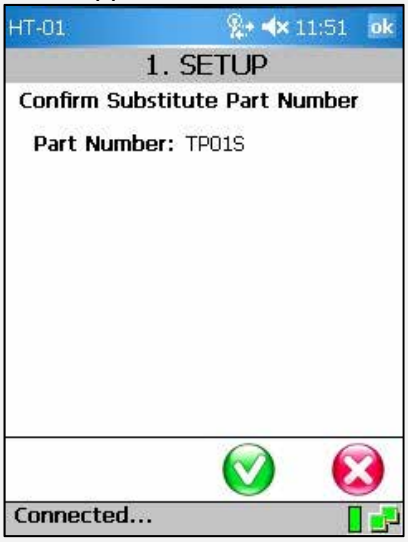
4	Select green check mark  and finish verification. 
5	Press supply finish button on tray and finish tray part supply.

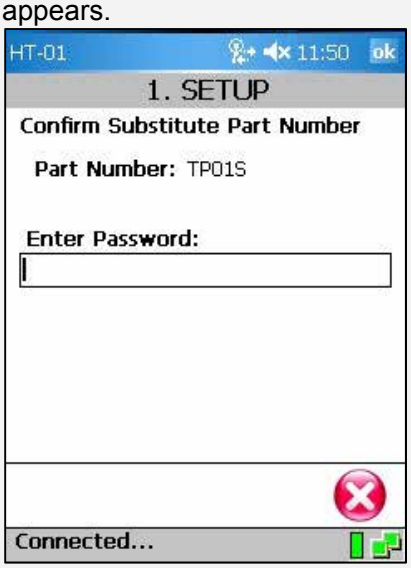
■ Substitute parts number

When parts are not available, you can use other parts. Substitute parts are normally set in PanaPRO system parts data. They can also be added manually to Product Setup or imported from a file. Select a check method for scanning substitute parts on Route Editor.



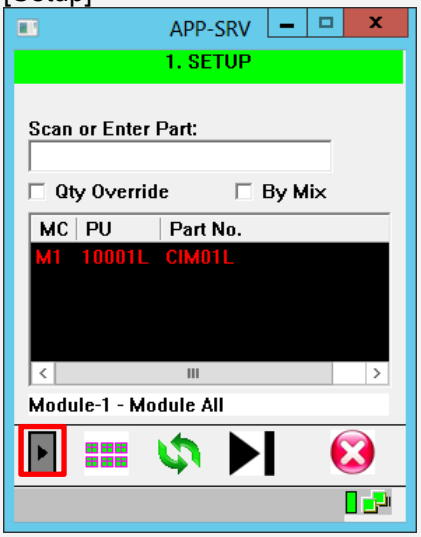
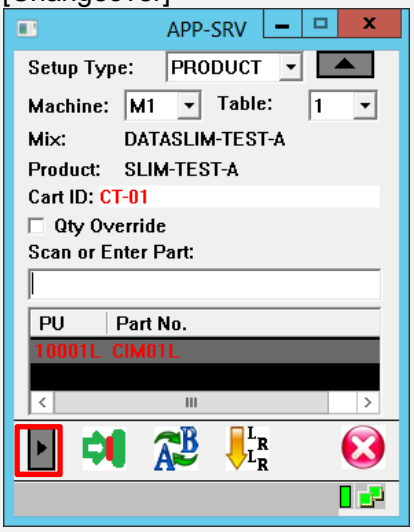
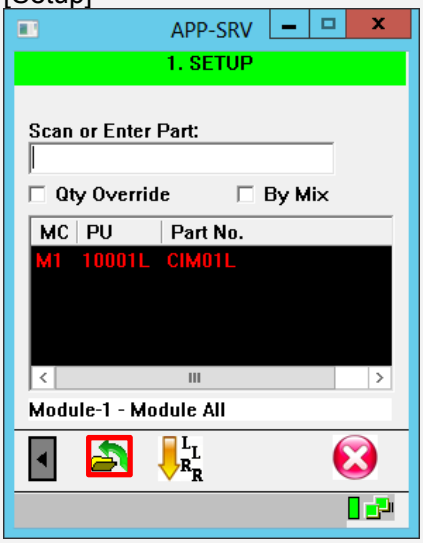
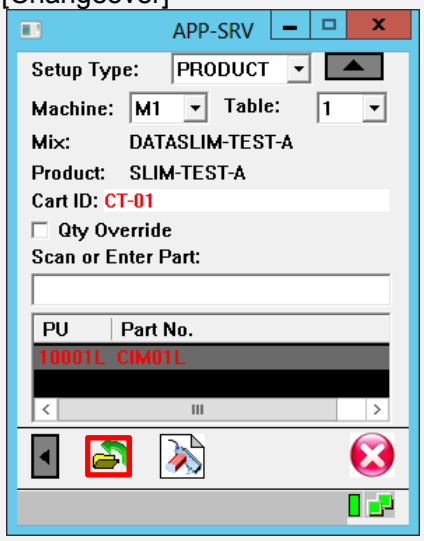
- Non: Allow substitute parts after parts scan
- Password: Allow substitute parts after password
- Yes/No: Allow substitute parts after operator check

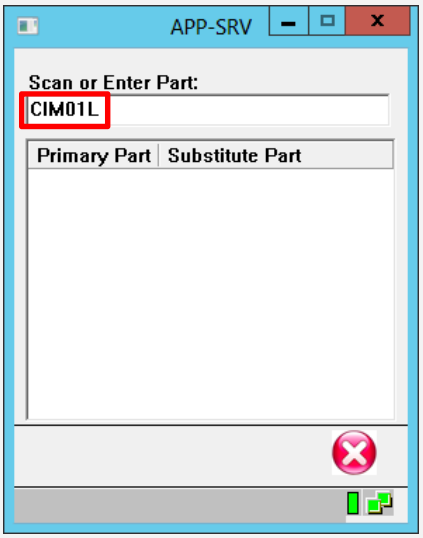
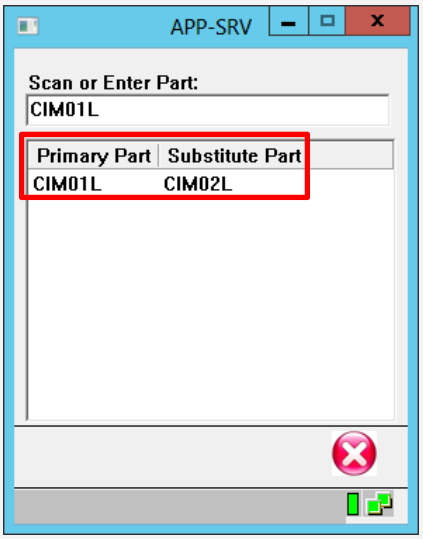
Step	Procedure
1	If scanned parts are set as substitute parts for multiple parts in the setup, select an appropriate location from the list. 
2	If substitute parts scan check method is set to **Yes/No** on Route Editor, the following menu appears.  Select green check mark to allow using substitute parts. Select red × not to allow them.

3	If substitute parts scan check is set to **Password** on Route Editor, the following menu appears.  To use substitute parts, enter the password set on Route Editor.

■ Display of substitute parts on the scanner

It is possible to check the scanner for which substitute parts are included with the desired major part.

Step	Procedure
1	Open the setup or changeover screen. Press the arrow icon at the bottom of the screen to scroll the menu sideways. *The same operation applies to the part supply screen. [Setup]  [Changeover] 
2	Press the  icon. [Setup]  [Changeover] 

3	Enter or scan the desired major part name. 
4	The substitute part information of the desired major part is displayed. 

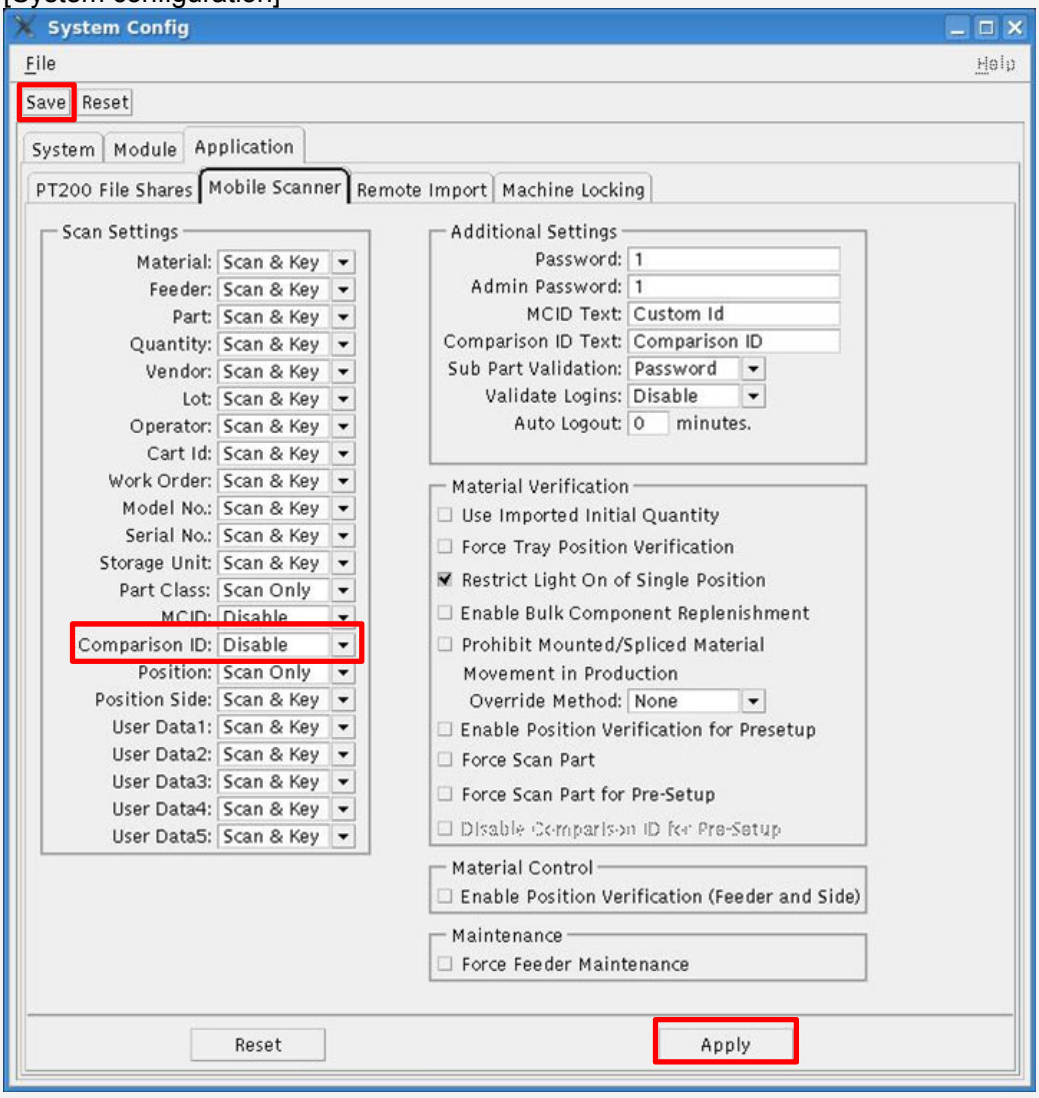
■ Scanning procedure using comparison ID

During setup or changeover using the comparison ID, if the comparison ID does not match the part number, verification will fail. This function avoids verification of parts which are not listed in the reel list (part list in use).

This function can be used in the following case:
*Verification mode of either MV or MC mode is acceptable.

- Setup
- Replenish
- Recheck
- Changeover
- New/old reel comparison

As an example, the step below shows splicing in MV mode.

Step	Procedure
1	Go to **System Configuration**-> **Application** -> **Mobile Scanner**, and change **Comparison ID** to "**Scan Only**". After click **Apply,** click **Save**. [System configuration] 

2	Log in the scanner again and open the SPLICE screen. Scan the comparison ID from the Reel List. *The same operation will be applied on the setup or changeover screen. [Splice]  
3	Scan the Part No corresponding to the comperision ID. *On the part input screen, even if you accidentially scaned the wrong comparision ID, you can scan the correct comperision ID again without doing a screen transition.  
4	Scan supplemental information such as a vendor or lot.
5	Press the feeder button.

= MEMO =

4. FUNCTION - PREVENTION FROM AN INCORRECT SET - PARTS SUPPLY

This chapter explains the Part Supply.

4.1 Overview

There are the following two modes available for the verification system: non-integrated and integrated versions.

Operations for replenish include part exhaust and splicing, and operations for recheck include feeder clog clear and no splicing check error.

Each operation is explained below.

- Part exhaust: Operation done while machine stopped after part exhausted during production. Standard machine setting judges part exhaust after three pickup errors and stops operation. The screen shows the machine slot and parts number.
- Feeder clog clear: Operation to clear part exhaust as feeder clogging when machine stopped after pickup failures. At this point, machine stopped due to the same status as part exhaust.
- Splicing: Operation to join a new reel to the current after splicing or other warning.
- No splicing check error: Operation to clear error that occurs when a machine detected a splicing tape and the memory's splicing area on the feeder does not have information of the new parts. At this point, the machine stopped as it cannot read information of the new parts from the splicing area. This error may occur also when splicing verification has not completed because of operation error or other reasons. Methods are provided to resume production promptly.

4.2 Settings

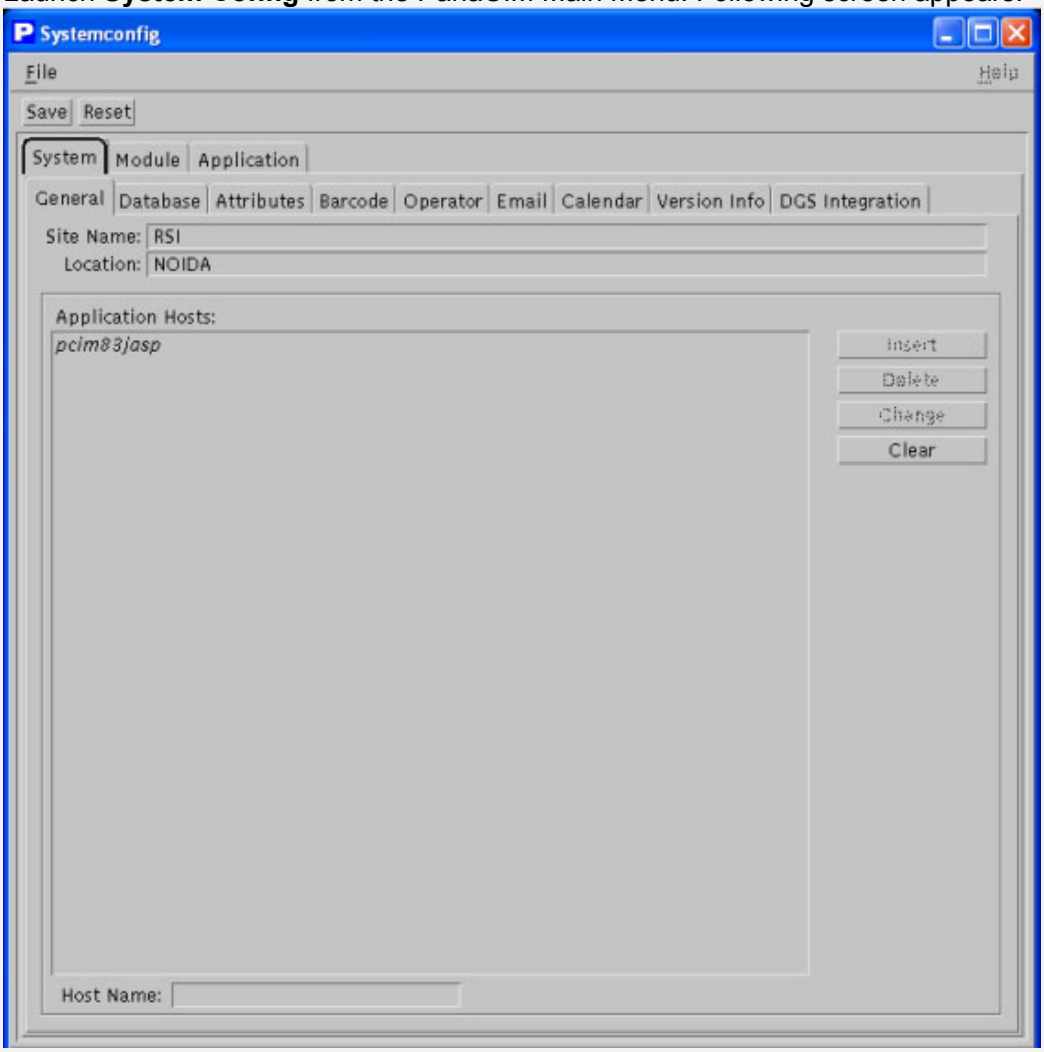
4.2.1 Old Part/New Part Comparison Settings

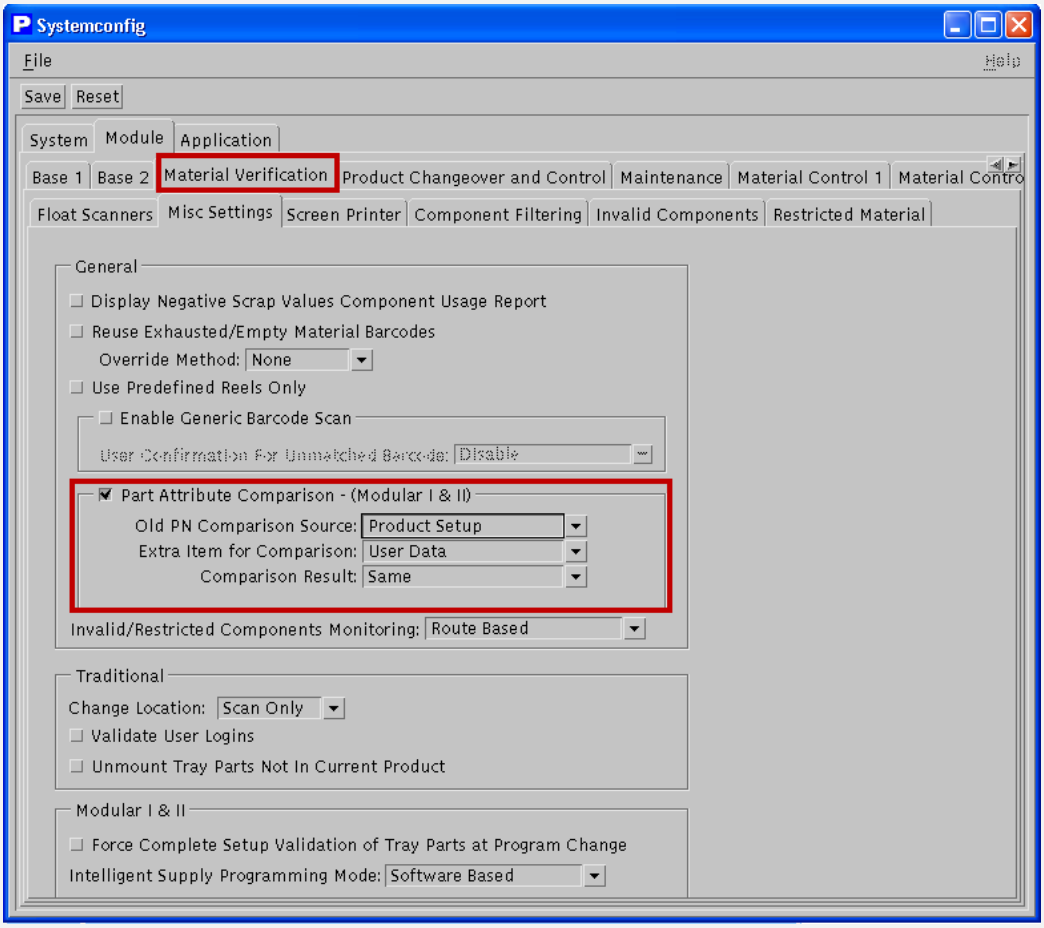
PanaCIM implements a Part Attribute Comparison processing in the form of old part / new part comparison to assist in part validations when performing replenishments. Configuring Part Attribute Comparison settings provides operators with the assurance that they are in fact scanning a new reel rather than an old reel. Also, scanning the old part and then the new part from each reel and comparing this data with the selected position provides confirmation to operators that they are working in the correct position.

If comparison of old reel with new reel is enabled, it can be used during part shortage or splicing (part shortage warning and optional splicing) on the CM or NPM (including AM100) series machines. Also the verification mode can be used regardless of MV/MC. However, it cannot be used on the screen printer and BM series machines.

Comparison of old reel with new reel is applicable for both Tray supply and Feeder supply. Old Part New Part validation supports scanning of single barcodes for each data element or concatenated barcode (QR code) for all data elements.

Refer to the action table below for enabling this option:

Step	Procedure
1	Launch **System Config** from the PanaCIM Main Menu. Following screen appears: 

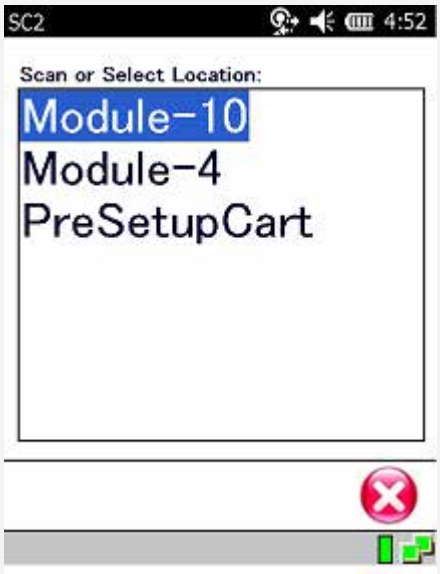
2	Click **Module** tab and then activate **Misc Settings** tab within **Material Verification** tab. Following screen appears: 
3	Check the **Part Attribute Comparison** checkbox. Checking / unchecking this checkbox enables/disables old part /new part comparison and the associated configuration items.
4	Configure the following items for old part / new part comparison: **A) Old PN Comparison Source:** Allows user to select the source of the comparison with respect to Part Number only. This combo consists of following drop-down options: • **Current Setup:** It contains part numbers belonging to product setup that is currently mounted on the machine. • **Product Setup:** It contains part number as defined in the current running product feeder setup. **B) Extra Item for Comparison:** This combo allows the user to select an additional attribute that will be used in conjunction with the part number comparison. The extra attribute is not compared against the Product Setup or Current Setup. The combo consists of following options: • **None** - No additional part attribute is used • **Vendor** - The vendor number will be used as an additional part attribute for comparing old and new part data • **Lot** - The lot number data will be used as an additional part attribute for comparing old and new part data • **User Data** -The user data will be used as an additional part attribute for comparing old and new part data.

	C) Comparison Result: - This is the result that must occur when comparing the extra data item. Available options are as follows: • **Same:** Value of the extra data item for both old and new part must be identical. It is not necessary that this data item should exist in the Product Setup or Current Setup. • **Different:** Value of the extra data item for old and new part can be different from one another. It is not necessary that either of this should exist in Product Setup or Current Setup.
5	Click the **Save** button at the top left corner of **System Config** dialog to save the changes made for Part Attribute comparison.

4.3 Operation

4.3.1 Common

Operations common to part exhaust, splicing, feeder clog clear and no splicing check error are described below.

Step	Procedure
1	From the main menu screen, select **Settings**. 
2	After **Settings** is selected, select a location. 

3	After a location is selected, select a work area for supplying parts. 
4	After the work area is selected, **1.SETUP** screen appears as default. 

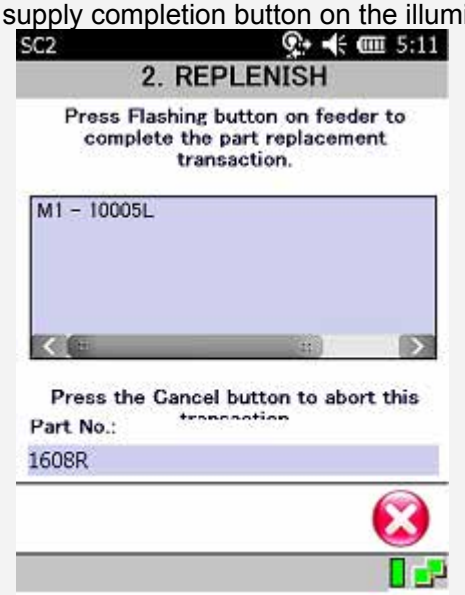
Tap **1. SETUP** highlighted in green color to display the following menu screen.



4.3.2 Part Exhaust/Splice

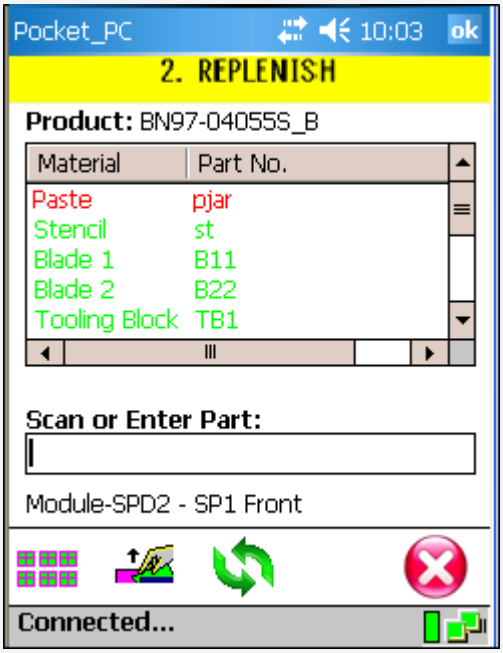
This section describes the procedures for operation of supplying parts to a feeder whose part is exhausted, and operation of splicing a new reel into a reel in a splicing warning state.

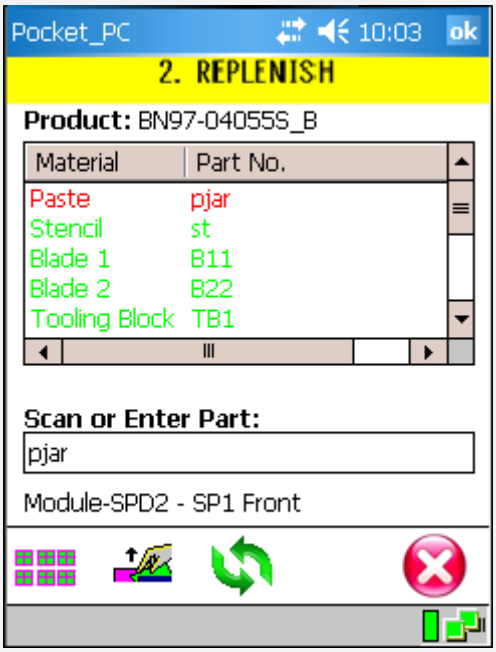
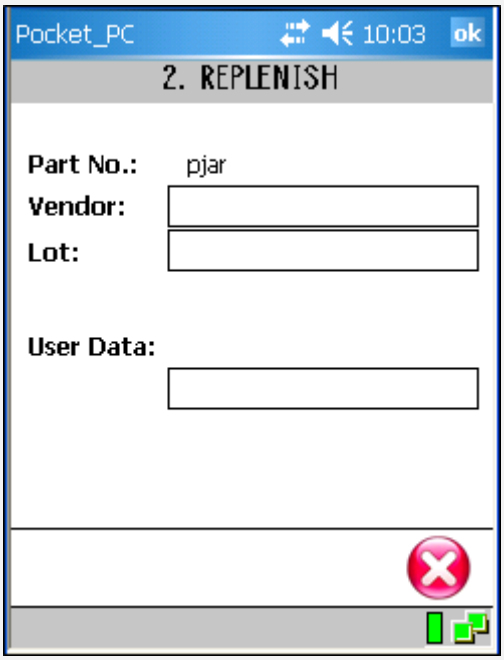
Step	Procedure
1	From the menu screen, select **2. REPLENISH** to perform operation corresponding to the warning state of either part exhaust or splicing. A feeder / part exhausted the part or a part in a splicing warning state is displayed, scan or enter the part number to be replenished. 
2	After scanning or entering the part number, enter part attributes such as Vendor, Lot, Quantity and User Data. 

3	After part attributes are entered, a message appears, telling you to press the supply completion button on the illuminated feeder.  After the button is pressed, part information is written into the feeder memory. Once the part is replenished, it is cleared from the list. 
4	<Machine operation procedure in case a part is exhausted> 1. Check the machine screen for information on a part to be exhausted. 2. Remove a table from the feeder and replenish a new reel. 3. Set the feeder to the machine again. 4. Scan the parts and press the feeder completion button.
5	<Machine operation procedure in case splicing warning occurs> 1. Check the machine screen for splicing warning or stop message information. 2. Use a splicing jig to splice a new reel into a currently used reel end. 3. Scan part information of a new reel and press the feeder completion button. 4. After the splicing sensor detects splicing, a reel moves from the splicing area to the new reel.

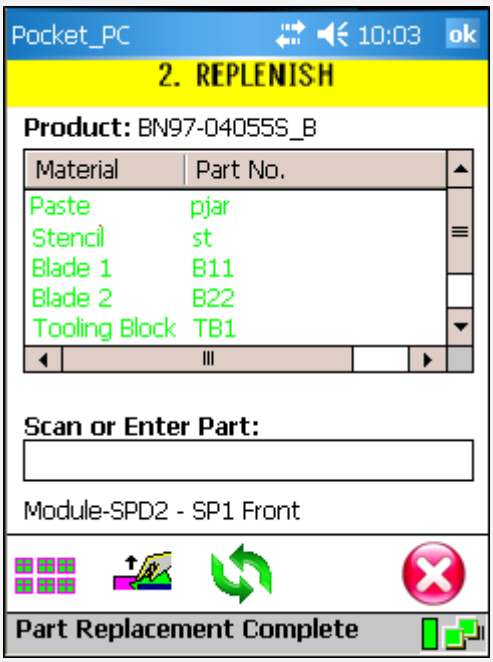
4.3.3 Part Exhaustion (SPV/SPG/SPD/SP70)

This section describes the procedure for supplying parts to be exhausted in the screen printer (SPV/SPG/SPD/SP70).

Step	Procedure
1	From the main menu screen of the mobile scanner, select **REPLENISH** and scan or enter a work area of the screen printer (SPV/SPG/SPD/SP70). The following screen appears.  **Note)** Note that the names of all materials set up in the screen printer appear on the screen. <SPV/SPG with dual solder pod> If two solder pods have been set, two solders are displayed as materials to be verified. ・As the first solder pod, [PASTE] appears under Material ・As the second solder pod, [PASTE2] appears under Material.

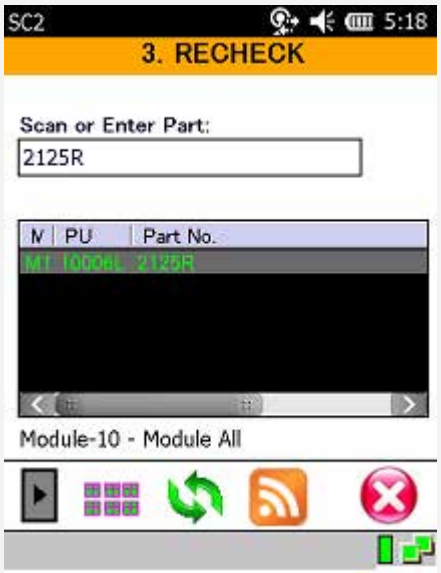
2	Scan or enter the number of a part to be exhausted. 
3	After scanning or entering part number, you are prompted to specify part attributes like **Vendor**, **Lot** and **User Data**. 

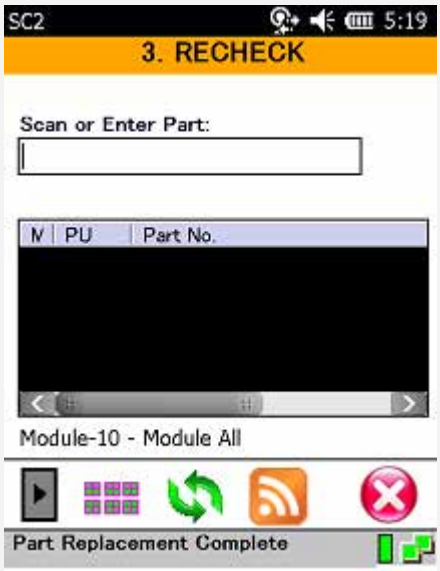
4	After all parts to be exhausted in the screen printer are successfully scanned, the screen returns to the step 1 screen.



4.3.4 Feeder Clog Clear and No Splicing Check Error

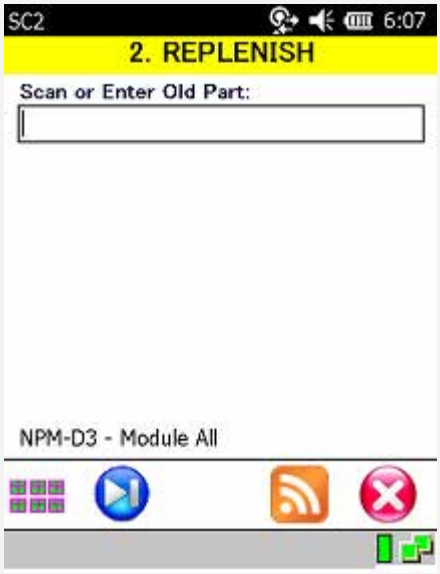
This section illustrates the operation methods to clear part exhaustion as feeder clogged and to clear an error in case of the machine detects a splicing tape while new part information is not written in the splicing area of the memory installed to the feeder.

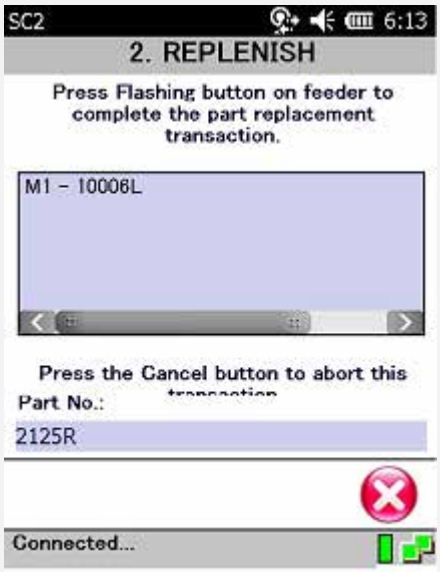
Step	Procedure
1	From the menu screen, select **3. RECHECK**. A list of feeders/parts to be exhausted in the selected work area, or a list of feeders/parts in a 'no splicing check error' state is displayed. 
2	After scanning or entering the part number, enter part attribute such as Vendor, Lot and User Data. Entered information must match the installed reel data. 

3	After part attributes are entered, a message appears, telling you to press the supply completion button on the illuminated feeder. After the button is pressed, part information is written into the feeder memory.
4	After feeder clog or no spicing check is cleared, a message appears, telling you that the material has been replaced. 

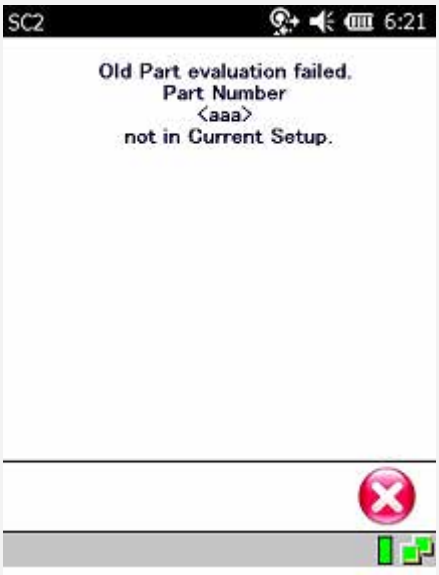
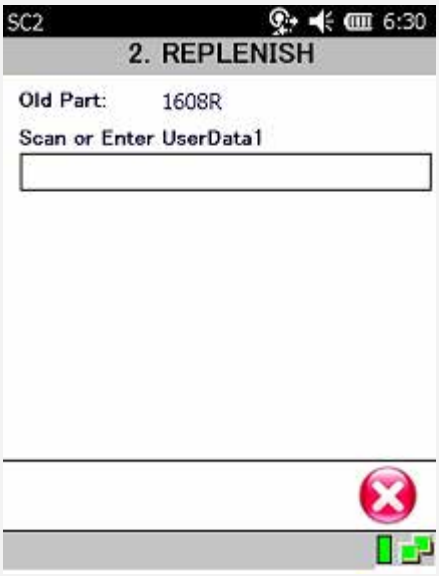
4.3.5 Comparison

This section describes scanner operation in the case of part exhaust and splicing when comparison of new and old reel is enabled.

Step	Procedure
1	After a work area is selected, the part scan screen for the following old part reel appears. The old part reel means a part that has already been installed in the machine and is target for splicing a new reel.  **>See the step 6 if you want to skip scanning/entering an old part.** **>See the step 7 if you failed to scan/enter an old part.** **>See the step 8 if additional comparison information exists after scanning/entering an old part.**
2	After scanning or entering an old part, the following screen appears. Scan part information from the barcode on a new reel.  **>See the step 10 if you failed to scan/enter a new part.**

3	After the part number is scanned or entered, enter information of a part to be added. 
4	If additional comparison items such as Vendor, Lot, Quantity and User Data conform to the setting of the comparison result, the following screen appears. 
5	When the supply completion button on the illuminated feeder is pressed, part information is written into the feeder memory. Repeat this step for all parts to be verified.

6	<skip operation for scanning or entering old part> Clicking  and entering the password on the following screen allows you to skip scanning an old part. Enter the password defined in **System Configuration->Application->Mobile Scanner Tab**. 
7	<Failure of old part verification> ・If scanned or entered old part number does not exist in Setup, but **Product Setup** is selected in the setting of **System Configuration-Old PN Comparison Source,** the following message appears. 

	・If scanned or entered old part number does not exist in Setup, but **Current Setup** is selected in the setting of **System Configuration-Old PN Comparison Source,** the following message appears. 
8	<Entering additional comparison information> After an old part is scanned or entered, the following screen appears.  **>See the step 9 if you failed to enter additional comparison information.**

9	<Failure of entering additional comparison information> If an additional comparison item (in this case, **User Data**) that is entered after old part is entered matches the condition of **System Confirmation-Comparison Result**, the following message appears. 
10	<Failure of new part verification> The part number of a scanned or entered new part must match the old part number. If not, a message appears on the center of the screen, telling you “Part Number Evaluation Failed”. 

5. FUNCTION - PREVENTION FROM AN INCCORECT SET - PRESETUP

This chapter explains the PreSetup.

5.1 Overview

■ Functional overview

An efficient way to improve machine productivity is to preset the feeders away from the machine using an Offline Programming Station. When changeover is required, Feeder Carts with pre-loaded feeders from the Offline Programming Station are loaded into the machine. Assuming that the feeder programming is correct and complete for the current product, all of the intelligent feeders will have the part information programmed into their memory, and the machine is able to go into production with no additional part scanning required.

The feeders on the cart can be programmed in one two ways, Free-Form or Product BasedPresetup.

* Splice verification is not available in the presetup regardless of the model of the placement machines (NPM or CM).

- **Free-Form**

  Components or feeders can be set up without any restriction. Select the **Part Scan** icon from the main Material Verification menu on the scanner attached to the offline programming station.

- **Product Based PreSetup**

  The cart is setup based on the selected product or mix. Select the **Pre Setup** icon from the main Material Verification menu on the scanner attached to the offline programming station.

  The feeder memory can be cleared by either of the following methods.

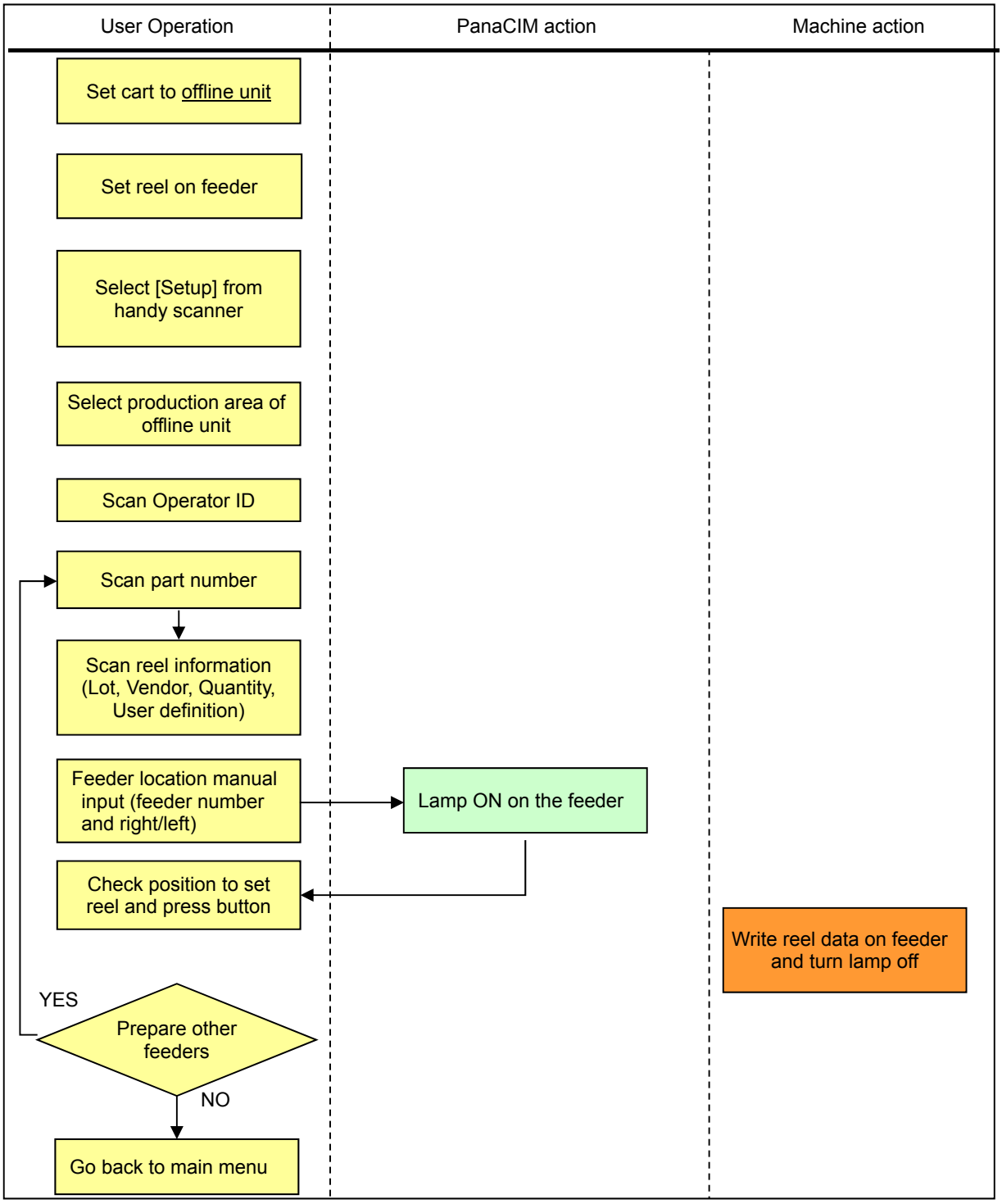
- **Cart Feeder Clear**

  Clears the memory of all feeders loaded on the cart at a time.

- **Single Feeder Clear**

  Clears the memory of the feeder one by one.

■ Prep Cart Setup Flow

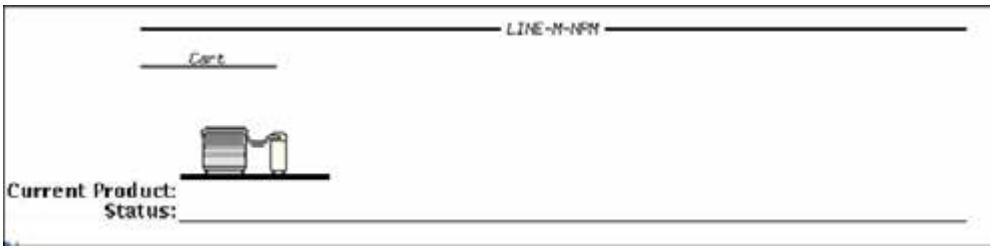


5.2 Settings

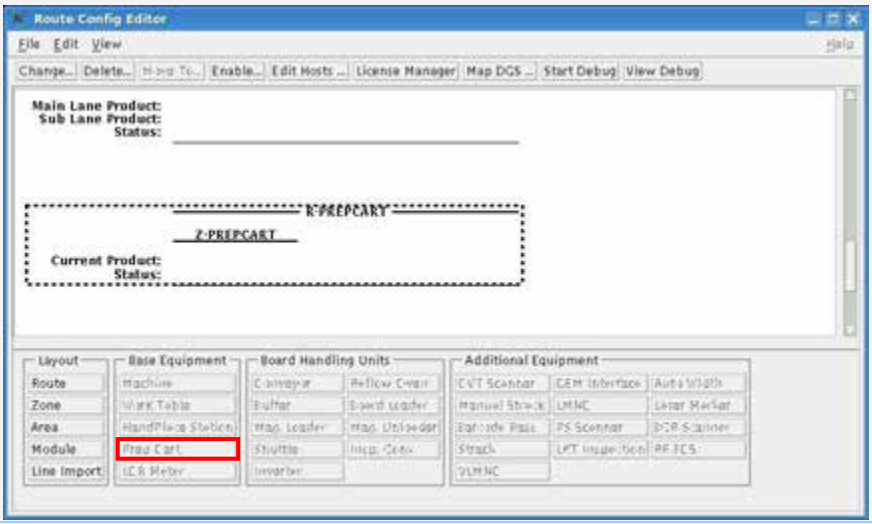
5.2.1 PreSetup Unit Settings

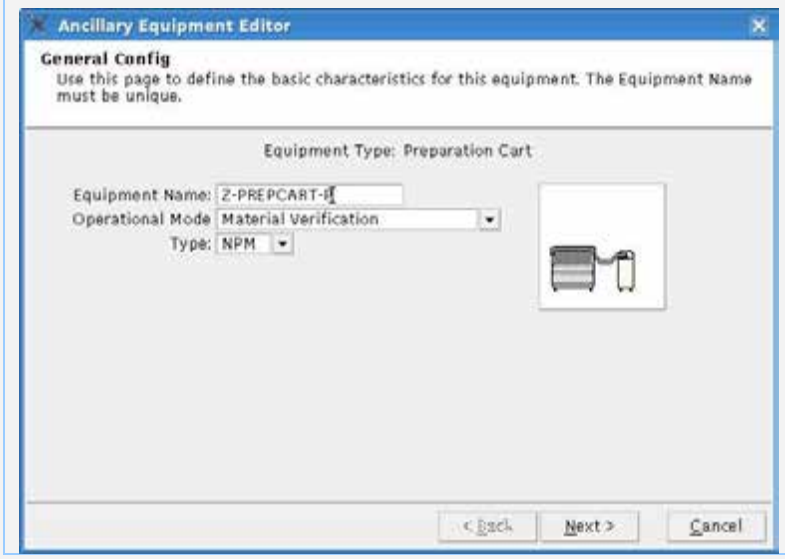
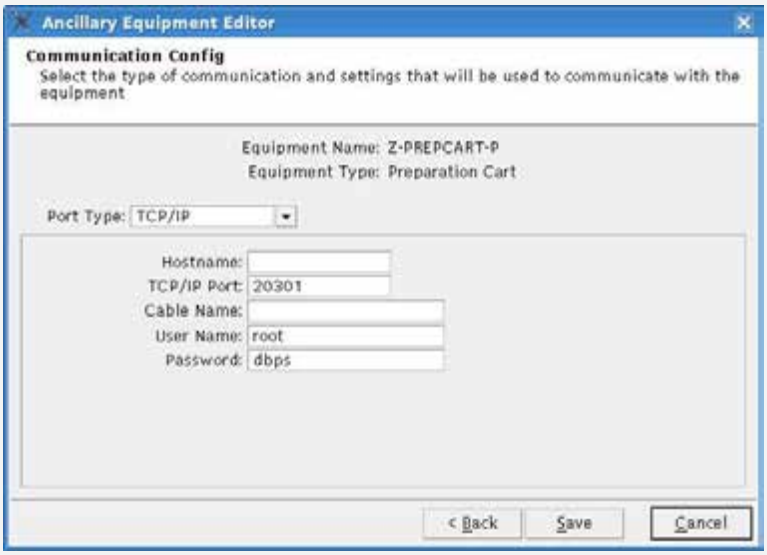
Setting up the feeders using a Panasonic Preset Unit reduces the time required for model change at the machine.

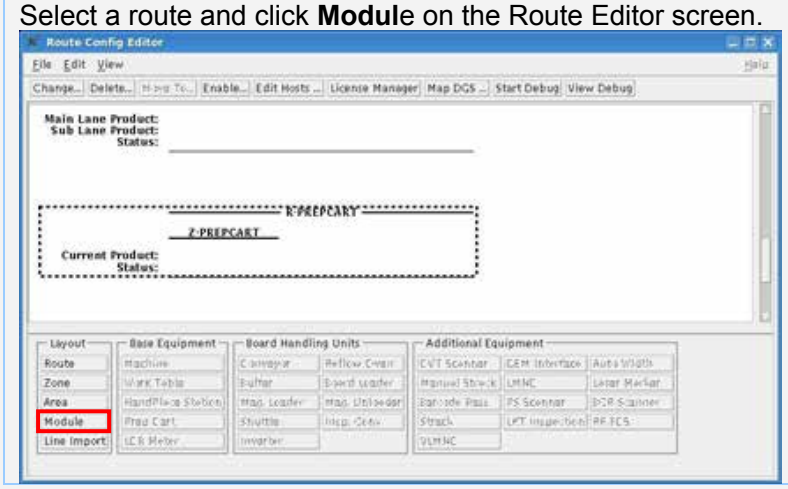
The presetup unit is treated almost like a modular machine in that it will become part of a Module in a Zone, which is part of a Route. The presetup unit can be entered into its own Route, or it can be added to an existing Route. In either case, the feeder carts verified using the presetup unit are available to all machines in the factory that are capable of using them.



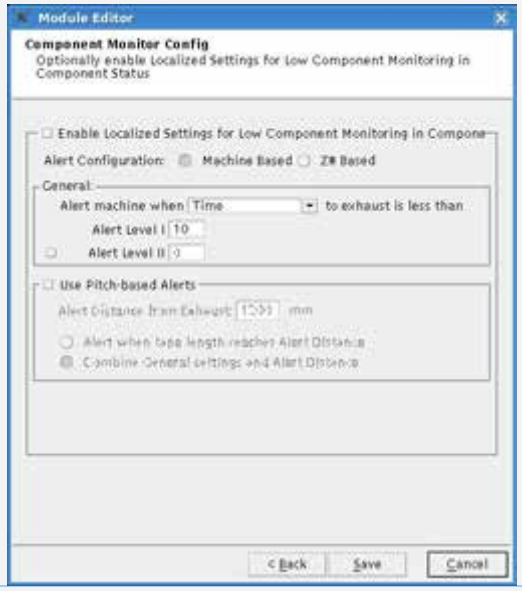
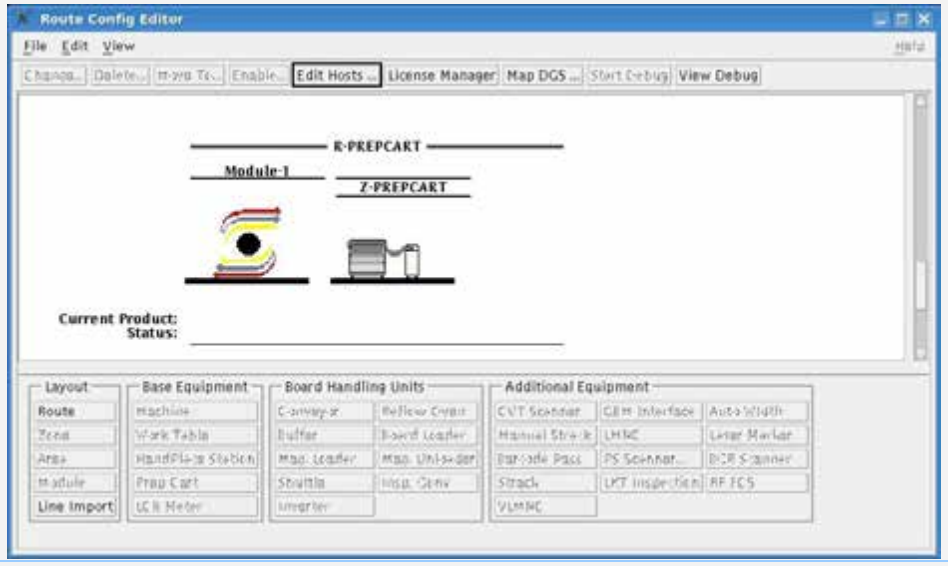
Refer to the table below to add the presetup unit in a single route.

Step	Procedure
1	On the Route Editor screen, click **CIMC stop** to exit the PanaCIM system.
2	Create a new route and zone.
3	Select a zone and click **Cart**. 

4	On the General Config screen, set the following: • Change the equipment name if necessary. • Click **Next**. 
5	On the Communication Config screen, set the following: • Change **Port Type** to **TCP/IP**. • Enter a host name of the offline unit in the **Hostname** field. The host name must be the same as the one in the [/etc/hosts] file containing IP address of the communication unit in the specified unit. • Click **Save**. 

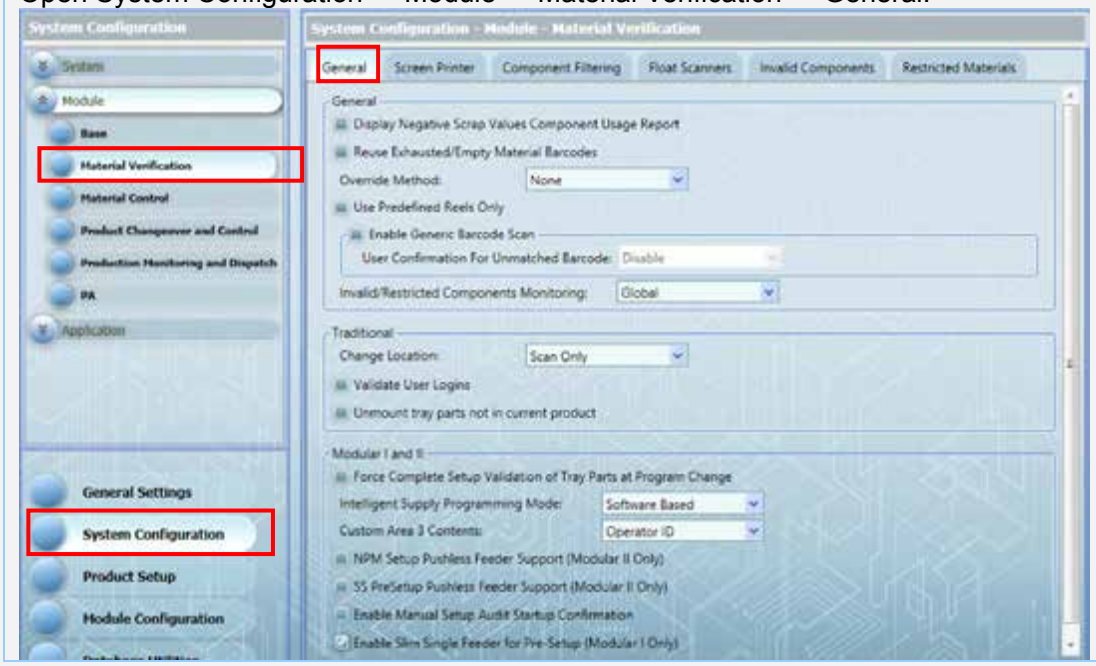
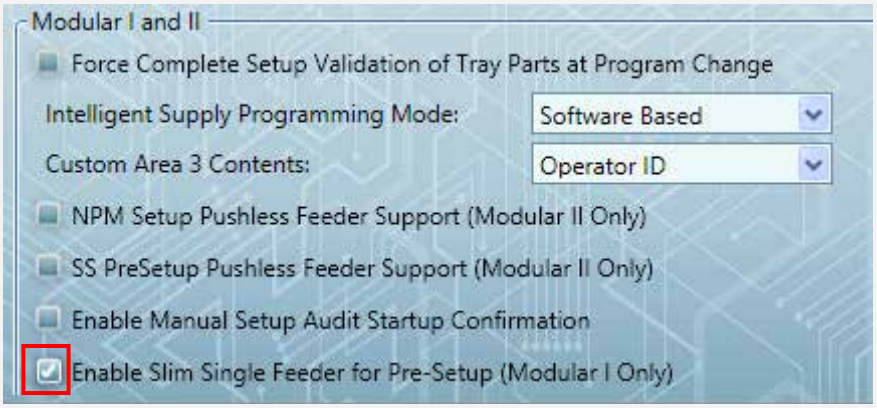
6	Select a route and click **Module** on the Route Editor screen. 
7	On the General Config screen, set the following: • Populate the **Module** field. • Click **Next**. 

8	On the LM Module Configuration screen, set the following: • Click **Next**. 
9	On the Module Equipment Setup screen, set the following: • Select **PreAreaCart** for the **Available for Module** field. Clicking **Add** moves it to the **Configured for Module** field. • Click **Next**. 

10	On the Part Monitor Setup screen, set as follows: • Click **Save**. 
11	Click **CIMC start**. 
12	Select a module zone.
13	Click **Enable**.

5.2.2 Thin Type Single Feeder – CM Pre-setup Settings

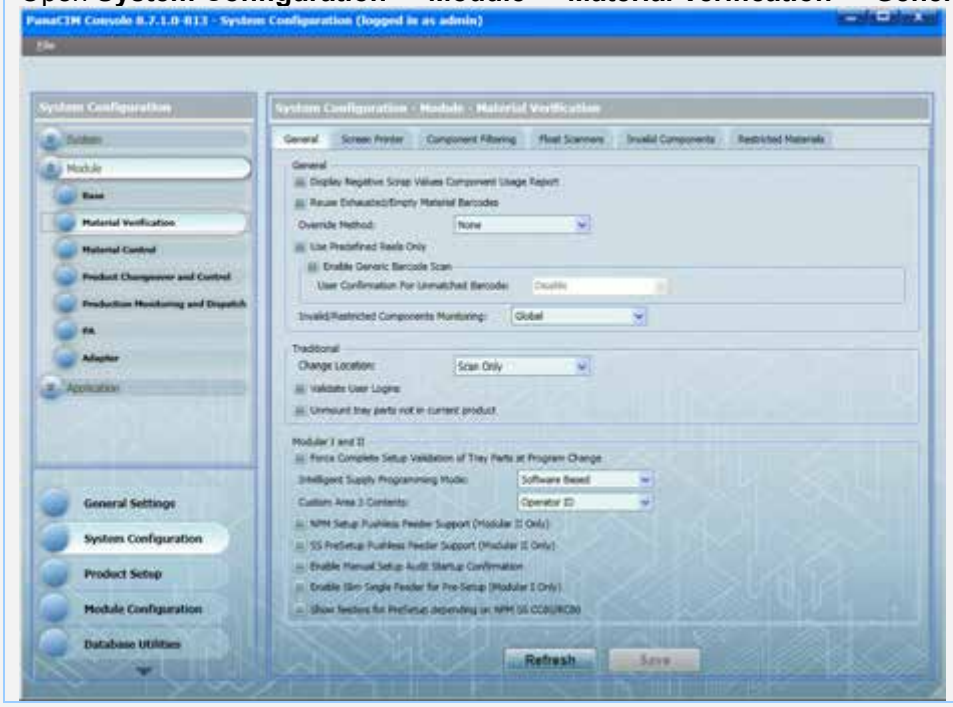
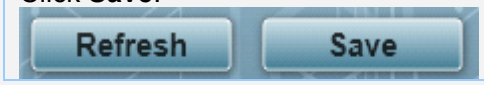
CM can be used to perform a thin type single feeder pre-setup.
The following describes the procedure for enabling CM's thin type single feeder pre-setup.

Step	Procedure
1	Open PanaCIM Console, and enter your user name and password to log in.
2	Open System Configuration -> Module -> Material Verification -> General. 
3	Enable the "Enable Slim Single Feeder for Pre-Setup (Modular I Only)" checkbox for NPM/CM series machines. 
4	Click **Save**. 

5.2.3 Fixing Feeder during Pre-setup

You can configure the display method of pre-setup of verification information on a feeder fixing to the machine such as NPM-TT, NPM-W.

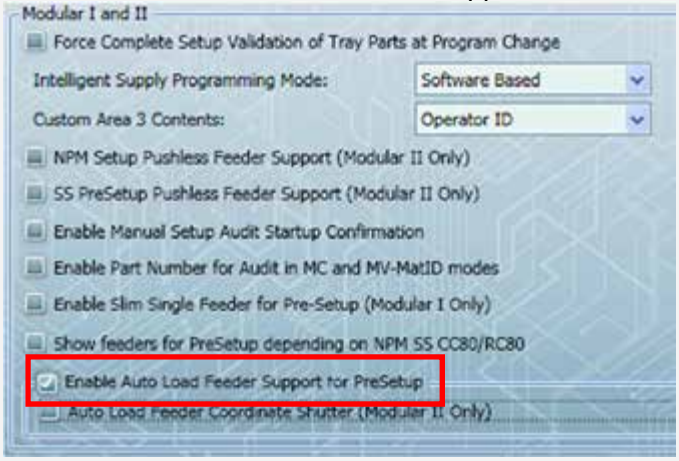
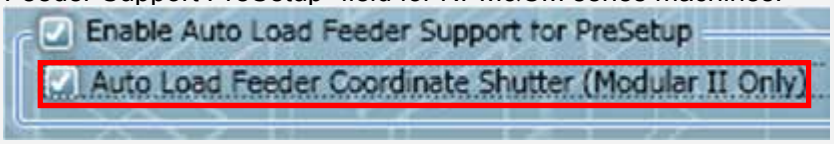
This section describes how to set up the display method of verification information on a fixing feeder during pre-setup verification.

Step	Procedure
1	Open **PanaCIM Console**, and enter your user name and password to log in.
2	Open **System Configuration** -> **Module** -> **Material Verification** -> **General**. 
3	Set “Show feeders for PreSetup depending on NPM SS CC80/RC80”. Without check: Both cart feeders and fixing feeders are displayed during pre-setup verification. With check: Cart feeders and fixing feeders are separately displayed during pre-setup verification. 
4	Click **Save**. 

5.2.4 AL (Auto Load) Feeder – NPM Pre-setup Settings

Pre-setup verification of the AL (auto load) feeders used for NPM can be made.
The following describes the procedure for enabling pre-setup of the AL feeders for NPM.

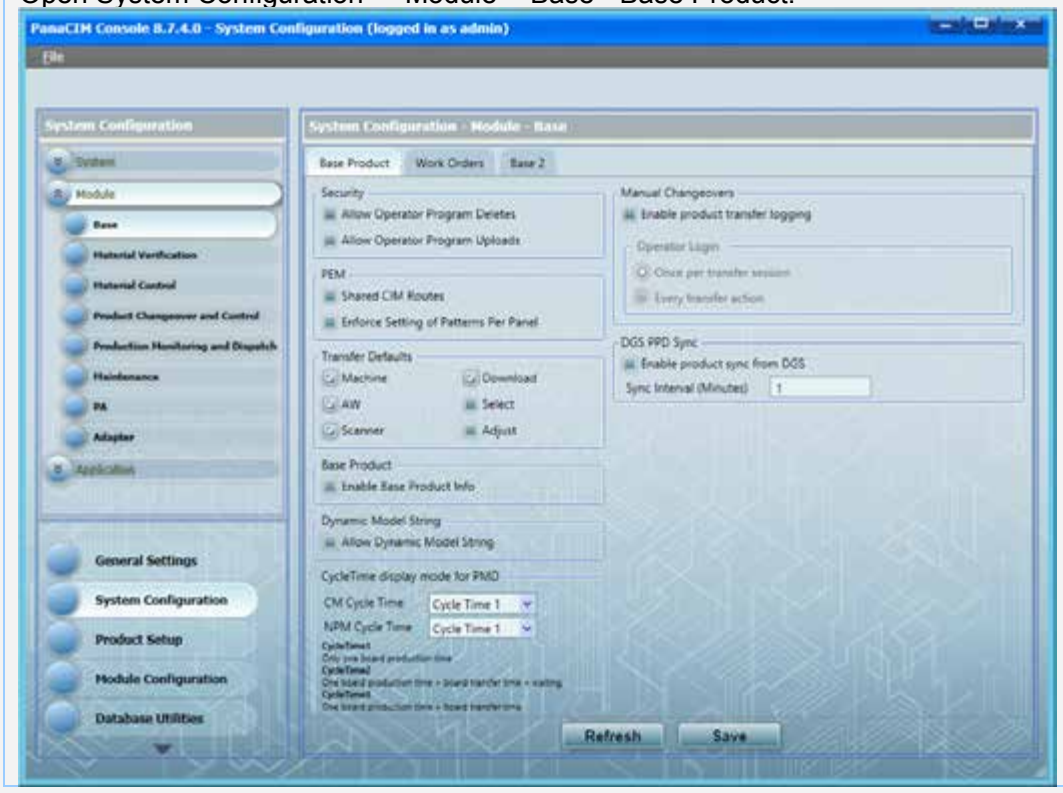
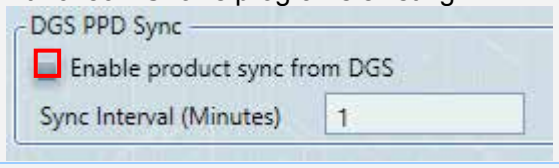
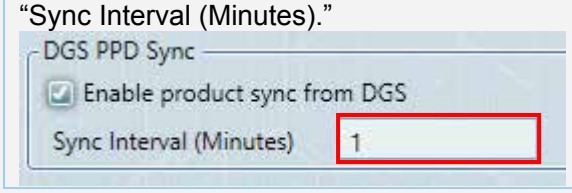
***For NPM, you can specify the timing to enable the shutter opening/closing button (white) on its AL feeder.**

Step	Procedure
1	Open the PanaCIM console, and enter your user name and password to log in.
2	Open System Configuration -> Module -> Material Verification -> General. *The same screen as the one for AL Feeder – CM Pre-setup Settings is displayed.
3	Enable “Enable Auto Load Feeder Support for PreSetup” for NPM&CM series machines. 
4	Set “Auto Load Feeder Coordinate Shutter (Modular II Only)” in the “Enable Auto Load Feeder Support PreSetup” field for NPM&CM series machines.  **<Auto Load Feeder Coordinate Shutter (Modular II Only)>** You can determine the extent to which the shutter button (white) on the auto load feeder is enabled. ・ With a check mark: enabled only when the message saying “Press the confirmation button on the feeder” is displayed on the scanner. ・ Without a check mark: enabled when the setup screen is being displayed by going to [PreSetup] - [Setup Cart] on the scanner.
5	Click [Save]. 

5.2.5 Display Settings of Pre-setup Program (NPM)

When pre-setup of NPM machine is performed, you can select the display area of programs selected in the program-based pre-setup.

This section describes the setting method of the display rea of programs displayed during program-based pre-setup.

Step	Procedure
1	Open PanaCIM Console, and enter your user name and password to log in.
2	Open System Configuration -> Module ->Base->Base Product. 
3	Set the “Enable product sync from DGS” for DGS PPD Sync Without check: Shows all programs imported to PanaCIM during pre-setup verification. With check: Shows programs existing in PPD folder of DGS during pre-setup verification. 
4	If you enable the “Enable product sync from DGS” checkbox for DGS PPD Sync, set “Sync Interval (Minutes).” 
5	Click **Save**. 

5.3 Operation

5.3.1 Free-Form

■ Scan

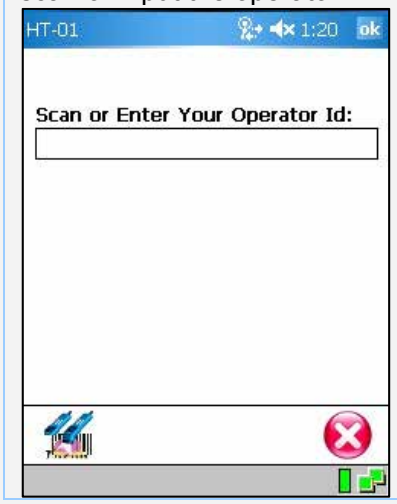
- **Scan Procedure**

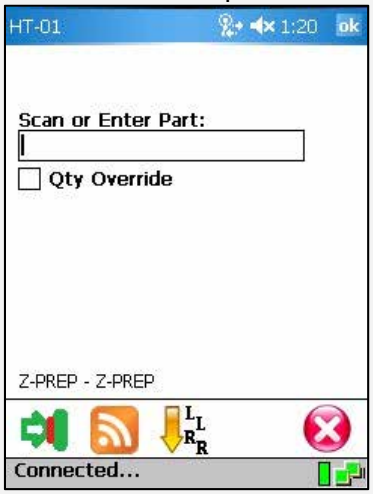
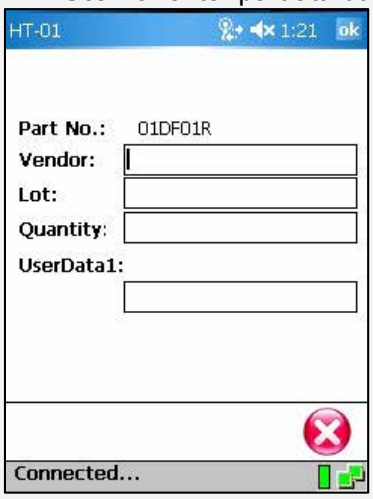
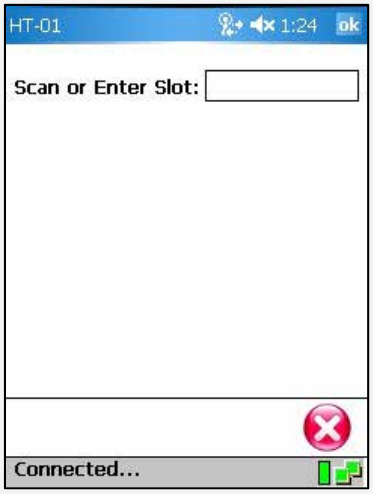
Free-form includes the following setups.

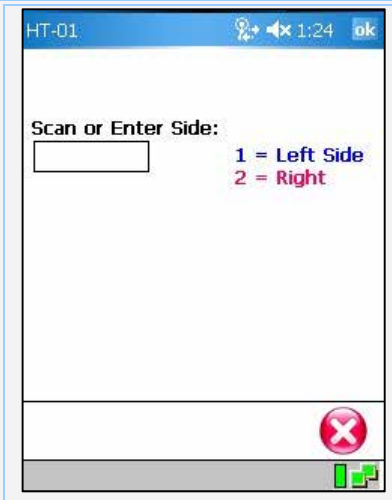
- Scanning/entering the part attributes such as part number, etc.
- Scanning/entering the feeder slots on the cart
- Pressing the program button for the feeders

No verification will be performed until the feeder cart is attached to the machine. Components required for the program are checked on the machine side to make sure they are arranged in the correct feeder slots.

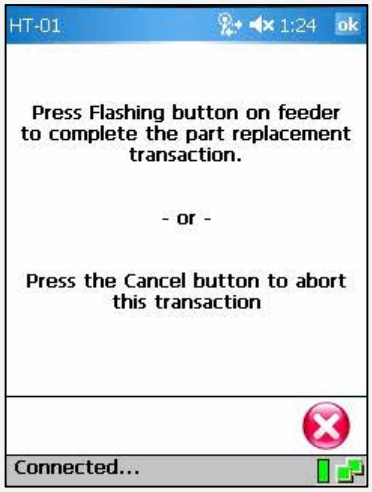
- There is no restriction on the components or feeders to be set. Not intended for any particular program or mix.
- Since this feature is not associated with programs, a paper-based list of table setup is required.
- No feeder carts or feeders on the cart will be tracked.

Step	Procedure
1	Make sure the feeder cart is connected to the offline unit.
2	Select **Part Scan** from the Production main menu of the scanner on the offline unit. 
3	When the operator ID has been enabled, the following screen appears to prompt you to scan or input the operator ID. 

4	Scan or enter the part number assigned to the feeder. 
5	The entered part number is shown accompanied by other attributes for which On has been set in the component filtering for the group including the entered part number. • Scan or enter part attributes. 
6	Load the physical feeder to the desired slot. Scan or enter the **Slot** number on the feeder cart. 
7	Scan or enter the feeder side. Enter 1 if the reel is in the left side of a double feeder or in the thin-type single feeder loaded on the left side. Enter 2 if the reel is on the right side of a double feeder. If the reel is located in the single feeder, enter 1.

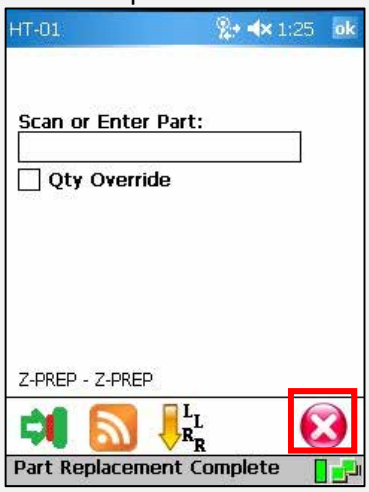


8 After all of the information has been entered or scanned, scanner shows the following screen.



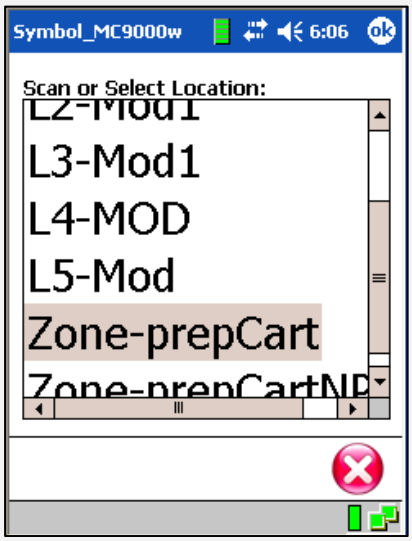
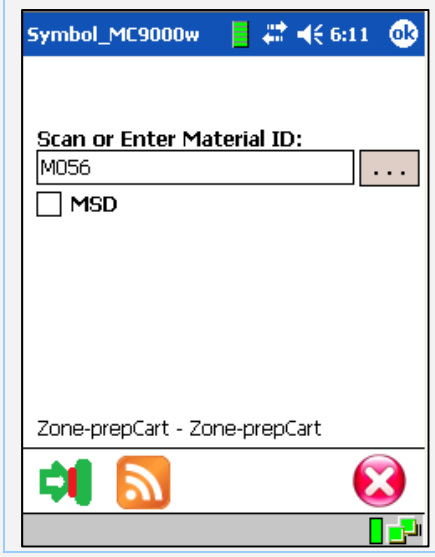
Correspondingly, the green light on the physical feeder will start flashing. Press the **white** button on the feeder to write the part information to it.

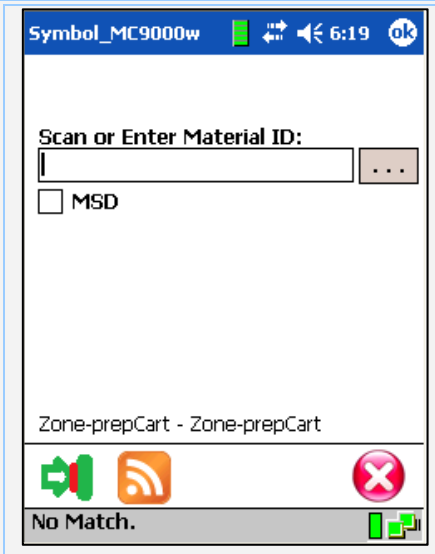
9 Repeat the process for all desired components. Since there is no selected Product, users must determine when the Free Form PreSetup process is completed. Select  when completed.



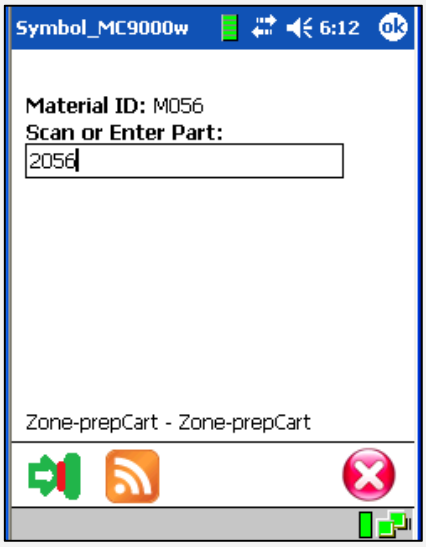
- **Scan Procedure with Comparison ID Enabled**

Refer to the action table below to perform free form PreSetup programming using the Mobile Scanner if Comparison ID scan has been enabled in the System Configuration application.

Step	Procedure
1	Ensure the feeder cart is connected to the Offline Programming Station.
2	Select  .
3	Select the zone that contains your PreSetup Cart. 
4	After selecting **Pre-Setup PrepCart** zone, following screen appears: 
5	Scan the **Comparison ID** if your System Configuration settings have been enabled to do so. **Note:** Be sure to scan the comparison ID. Otherwise, key input is disabled.

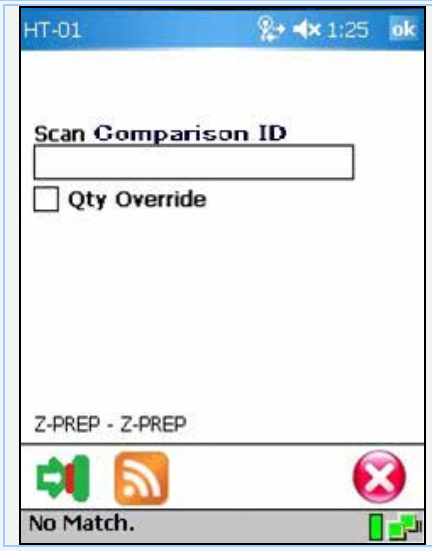


6 Scan or Enter the **Part Number** of the component mounted on the feeder.

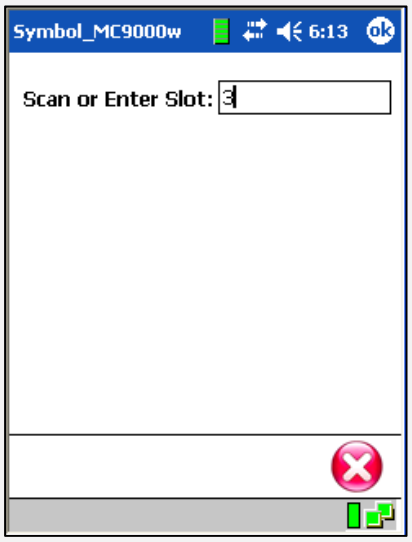


Notes:

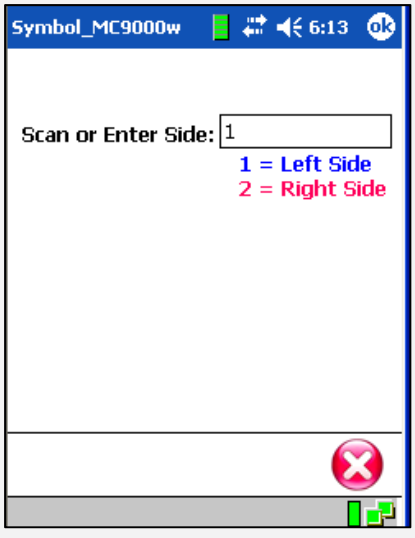
- If the scanned part is detected as an Invalid or Restricted component, an error message will appear.
- In some cases the Part Number may be longer than the Comparison ID (i.e. substitute parts), but will validate if the ID matches as a substring of the Part Number. If the Part Number and Comparison ID is not a match, the message 'No Match' will appear. In such a case, the attributes need to be entered again.



7 Load the physical feeder to the desired slot. Scan or Enter the **Slot** number on the feeder cart.



8 Enter **1** if the reel is on the left side of a double feeder or on the thin-type single feeder in the left subslot. Enter **2** if the reel is on the right side of a double feeder.



9 After all of the information has been entered or scanned, the green light on the

	physical feeder will start flashing. Press the **white** button on the feeder to write the part information to it.
10	Repeat the process for all desired components. Since there is no selected Product, users must determine when the Free Form PreSetup process is completed. Select  when completed.

5.3.2 Product Based PreSetup

The Product Based PreSetup is enabled by selecting the icon from the main menu of the Material Verification.

This feature sets up and maintains the feeder cart based on the presetup IDs. Presetup is programmed according to the programs including product concerned or mix.

* Take the following steps to perform the Product Based PreSetup with the offline programming station:
  - Enable the cart ID in the scanner setup on the Route Editor.
  - Enter a barcode formula representing the cart ID barcode under the **Barcode** tab on the **System Configuration** screen.

Setup Cart is used to create feeder cart setup according to the requirements of a feeder table for a Product or Mix. The setup is controlled by the cart IDs. The cart IDs will be scanned or entered when the feeder is set up using the offline programming station. **Cart Edit** is used to modify the existing feeder cart setup. **Product Check** is used to verify that the PreSetup matches the requirement of a specific product or mix.

In the presetup of the feeder cart, the feeder setup required for Product is performed on the cart, rather than on the machine table. Presetup can be performed for all the machines in the factory, but the contents of the setup of the feeder cart must agree with the requirements of the program setup of the machine concerned.

When the cart is physically connected to the machine, use **Mount Cart** to tie the setup to the machine. While the cart is mounted, the setup and feeders are not available to the PreSetup operations. Once the cart is disconnected from the machine via **Unmount Cart**, the setup and feeders are available again for mounting, editing and product check.

■ Presetup overview

Select ‘Presetup’ icon from production main menu. <Presetup> menu appears.



Presetup function performs presetup on offline changeover unit and trace. This presetup icon appears only when offline changeover unit is set on Route Editor. ‘Setup cart’ icon appears only on scanners used at offline program station.

‘Mount Cart’ and ‘Unmount Cart’ icons appear only on the scanners used for the target machine.

- Select ‘Setup Cart’ to verify if the setup matches with program or mix by offline changeover unit’s scanner.
- Select ‘Mount Cart’ to set verified cart to machine. Scan or enter cart’s ID number.
- Select ‘Unmount Cart’ to remove cart from machine. Scan or enter cart’s ID number. Remove work means removing all feeders from machine. No change with the setup on the cart.
- Select ‘Edit Cart’ to change the setup on cart.
- Select ‘PCB Check’ to check if the setup matches with program.

■ Offline changeover unit scanner menu

The following production menu appears on the scanner set for an offline changeover unit.

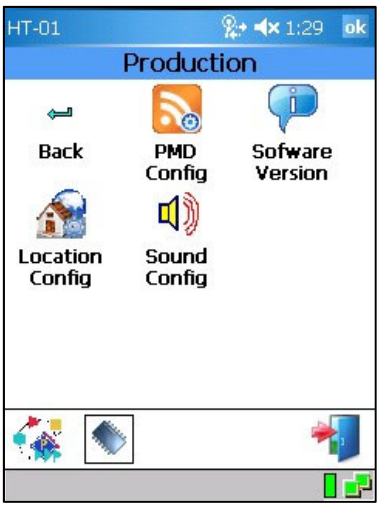


You cannot set the scanner which is set for offline changeover unit to machine's production area.

'Part Scan' is used in free presetup. In this case, part information and slot location are set for feeder. Program or mix is not linked.

'Presetup' is used in presetup on program base. Program, mix, module, machine and table are to be selected. Part information, specified program or mix of the feeder is verified.

Setting menu includes 'Software Version', 'Location Configuration' and 'Operation Monitor'.



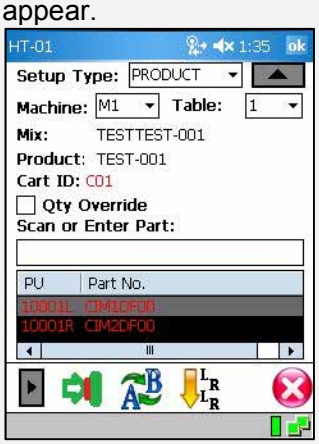
■ Scan

● Display presetup scan screen

Operate scanner as follows to set the feeders needed for the selected program or mix.

Step	Procedure
1	Select **Pre Setup** on scanner's Main Menu screen. 
2	If Operator ID is set to on, the following menu appears. Scan or enter operator ID.  If entered operator is not registered in the system configuration application, a message appears to inform that the operator ID is not valid.
3	Production main menu appears. • Select **Setup Cart** and verify the setup on cart at offline changeover unit. 

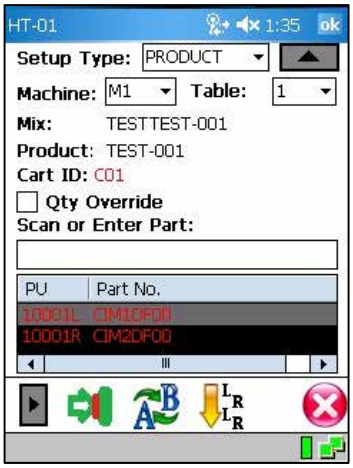
4	Set the following. • Select **Route** and **Module**. Related product setup appears. • Select an appropriate program from **Select Program** list. • Select  and continue.  
5	• To show only the part numbers of the selected program, select **PCB** as **Changeover Type**. To show part numbers of all programs in the mix, select **Mix**. • Select machine and table where the feeder will be connected. If the checkbox of “Show feeders for PreSetup depending on NPM SS CC80/RC80” is marked in 5.2.3, FX-O (a table having the fixing unit) can be selected for the fixing feeder. • If Cart ID is set to ON on Route Editor, scan or enter cart ID. Cart ID is used for trace or verification. 

6	Presetup scan menu appears. Programs, machines, table feeder location and part list appear. 

Presetup scan menu

Presetup unit shows part numbers allocated to feeder slots on cart. It also checks if the parts necessary for the program or mix are allocated properly on slots.

Presetup scan menu is shown as below.




Connected with Support Station


Back to program list


Part display order changeable


Cancel current operation


Go to cart select screen


Delete parts from the selected feeder location


Back to part sort menu



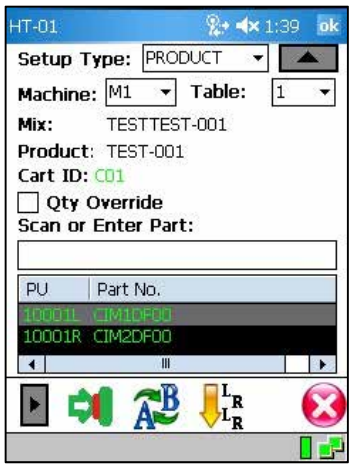




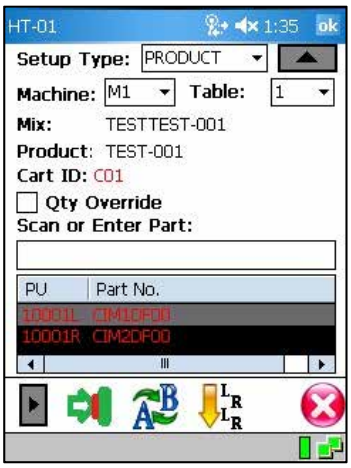
Select ‘PCB’ or ‘Mix’ as ‘Changeover type’. Selecting ‘PCB’ shows only the parts of current program. Selecting ‘Mix’ shows all parts for combined setup.

‘Machine’ or ‘Table’ indicates the location of table where feeder cart will be set.

When ‘Cart ID’ is displayed in green, all the parts on the cart have been scanned. When it is red, some parts on the cart have not been scanned. See the examples below.



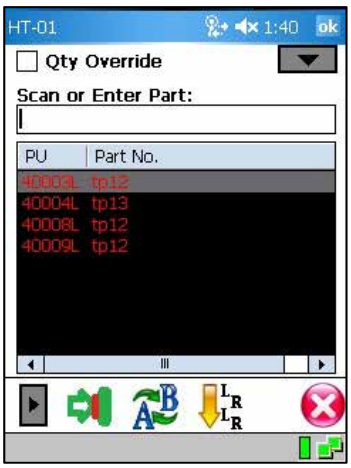
Example: Cart ID is red



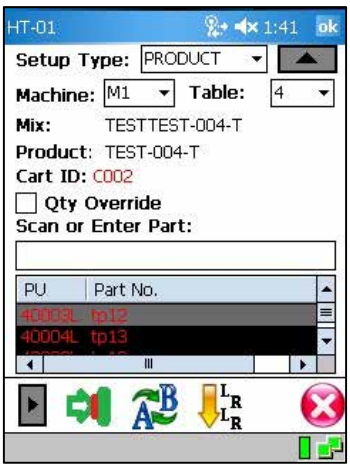
Example: Cart ID is green

'Part scan/enter' method is determined by offline changeover unit setting on Route Editor.

Click [▲] upper screen to view the list of parts to be scanned. Click [▼] again to go back to presetup menu screen.



Presetup scan menu screen



Scanning part list screen

Color of slot and part number shows respective status of verification or if they are used in another program within the same mix or another. The meaning of each color is shown below.

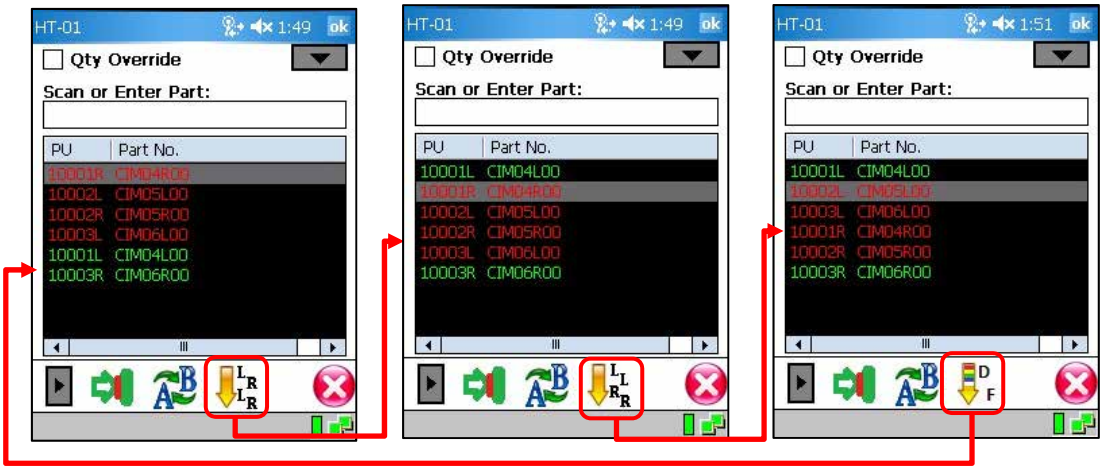
- **Red:** Part number used in the program but not verified yet
- **Green:** Part number that has been verified and used in the program
- **White:** Part number on cart that is not used in the program

If 'Changeover type' is set to 'Mix', feeders that are not used in a program but used in the mix also appear.

- **Yellow:** Part number used in the mix but not verified yet
- **Blue:** Part number that has been verified but not used in the programs in the mix.

The order of parts shown on the scanning list can be changed as follows.

- **DF:** In the order of programs (order of feeder number and left to right).
  Parts that are already setup are shown under those that have not finished setup.
- **LRLR:** In the order of feeder number and left to right
- **LLRR:** In the order of feeder number and left to right



Screen display in DF order Screen display in LRLR order Screen display in LLRR order

- **Presetup scan method**

The following table shows how to set feeders according to the selected program or mix table.

Step	Procedure
1	Scan or enter part numbers needed in the program or mix. 
2	If part filtering is set to the group to which the parts belong and necessary information cannot be obtained from the barcode, the following message appears. 
3	After all information is entered or scanned, the following message appears. 
4	Green light on the feeder starts blinking. Press white button on the feeder to write the information onto the feeder.

5	After writing on feeder memory finished, the screen goes back to Presetup scan menu. Next reel can be selected. If there is no parts in red, setup has finished for the table. When changeover type is mix, verify parts displayed in red or yellow first and then combined setups in the mix. 
6	If scanned part number is included in the program or mix but does not exist on the table, part sort menu appears automatically and shows its correct location in the module. Press  to go back to the selected table's part list. 
7	If already verified parts are scanned and there is no unverified feeder for the part, message "Delete feeder before verifying again" appears. Use cart edit function and delete part information from PanaCIM to verify feeder again. 

■ Trace of PreSetup

After completed presetup, PanaCIM traces setup information based on cart ID number. The following operations are also available for presetup trace.

- **Mount Cart:** Changes cart or feeder status from presetup to set on PanaCIM.
- **Unmount Cart:** Remove cart while holding feeder information to use it again for presetup.
- **Edit Cart:** Delete feeder part information from PanaCIM to verify the feeder again.
- **Product Check**: Show the ID number of the cart that has the optimal setup for the selected program.

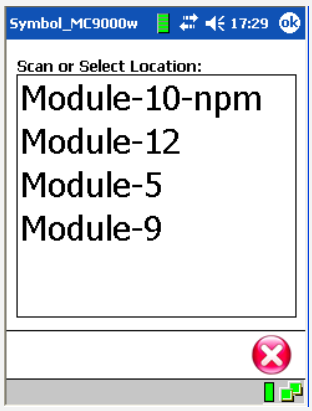
● Mount a PreSetup on a Machine

The Offline Programming Station operator will enter or scan the Cart ID for the setup to the feeder cart as part of creating the PreSetup. This Cart ID number then identifies the setup on the cart until it is mounted in a machine. When the cart is Unmounted, the Cart ID is once again available.

The Mount Cart operation is performed at the placement machine when a feeder cart is loaded to the machine. The machine operator enters or scans the Cart ID number of the feeder cart so that the PreSetup is no longer available for editing, part checking, or mounting.

Refer to the action table below to mount a PreSetup on a Machine using the Mobile Scanner.

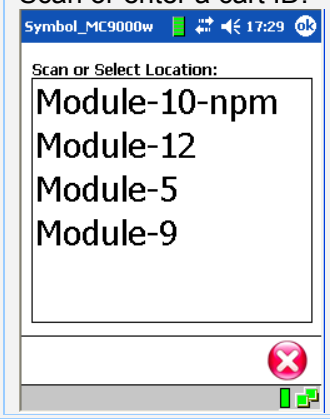
Step	Procedure
1	Select **Pre Setup** on scanner's Main Menu screen. 
2	Select **Mount Cart** on the Production main menu screen. 

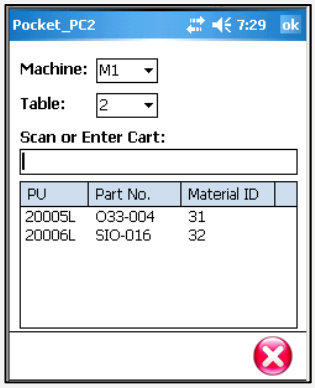
3	Select the **Machine** and the **Table** that is being mounted from the drop-down selections. Next, scan or enter the **Cart ID** of the Feeder Cart setup. 
4	
5	

Unmount a PreSetup from the Machine

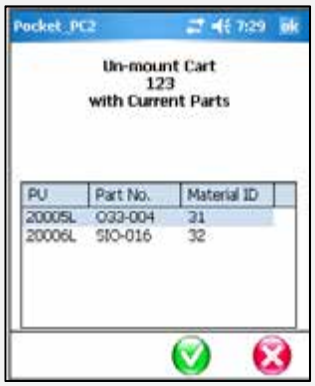
The Unmount Cart operation is performed at the placement machine when a feeder cart is removed from the machine. The machine operator is required to enter or scan the Cart ID number. This operation provides verification that the feeders on the cart are no longer allocated to the machine and it frees up the PreSetup on the cart to be re-mounted on this or another machine.

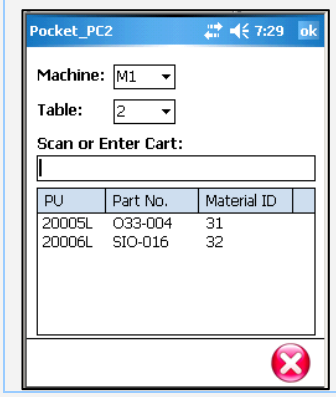
Refer to the action table below to unmount a PreSetup from the Machine using the Mobile Scanner.

Step	Procedure
1	Select **Pre Setup** on scanner's Main Menu screen. 
2	Select **Unmount Cart** on the Production main menu screen. 
3	Select **Machine** and **Table** from which a cart is removed. Scan or enter a cart ID. 



4 Click the green check button
Cart removal is completed.





- **Edit a PreSetup**

  The **Edit Cart** option is used to remove one or all feeders from a setup. The removal is required in order to reprogram the memory of a feeder in the setup. The **Edit Cart** icon is available for scanners used at the machine, or the Offline Programming Station.

Refer to the action table below to perform edit a PreSetup using the Mobile Scanner.

Step	Procedure
1	Select **Pre Setup** on scanner's Main Menu screen. 
2	Select **Cart Edit** on the Production main menu screen. 
3	Enter an operator ID if necessary. Scan or enter the **Identification Code** of the Feeder Cart setup for the verified feeders to display. 



 Removes one feeder from the setup.

 Removes all feeders from the setup.

 Cancels the setup.

4 If the 'Remove One Feeder' icon is selected, the following menu appears.

- To remove the selected feeder from the setup, click  .



5 If the 'Remove All Feeders' icon is selected, the following menu appears.

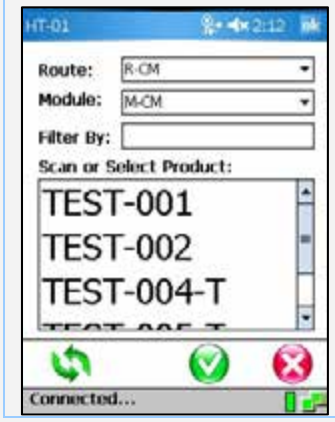
To remove all the feeders from the setup, click  .

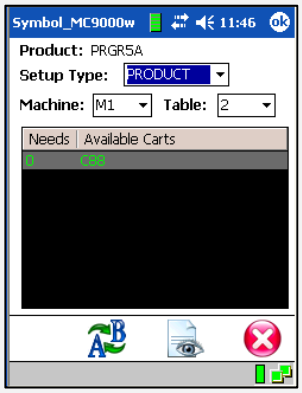
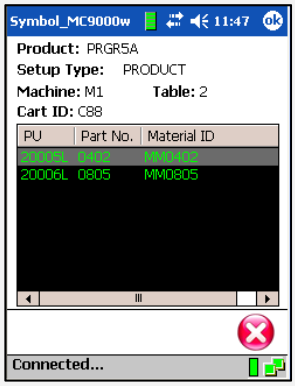


Perform a Product Check

The Product Check function is used to show the Cart ID number of PreSetups that are a best fit of cart contents for a selected Product. A cart with the closest match is displayed over carts with fewer matches. The current feeder setup on the cart can be displayed as desired. The **Product Check** function is available at the machine or at the Offline Programming Station.

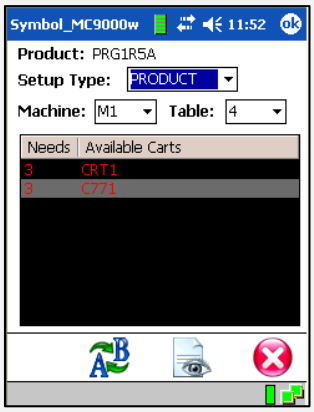
Refer to the action table below to perform a product check using the Mobile Scanner.

Step	Procedure
1	Select **Pre Setup** on scanner's Main Menu screen. 
2	Select **Product Check** on the Production main menu screen. 
3	Set the following: • Select **Route** and **Module**. The product setup associated with them appears. • Select a program from the **Select Program** list. • Select the green check mark  to continue the product check. 

4	All Cart IDs for Presets that use the selected Product will appear. • If Setup Type is set to **Product**, the **Needs** column indicates the number of feeders that require verification in order to complete the selected Product setup. • If Setup Type is set to **Mix**, the **Needs** column indicates the number of feeders that require verification in order to complete the setup for all the Products in the Mix containing the selected Product. Perform one of the following: • Select a **Cart** from the list.   Return to Product List menu.  List feeders in the setup on the selected cart.  Cancel the operation.
5	Consider a product for which all feeders are mounted on the selected feeder cart.  As shown above, the row is colored green since presetup is completed for all the feeder locations and hence **Needs** column contains 0. Upon clicking  icon, following screen is displayed:  Consider following scenario where presetup is completed for only one location out of four feeder locations.

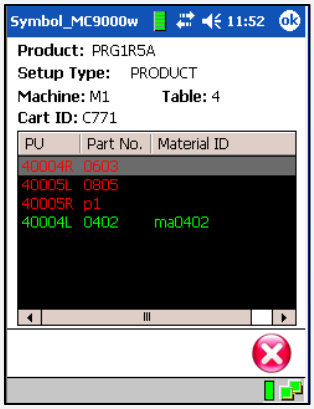


On doing **Product Check** for this Presetup, following screen appears:



As shown above, there are two presetup carts and each cart has three feeder locations for which presetup is not completed. Hence **Needs** column contains value **3** for both the presetup carts.

Select a presetup cart, and click  . Following screen appears:



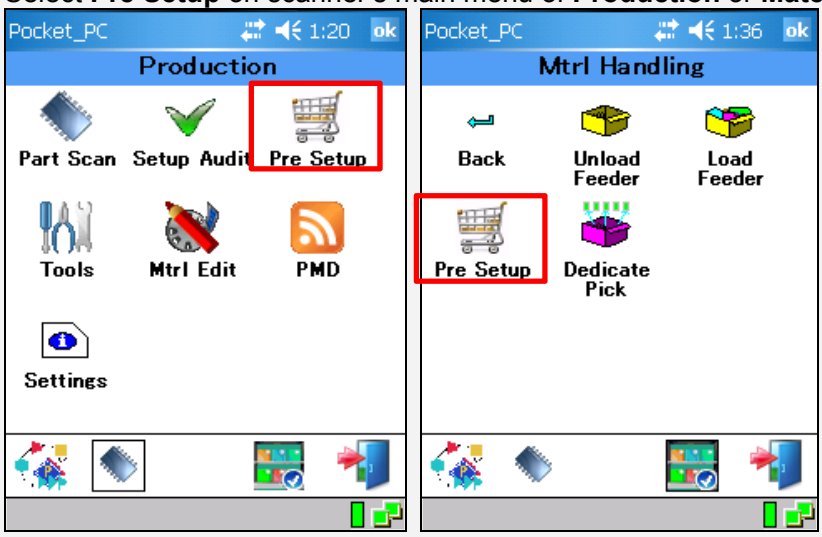
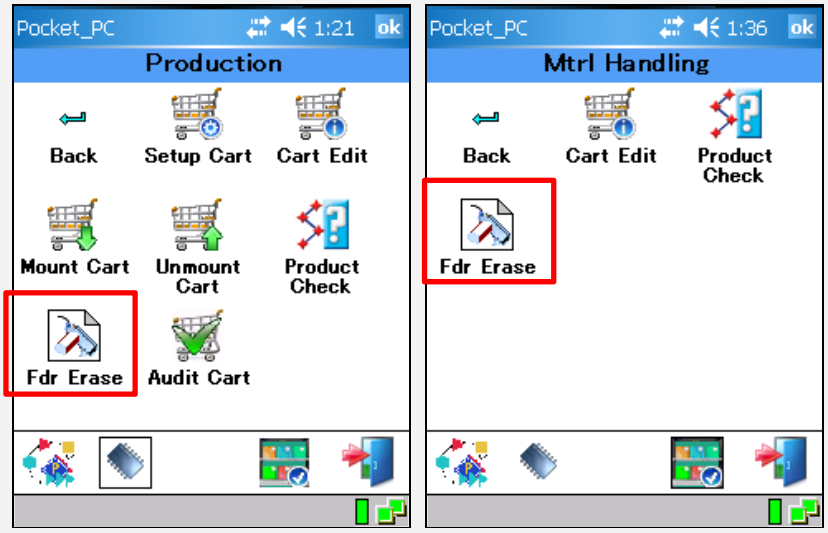
■ Removing Feeders

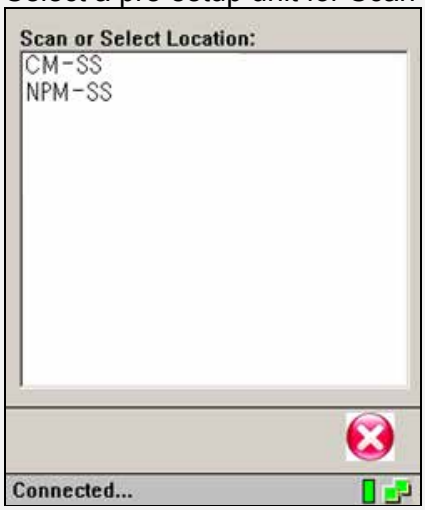
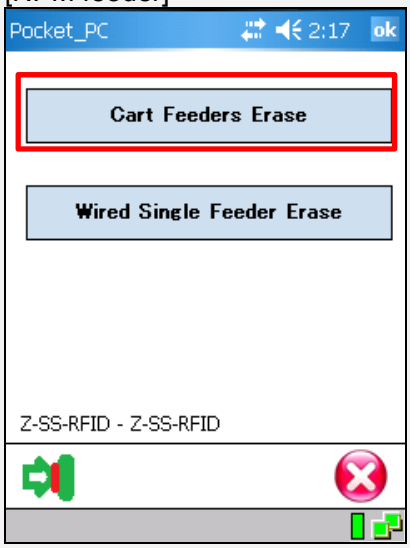
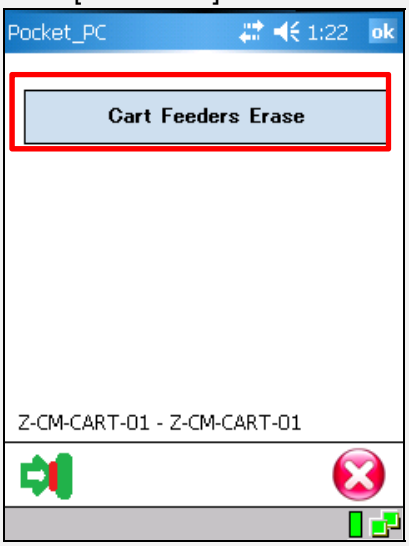
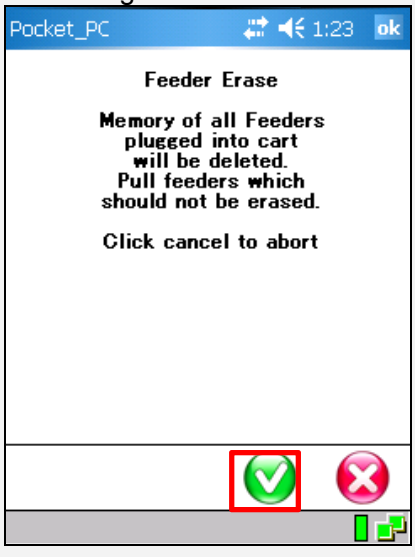
Feeder memories can be cleared collectively or individually by selecting the **Pre Setup** - **Remove Feeder** icons.

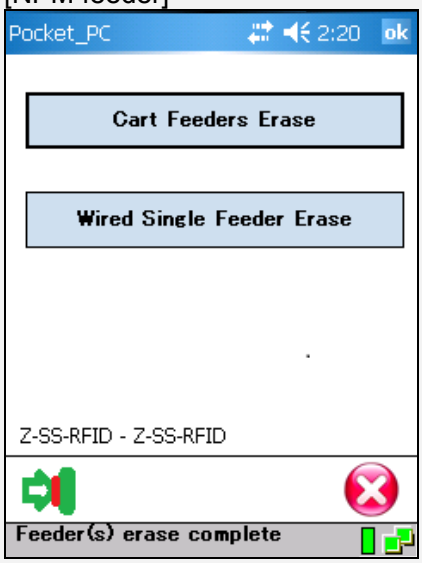
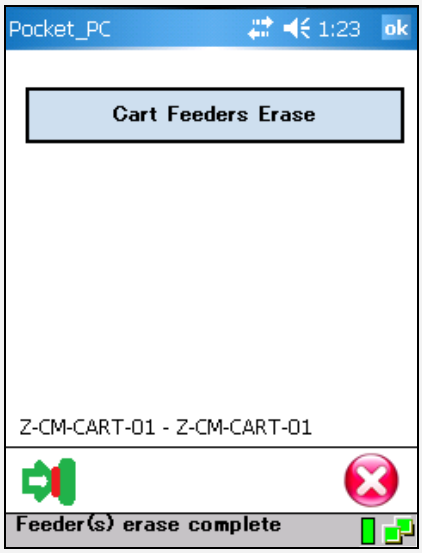
Take the steps described below.

• Remove Cart Feeder (Clear all)

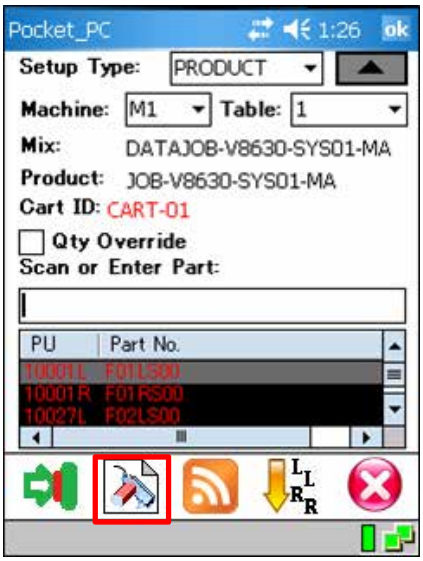
This feature allows the memories of all feeders loaded on the cart to be cleared at once. Refer to the table below to clear all feeder memories. (To keep the memory intact, remove the feeders from the cart in advance.)

Step	Procedure
1	Select **Pre Setup** on scanner's main menu of **Production** or **Material Control**. 
2	Select **Remove Feeder**. 

3	Select a pre-setup unit for Scan or Select Location. 
4	Select **Cart Feeders Erase**. *Only **Cart Feeders Erase** is available for CM. [NPM feeder]  [CM feeder] 
5	Click the green check button. 

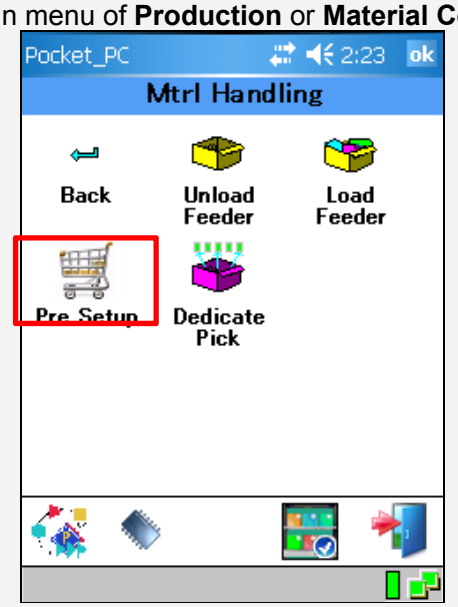
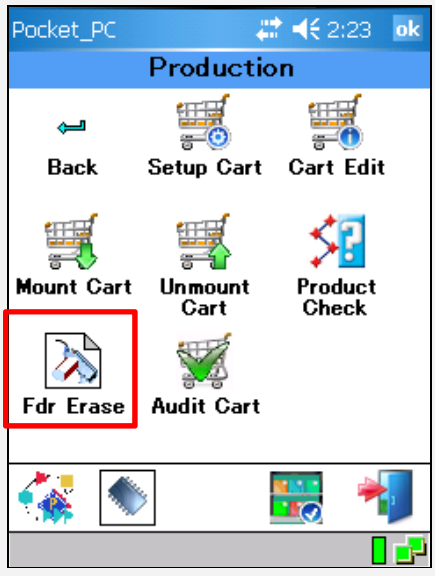
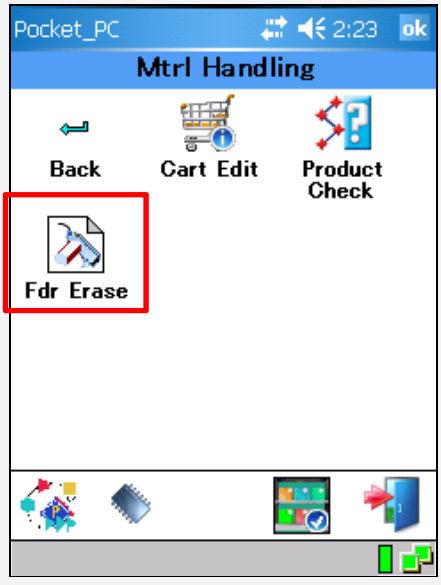
6	The memories of all feeders on the cart will be cleared. [NPM feeder]  [CM feeder] 

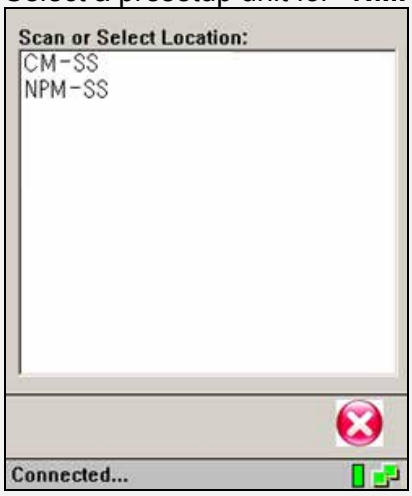
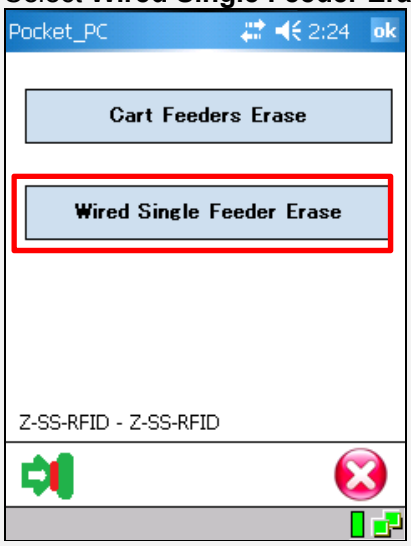
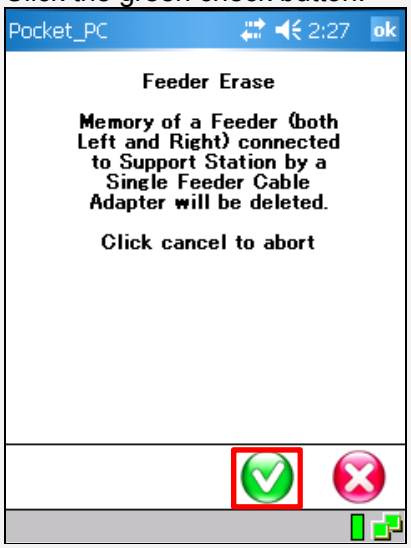
*You can also reach the screen in step 4 by pressing the button in the frame below on the Setup Cart screen.

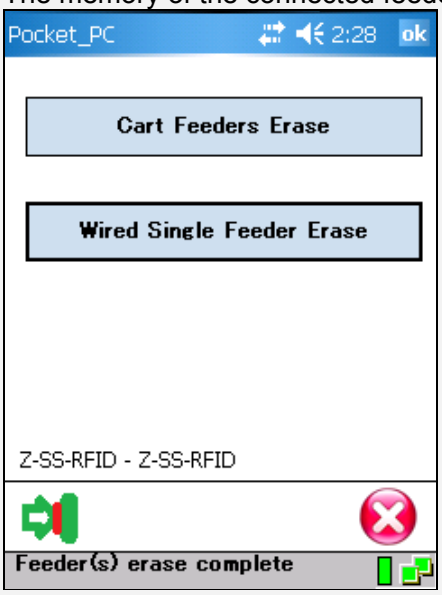


• Remove a Single Cart Feeder (Clear individually)

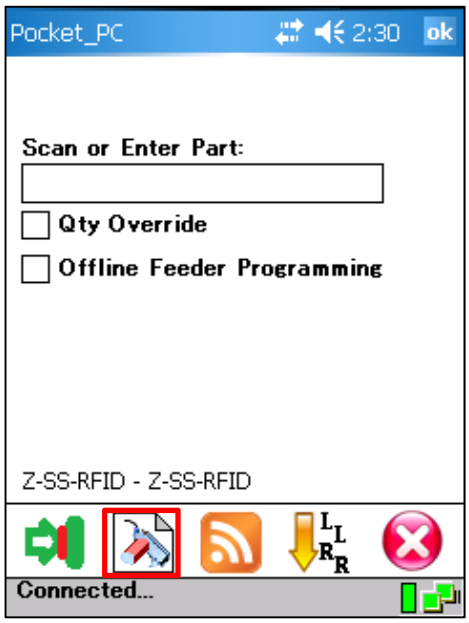
The memory of a feeder can be cleared by connecting the support station and the feeder via the feeder cable. (A presetup unit for CM is not supported.)
Refer to the table below to clear the memory of a feeder.

Step	Procedure
1	Select **Pre Setup** on scanner's main menu of **Production** or **Material Control**.  
2	Select **Remove Feeder**.  

3	Select a presetup unit for **"NMP"** by selecting a location. 
4	Select **Wired Single Feeder Erase**. 
5	Click the green check button. 

6	The memory of the connected feeder will be cleared. 

*You can also reach the screen in step 4 by pressing the button in the frame below on the Free-Form screen.



= MEMO =

6. FUNCTION - PREVENTION FROM AN INCORRECT SET - SETUP AUDIT

This chapter explains the Setup Audit.

6.1 Overview

Setup Audit allows you to revalidate Part Numbers in the currently running setup. Periodically validating the setup reduces the likelihood of building products with incorrect Part Numbers.

6.2 Settings

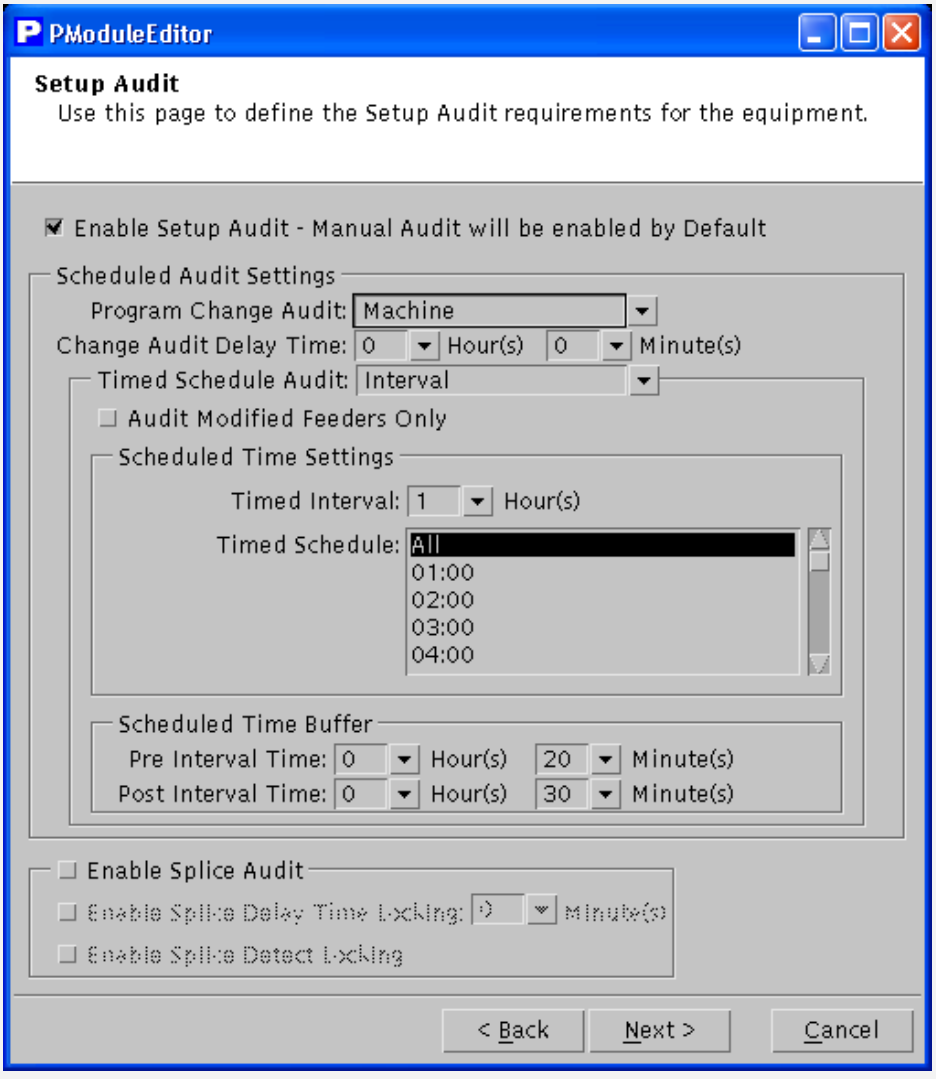
6.2.1 Setup Audit Settings

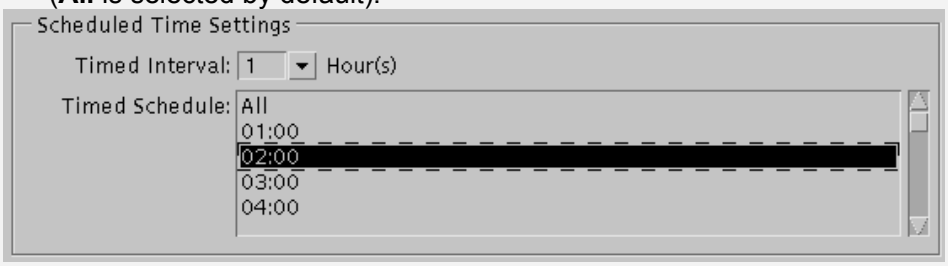
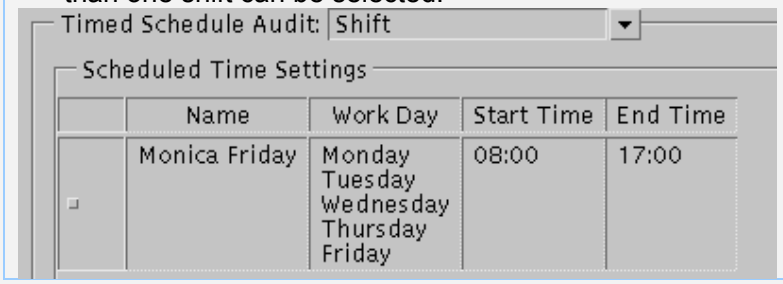
■ Configure Setup Audit for Placement Machines

Periodically validating the setup reduces the likelihood of building products with incorrect Part Numbers.

Configure the following if the Equipment Type is CM/DT/NPM series. Setup Audit configuration for Screen Printers is achieved using the Machine Editor.

Refer to the action table below to configure Setup Audit for Placement machines.

Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the machine to configure to invoke the **Module Editor**.
3	Continue through each screen by clicking **Next** until the **Setup Audit** screen is reached. 
4	Select the **Enable Setup Audit** checkbox. **Note: Manual Auditing** requires the configuration of this step only. Manual Audit

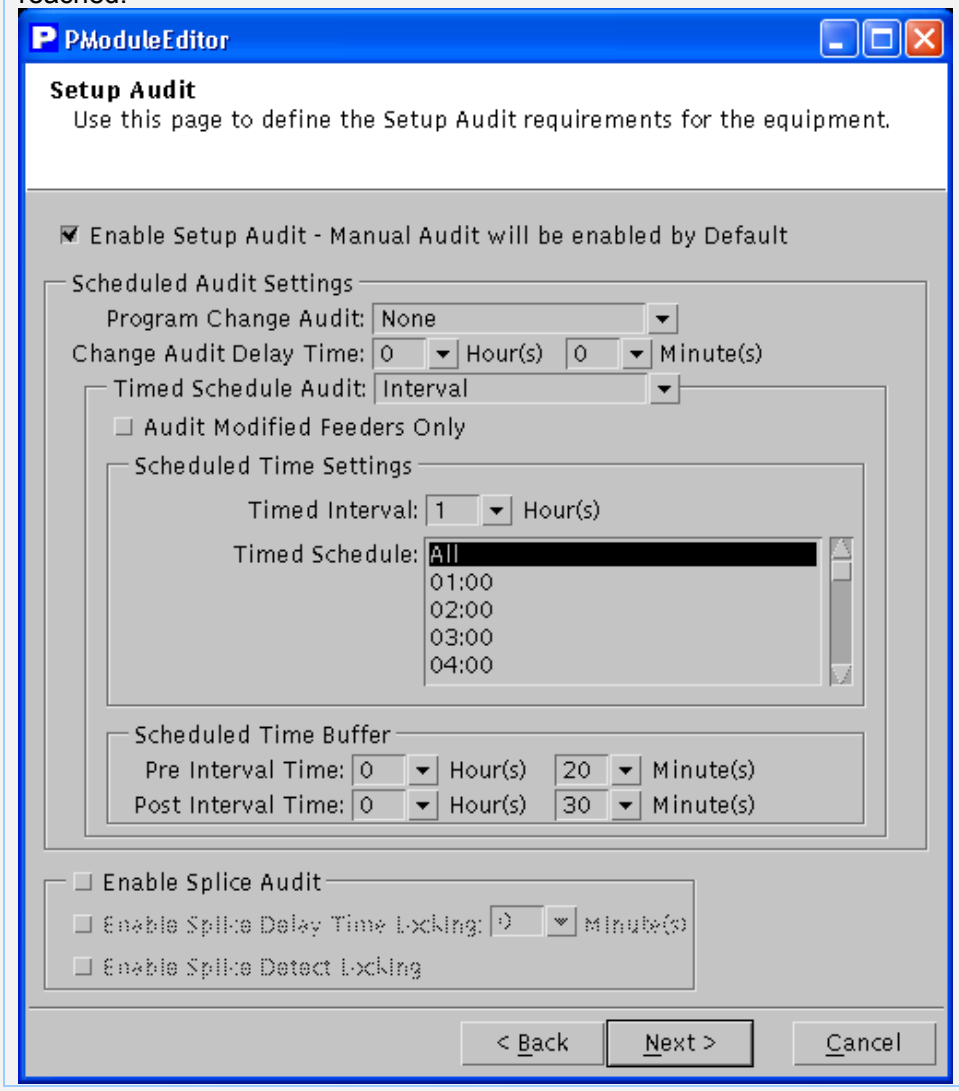
	allows users to audit a few key material positions at any time.
5	Select a scheduled **Program Change Audit**. Initiates audits when a program change has been detected. Choose one of the following drop-down options: • **None:** Program change auditing will not be done. • **Position Based:** Evaluate previous program and current program feeder setups, and require the audit scanning of feeder position differences for any machine. • **Machine Based:** Evaluate previous program and current program feeder setups and require the audit scanning of all product feeder positions for that machine. You can configure a Change Audit Delay Time which is essentially a time lag after product changeover. Within this grace period, setup audit should be completed else machine will be interlocked.
6	Select a **Timed Schedule Audit**, which allows for Interval or Shift auditing. • **Interval Auditing:** Uses timed interval scheduling. Time based audits contain time intervals that are divisible into 24 hours a day, meaning audits can be performed every 1, 2, 3, 4, 6, 8, and 12 hours. The schedule displays in the scrolling list where you may selectively disable a given scheduled audit. Unselect a scheduled interval to exclude it from the list of scheduled audits. (**All** is selected by default).  • **Shift Auditing:** Makes use of shift schedules configured in *PanaCIM Web Application* > *Administration* > *Shift Administration*. Shift schedules display where you may select specific shifts to be used for scheduled audits. More than one shift can be selected. 
7	Select a **Scheduled Time Buffer**. Scheduled Time Buffer is a grace period by which an audit must be performed. Default pre and post buffer times are set to 10 minutes. If you set the buffer time to zero, the audit time will start and end at a single point of time in which the machine will lock immediately since no buffer is configured. **Note:** Buffer pre and post times cannot overlap (i.e. An interval of 1 hour cannot have a pre-or post-interval greater than 30 minutes each, or any combination that would constitute an hour of time).

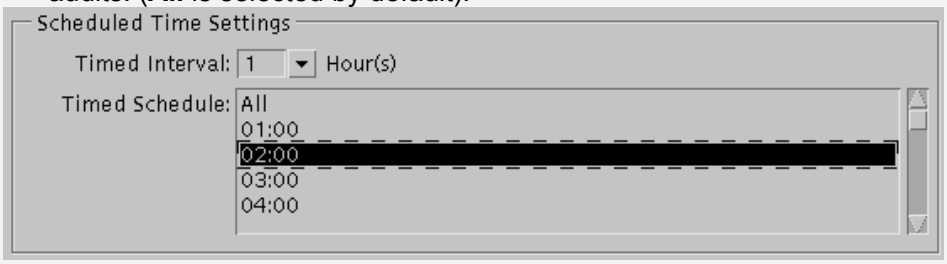
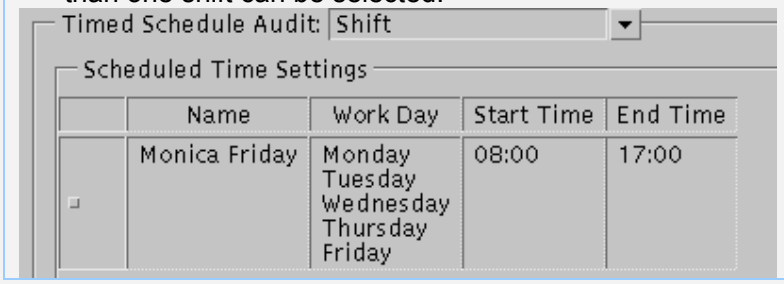
■ Configure Setup Audit for Screen Printers

Periodically validating the setup reduces the likelihood of building products with incorrect Part Numbers.

Configure Setup Audit if the Equipment Type is a Screen Printer SPV, SPG, SPD, SP70, SP60 or SP80.

Refer to the action table below to configure Setup Audit for Screen Printers.

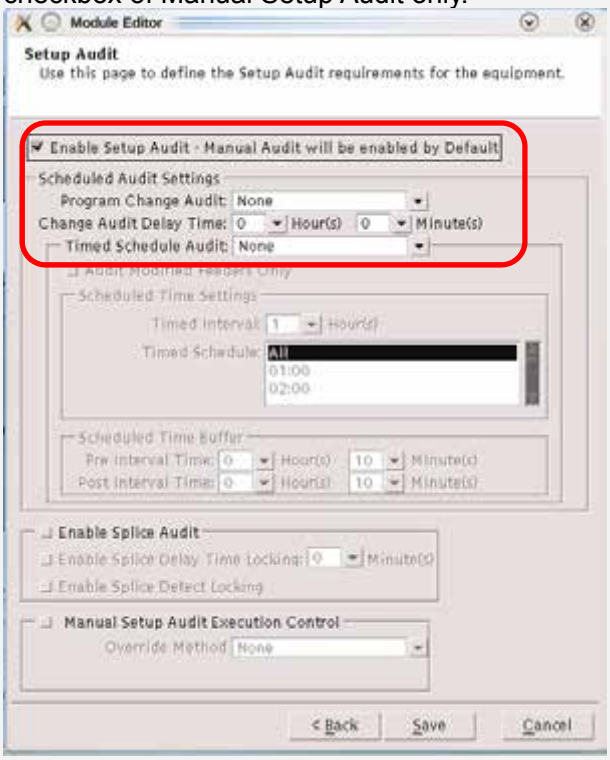
Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the machine to configure to invoke the **Module Editor**.
3	Continue through each screen by clicking **Next** until the **Setup Audit** screen is reached. 
4	Select the **Enable Setup Audit** checkbox. **Note: Manual Auditing** requires the configuration of this step only. Manual Audit allows users to audit a few key material positions at any time.
5	Select a scheduled **Program Change Audit.** Initiates audits when a program change has been detected. Choose one of the following drop-down options: • **None:** Program change auditing will not be done. • **Position Based:** Evaluate previous program and current program, and require the audit scanning of printer material position differences for any machine. • **Machine Based:** Evaluate previous program and current program, and

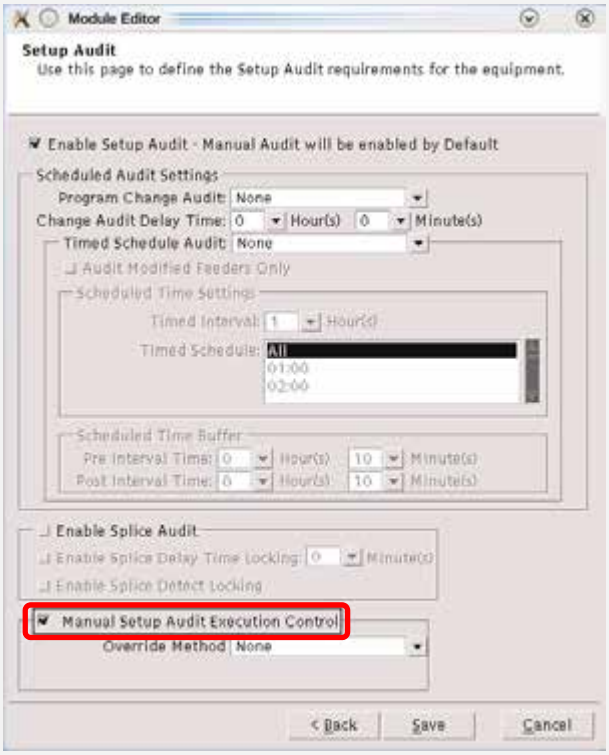
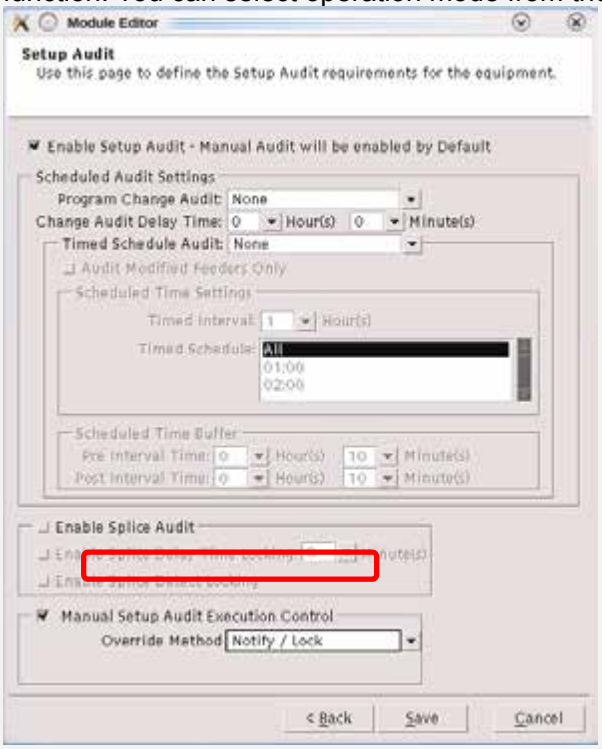
	require the audit scanning of all printer material positions for that machine.. You can configure a Change Audit Delay Time which is essentially a time lag after product changeover. Within this grace period, setup audit should be completed else screen printer will be interlocked.
6	Select a **Timed Schedule Audit**, which allows for Interval or Shift auditing. • **Interval Auditing**. Uses timed interval scheduling. Time based audits contain time intervals that are divisible into 24 hours a day, meaning audits can be performed every 1, 2, 3, 4, 6, 8, and 12 hours. The schedule displays in the scrolling list where you may selectively disable a given scheduled audit. Unselect a scheduled interval to exclude it from the list of scheduled audits. (**All** is selected by default).  • **Shift Auditing**. Makes use of shift schedules configured in *PanaCIM Web Application* > *Administration* > *Shift Administration*. Shift schedules display where you may select specific shifts to be used for scheduled audits. More than one shift can be selected. 
7	Select a **Scheduled Time Buffer**. Scheduled Time Buffer is a grace period by which an audit must be performed. Default pre and post buffer times are set to 10 minutes. If you set the buffer time to zero, the audit time will start and end at a single point of time in which the machine will lock immediately since no buffer is configured. **Note:** Buffer pre and post times cannot overlap (i.e. An interval of 1 hour cannot have a pre-or post-interval greater than 30 minutes each, or any combination that would constitute an hour of time).

6.2.2 Setting of Setup Audit (Automatic Interruption)

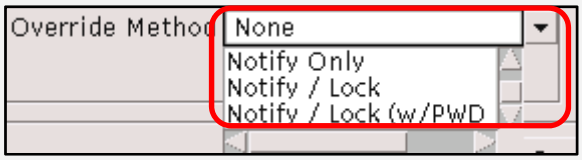
Setup audit (automatic interruption) is the function that setup audit (manual audit) operation will be interrupted once production stops due to error occurred in the machine which is on the same line of setup audit performed (manual)
The procedure for how to continue setup audit (manual audit) operation which is temporarily interrupted is described below.

Note that this function does not support the printing machine.

Step	Procedure
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the machine to configure to invoke the **Module Editor**.
3	Continue through each screen by clicking **Next** until the **Setup Audit** screen is reached.
4	**Enable Setup Audit:** Mark the checkbox if the setup audit function is enabled. **Note)** If you want to use the setup audit (automatic interruption) function, mark the checkbox of Manual Setup Audit only. 

Step	Procedure
5	**Manual Setup Audit Execution Control**: Mark the checkbox to enable the setup audit (automatic interruption) function. 
6	**Override Method**: Select operation mode for the setup audit (automatic interruption) function. You can select operation mode from the following four types as follows. 

<Operation mode selection>



- **None:**
  The setup audit (automatic interruption) function is not used.
- **Notify Only:**
  The setup audit (automatic interruption) function is used.
  If an error occurred on the machine while setup audit is performed, a message and the machine number that the error occurs appears on the pop-up screen.
  You can continue setup audit, click the Confirm icon on the pop-up screen.
- **Notify / Lock:**
  The setup audit (automatic interruption) function is used.
  If an error occurred on the machine while setup audit is performed, a message and the machine number that the error occurs appears on the pop-up screen.
  You cannot continue setup audit unless you reset the error on the machine.
- **Notify / Lock(w/PWD Override)**:
  The setup audit (automatic interruption) function is used.
  If an error occurred on the machine while setup audit is performed, a message and the machine number that the error occurs appears on the pop-up screen.
  You can continue setup audit by entering the password on the pop-up screen.

6.3 Operation

6.3.1 Manual Audit

Manual auditing is an on demand type of auditing and does not require scheduling as a prerequisite performing an audit. Manual Auditing only requires the enabling of Setup Audit functionality for the machine/module.

In PanaCIM 8.4, current state and progress of manual setup audit is maintained. This assists in continuing an existing manual setup audit when interrupted for other operations on the scanner. Hence, operator can abandon existing manual setup audit by going out from the audit screen, fulfill a pending feeder jam or part exhaust and then resume manual audit from where he previously left off. This implementation of Manual Setup Audit is done for both placement machines and screen printers in both Material Verification and Material Control modes.

Also, manual setup audit can be reset in order to perform full audit scan of all feeder locations. Manual setup audit is reset only for the current working area. Manual Audit reset does not clear parts scan mismatch errors.

Apart from this, type of manual audit currently being performed is displayed on the setup audit screen of mobile scanner.

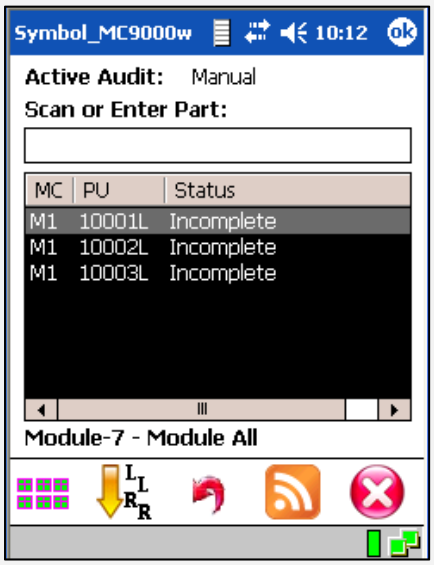
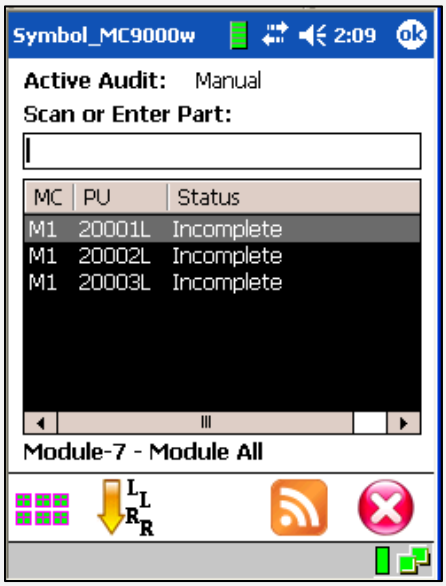
The following steps show you how to perform a Manual Audit using the Mobile Scanner. Actions performed at the scanner consequently the Route Status Application. Route Status will display the current Setup Audit feeder status for a given machine.

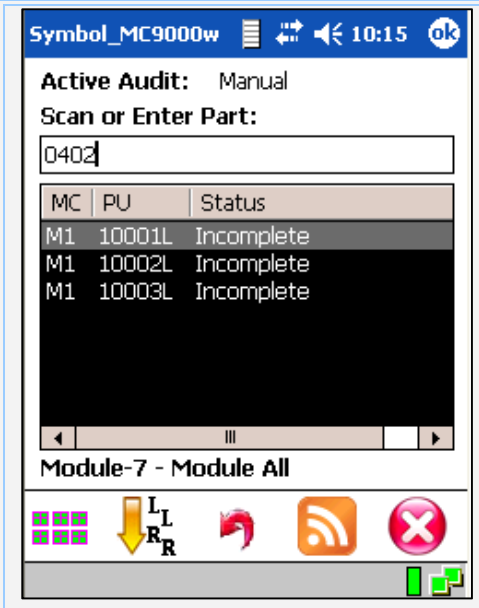
Manual Audit is designed to be used to audit only a few key components, not the entire setup.

To access the Route Editor display:
a. Select the appropriate **Zone** in the Route.
b. Click the **Setup** tab.
c. Click the appropriate **Table** tab.
d. Click **Setup Audit Status** and **Last Audit Scan Time** checkboxes in order to display those columns.

* If you encounter any error messages when performing your audit, please reference the Setup Audit Error table at the end of this section.

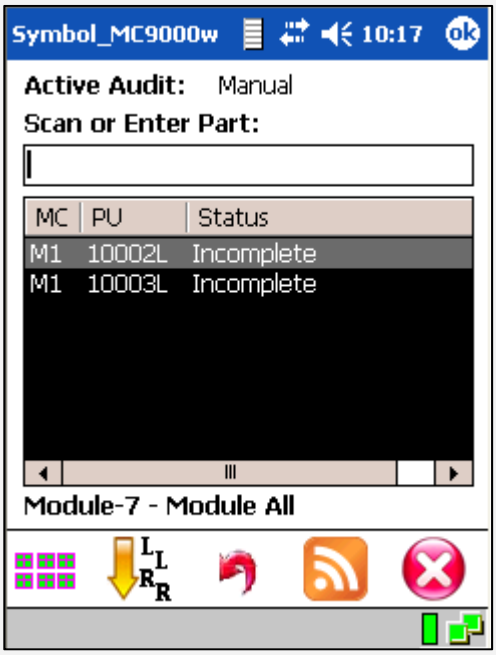
Refer to the action table below to perform a manual audit using the Mobile Scanner.

Step	Procedure
1	From the Mobile Scanner, select  from the Production main menu display.
2	Select the **Machine/Location** (if not already highlighted), and then scan the **Part**.  If Setup Audit is being performed for the first time for a product,  icon does not appear as shown below: 
3	Scan the part as shown below:



4

After a part has been successfully validated, the machine/location item will be removed from the display list on the Mobile Scanner.



After auditing a single location of a program for which Manual Audit is being initiated for the first time, audit reset icon  is displayed at the bottom of the screen.  icon is also displayed when user again initiates setup audit after closing the previous manual audit either by clicking  or after automatic logout. Reset works only for the currently selected work area.

After automatic logout or clicking  , when 

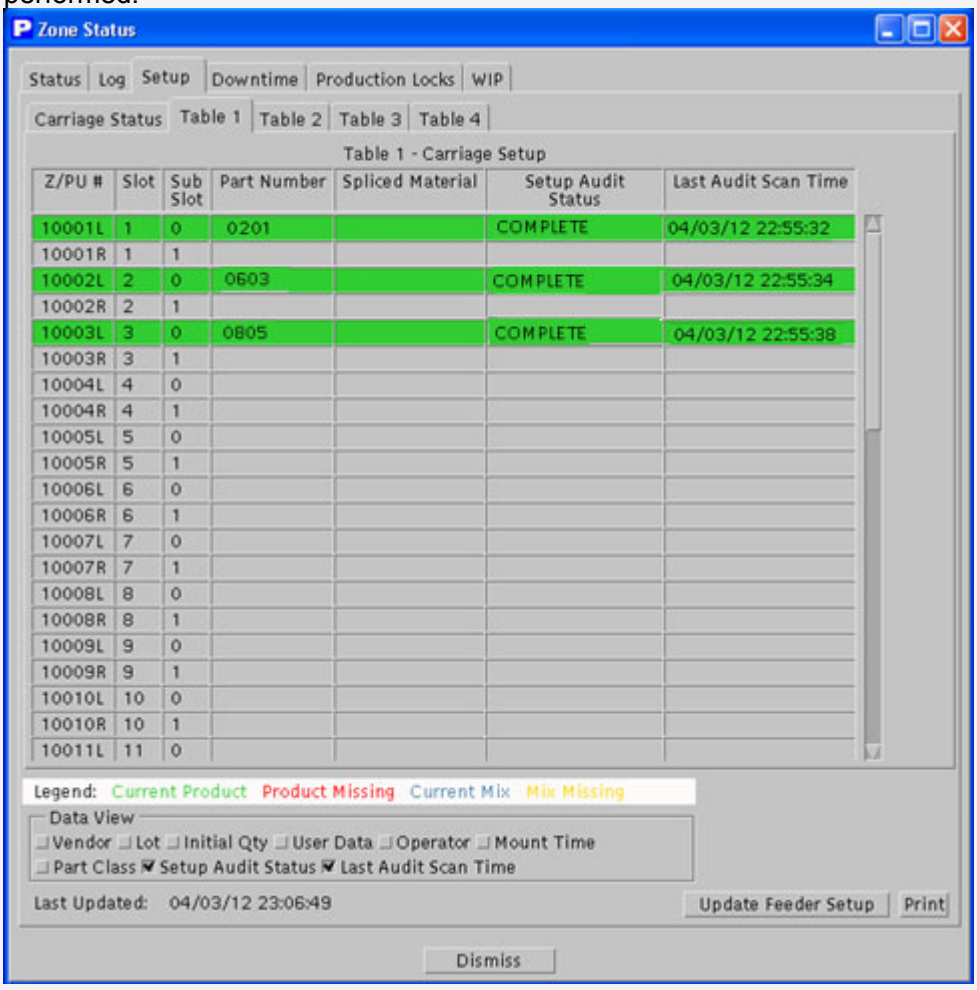
is clicked in the production main menu screen, above screen shown in step 3 appears. Thus the current state of currently active manual setup audit is maintained every time it is launched.

5

If the user manually audits all positions, the following confirmation will appear on the Mobile Scanner.



The Route Status Application displays COMPLETE in the Setup Audit Status column, along with corresponding timestamps displaying when the audit was performed.



6.3.2 Program Changed Audit

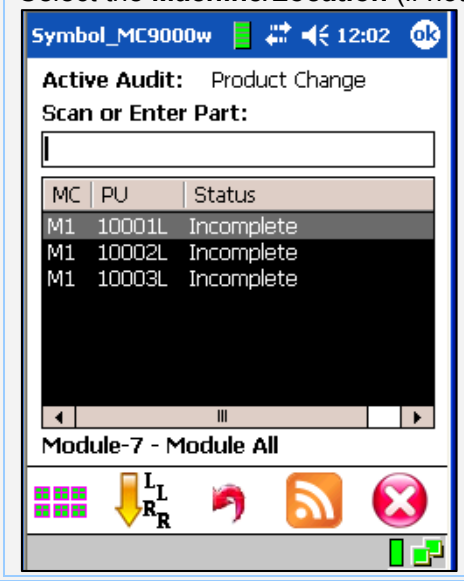
A Program Change Audit initiates audits when a program change has been detected at either the Position or Machine.

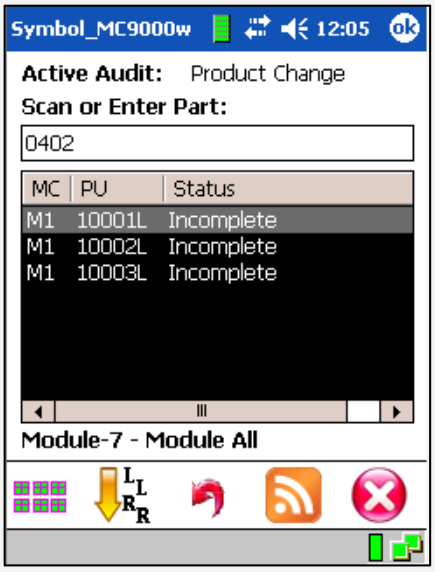
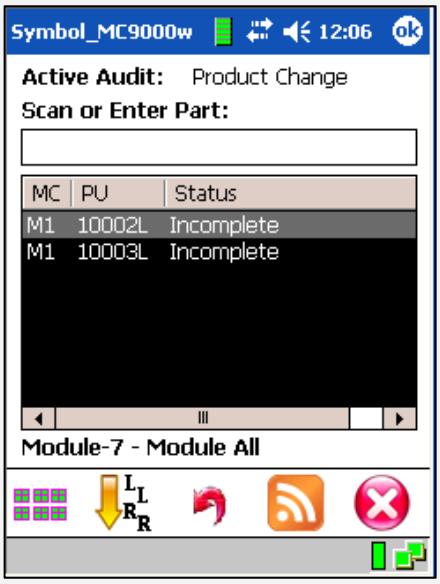
The following steps provide an example of how to perform a Program Change Audit by Machine using the Mobile Scanner. These procedures also incorporate how the actions performed on the scanner affect the screens in the Route Status Application. To access the Route Editor display:

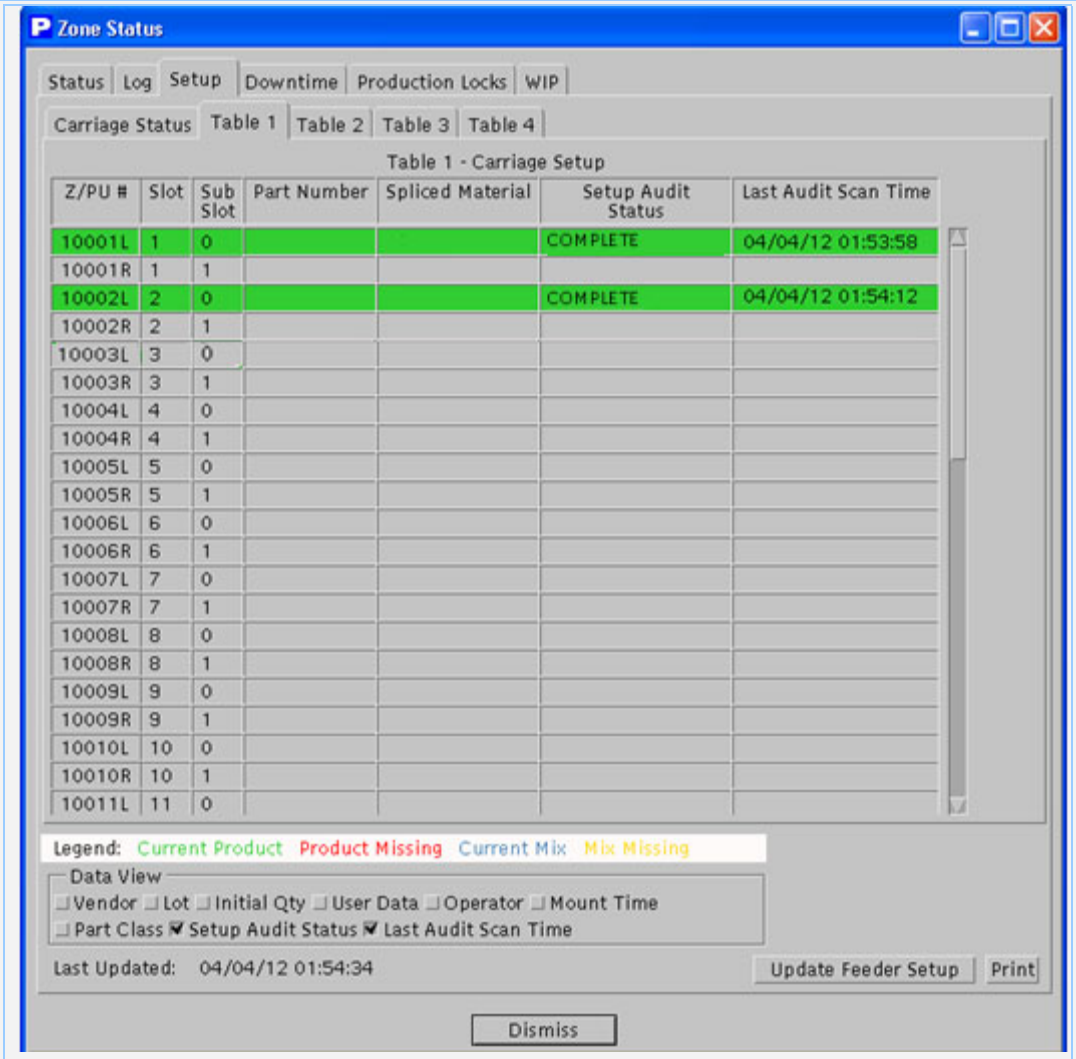
a. Select the appropriate **Zone** in the Route.
b. Click the **Setup** tab.
c. Click the appropriate **Table** tab.
d. Click **Setup Audit Status** and **Last Audit Scan Time** checkboxes in order to display those columns, which will be blank by default.

* If you encounter any error messages when performing your audit, please reference the Setup Audit Error section for guidance.

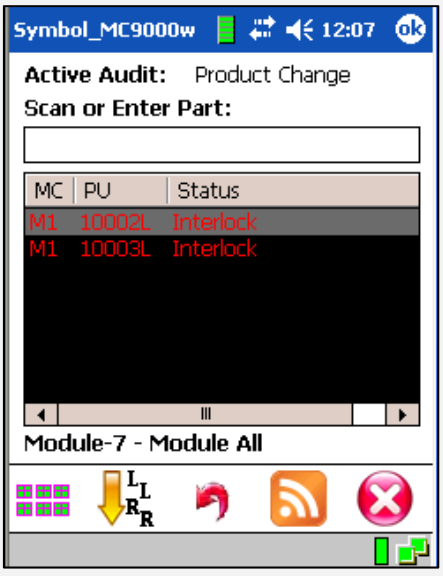
Refer to the action table below to perform a program change audit using the Mobile Scanner.

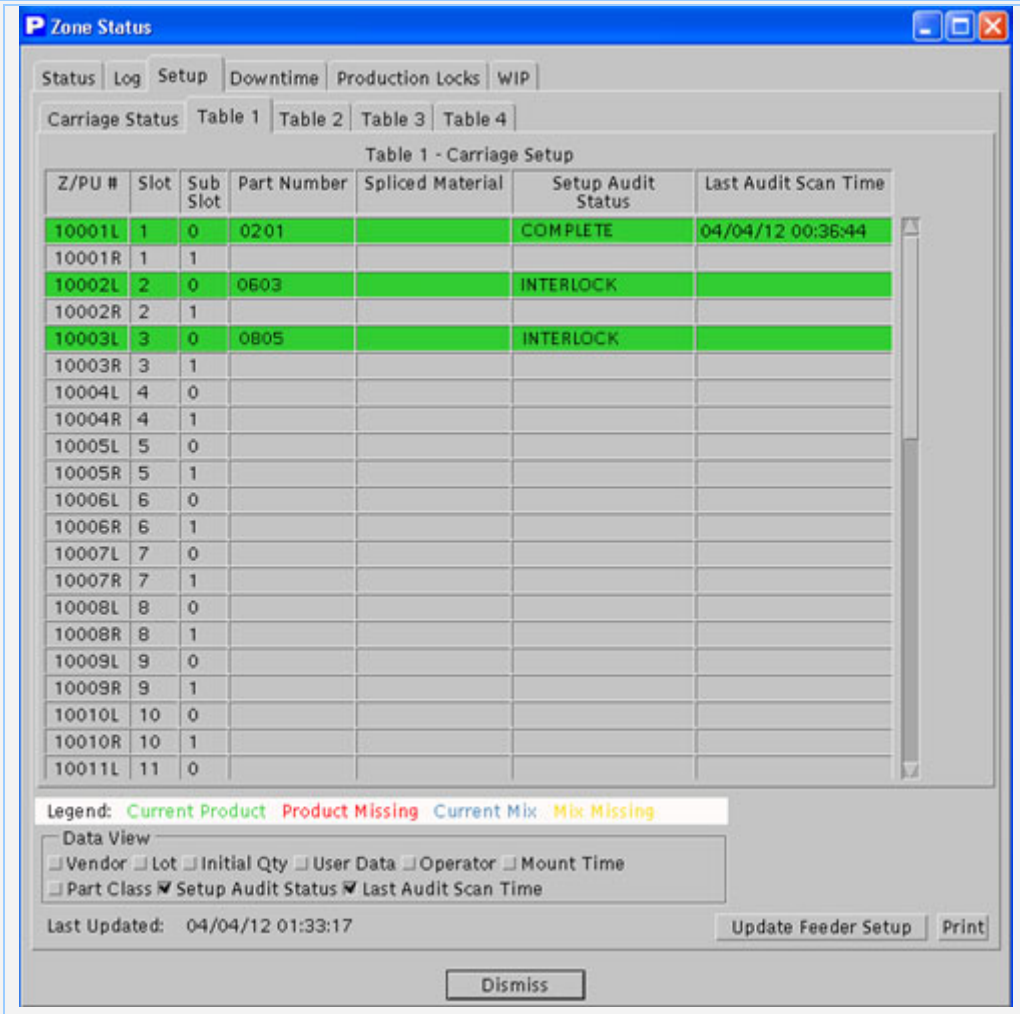
Step	Procedure
1	Setup Audit cannot begin until the Product Setup for the changeover product has been fully mounted. Mounting the setup can occur in 3 ways: 1. PreSetup cart mount operation of known Cart ID and contents. 2. Cart mount without PreSetup via Feeder Set Check. 3. Part Scan operation for each feeder position. Feeders mounted using option 1 or 2 require subsequent setup audit since the components have not been recently scanned/validated. Choosing option 3 will not require setup audit since the components were just scanned/validated during Part Scan operation.
2	From the Mobile Scanner, select  from the Production main menu screen.
3	The machine will be locked in preparation for the Program Change Setup Audit, therefore the **Interlock** message displays. Select the **Machine/Location** (if not already highlighted), and then scan the **Part**. 

4	Scan the part as shown below: 
5	After a part has been successfully validated, the machine/location item will be removed from the list.  Feeder locations for which Product Change audit is successfully completed are marked **"COMPLETE"** in the **Route Status** application:

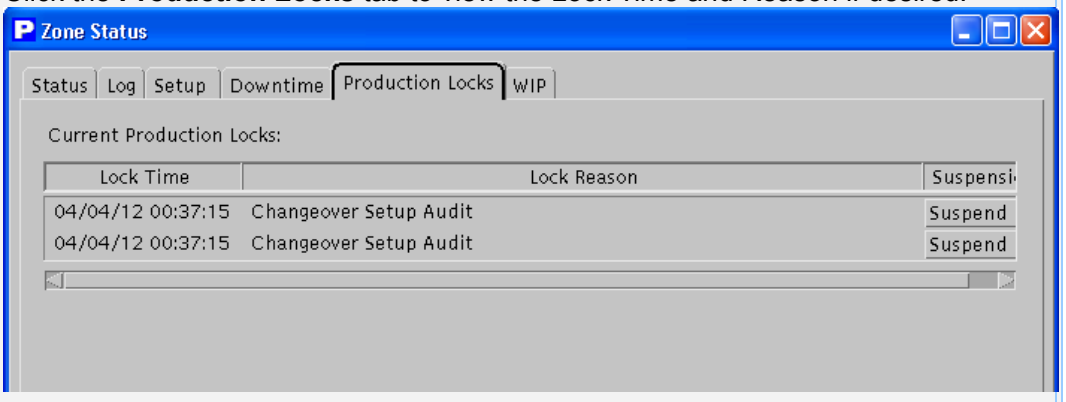


6 Mobile scanner and Route Status indicate an **Interlock** Status, if Program Change Audit is not completed within the scheduled time. This is shown below:

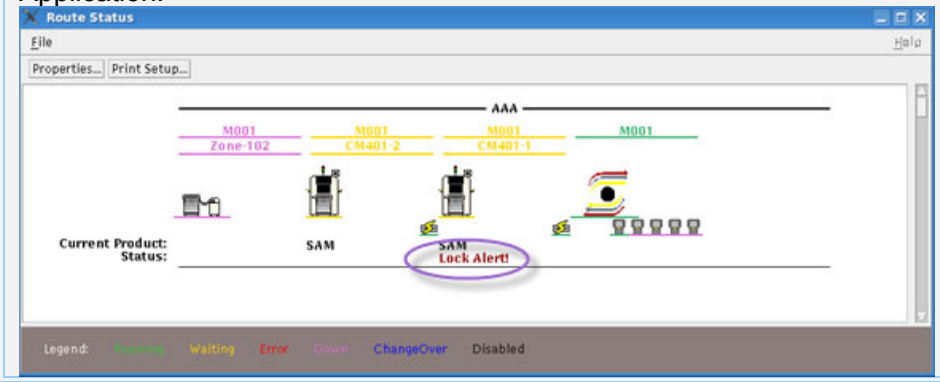




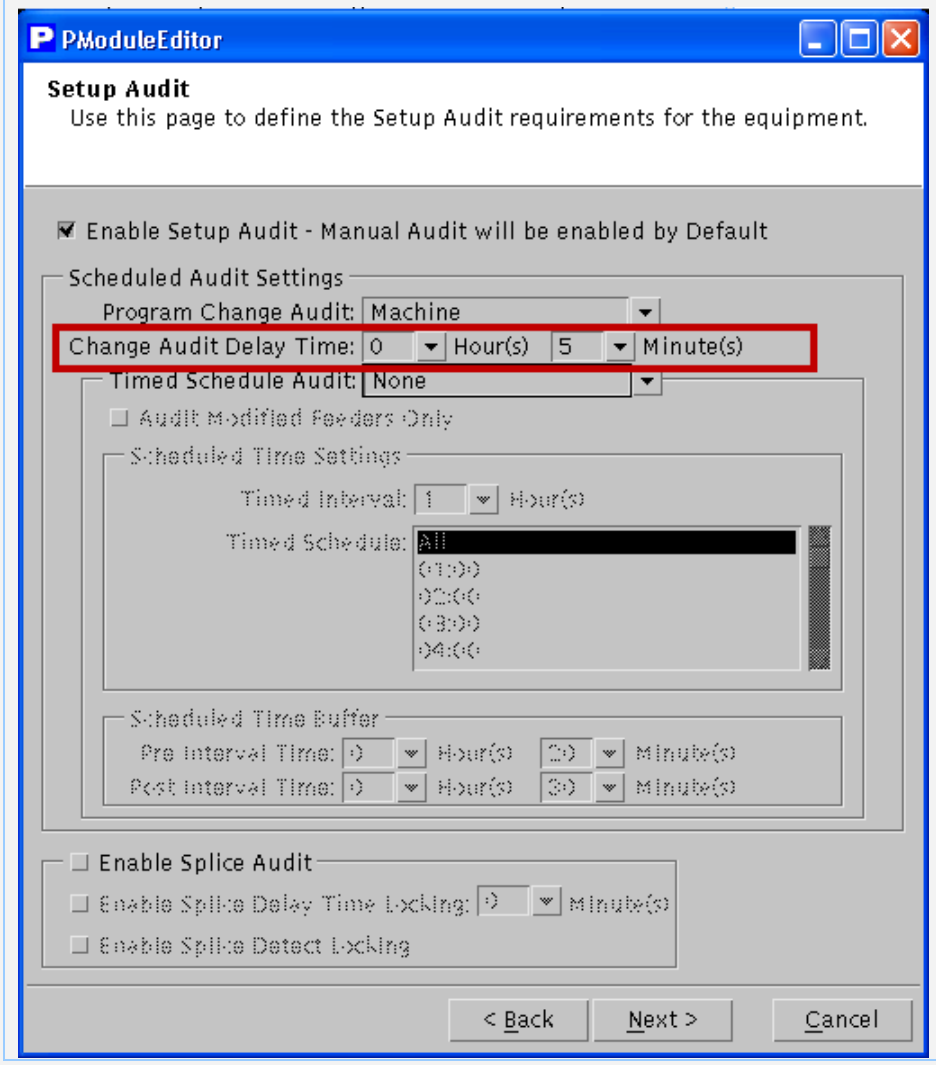
Click the **Production Locks** tab to view the Lock Time and Reason if desired.



You may also view the **Lock Alert** on the main screen of the Route Status Application.



Change Audit Delay Time by which product change should be completed is set when configuring Setup Audit as shown below:



7 After Setup Audit is completed, the following message appears on the Mobile Scanner.



* If you are performing Product Change Audit, the Reset icon becomes available. Select  to invoke **Setup Audit Reset** processing dialog. An audit reset results in all current feeder locations for the current work area for the given Product feeder locations to be set to incomplete. Essentially it resets an audit and begins a new one.

6.3.3 Timed Audit

A Timed Audit allows for Interval or Shift auditing. Interval Auditing uses timed interval scheduling whereas Shift Auditing makes use of shift schedules configured in Production Analysis.

The following steps provide an example of how to perform a Timed Audit using the Mobile Scanner. These procedures also incorporate how the actions performed on the scanner affect the screens in the Route Status Application.

To access this display:

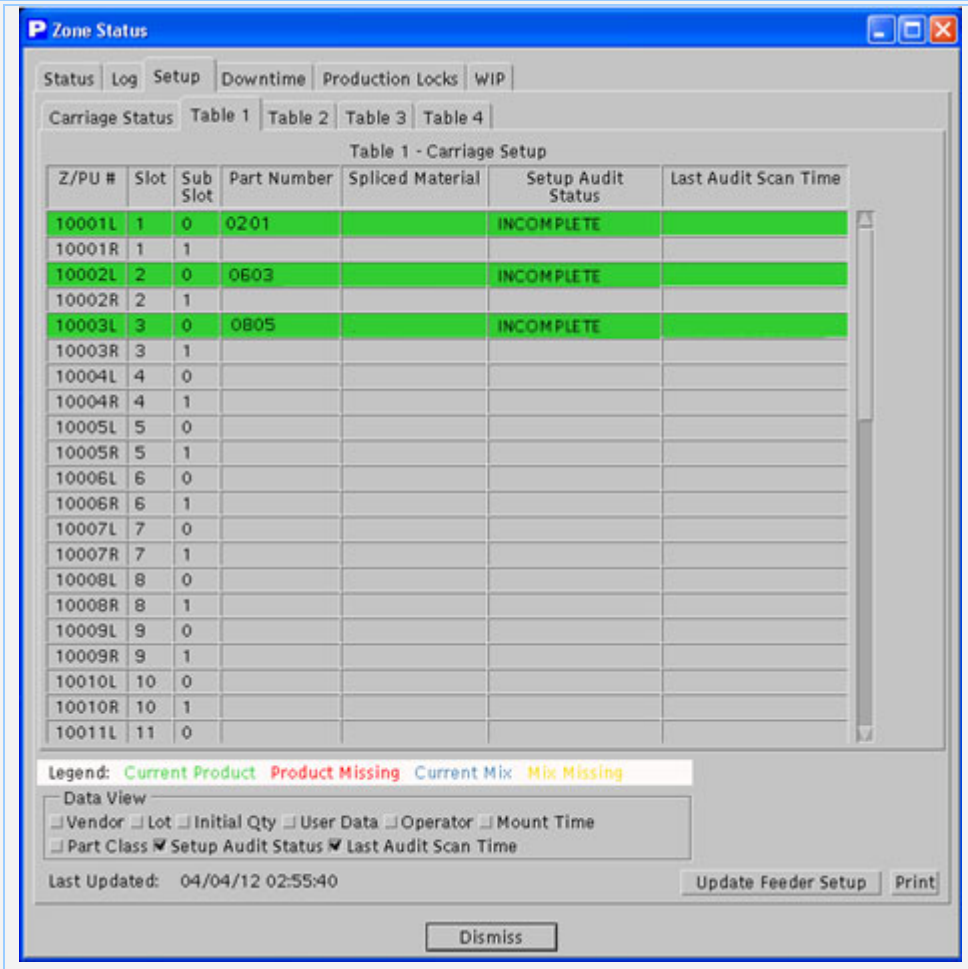
a. Select the appropriate **Zone** in the Route.
b. Click the **Setup** tab.
c. Click the appropriate **Table** tab.
d. Click the **Timed Audit Status** and **Times Audit Time** checkboxes in order to display those columns, which will be blank by default.

Important: The machine is still running during the scheduled buffer time. If you do not complete the Timed Setup Audit successfully within the scheduled buffer time, the Setup Audit Status will change to INTERLOCK, thus locking the machine. Setup Audit will have to be completed successfully in order to unlock the machine.

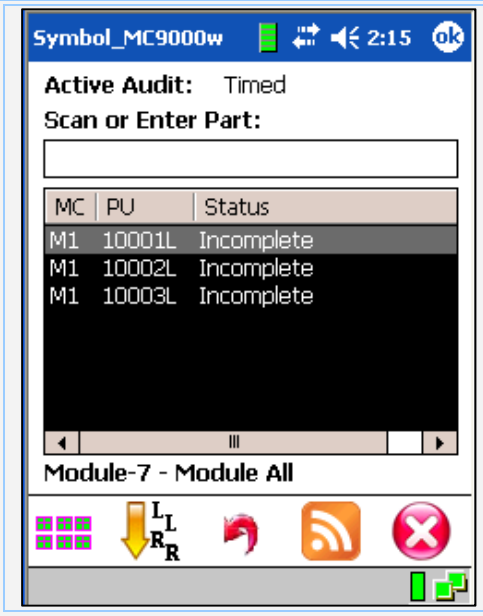
* If you encounter any error messages when performing your audit, please reference the Setup Audit Error section for guidance.

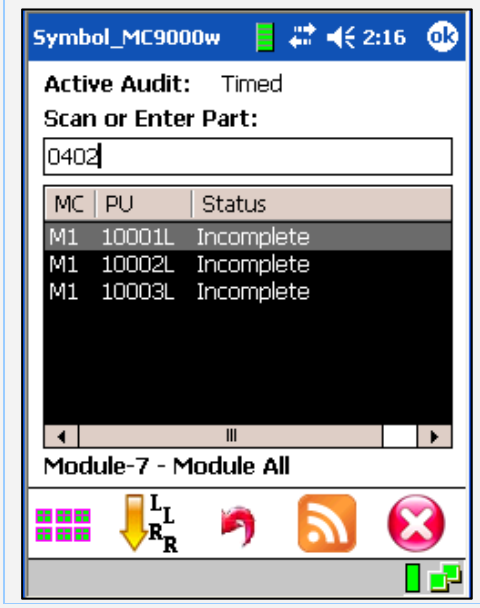
Refer to the action table below to perform a timed audit using the Mobile Scanner.

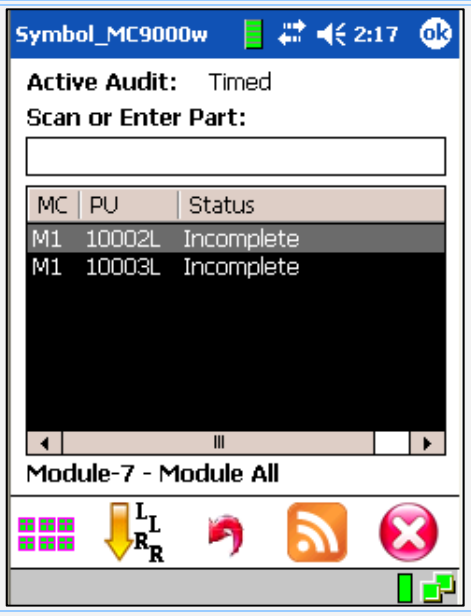
Step	Procedure
1	The Route Status Application displays the feeders as **INCOMPLETE** when the buffer time begins.



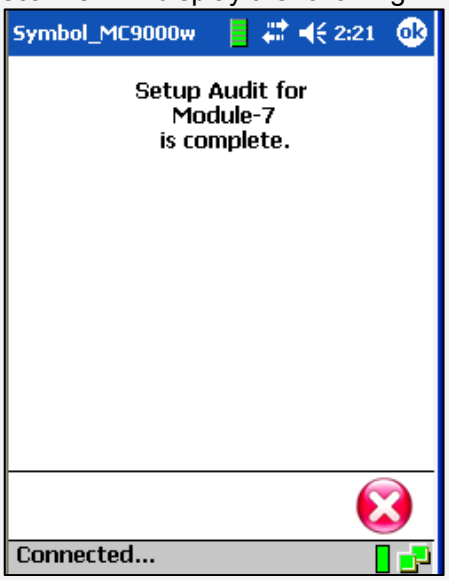
2	From the Mobile Scanner, select  from the Production main menu screen.
3	The Status for the machine will be displayed as **INCOMPLETE** on the Mobile Scanner until Setup Audit is completed. Select the **Machine/Location** (if not already highlighted), and then scan the **Part**.



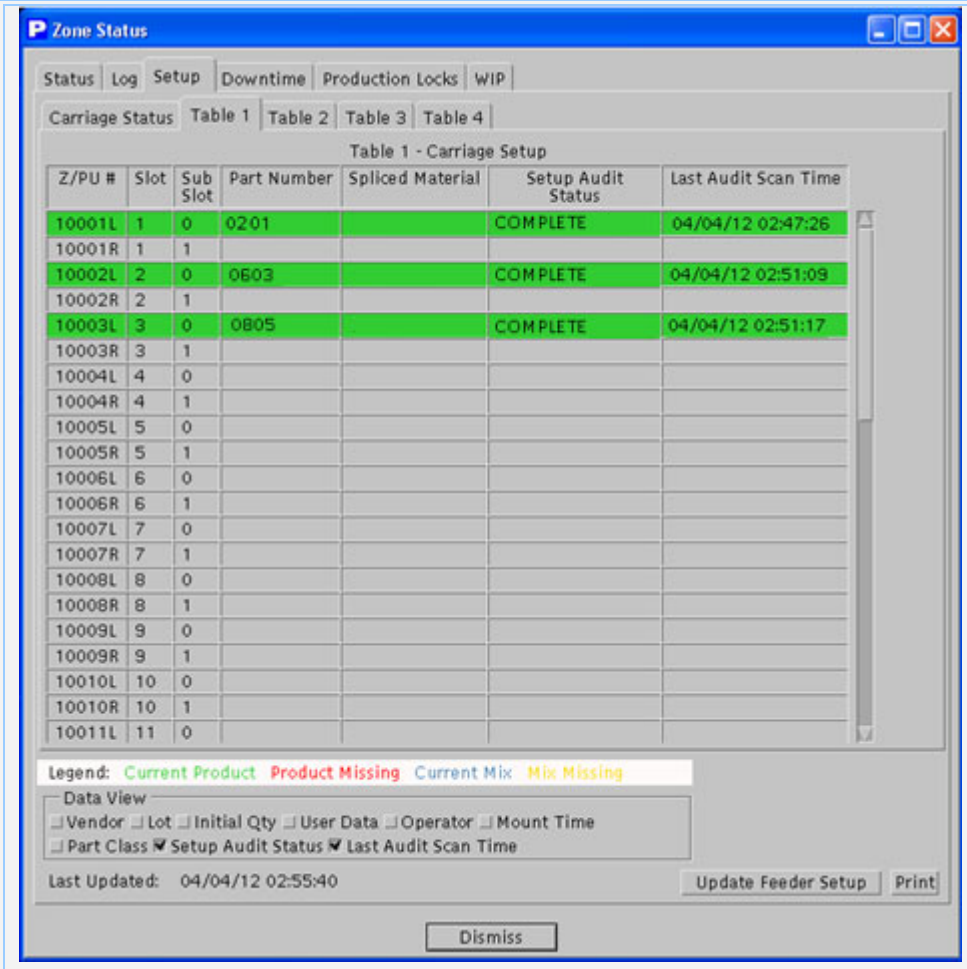
4	Scan the part as shown below: 
5	After a part has been successfully validated, the machine/location item will be removed from the list.



6 After successful part scan, and the Timed Setup Audit is completed, the scanner will display the following.



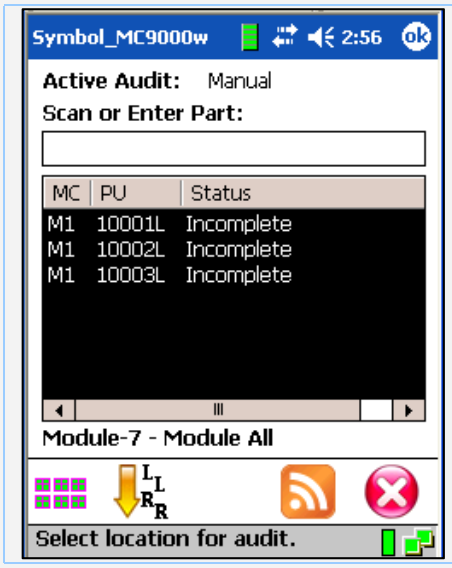
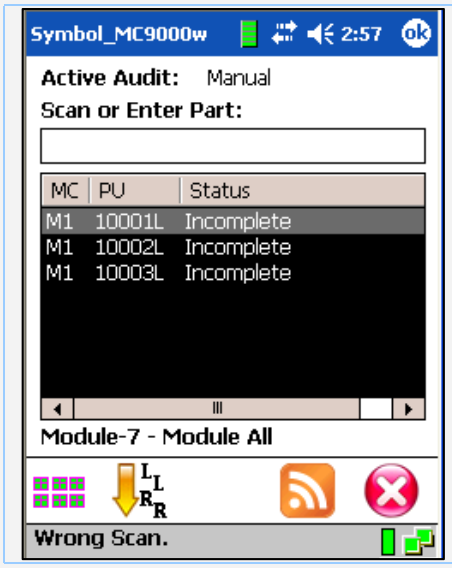
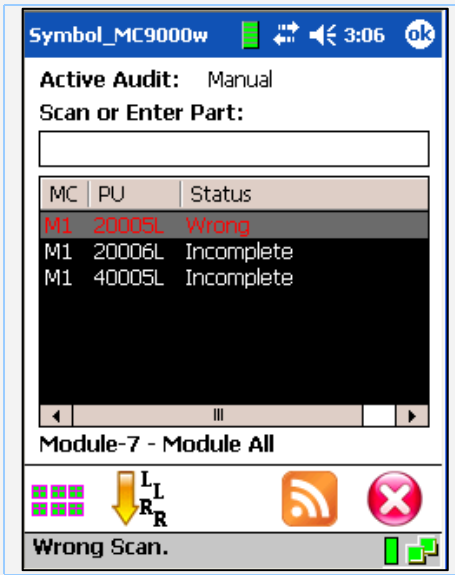
Route Status will also display a **COMPLETE** status with corresponding timestamps.

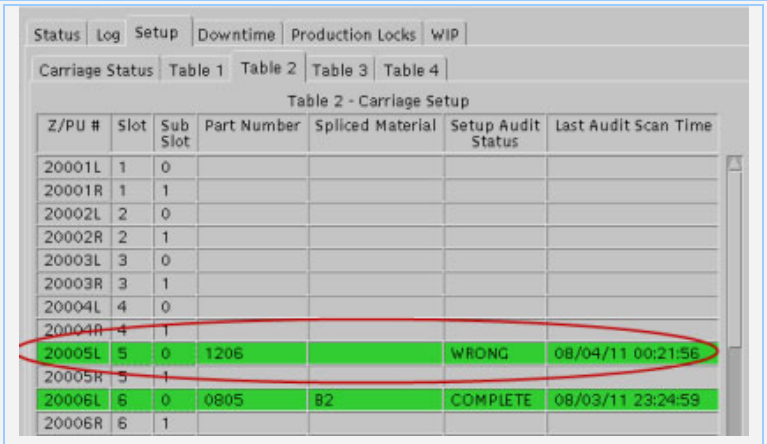


* If you are performing a Timed Audit, the Reset icon becomes available. Select  to invoke **Setup Audit Reset** processing dialog. An audit reset results in all current feeder locations for the current work area for the given Product feeder locations to be set to incomplete. Essentially it resets an audit and begins a new one.

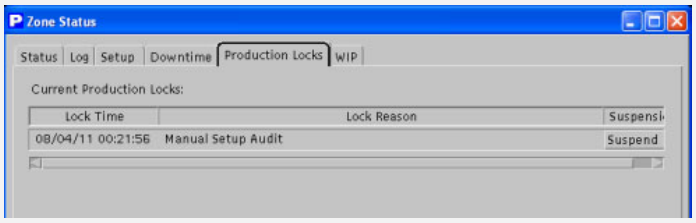
6.3.4 Setup Audit Errors

The following table illustrates common errors that may occur when performing a Setup Audit.

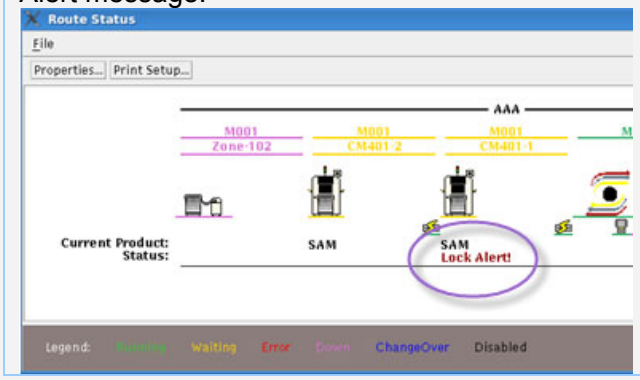
Screen	Description
	If you fail to select a location before scanning the part, you will be prompted to do so. Select a location to proceed. **Note:** As long as there is at least one item in the list, an item must be selected. It is not possible to leave an item unselected. This process will continue until you exit out of the dialog (click the red X), or all locations are successfully validated.
	If you perform an incorrect part scan on your first attempt, you will receive the following message. Rescan the correct part to proceed.
	If, on the *second* attempt, the part is scanned incorrectly, you will receive the following message on the scanner and the machine will lock. The machine cannot be unlocked until Setup Audit is completed successfully. Scanning the incorrect part on the second attempt also affects the information in the Route Status Application: The Table Setup tab will display WRONG for the incorrectly scanned feeder:



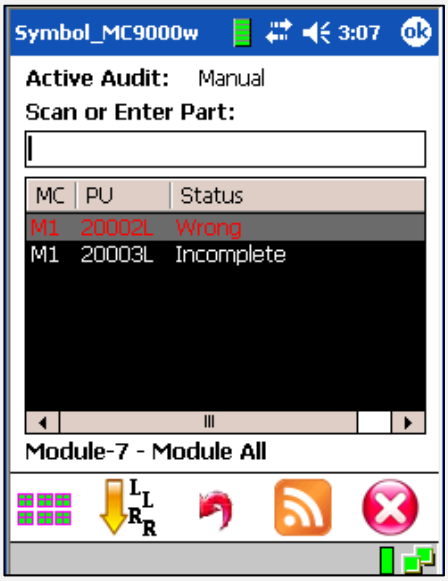
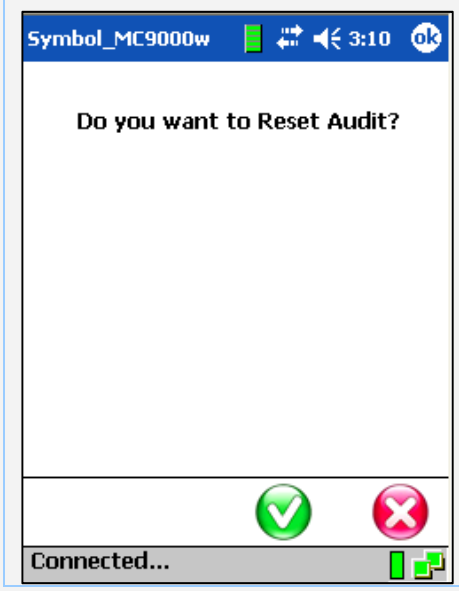
The **Production Locks** tab in Route Status displays the Lock Time and Lock Reason:

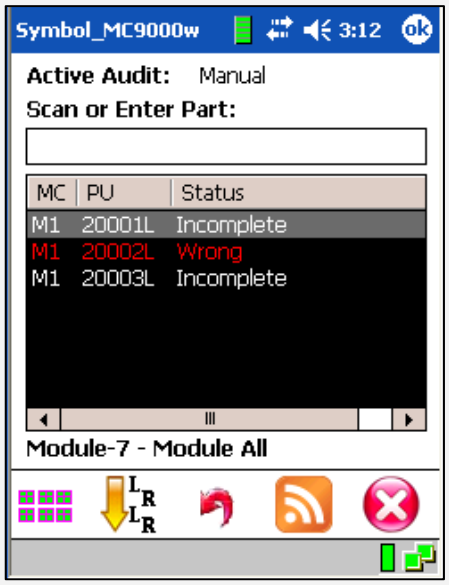


The Route Status Application main screen displays a Lock Alert message.

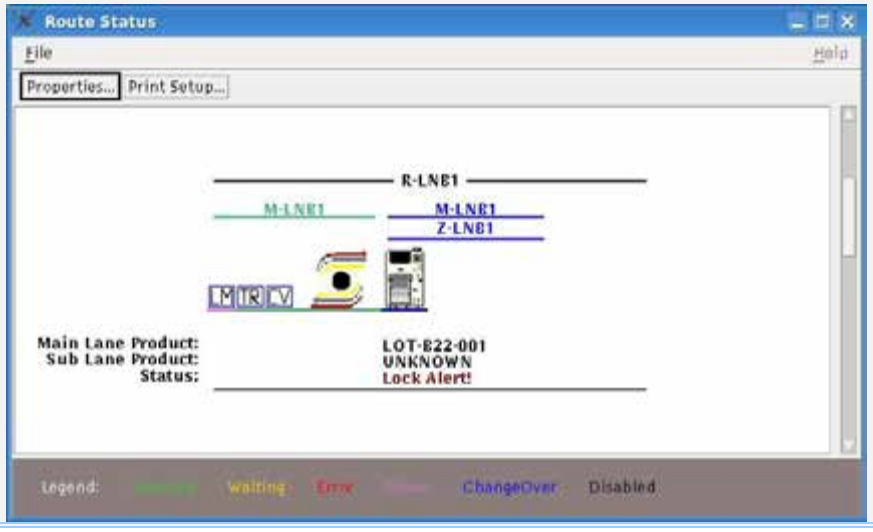


6.3.5 Resetting Manual Audit

Step	Procedure
1	Consider the following scenario in which out of three feeder locations, first one is successfully audited, second has a wrong scan and third is yet to be audited.  **Note:** There will be no machine locking for the Manual Audit except for a wrong scan which is two incorrect audit scans for any single feeder location. When machine is locked, **"Lock Alert"** appears on the **Route Status** screen. This lock can be cleared by suspending the manual audit from the **Route status** screen. When suspended manual audit is again resumed, previously maintained status of wrong scans is again displayed. Manual audit can only be suspended by operator who has administrative privileges.
2	Click the  icon to reset the manual audit. After doing so, a confirmatory message appears to reset the audit. 
3	Click the  in the above shown confirmatory message. This resets the manual audit

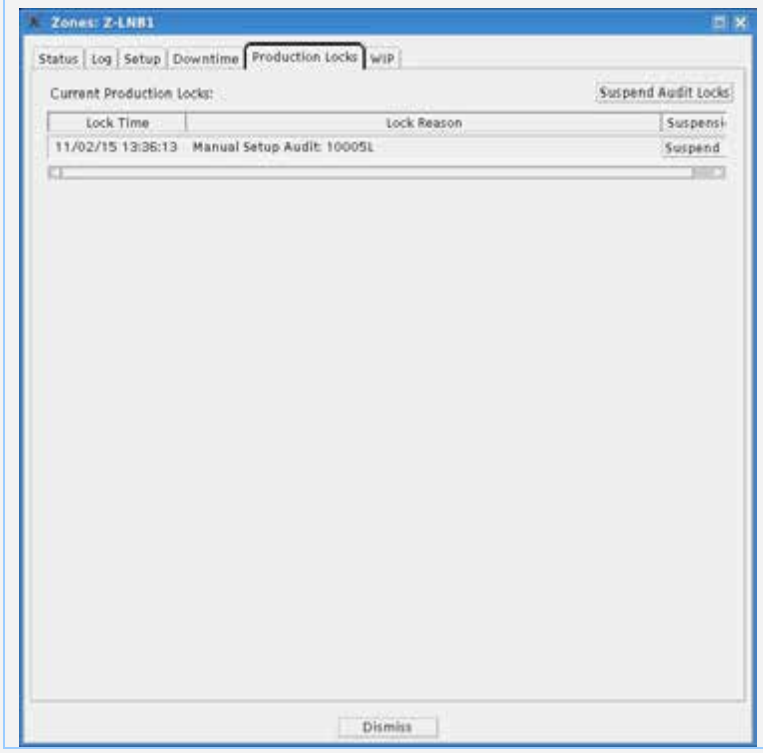
	successfully performed for all the feeder locations. However, status of wrong part scans is not reset. Also, production lock that was set on account of wrong scan is also not cleared by resetting the manual audit. 
4	Now the user has to perform audit for all the feeder locations in the currently selected work area. Note that feeder locations that were successfully audited in the current selected work area are also displayed.
5	If all the feeder locations show wrong scan,  icon is displayed only if you initiate setup audit again by clicking  icon in the production main menu screen of mobile scanner.

Suspending manual audit

Step	Procedure
1	When the machine is locked due to wrong scan during manual audit, use the steps below to suspend the audit in progress. 

2

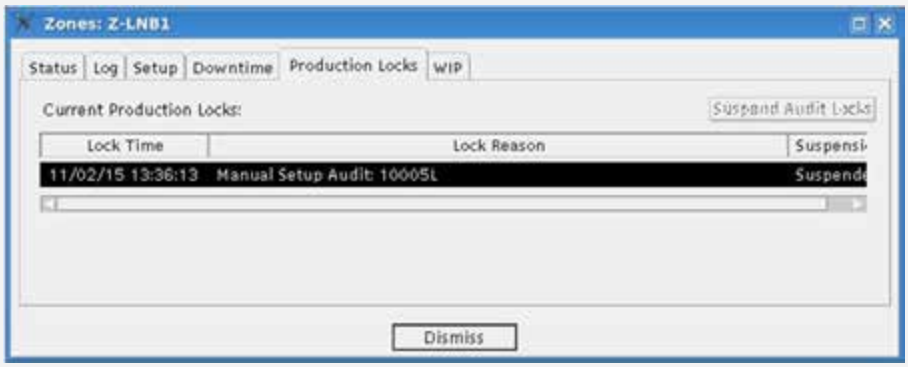
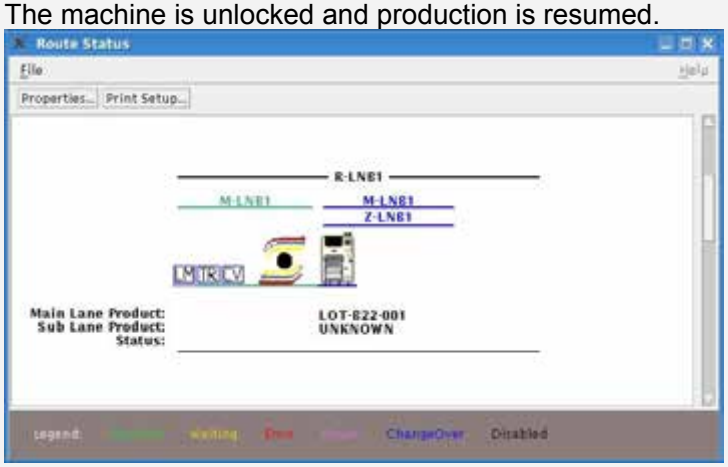
Double-click the machine icon  within the production zone (**Zone 8 in this case**) and click the **Production Locks** tab. The following screen appears.



3

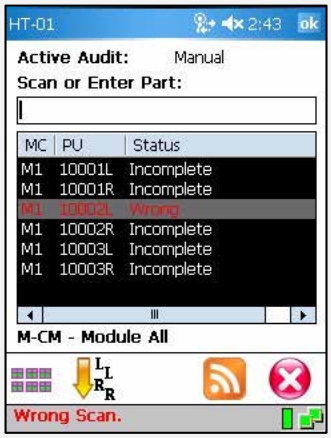
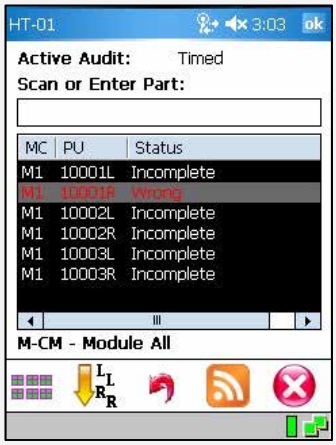
Click **Suspended** under **Suspension**. The Operator Login dialog pops up.

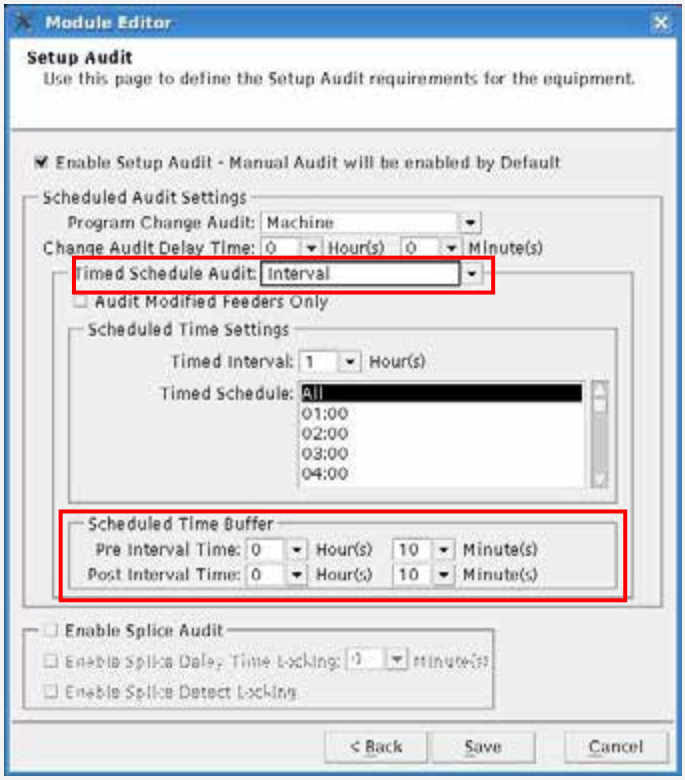
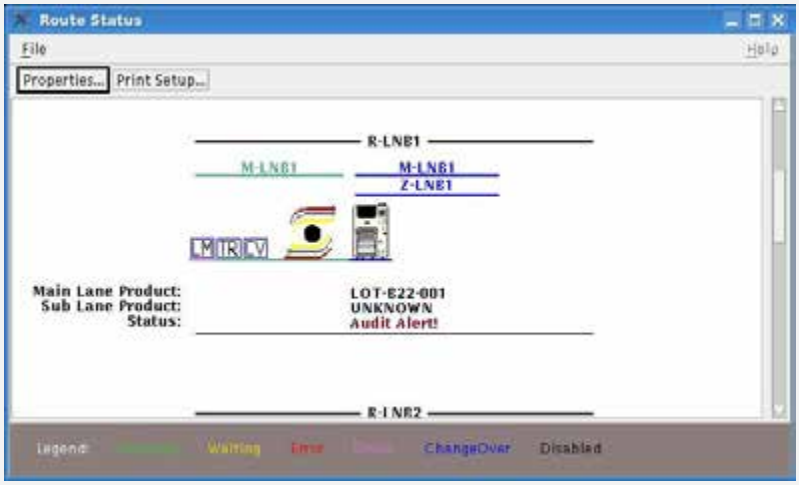


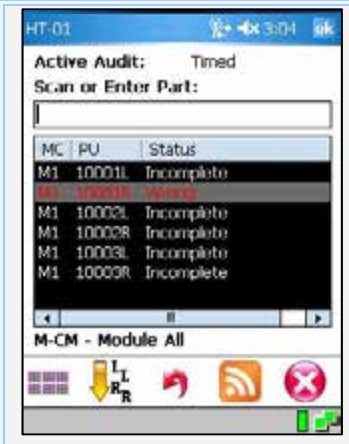
4	Select the operator having administrative privileges and enter the barcode, then click **Apply**. A dialog appears indicating the manual audit has been suspended.  The machine is unlocked and production is resumed. 

Note:
The PanaCIM holds the status of the suspended manual audit.
If audit is resumed by the **Setup Audit** icon on the Production main menu screen of the mobile scanner, the last saved status (Wrong, Incomplete) of the feeder locations will be shown again.
This lets you confirm the feeder locations where trouble occurred in the audit or where audit is required.
The feeder locations that were successfully audited before suspension are not shown.

Changing the audit mode from Manual to Timed

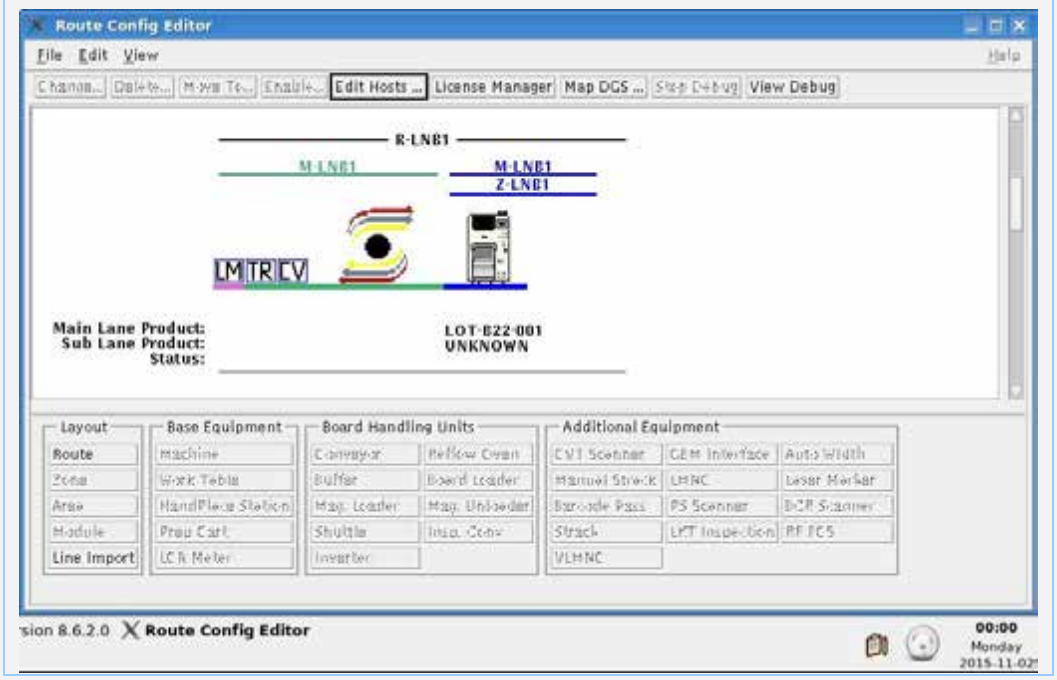
Step	Procedure
1	The screen below shows the manual audit is running. 
2	If **Time Scheduled Audit** on the Setup Audit screen on the Modular Editor is set or the scheduled end time for setup audit is reached, the following message appears. 
3	Click  on the screen in step 2. The following dialog is displayed.  When the scheduled time for setup audit is reached, the manual audit in progress is finished and the mode is switched to the timed audit. In such a case, the manual

	audit has been finished. Note that the checked data in the manual audit will be lost accordingly. When the timed audit is launched, the status of all feeder locations will be reset to **Incomplete**, indicating all feeder locations for the selected work area need to be audited.
4	In this case, the message in step 2 will be shown for five minutes until the next audit begins. This is because the timed audit is supposed to be completed within five minutes (interval till the next audit).  
5	It is necessary to complete the timed audit within the specified time (five minutes). If not, the machine will be locked. In the example of the timed audit shown below:

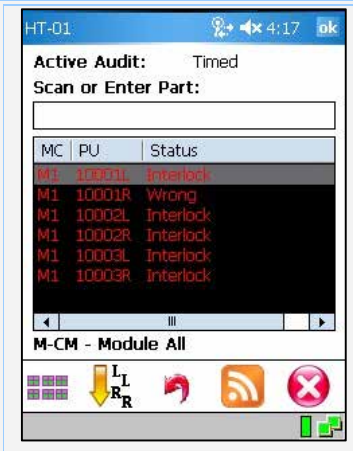


If the timed audit fails to be completed within the specified time, the following message appears on the mobile scanner in five minutes.

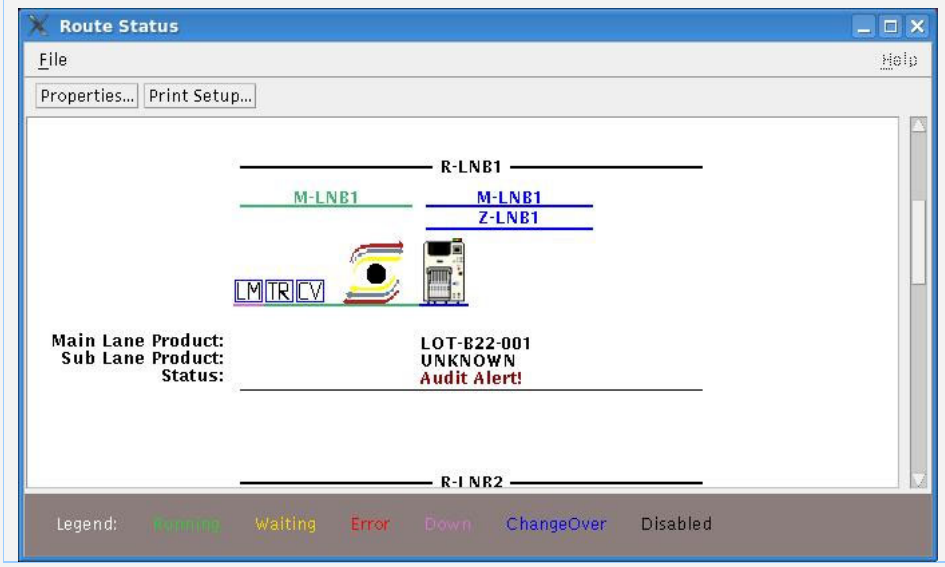




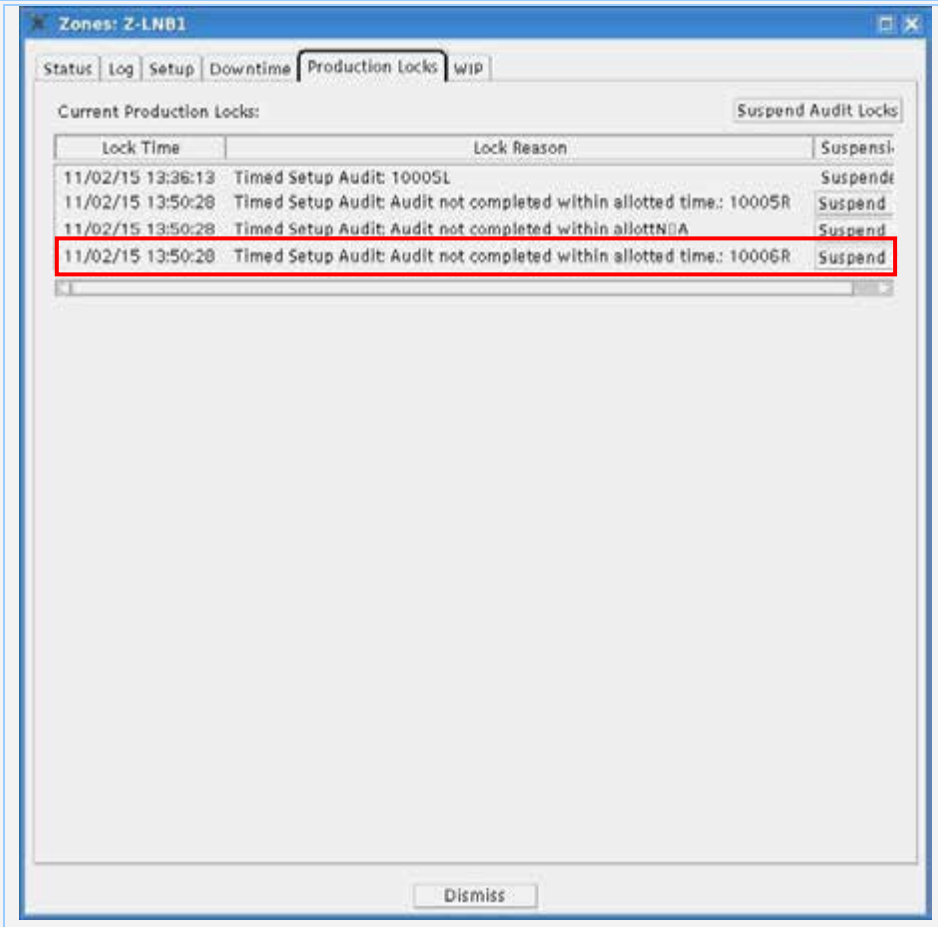
6	Click  on the screen containing the message. The screen changes as follows:

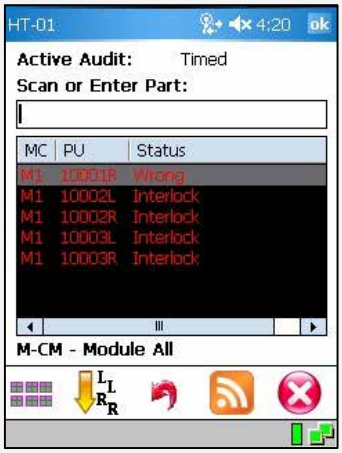


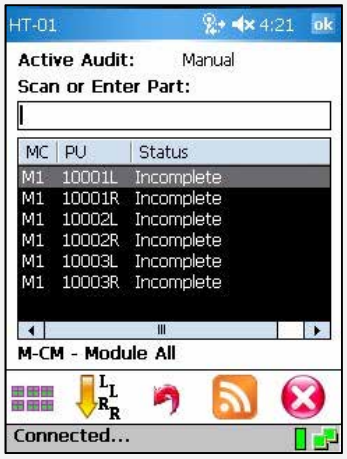
In the above screen, **Interlock** is displayed for the feeder locations where the audit failed to be completed within the specified time. The machine with such feeder locations will be locked as shown below.



7	If one or more feeder locations are interlocked in the timed audit, the information will be reflected in the **Production Locks** tab on the Zone Status screen.


Zones: Z-LNB1
Status
Log
Setup
Downtime
Production Locks
WIP
Current Production Locks:
Suspend Audit Locks
Lock Time
Lock Reason
Suspensi
11/02/15 13:36:13
Timed Setup Audit: 10005L
Suspende
11/02/15 13:50:28
Timed Setup Audit: Audit not completed within allotted time.: 10005R
Suspend
11/02/15 13:50:28
Timed Setup Audit: Audit not completed within allottN□A
Suspend
11/02/15 13:50:28
Timed Setup Audit: Audit not completed within allotted time.: 10006R
Suspend
Dismiss

8	Check the feeder locations to clear the **Lock Alert**.  
9	After all feeder locations have been audited, the following message appears. 

10	If audit is run again manually, manual audit begins as default. 

Setting Program Change Audit

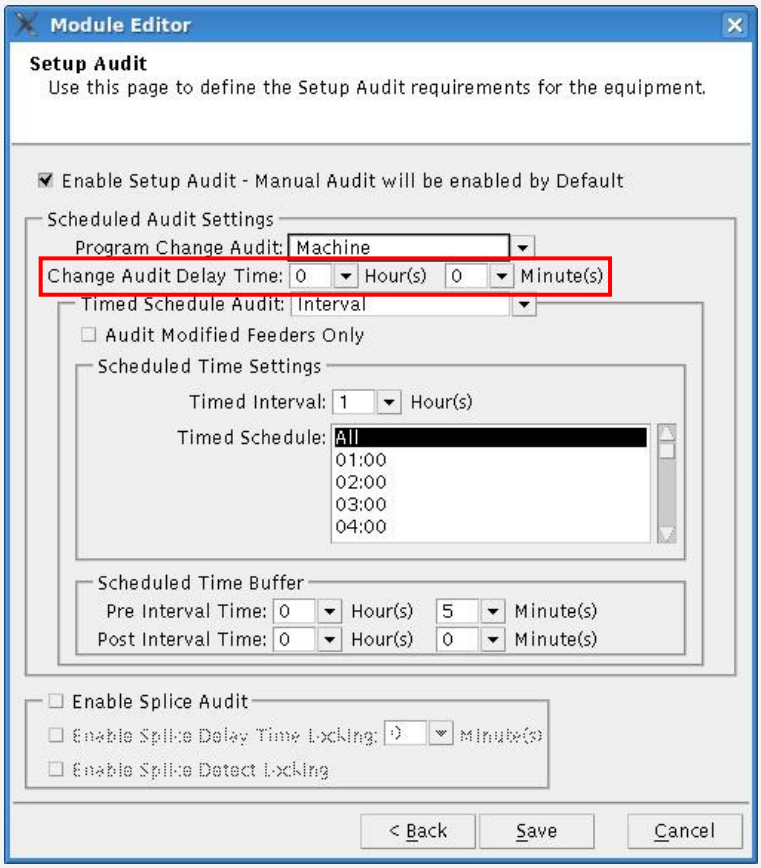
The update feature permits you to specify the delay time after the program has been changed until when the machine is locked by PanaCIM.

The previous version of Material Verification required setup audit for feeder changeover to be done before starting production, because the machine would be locked after the product changeover had been completed.

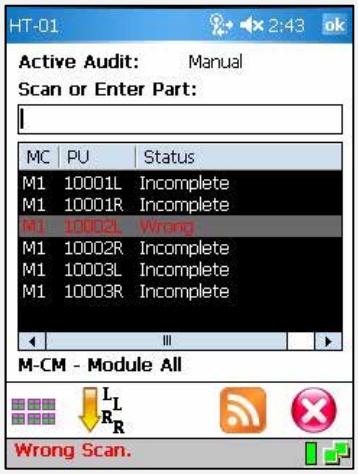
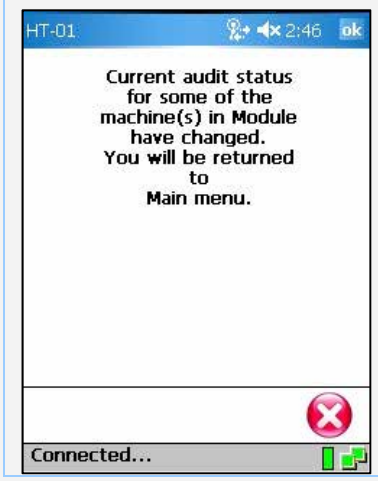
In contrast, this version starts the audit delay timer once the changeover has been completed, as long as the delay time has been set. If the audit fails to be completed within the delay time, the machine will be locked. This timer, in a practical sense, is a delay timer to allow the machine to be locked after the program changed audit. It secures the time for auditing and prevents possible delay in production.

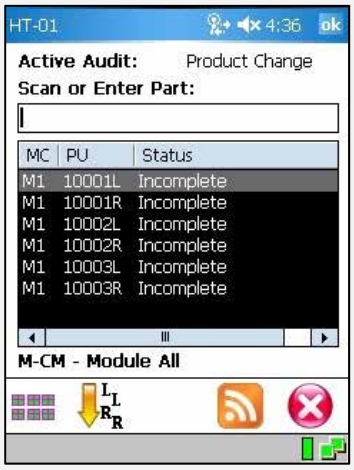
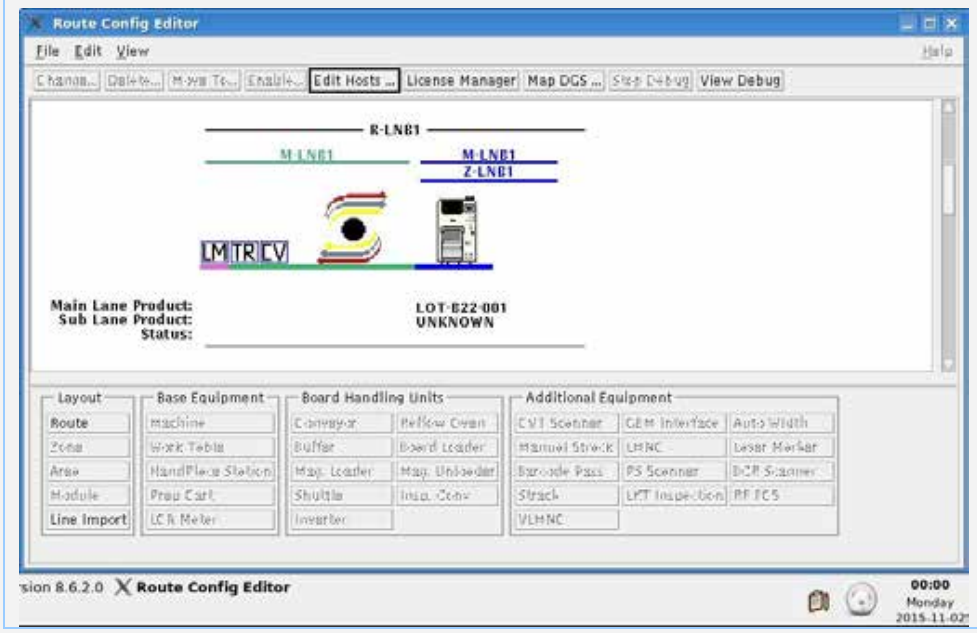
Refer to the following machine lock matrix table when setting set the Program Change Audit.

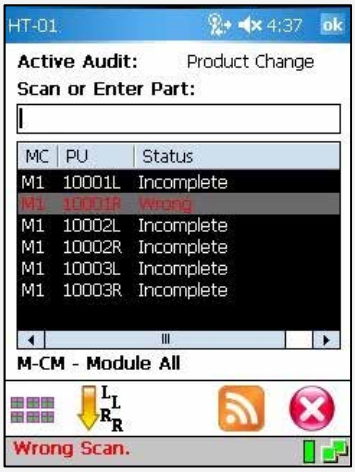
Set Audit Delay Time?	**Changeover setting**	**Start Audit Delay timer?**	**End Audit Delay timer?**	**Lock the machine?**
Yes	Incomplete	No	No	No
Yes	Completed	Yes	No	No
Yes	Completed	Yes	Yes	Yes
No	Incomplete	N/A (Not set)	N/A (Not set)	No
No	Completed	N/A (Not set)	N/A (Not set)	Yes

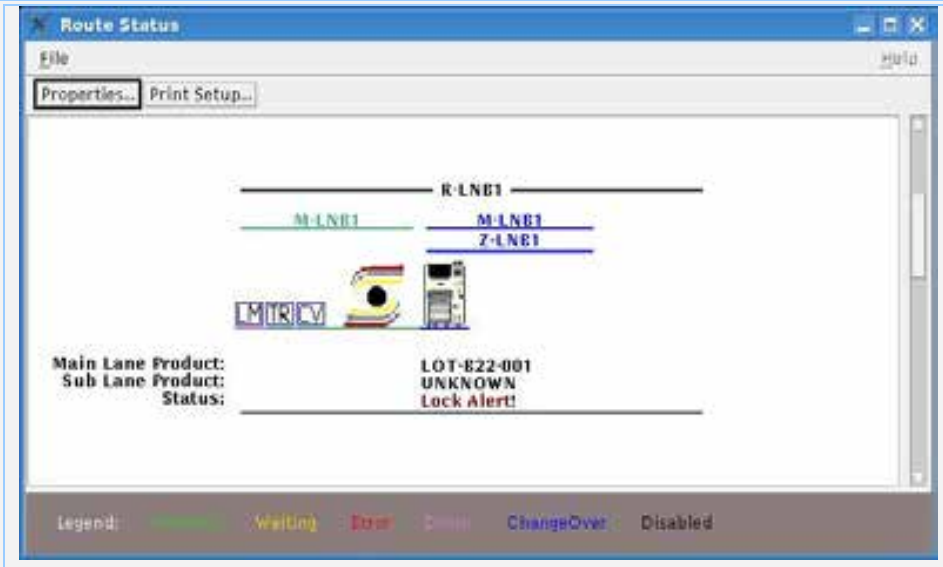
Step	Procedure
1	Double-click the module editor icon and click **Next** to display the Setup Audit page. 
2	Set **Change Audit Delay Time** (hh mm) with the drop-down lists.
3	Click **Next** to advance to the Part Monitor Setup page. Click **Save** on that screen to save the changes made to the program in step 2.

Changing the audit mode from Manual to Program Change

Step	Procedure
1	The screen below shows the manual audit is running. 
2	Once the product changeover is started, the following message appears on the mobile scanner and the manual audit stops. 

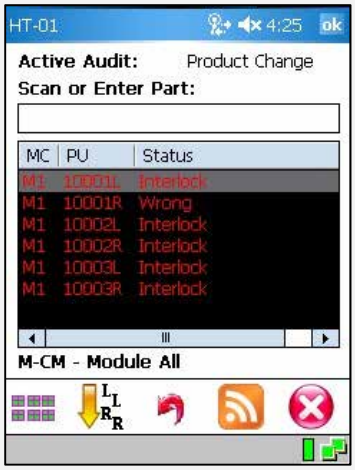
3	Clicking  on the above screen displays the screen below indicating that the mode is changed to the program changed audit.  Program changed audit must be completed within the specified **Audit Delay Time** on the Setup Audit page on the Modular Editor.
4	Program changed audit is started when the CIMC server is at "**00:00**". 

5	While the program changed audit is running at the respective locations, the rows of successful audit positions will disappear. In case of a wrong scan, the row turns red and **Wrong** is shown under **Status** column. The following describes the procedure if the program changed audit fails to be completed within the delay time. It assumes, as shown in the screen below, that the first feeder location has been audited successfully, but the second feeder has not been audited yet. 
6	If the preset delay time has elapsed, the following message appears on the mobile scanner.  Suppose the delay time is 5 minutes and the program changed audit started at **00:00.** In this case, if the audit fails to be completed even after 00:05 on the CIMC server clock, the following screen appears.



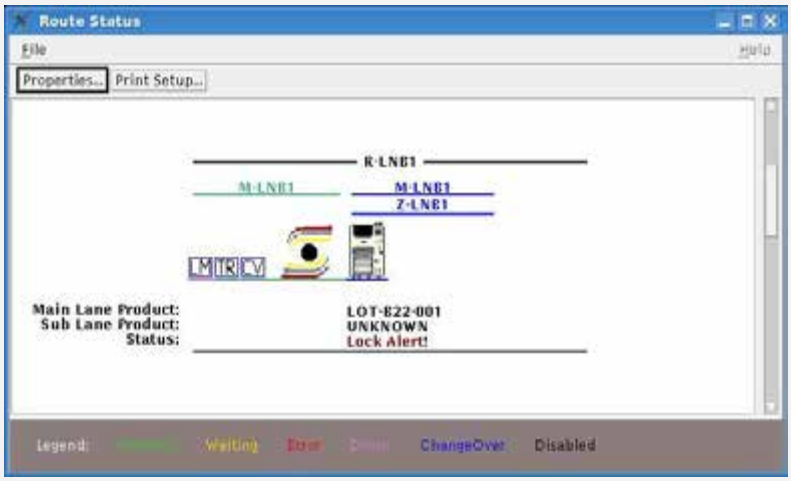
7

Clicking  on the message screen on the mobile scanner displays the screen shown below.

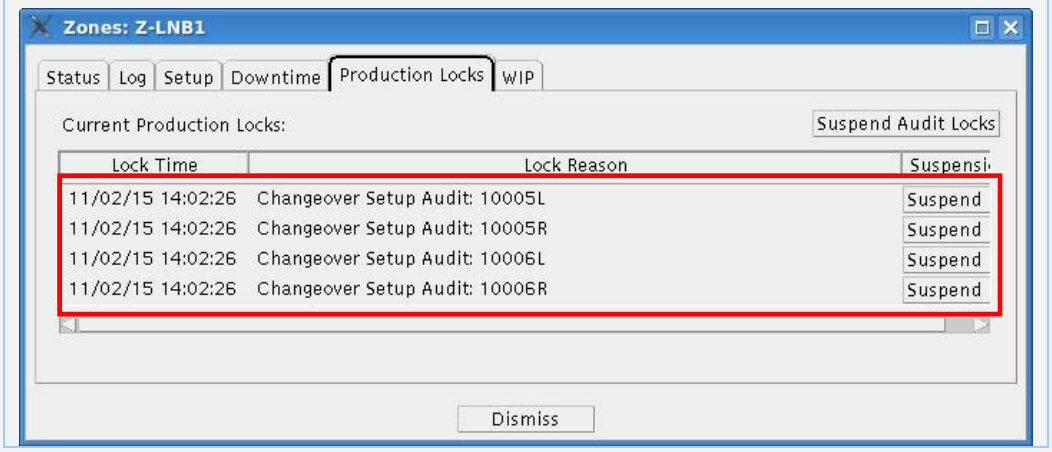


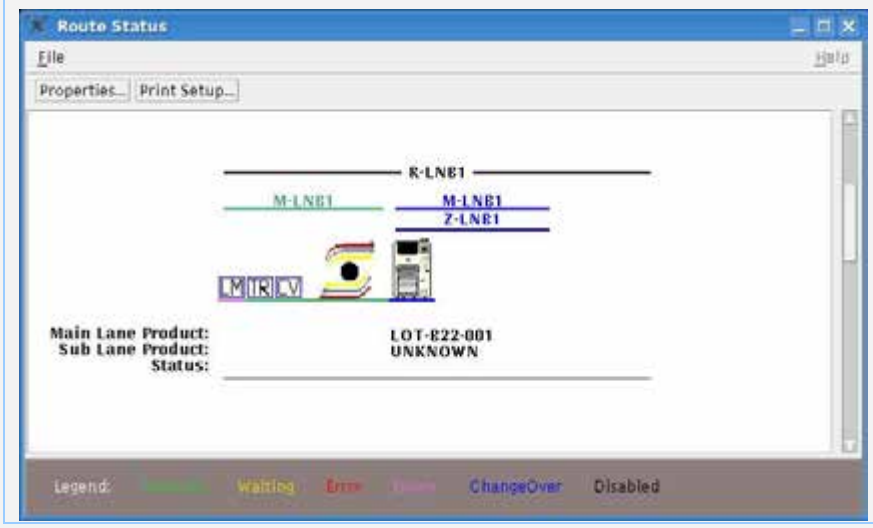
The feeder locations yet to be audited (**Incomplete** is shown in the **Status** column) will be shown with **Interlock** in red. In such a case, production cannot be resumed unless these feeder locations are audited.

8	If the program changed audit fails to be completed within the specified delay time and production is stopped, **"Lock Alert"** appears on the **Route Status** screen.



Double-clicking the machine and clicking the **Production Locks** tab displays the cause of the stopped production under the **Lock Reason** column.
(In the example below, **Changeover Setup Audit** is shown.)

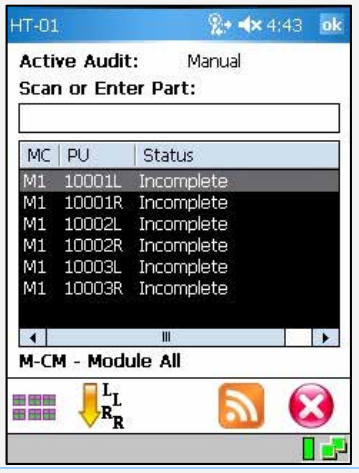


9	After the audit for all locations has been completed successfully as a part of program changed audit, the following message appears.  The machine will be unlocked as shown below. 

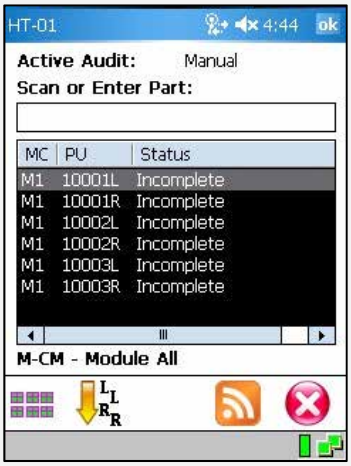
10

Click the 

icon on the Production main menu screen to start the audit. The manual audit is launched as shown below.



Note:
The delay time for the program change audit has been set to 0:00 by default. This means all feeder locations are interlocked immediately after production is changed, causing the machine to be locked. The below screen shows such a condition.



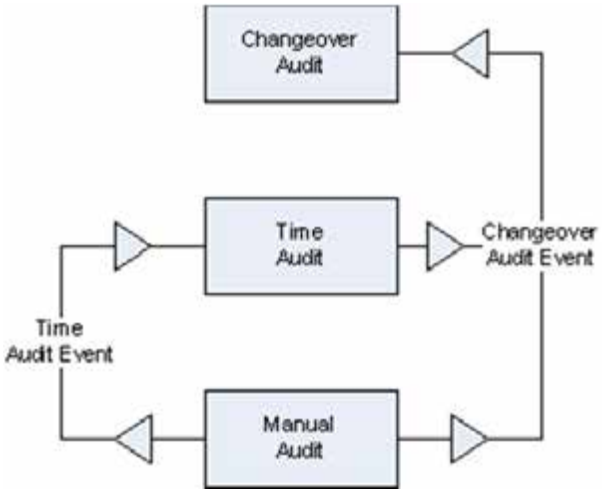




Note: Any of these three audits can be reset only within the work area you logged in. Resetting will not set all feeder locations or create a new audit. Only the feeders within a work area where setup audit is running are applicable.

The program changed audit and timed audit can also be stopped using the same procedure as the manual audit to go back into production.

Priority of setup audits



As shown in the figure above, the program changed audit has the highest priority, followed by the timed audit and manual audit. For example, if a timed audit is started while a manual audit is running, the manual audit will be finished. After that, a timed audit is started and then completed within about ten minutes until the next interval begins. Production changeover will be performed in a few seconds before the ten-minute interval is finished. Then a program changed audit is run and the timed audit is finished. When the program changed audit is finished within ten minutes before the next interval begins and setup audit is started again, a manual audit will be run by default. Whenever an audit is stopped, overridden by the one with higher priority, the current status will be lost.

Audit to run	**Audit in processing**			
	None	Manual audit	Timed audit	Program changed audit
Manual audit	Manual audit is started	Manual audit is continued	N/A	N/A
Timed audit	Timed audit is started	Manual audit is stopped and timed audit is started	Timed audit is continued	N/A
Program changed audit	Program changed audit is started	Manual audit is stopped and program changed audit is started	Timed audit is stopped and program changed audit is started	Program changed audit is stopped and restarted

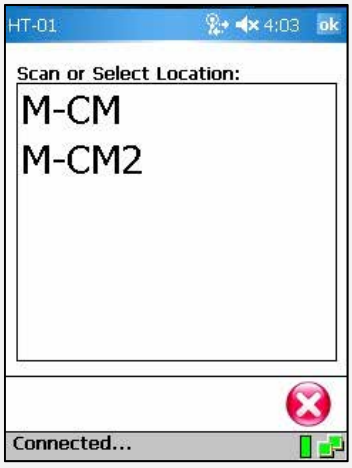
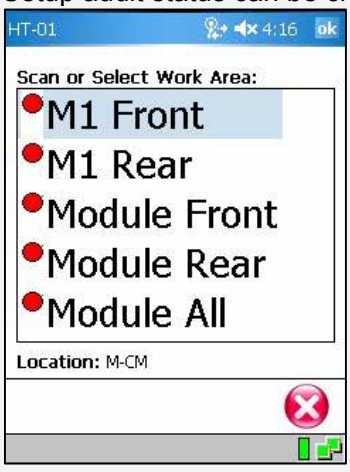
6.3.6 Confirmation of Setup Audit Status

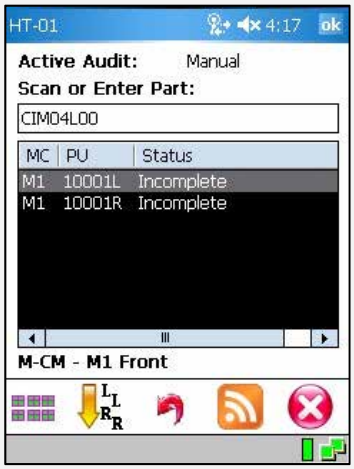
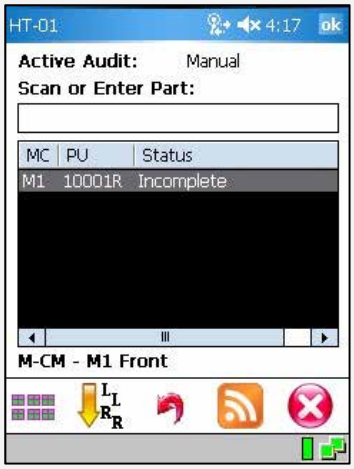
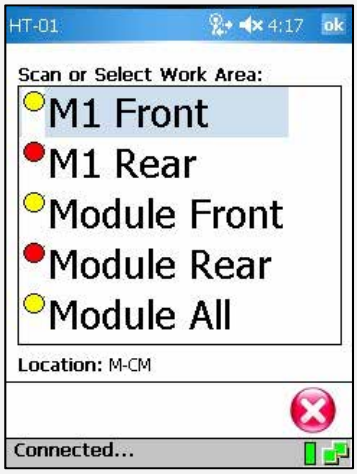
Setup audit status can be checked on “Scan/Select work area” screen that appears after Setup Audit was selected.

The icons displayed next to work area shows setup audit status.

<Icon color and audit status>
- ✧ Red: Audit is [Undone] for the target area.
- ✧ Yellow: Audit is [Running] for the target area.
- ✧ Green: Audit is [Finished] for the target area.

The following steps show how to check setup audit status on a mobile scanner.

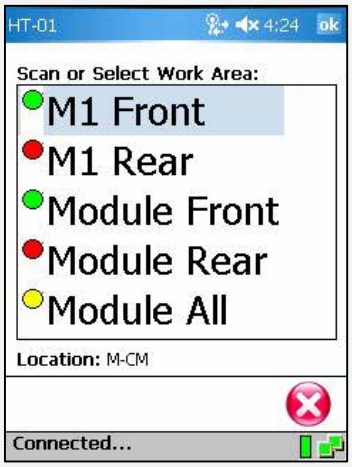
Step	Procedure
1	Select  from Production main menu on mobile scanner.
2	On “Scan/select location” screen, select a location for setup audit. 
3	“Scan/select work area” screen appears. Select “M1 Front”. Setup audit status can be checked from here.  The screen above shows that no setup audit has been done for the loaded parts.

4	Scan/enter material ID of target parts. 
5	Setup completes for the target parts, and the details disappear from the screen. 
6	Press  icon on the screen in step 5.  Setup audit status of “Work area” ”M1 Front / Module Front / Module All” status changes from Undone→Running.
7	After setup finished for all parts loaded at “M1 Front”, the following message appears.



8

Make sure setup audit for “M1 Front” finished and press  .

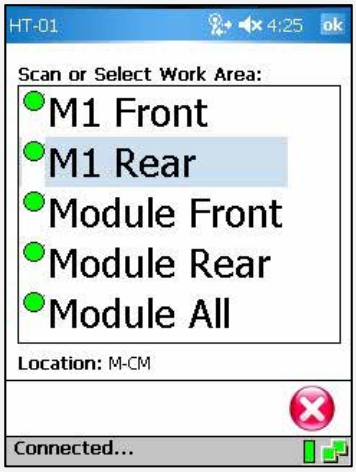


“M1 Front / Module Front” setup audit status changes from Running→Finished in “Work Area”.

9

After setup finished for all parts loaded at “M1 Rear”, the following message appears.

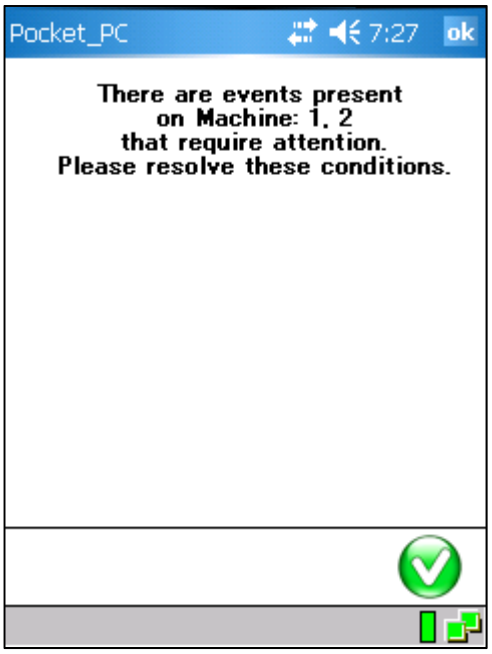


10	Make sure setup audit for all parts finished and press  to go back to production menu.  "M1 Front / Module Front" setup audit status changes from Running→Finished in "Work Area".
11	Select  again to select a location. You can check if setup audit has finished from this screen.  "M1 front / Module Front" setup audit status changes from Running→Finished in "Work Area", and you can check setup audit for all parts has been finished.

6.3.7 Operation when Setup Audit (Automatic Interruption) Starts

If you enable setup audit (automatic interruption), the pop-up message appears on the mobile scanner telling the following notification. Since the operation procedure differs depending on the setting configured on the module editor screen, the operation procedures for each operation mode are described below.

<Notification message>



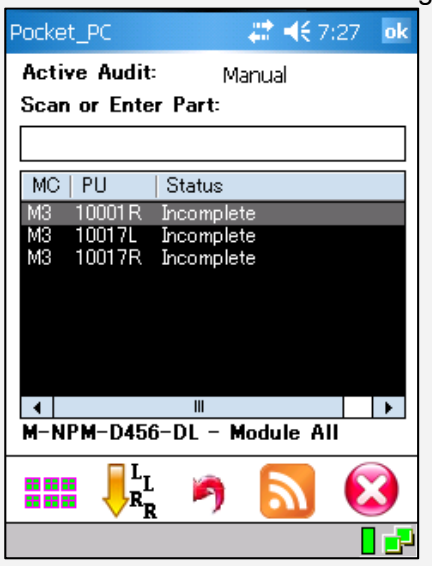
<Operation mode>

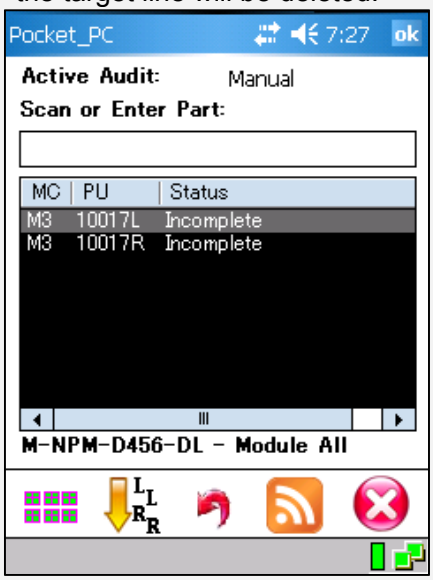
- ✧ Notify Only
- ✧ Notify / Lock
- ✧ Notify / Lock(w/PWD Override)

This shows an example of the operation procedure to follow when an error occurs in the first or second machine of three coupled machines during a setup audit on the third machine upon completion of setup audits on the first and second machines.

■ Operation mode: “Notify Only”

The following is an operation procedure of the mobile scanner when setup audit (automatic interruption) starts.

Step	Procedure
1	The pop-up screen below appears during performing setup audit via the mobile scanner. Click  .  *Because operation mode is “Notify Only”,  is displayed.
2	The screen on the scanner returns to setup audit (manual) operation screen. Continue to scan or enter the target part number. 

3	Setup audit of the target part completes and detailed information is deleted from the screen. *As an example, if you enter the part number which is arranged on PU：1001R, the target line will be deleted. 

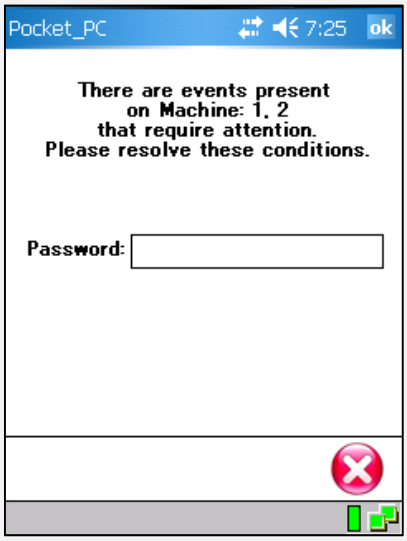
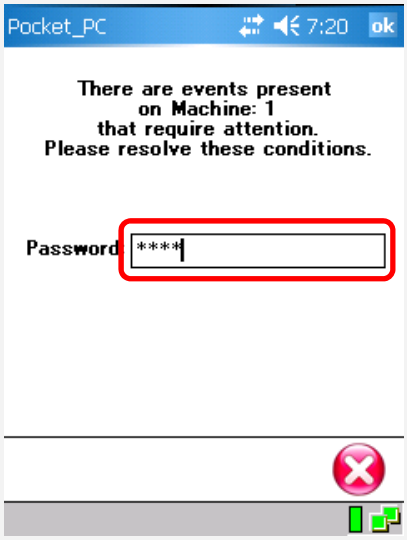
■ Operation mode: “Notify / Lock”

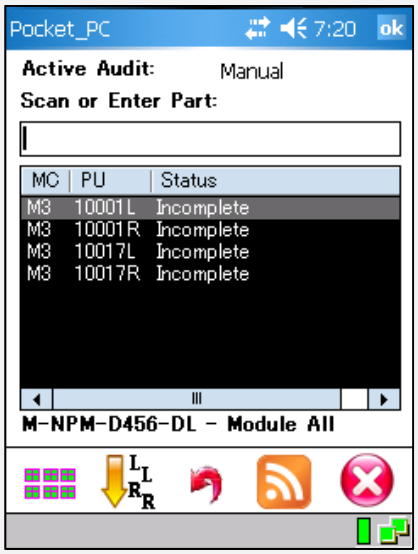
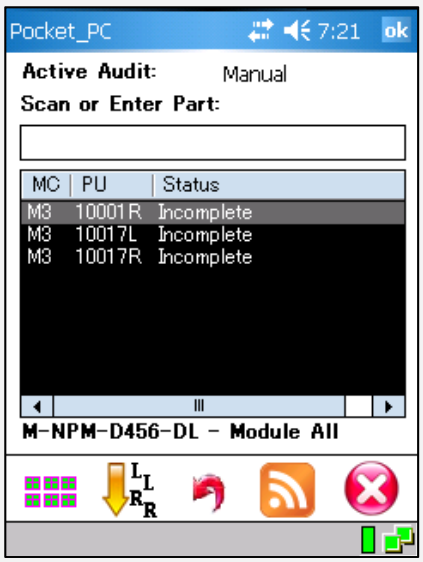
The following is an operation procedure of the mobile scanner when setup audit (automatic interruption) starts.

Step	Procedure
1	The pop-up screen below appears during performing setup audit via the mobile scanner. Click  .  *Because operation mode is “Notify / Lock”,  is displayed.
2	Setup audit (manual) screen ends and the “Production” menu screen appears.  *If operation mode is “Notify / Lock”, even though you select **Setup Audit** again, the screen in step 1 is displayed and you cannot continue setup audit operation until you reset error on the machine.

■ Operation mode: “Notify / Lock(w/PWD Override)”

The following is an operation procedure of the mobile scanner when setup audit (automatic interruption) starts.

Step	Procedure
1	The pop-up screen below appears during performing setup audit via the mobile scanner.  *Because the operation mode is “Notify / Lock(w/PWD Override)” the password input field is displayed. And also  is displayed as well.
2	If you want to continuously perform setup audit (manual), type the password entered in **Additional Settings** in **System Configuration** – **Application** – **Mobile Scanner** tab into the password field. If you want to quit setup audit (manual) click  . 

3	The screen on the scanner returns to setup audit (manual) operation screen. Continue to scan or enter the target part number. 
4	Setup audit for the target part completes and detailed information is deleted from the screen. *As an example, if you enter the part number which is arranged on PU：1001L, the target line will be deleted. 

7. FUNCTION - PREVENTION FROM AN INCORRECT SET - INTERLOCK

This chapter explains machine Interlock.

7.1 Overview

This chapter describes how the PanaCIM software can be configured to recognize conditions where it is necessary for factory production to stop, subsequently prohibiting Production to continue before these conditions have been cleared. You will learn what times during Production a lock can be trigged, as well as how to configure and then consequently clearing the locks.

The following table illustrates the Machine Types that support Production Locks, the supported conditions that trigger the lock, and how the Production Lock is deployed for a Machine Type.

Machine Type	Interlock Conditions	Deployment
CM/DT	• Expired MSD Parts • Invalid/Restricted Components • Setup Audit • Expired Control Parts	Machine Lock
NPM	• Expired MSD Parts • Invalid/Restricted Components • Setup Audit • Expired Control Parts	Machine Lock
BM	• Expired MSD Parts	LMNC Panel Hold

To ensure operators are always kept informed of factory operations, a variety of notifications will be visible in the following locations as described in the table below.

Visibility	Alert
Route Status	• Each locked machine will display a “Lock Alert!” message on the main Route Status screen. • The Production Locks tab (Zone Status popup).
LMNC	• For CM/DT series machines, the LMNC display will provide a simple list of machine numbers “M1 M2” when locks are active. • The LMNC will display ‘Panel Hold’ on the LMNC for BM series machines.
PMD	The Machine Event display will provide a lock alert message.

■ Lock and Release Functionality

Lock and Release functionality recognizes conditions within a running PanaCIM system that require immediate suspension of production on a machine-by-machine basis for CM/DT, NPM and BM series machines.

Certain lock conditions will prohibit certain actions at certain times. For CM/DT and NPM series machines, the Navi MI Interface is responsible for evaluating the current system state and determining whether the machine or changeover should be locked, and whether a lock or release command is necessary to be sent to the machine. For BM series machines, the Lock Status Manager will produce Panel Hold lock requests and process lock related messages to the LMNC.

Locking Timeframes

The timeframes of when locking can occur include:

- Production **Start (S)**. Certain lock conditions may be detected at the start of production, due to some initializations that occur just after the machine has started. When one of these conditions arises where the machine locks, the current panel will be completed *before* machine stop. Clearing the lock conditions result in a release of the lock, production will resume, and the next panel will be loaded.
- **During (D)** production. When a condition arises where the machine locks during production, the current panel will be completed *before* machine stop. Clearing the lock conditions result in a release of the lock, production will resume, and the next panel will be loaded.
- Before **Changeover Completion (C)**. Certain lock conditions may cause the changeover process to halt until they are cleared. Detection of these conditions result in an LMNC message that will convey that the changeover cannot be completed, and the machine start command will be disabled. Clearing the lock conditions will result in the completion of the changeover, and production can be started.

The following table provides a list of lock conditions and their applicable timeframes.

Timeframe	Condition
During Production (D) Pre Changeover Completion (C) Production Start (S)	• Expired MSD Part • Dependent Part Rescan • Invalid/Restricted Component • Expired Control Parts
During Production (D) Pre Changeover Completion (C)	• Setup Audit Interlock

7.2 Settings

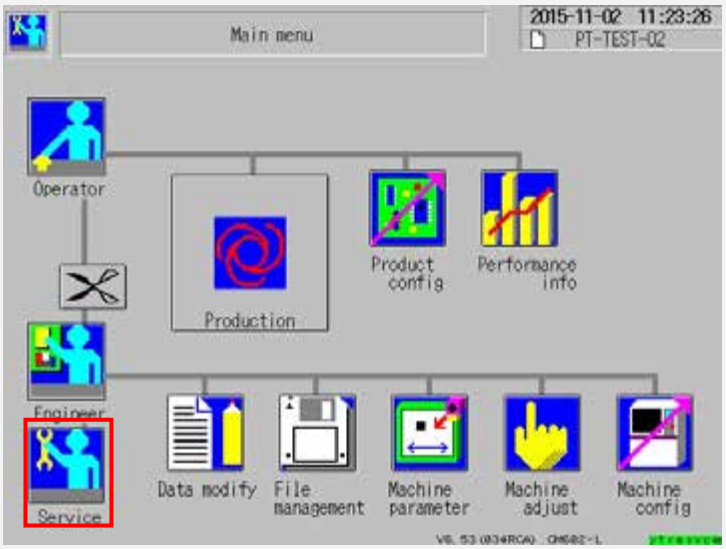
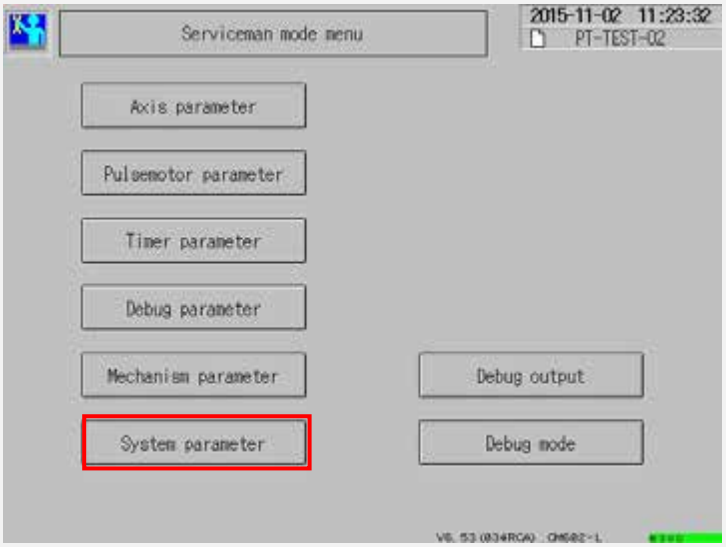
7.2.1 Tray Interlock Settings

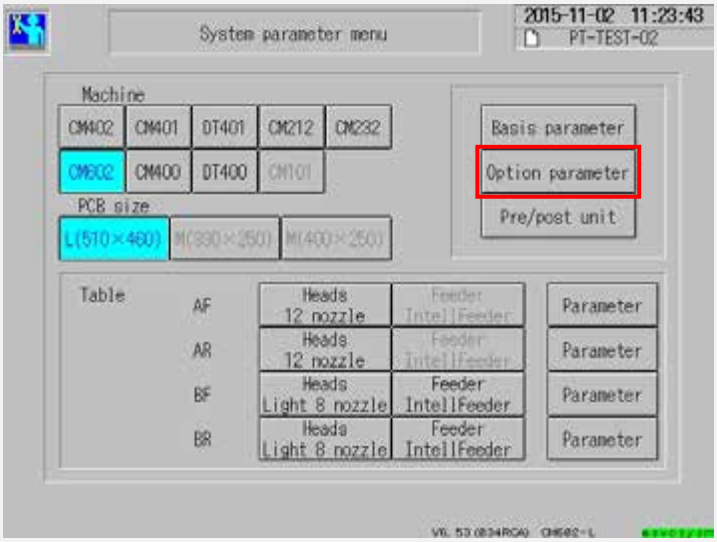
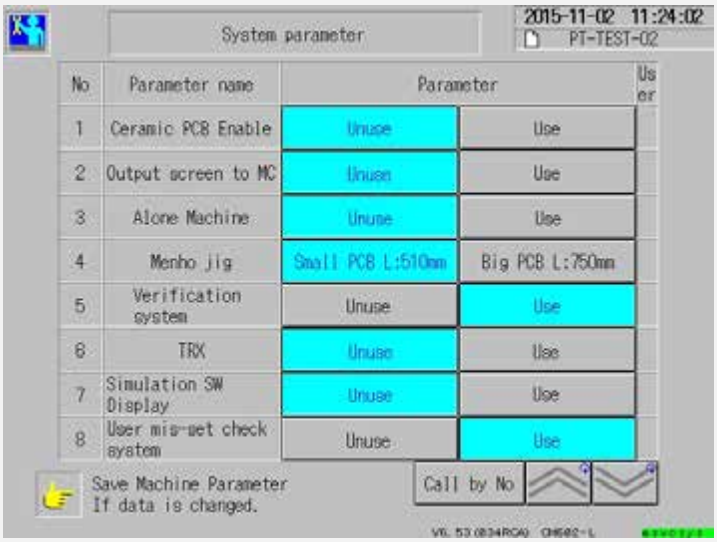
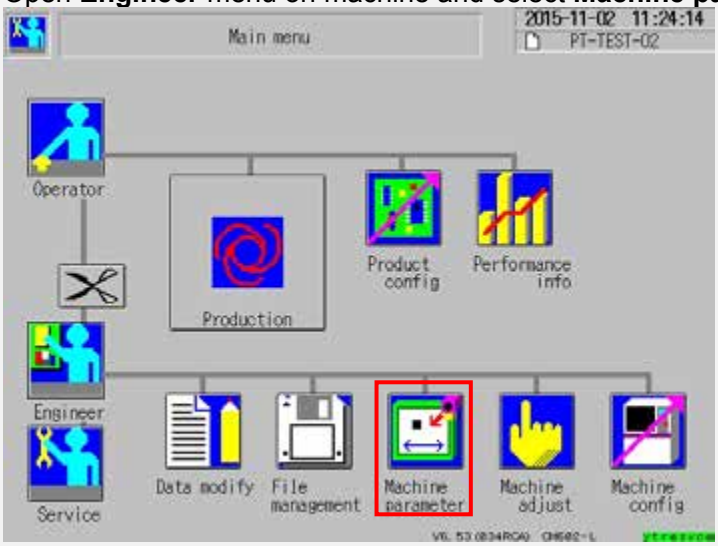
■ CM/DT Series

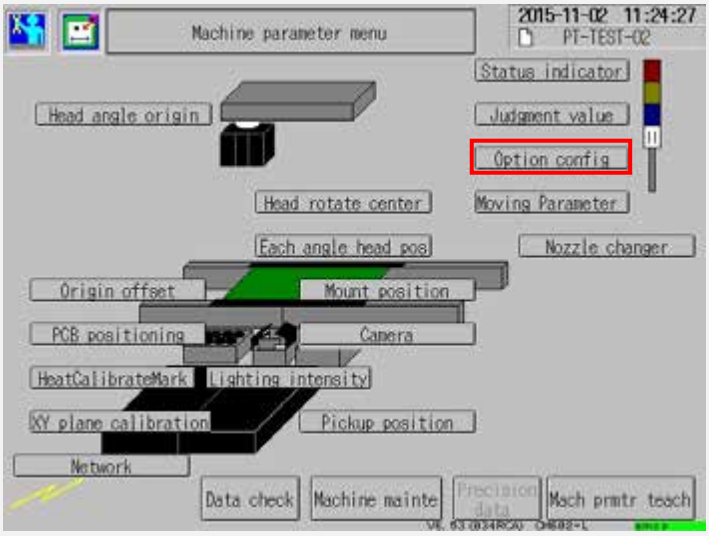
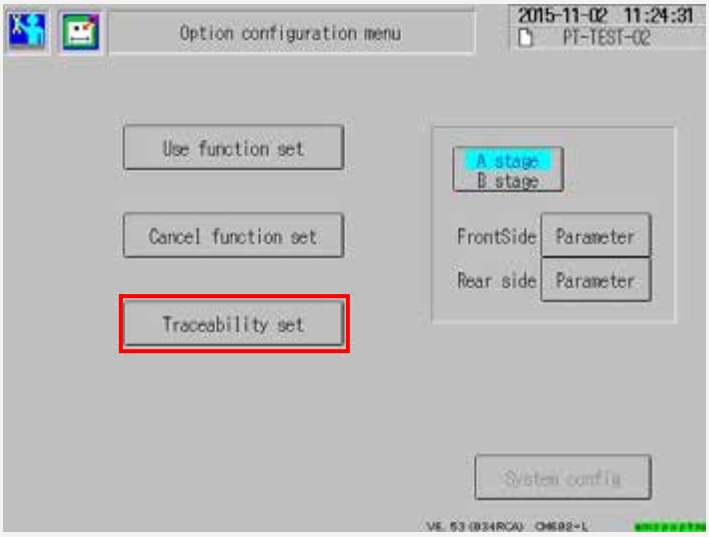
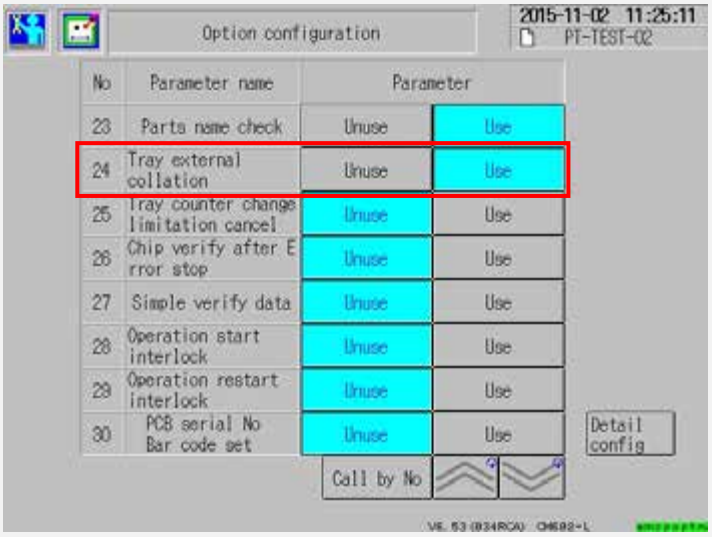
To use tray interlock on CM/DT machines, change the following settings on machine (NPM).

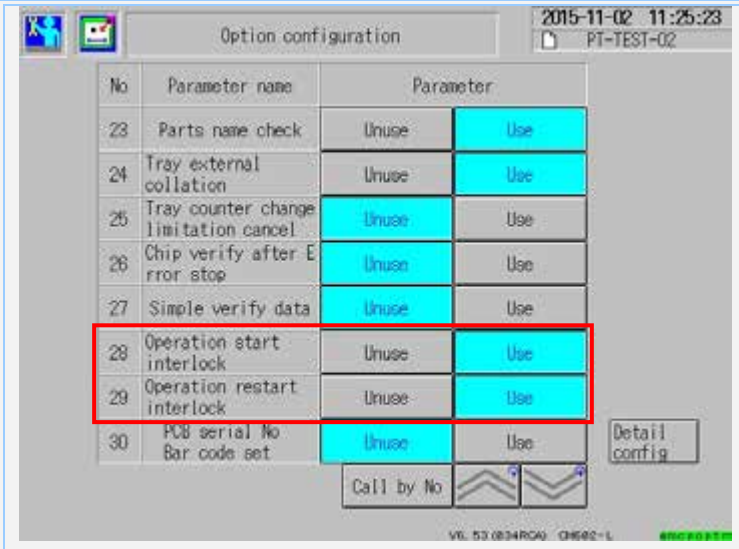
● Setting on machine

The following table shows the machine settings necessary to use material verification on CM/DT machine.

Step	Procedure
1	Open **Serviceman** menu on machine. 
2	Select **System parameter** from serviceman mode menu. 

3	Select **Option parameter** from system parameter menu. 
4	Set parameters according to “CM/DT machine system parameter setting” table below. 
5	After finished setting, go back to the main screen.
6	Open **Engineer** menu on machine and select **Machine parameter**. 

7	Select **Option setting**. 
8	Select **Traceability setting**. 
9	Set parameters according to “CM/DT machine option setting” table below. 



10	After finished settings, go back to the main screen.

- **CM/DT machine system parameter setting**

Number	Option name	Setting	Explanation
8	User wrong set check system	ON	Apply interlock on tray parts when starting production not to start production.

- **CM/DT machine option setting**

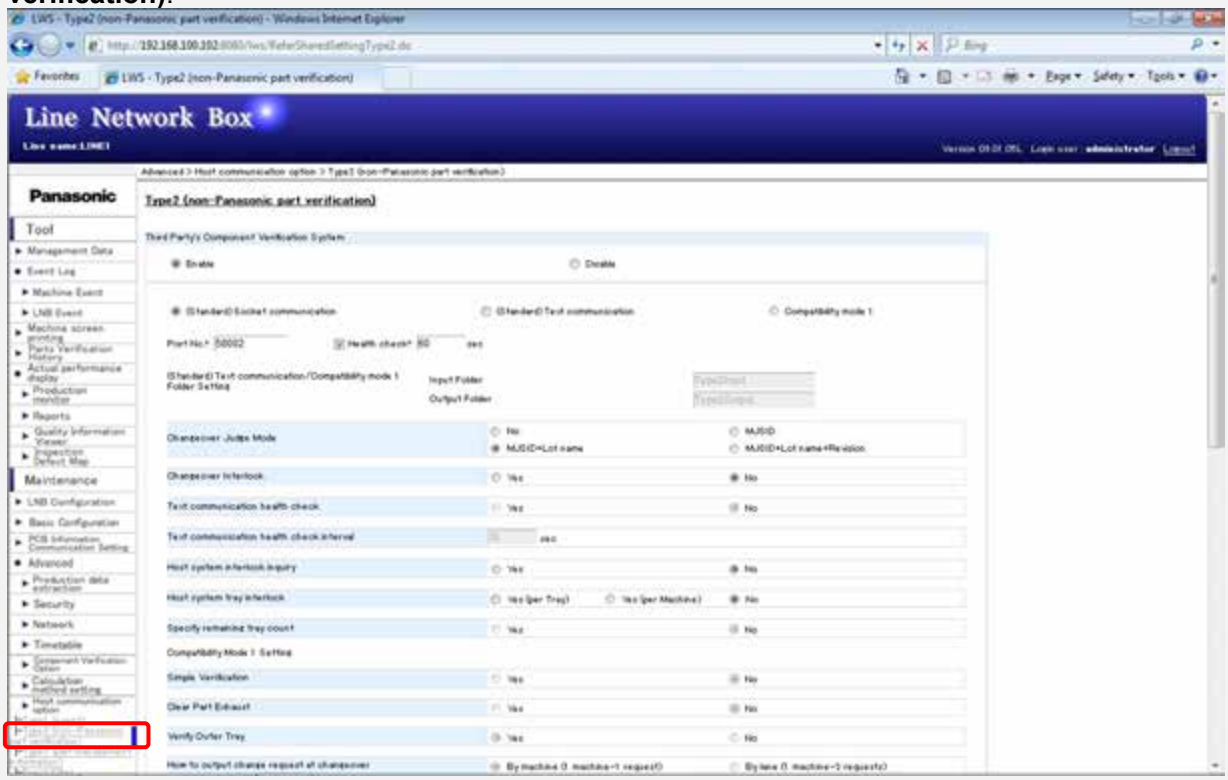
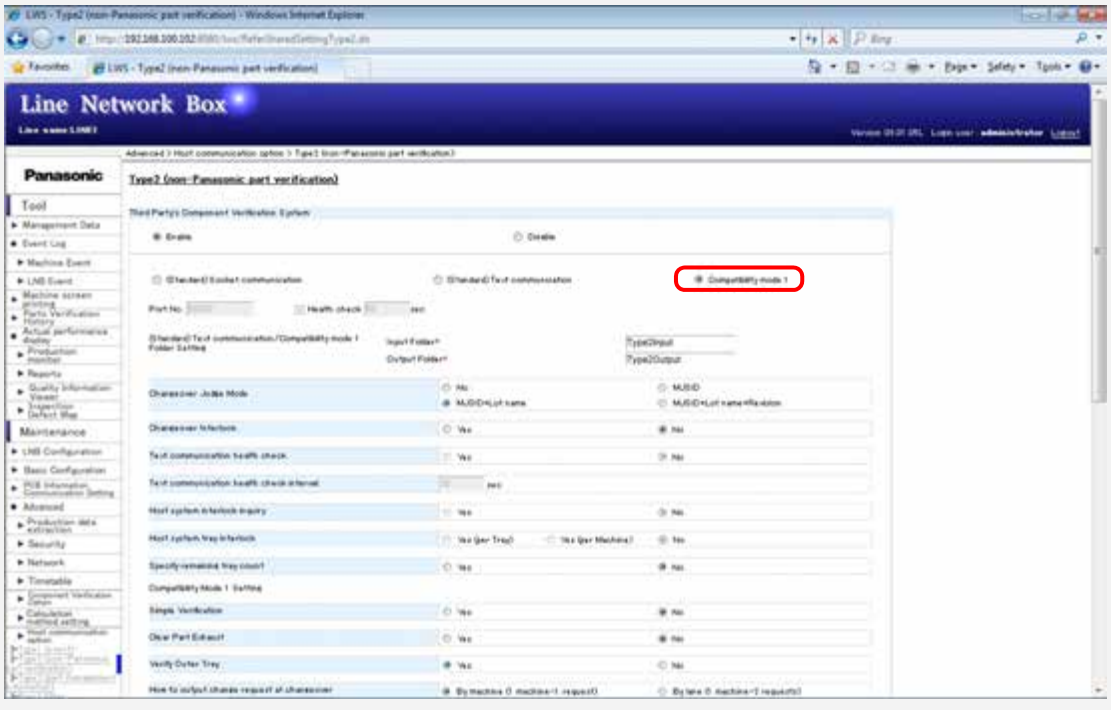
Number	Option name	Setting	Explanation
22	Verify Outer Tray	ON	Use [User wrong set check system] as tray parts verification check.
26	Operation Start Interlock	ON	Run interlock check at production start.
27	Operation Restart Interlock	ON	Run interlock check at production restart.

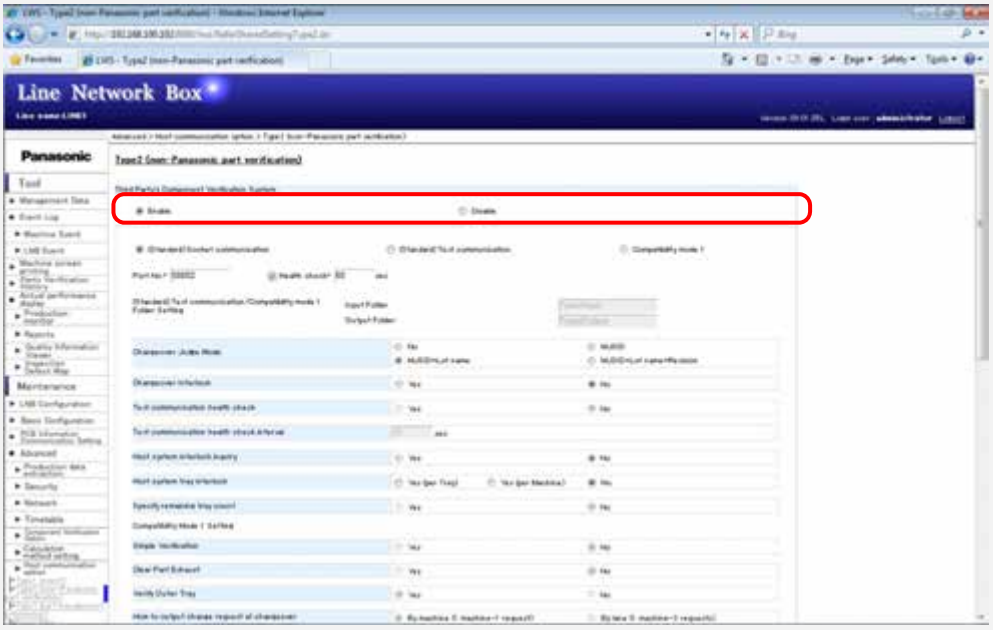
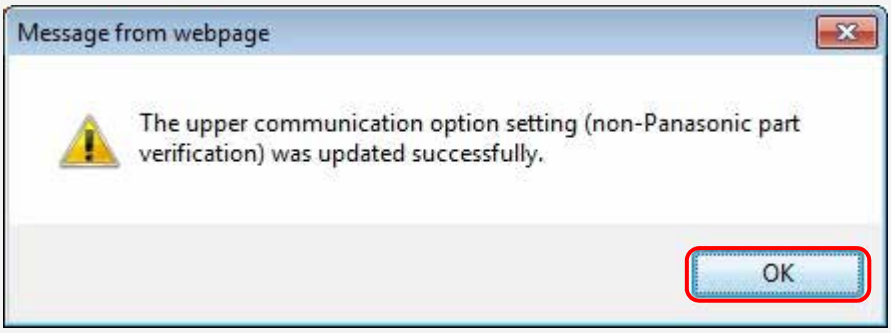
■ NPM Series

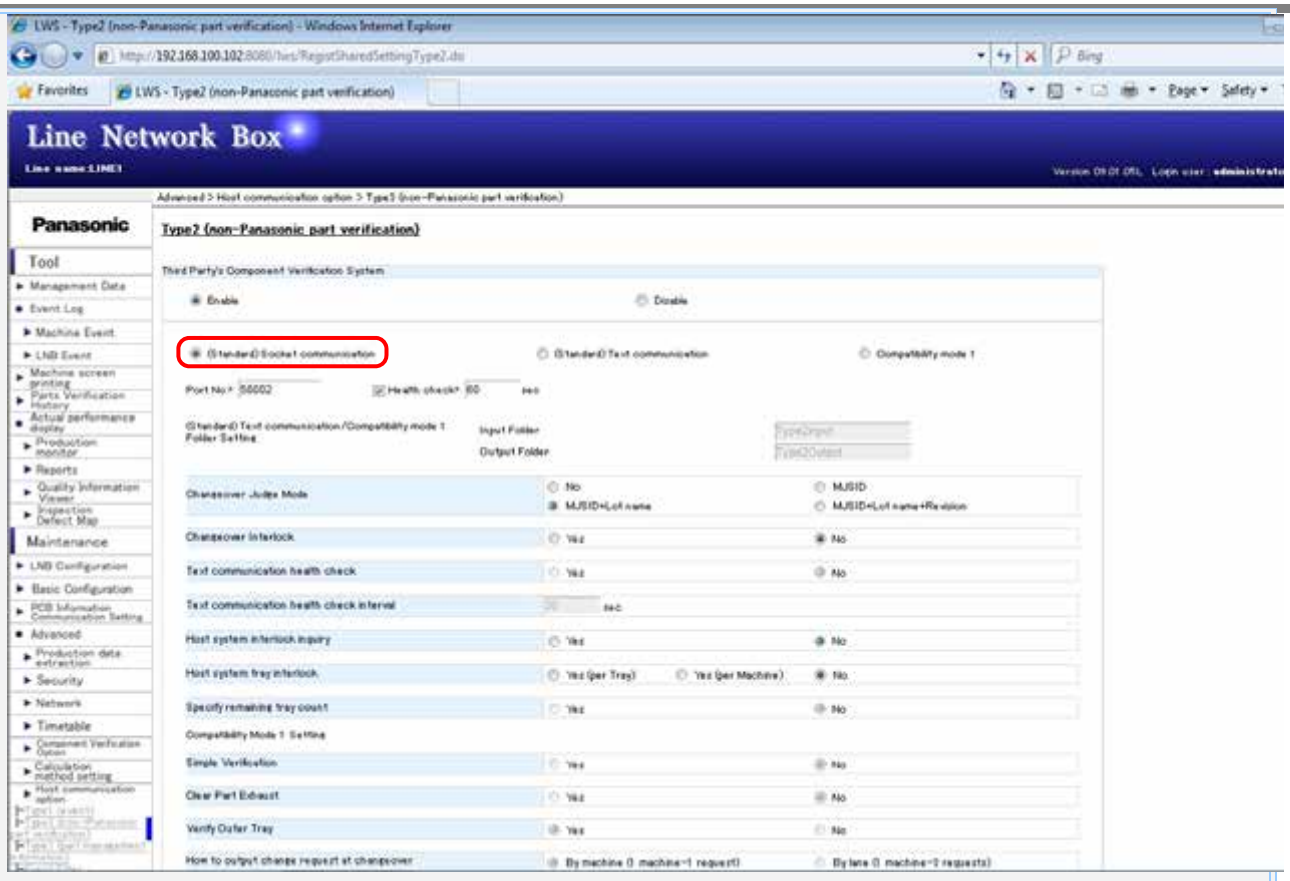
To use tray interlock on NPM machines, modify settings on Machine (NPM) and LNB.

● LNB setting

To use tray interlock on NPM machines, set as follows on LNB.

Step	Procedure
1	After login to LNB, select (1) **Upper communication option-type2 (non-Panasonic part verification)**. 
2	Check **Compatibility mode1** to allow parameters to be edited. 

3	Set **Verify Outer Tray** to **Yes** and press **Apply**. 
4	Pop-up message appears to inform the setting has been changed. Press **OK** button. 
5	Change **Compatibility mode1** to **(Standard) Socket communication**. Press **Apply** button.



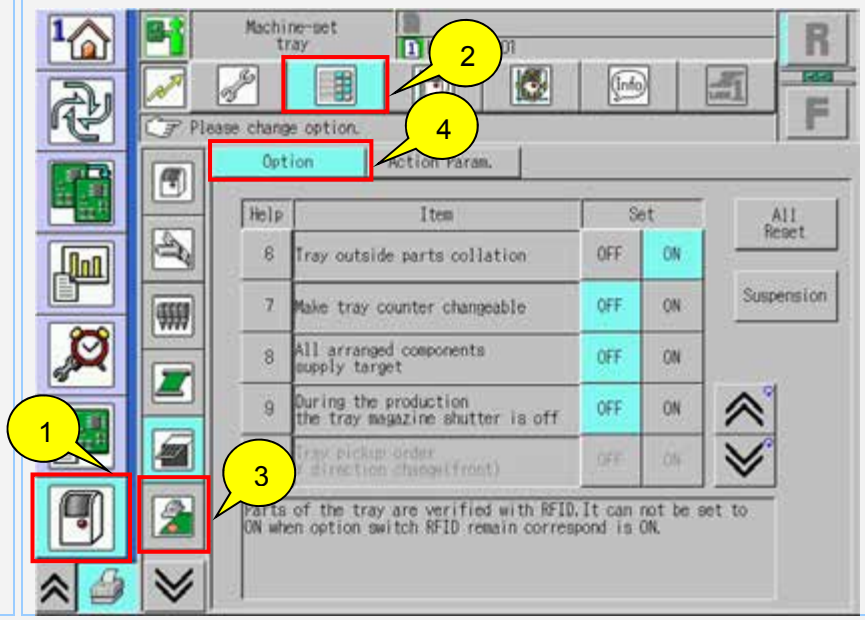
<Tray interlock setting items>
Set the parameters according to the following “NPM (LNB) tray interlock setting”.

- **NPM (LNB) tray interlock setting**

Setting item	Setting	Explanation
Verify Outer Tray	Enable	Check **Enable** to allow LNB to apply tray interlock

Machine setting

To use tray interlock on NPM, follow the steps below.

Step	Procedure
1	Select (1) Machine → (2) Setting → (3) Tray → (4) **Option** on machine. 
2	Set parameters according to “NPM machine tray interlock setting” below.

- **NPM machine tray interlock setting**

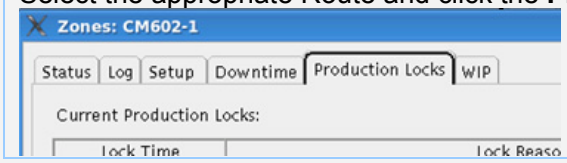
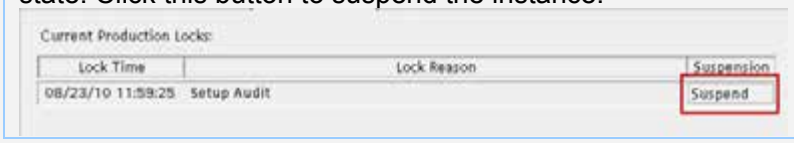
Option name	Setting	Explanation
Verify Outer Tray	ON	Stop production when tray interlock is received from LNB.

7.3 Operation

7.3.1 Suspend Production Locks

Configure Route Status to suspend a production lock. Changes applied here will immediately affect lock status for the entire machine.

Note: Lock types may only be ignored if they have already been enabled in System Configuration, and their specification indicates that they are eligible for suspension.

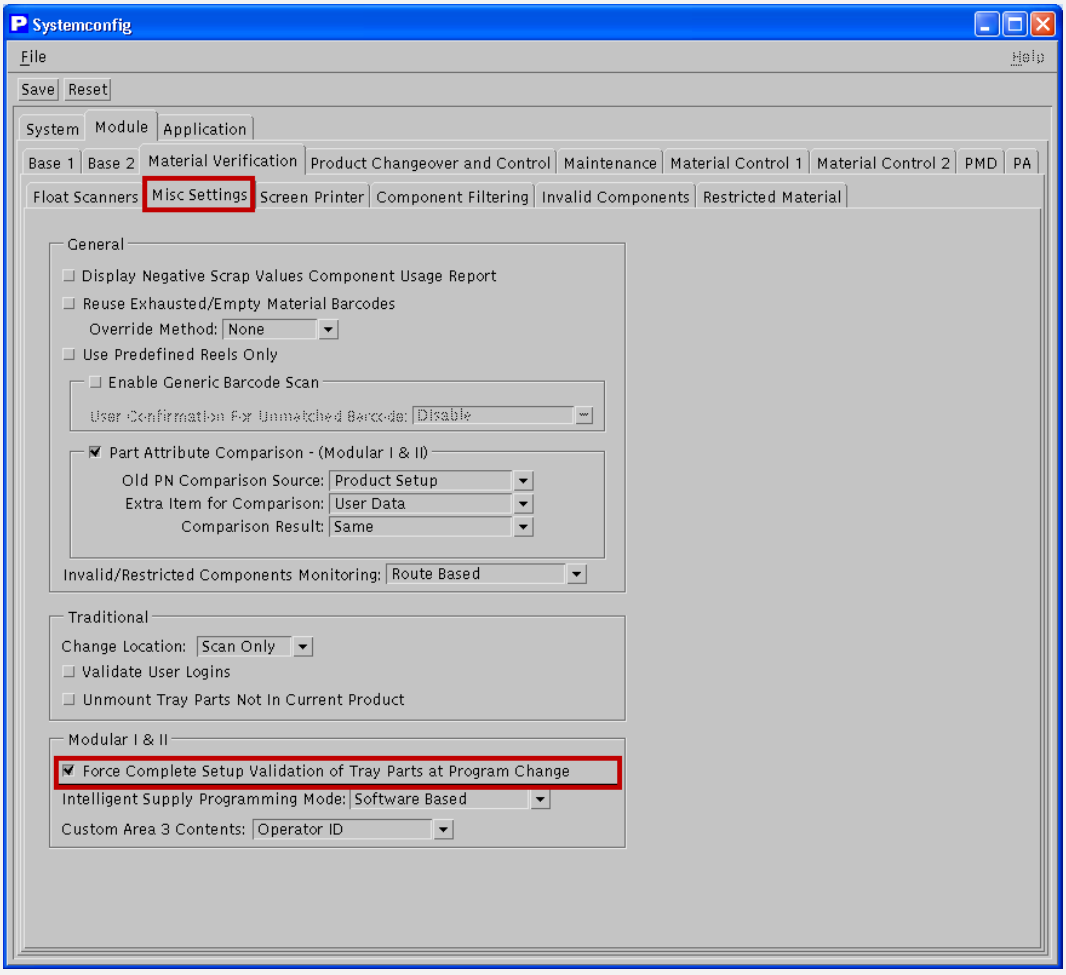
Step	Procedure
1	Select the **Route Status** Application from the PanaCIM Main Menu.
2	Select the appropriate Route and click the **Production Locks** tab. 
3	The Suspension column will display the **Suspend** button, or a Suspended state. Click this button to suspend the instance. 
4	The Suspension column will now display as Suspended. 

7.3.2 Automatic Removal of Tray Parts from Active Setup

Generally, Tray parts need to be removed on changeover when there are no common parts between currently running product and product to be activated after changeover. This helps to completely revalidate the tray feeder setup by rescanning certain parts ensuring that feeders are mounted at correct locations / tray parts are fed to correct tray slots as required by the newly activated product.

If there are common parts, you would want to retain them and hence would not want to remove them. This results in minimal changeover and prevents time loss in mounting new trays. Even if, there are common tray parts and tray part quantity has dwindled to low numbers, then you would want common tray parts to be removed as well so that new tray feeders are setup which have right amount of tray parts required for production of new product.

Refer to the action table below for configuring tray part removal in PanaCIM upon product changeover.

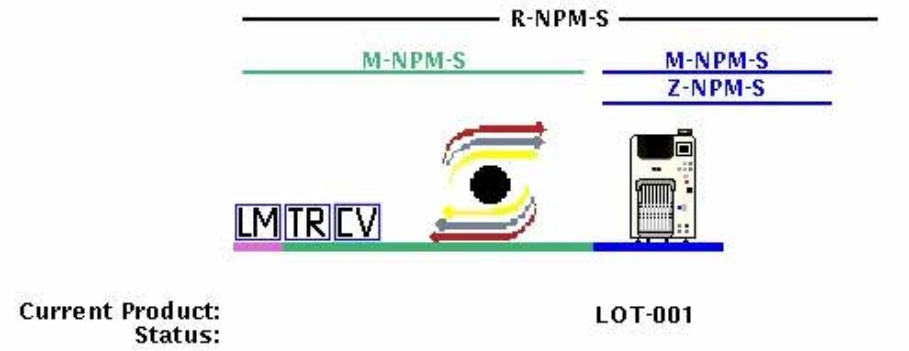
Step	Procedure
1	Click **System Config** on the PanaCIM Launcher Toolbar.
2	Click the **Module** Tab and then click the **Material Verification** tab. Following screen appears: 

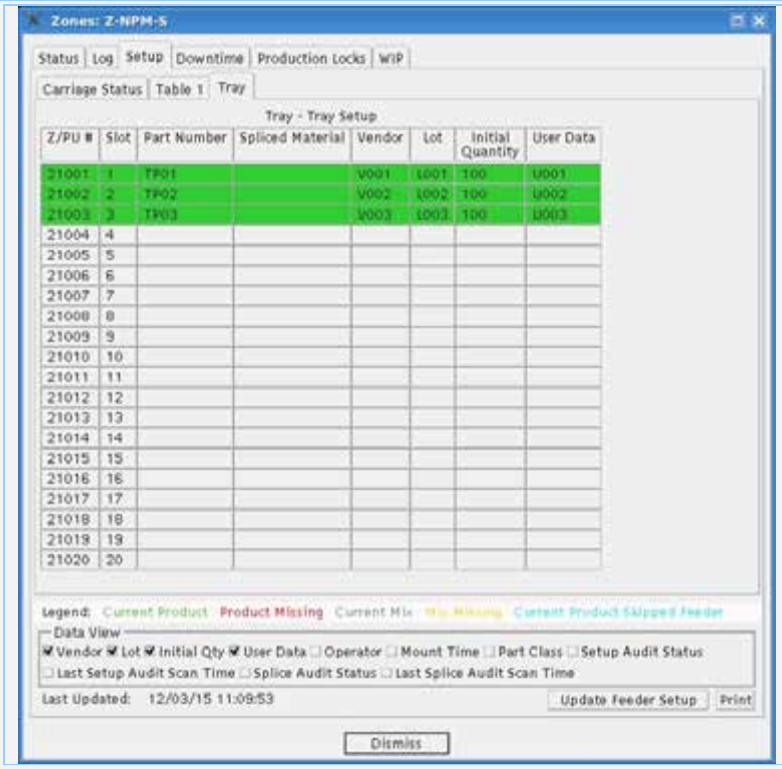
3	Check the **Force Complete Setup Validation of Tray Parts at Program Change** checkbox to enable the ability to remove tray product setup feeders at program change thus enforcing a setup validation. **When this checkbox is checked:** All setup information of tray parts are deleted at product changeover, which requires all tray parts to be verified. **When this checkbox is not checked:** The setup information of tray parts other than those of common tray parts (tray parts with the same part number and tray position) are deleted at product changeover. The setup information of the common tray parts are held.
4	Click the **Save** button.

7.3.3 Single Lane NPMs with Common Tray Parts

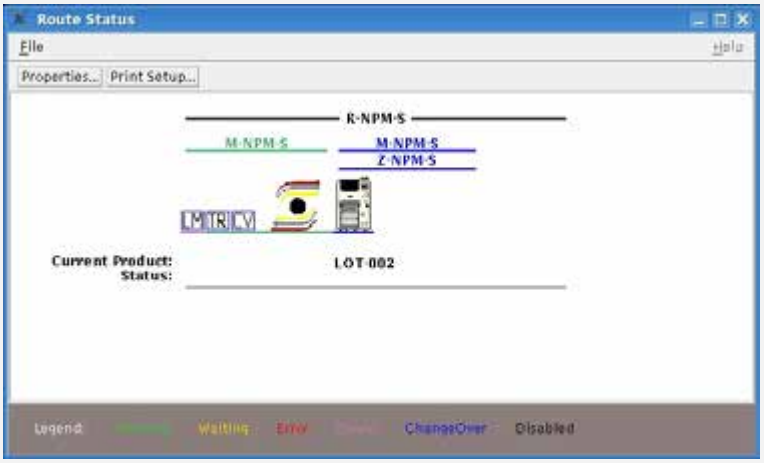
In a changeover when single lane NPMs have common tray parts and **Force setup verification at tray parts changeover** is selected, all the tray parts setup information used in the previous production will be deleted and tray parts must be verified**.**

The following table shows how program information is displayed on the scanner or route status before and after removing tray parts. The following operation can also be applied to dual-lane NPM in independent mode.

Step	Procedure
1	Product program used in NPM single-lane is as follows.  Tray parts setup status for this product program is shown below. <Setup status shown on mobile scanner> 



2 Download a different product program to a NPM single-lane and change machine status to **Changeover**.



All setup information will be deleted as **Force setup verification at tray parts changeover** is selected on **System Configuration dialog**. Even when programs before and after changeover have common tray parts and they are at the same location, the following setup information will be deleted. (It turns to unverified status on PanaCIM.)

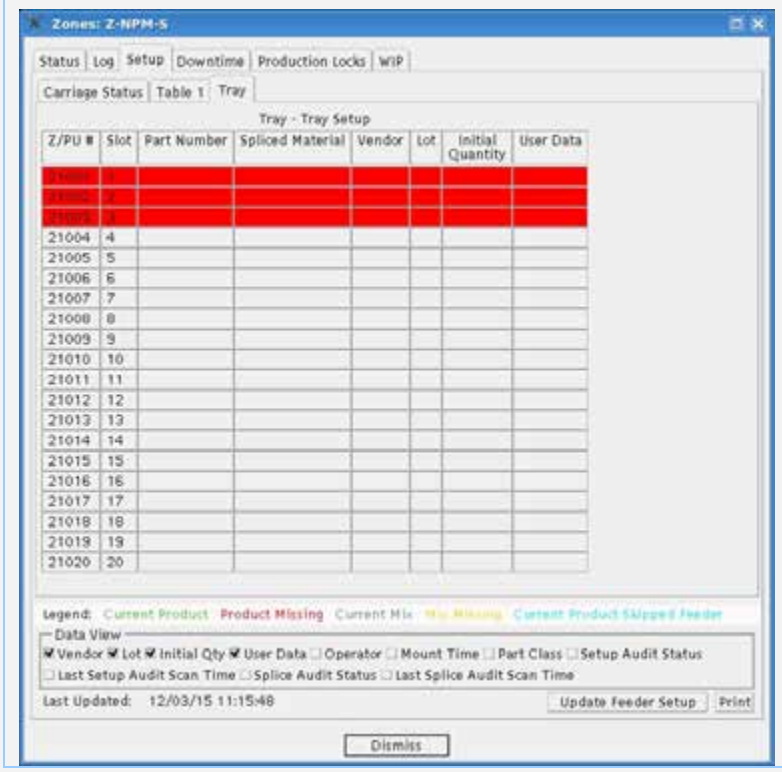
Tray parts setup status on this product program is as follows.

<Setup status shown on mobile scanner>

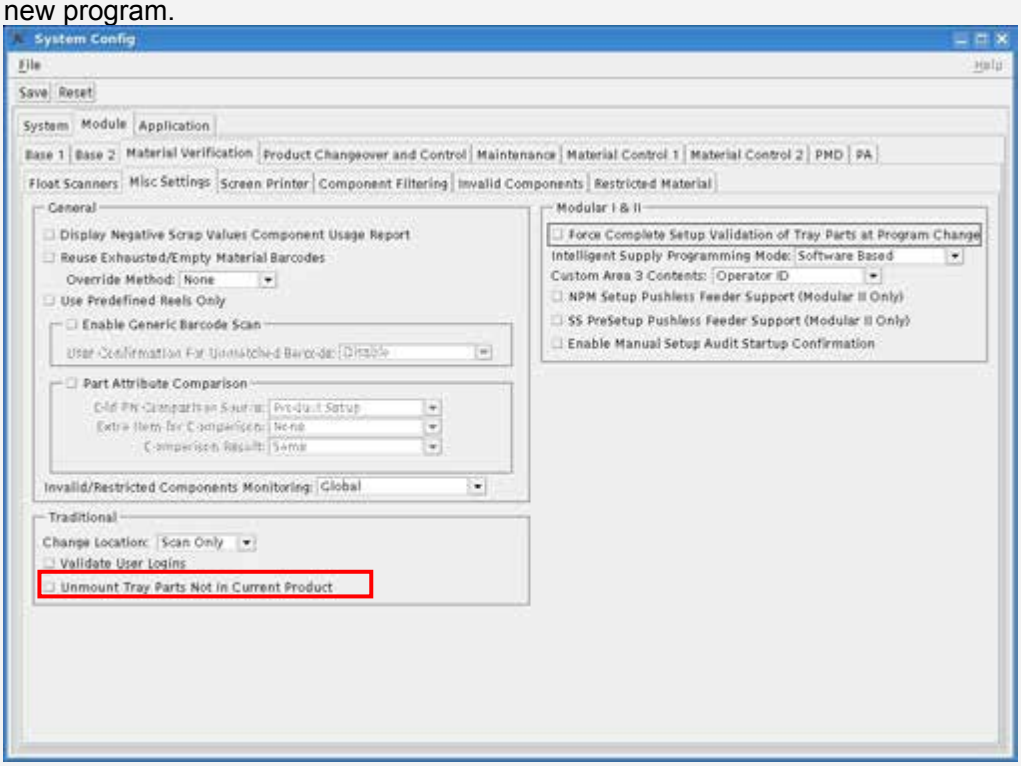
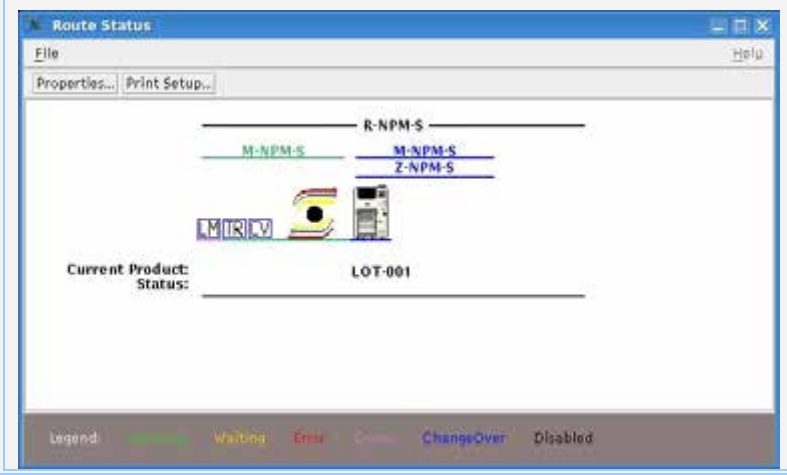


Note: Part number is used in the program before changeover and the one after changeover. However the common tray parts setup information will be deleted during changeover.

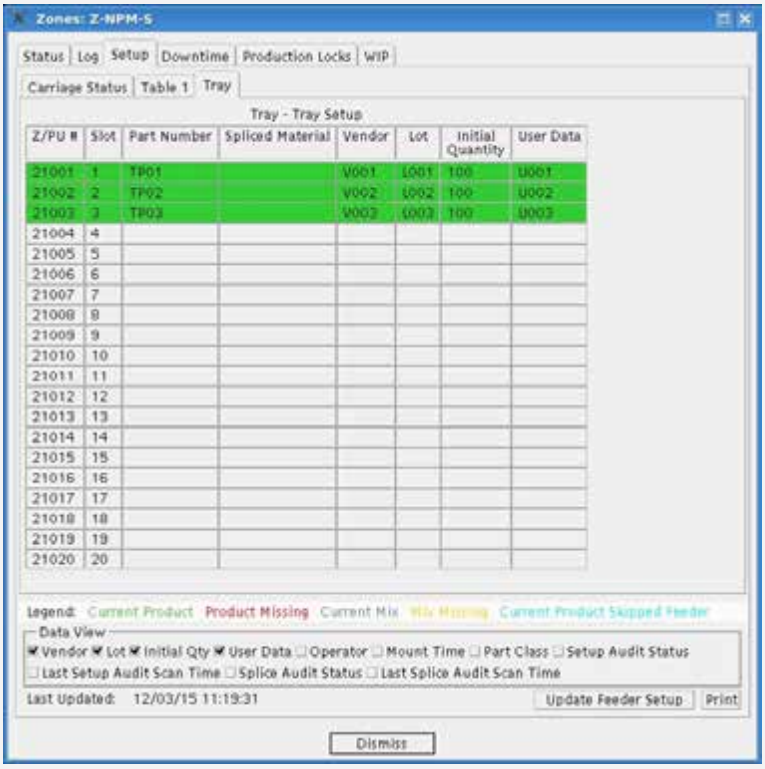
<Setup status shown on mobile scanner>



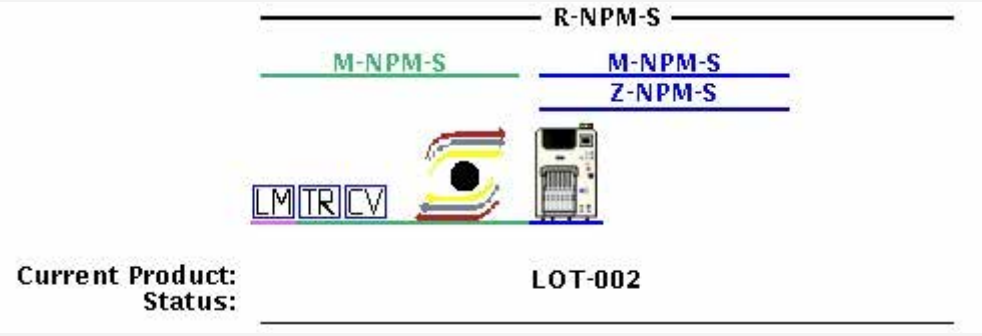
3	Use a mobile scanner to set up tray parts for all locations of the newly selected program.

Step	Procedure
1	If **Force setup verification at tray parts changeover** option is not selected above, the procedure will be as follows: To hold tray parts, do not select this option and use the parts for a new product. By doing so, it will not be necessary to change trays for the parts used in the old and new program. 
2	The program running in NPM single-lane is shown as follows. 





3 Changeover will be performed for NPM single-lane.



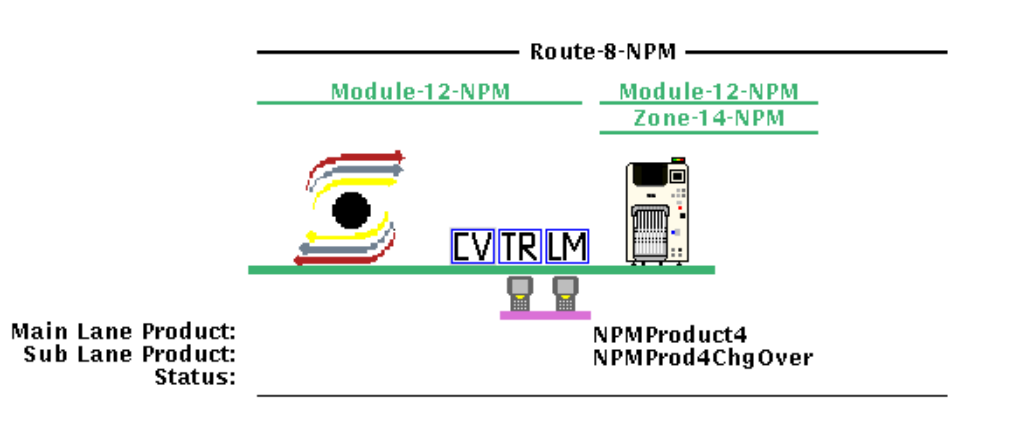
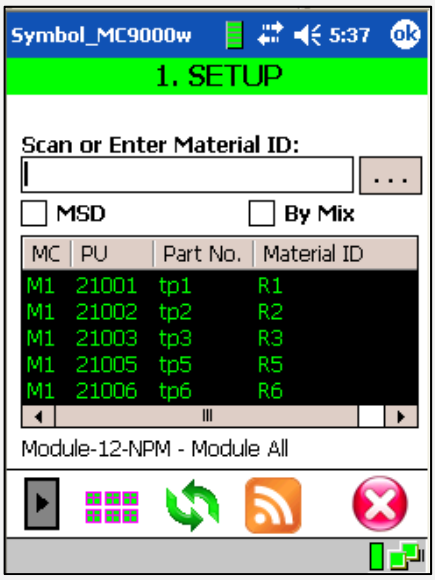
4	After changeover is finished, the tray parts that include common parts will be held, and setup information of the remaining tray parts will be deleted.  **Note:** Part t1 and t2 are set at tray position 21002 and 21003 in the old program. **t119p is a common tray part in the new and old program.** 
5	Green row shows that no setup is needed at these positions as tray parts stayed set after changeover. Red row shows that setup is needed as tray parts setup information was deleted after changeover. In the example above, tray parts were removed from tray positions 21002 and 21003 while tray parts on 21001 stays.

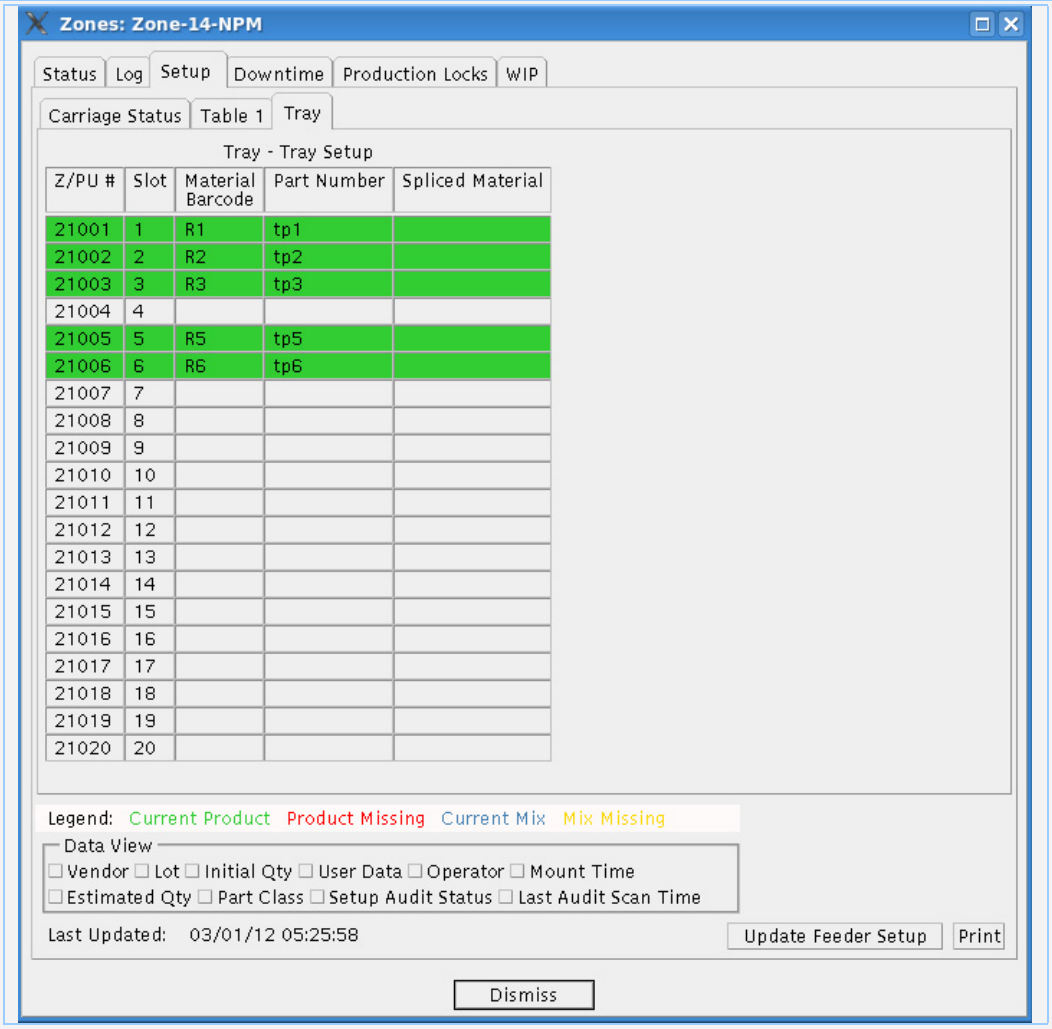
* In a changeover for NPM single-lane without common tray parts, all tray parts of the old program will be removed and setup all parts used in the new program. This is done regardless of the setting **Force Complete Setup Validation of Tray Parts at Program Change** on **System Config** dialog.

7.3.4 Dual Lane NPMs with Common Tray Parts

There could be a requirement that upon changeover on sub lane, all tray parts in the previously running program on this lane (except those used by the program running in the main lane) are removed. This includes any common parts that are in the previously running program and newly activated program on sub lane should be removed. As a result, tray feeders need to be setup for newly activated program on sub lane. To achieve this objective, **Force Complete Setup Validation of Tray Parts at Program Change** option is checked in the **System Config** dialog.

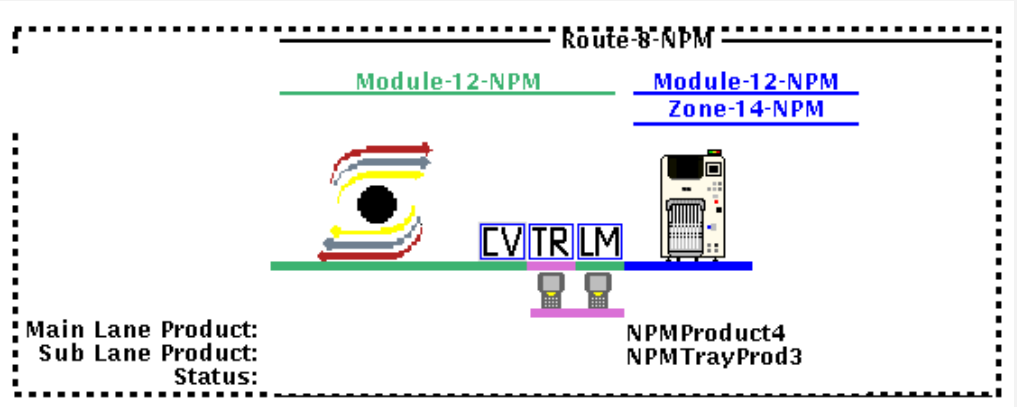
Refer to the action table below that shows how the scanner and route status present the program information before and after removing tray parts from sub lane for dual lane NPMs in shared mode.

Step	Procedure
1	Products currently running in the main lane and sub lane are shown below: 
2	Current setup is shown below. In this setup being considered, product running in the main lane has three tray parts with part numbers **tp1**, **tp2** and **tp3** present at feeder locations **21001**, **21002** and **21003**. Product running in the sub lane has four tray parts with part numbers **tp1**, **tp2**, **tp5** and **tp6** present at feeder locations **21001**, **21002**, **21005** and **21006** respectively. 



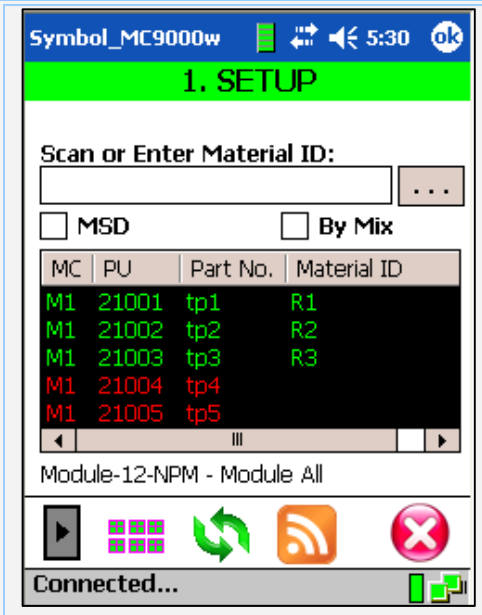
3 After changeover is completed on sub lane, it is reflected in the Route Editor as well as Route Status. The newly selected product on the sub lane has four tray parts with part numbers **tp1**, **tp2**, **tp4** and **tp5** present at locations **21001**, **21002**, **21004** and **21005**.

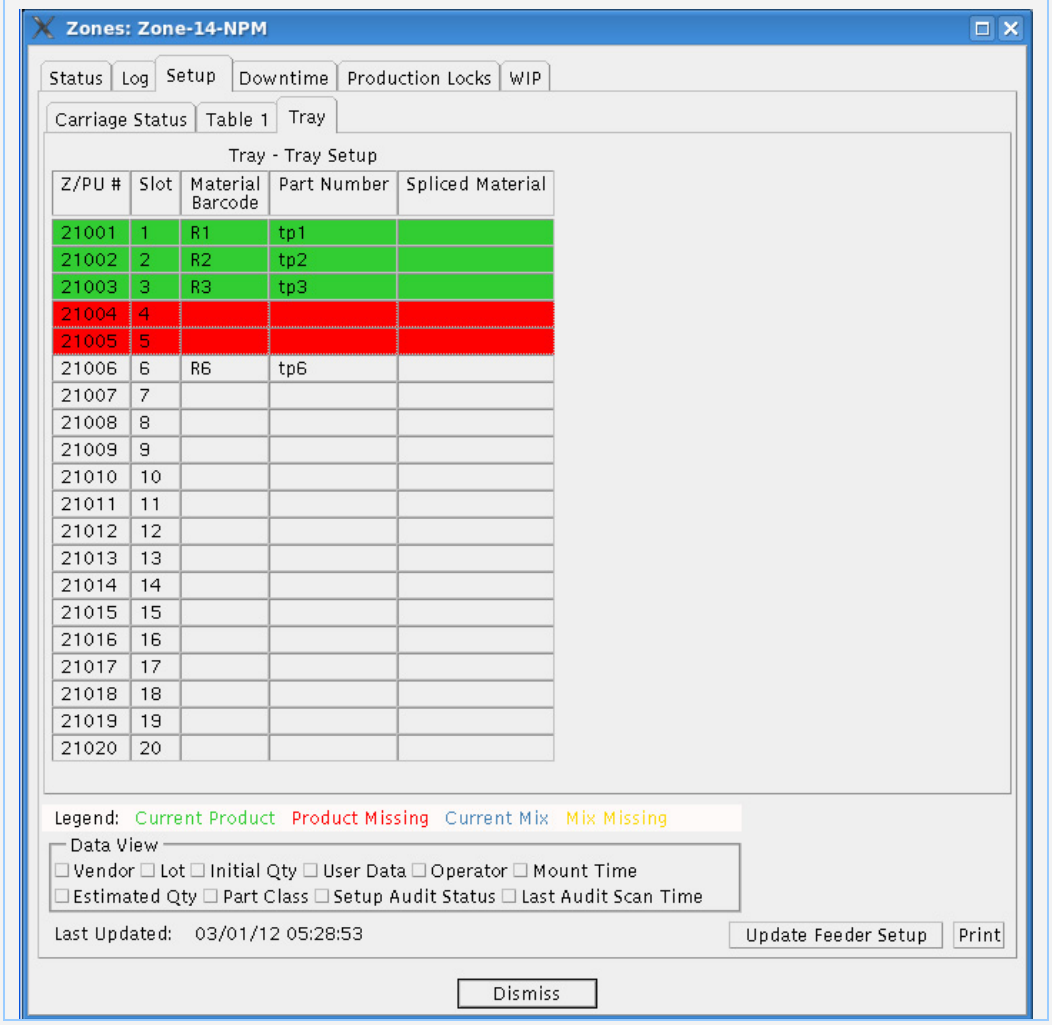
As shown in the **Route Editor** screen shot below, the color of the zone containing NPM machine turns blue after completion of changeover.



Note: Option **Force Complete Setup Validation of Tray Parts at Program Change** is checked in the **System Config** dialog.

4 The new setup after changeover is shown below:

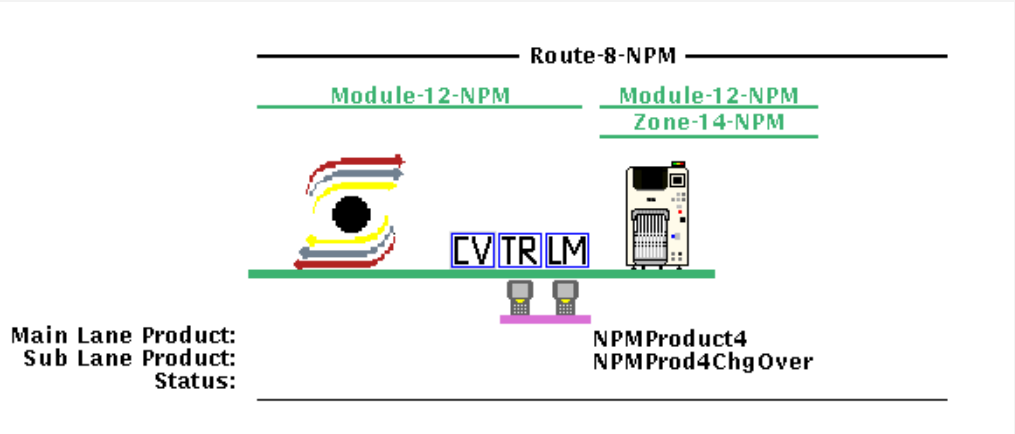
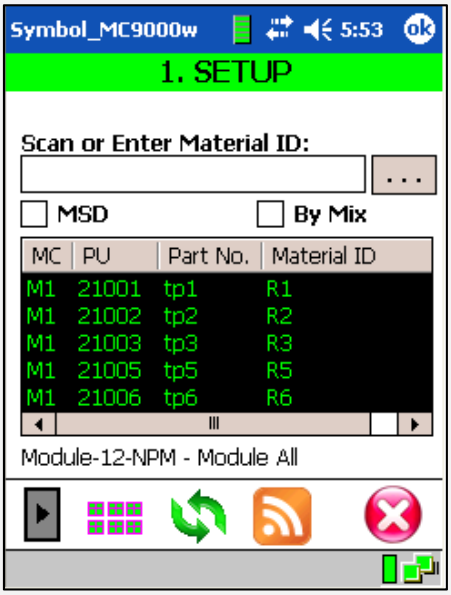


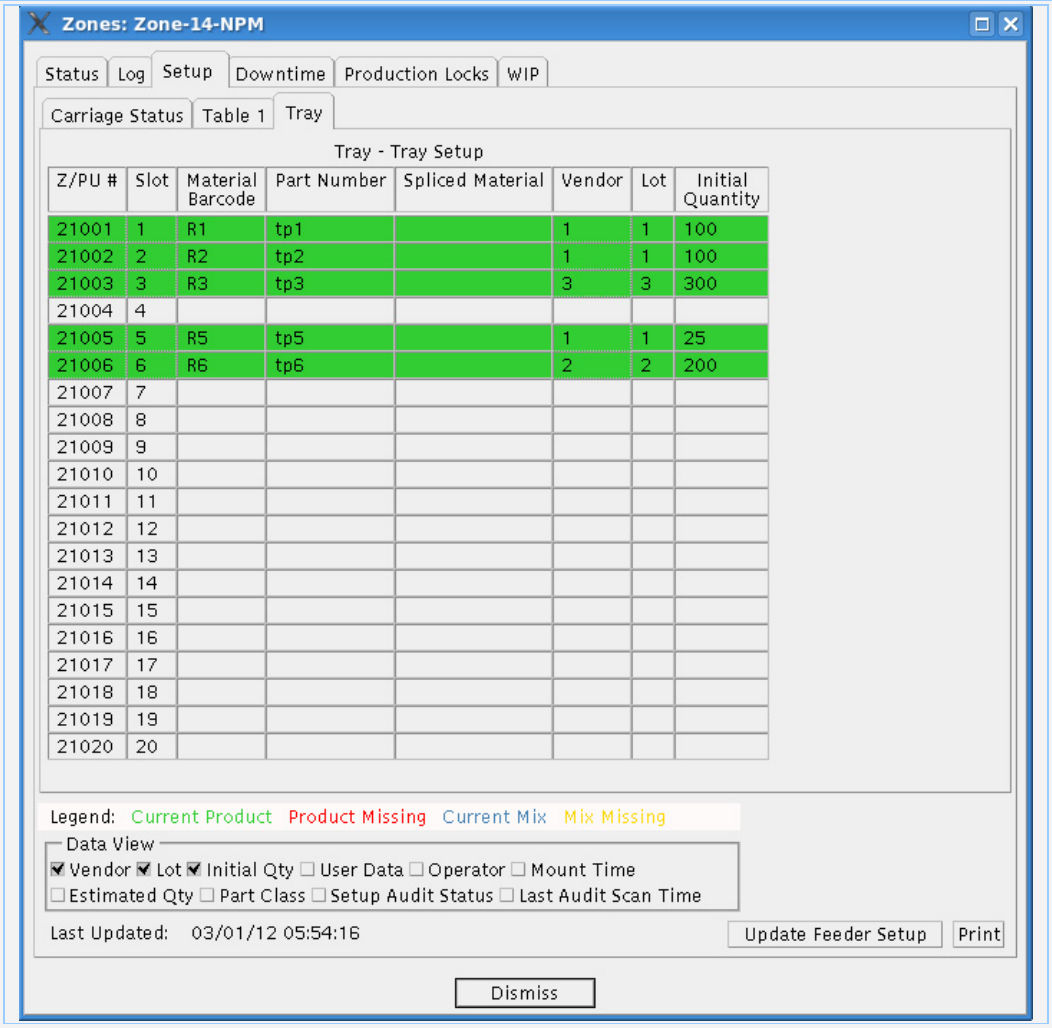


5	As shown above, all feeders from main lane stay mounted. Operator has to perform setup for tray parts of new program running in sub lane.

Refer to the action table below that considers the scenario mentioned above but with option **Force Complete Setup Validation of Tray Parts at Program Change** unchecked. This option is unchecked to fulfill a requirement that upon changeover on sub lane, all tray parts in the previously running program on this lane are removed. However, any tray parts that are common to previously running program and newly

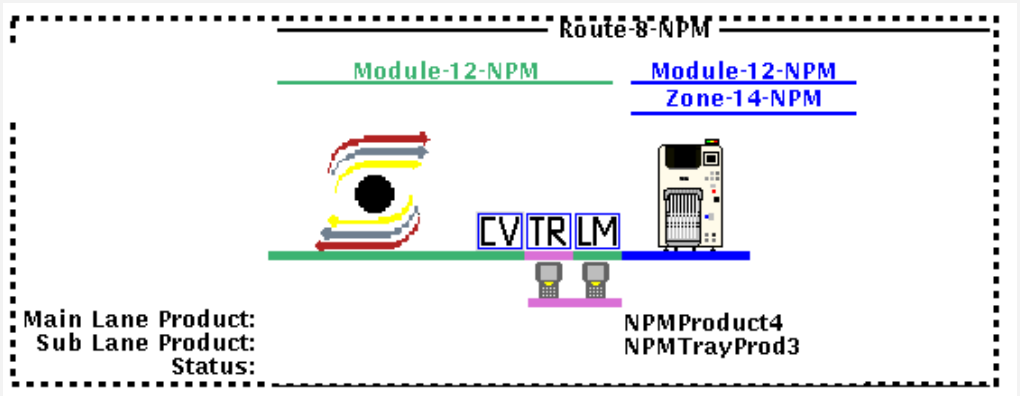
activated program on sub lane should not be removed. Tray parts used by the program running in the main lane would anyways be retained upon changeover since dual lane NPMs are running in shared mode.

Step	Procedure
1	Products currently running in the main lane and sub lane are shown below: 
2	Current setup is shown below. In this setup being considered, product running in the main lane has three tray parts with part numbers **tp1**, **tp2** and **tp3** present at feeder locations **21001**, **21002** and **21003**. Product running in the sub lane has four tray parts with part numbers **tp1**, **tp2**, **tp5** and **tp6** present at feeder locations **21001**, **21002**, **21005** and **21006** respectively. 



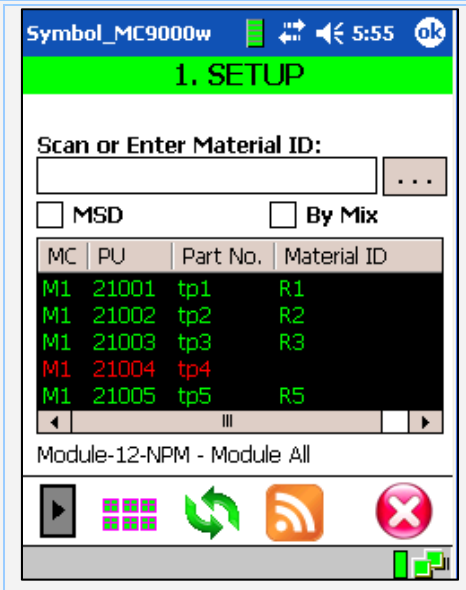
3 After changeover is completed on sub lane, it is reflected in the Route Editor as well as Route Status. The newly selected product on the sub lane has four tray parts with part numbers **tp1**, **tp2**, **tp4** and **tp5** present at locations **21001**, **21002**, **21004** and **21005**.

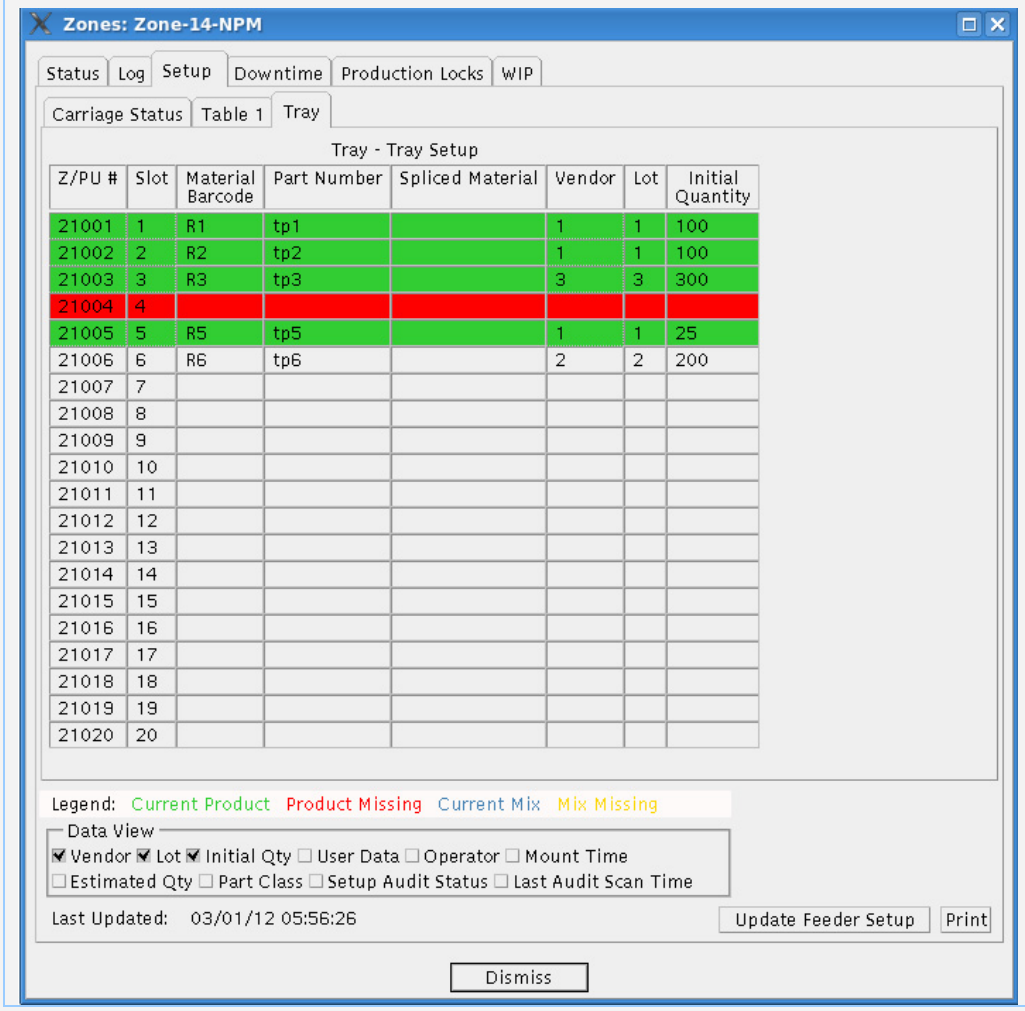
As shown in the **Route Editor** screen shot below, the color of the zone containing NPM machine turns blue after completion of changeover.



Note: Option **Force Complete Setup Validation of Tray Parts at Program Change** is unchecked.

4 The new setup after changeover is shown below:





5	As shown in the screen shots in step 4, All common feeders from main lane and sub lane stay mounted. Operator has to perform tray setup for new parts that are present in the newly selected program on the sub lane but were not present in the previous program running on the sub lane. In the program setup considered above, tray setup needs to be performed only for part **tp4** present at tray slot **21004**.

7.3.5 NPM Tray Interlocking Process Flow

■ Part Exhaust - Non Bulk

Refer to the tables below that highlight the NPM Tray Interlocking in the context of various operations like Part Exhaust replenish (Bulk and Non Bulk), Product changeover. It is to be noted that for tray interlocking feature to work at machine level, Interlock feature should be turned on at the NPM machine console.

Step	Procedure
1	Equipment report parts exhaust for a tray part. If there are no more trays with this part number, production is paused and exhaust alert message appears on the PanaCIM Route Status application.
2	Operator replenishes exhausted tray at the NPM machine by performing material verification on tray locations and attempts to put machine back into production.
3	NPM sends Interlock inquiry reply to LNB which in turn forwards it to PanaCIM.
4	PanaCIM sends Interlock Inquiry reply containing list of invalid tables. This list includes tray locations which do not contain any part as per the program present in PanaCIM and hence have no use in producing current product.
5	LNB evaluates if invalid slots are acceptable to the current program. If slots are acceptable to current program, LNB sends production start signal to NPM machine and production is started on the equipment.

■ Part Exhaust - Bulk Part Replenishment

Step	Procedure
1	Equipment report parts exhaust for a tray part. If there are no more trays with this part number, production is paused and exhaust alert message appears on the PanaCIM Route Status application.
2	Operator replenishes exhausted tray at the NPM machine by loading tray and pressing button assuming that this location was set up with bulk replenishment.
3	Each time a tray is replenished, PanaCIM updates bulk package count by deducting the tray size (components per tray) from it.
4	As long as bulk package quantity count is greater than zero, production resumes after tray is replenished. The moment bulk package count reaches zero, indicating that parts are not replenished, NPM machine will lock with exhaust and production is stopped because it can’t pick up parts. Production is stopped on NPM machine after interlock is set.

* Bulk part replenishment is valid only in the context of Material Control mode.

■ Product Changeover

Step	Procedure
1	Changeover happens at the NPM machine.
2	After changeover, interlock request is sent to LNB with list of all tray locations used in newly selected product.
3	LNB forwards interlock request to PanaCIM.
4	PanaCIM checks tray setup of the newly selected product and sends an interlock inquiry request.
5	As long as there is no issue in the tray setup of newly selected product, production is started on NPM machine. In case, PanaCIM determines that there is some fault in the tray setup of the newly selected product, interlock inquiry request sent by PanaCIM contains a list of invalid tables. NPM machine locks these tables thereby preventing production from being started.

8. FUNCTION - REPORT - MATERIAL VERIFICATION REPORTS

This chapter explains the Material Verification Report.

8.1 Overview

PanaCIM/Material Verification creates various reports. To create reports, access to PanaCIM application server from a browser (Internet Explorer 11 or later) and launch PanaCIM Web application to create them.

For more details on PanaCIM Web application, see Basic Operations (Web Application).



8.1.1 List of reports

The table below is the list of selectable reports.

■ Material verification-Base

Report	Description
Component Trace	Shows the time when the part reel was mounted on the machine and the operator involved.
Current Status Report	Shows the information of components that currently exist on the machine, such as Part Number, Slot number, etc.
History Report	Shows the history of material verifications.
Trasaction Trace Report	Shows the history of material verification. You can also check a lot name or a mix name for each lane during verification.
Work Order Trace Report	Shows traceability of a material and work order.
Loaded Feeders Report	Shows a list of feeders with reel installed..
Feeder Routine Check History	Shows the history of routine check result for each feeder.
Production Lock History	Shows the history of production stop at the specific machine.
Remaining Shelf Time	Shows a list of remaiming time of MSD parts.
LCR Verification History	(Not supported)
Routine Check History	Shows the history of routine check operation.

■ Material verification-Presetup

Report	Description
Presetup Detail	Shows a list of detailed information on presetup that have been performed so far
Presetup Summary	Shows brief information on all presetup performed in a target zone.
Modular Presetup Cart Mount History	Shows the mounted time and removed time of the cart that is filtered by the specified conditions.
Modular Presetup Cart Current Contents	Shows the part information on afeeder, installed in the removed cart, filtered by the specified conditions.
Modular Presetup Cart Feeder History	Shows the events of a feeder, unmounted from or mounted on the cart, filtered by the specified conditions.
Presetup Cart Mount Status Report	Shows whether a cart filtered by the specified conditions is being mounted on the machine or not. * The material control license is required separately to display the location and cart mounted time.
Presetup Cart Check Report	Shows a program in each machine and table to mount the selected cart. If the mounted parts are less than the required parts, scan other parts in the cart (presetup cart or the cart located in the machine) to start production.
Presetup Product Data Check Report	Shows all the carts set up for all programs in mix including the selected program. If the mounted parts are less than the required parts, scan other parts in the cart (presetup cart or the one located in the machine) to start production.
Presetup Cart Routine Check	Shows the routine check result of a selected cart.
Presetup Cart ID List	Shows a list of cart IDs.

8.2 Setting

8.2.1 Filter criteria

For filter criterial of reports, refer to Basic Operations (Web application 3.2.2 [Filter Criteria].

8.3 Operation

8.3.1 Operation Procedure

For how to output reports, refer to refer to Basic Operations (Web application) 3.3.1 [Operation Procedure].

8.3.2 Explanations of Transaction Trace Report Items

■ Explanations of transaction trace report items

Report	Description
Part Scan	Part was scanned to verify setup.
Splice Scan	Reel was scanned to verify splicing during production.
Splice	When the first pickup position was reached (machine event) after the spliced tape was detected by the feeder sensor
Part Exhaust	When part exhaustion verification was performed
Feeder Jam	Feeder jam was cleared via the PanaCIM scanner.
Clist Mount	Timing when the latest feeder setup information wasobtained from the machine * When the feeder setup screen is closed and at the start of production

<Verification result>
If you failed to scan, the following message appears.

Report	Description
INVALID/RESTRICTED PART SCAN FAILURE	When invalid or restricted part was scanned
INAPPLICABLE PART SCAN ERROR	When inapplicable part was scanned
UNUSED PART SCAN ERROR	When unused part was scanned
INAPPLICABLE SUBSTITUTE PART SCAN ERROR	When inapplicable substitute part was scanned
SPLICED PART SCAN ERROR	When already spliced part was scanned
UNMATCHED OLD/NEW PART SCAN	When unmatched old/new part information was scanned (When old/new reel comparison function is used)
EMPTY REEL PART SCAN ERROR	When used-up (empty) reel part was scanned
USED PART SCAN IN PRODUCTION	When part already used in other location was scanned
MACHINE IS NOT READY	When the setup screen is not displayed on the machine
MEMORY FULL	When feeder memory has not been cleared
CANCEL BUTTON ON SCANNER	When the cancel button on the scanner screen was pressed
CANCEL ON CM	When the cancel button on the setup screen of the CM was pressed
TIMEOUT	When scanner timeout has occurred
HOST COMM. ERROR/TIMEOUT	When communication disconnection with PanaCIM occurred or Host failed to respond within the specified time
LNB TIMEOUT	When LNB failed to respond within the specified time
FEEDER PULL OUT/IN	When af eeder was pulled out

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9. FUNCTION - REPORT - MULTIUSER DATA

This chapter explains MultiUser Data.

9.1 Overview

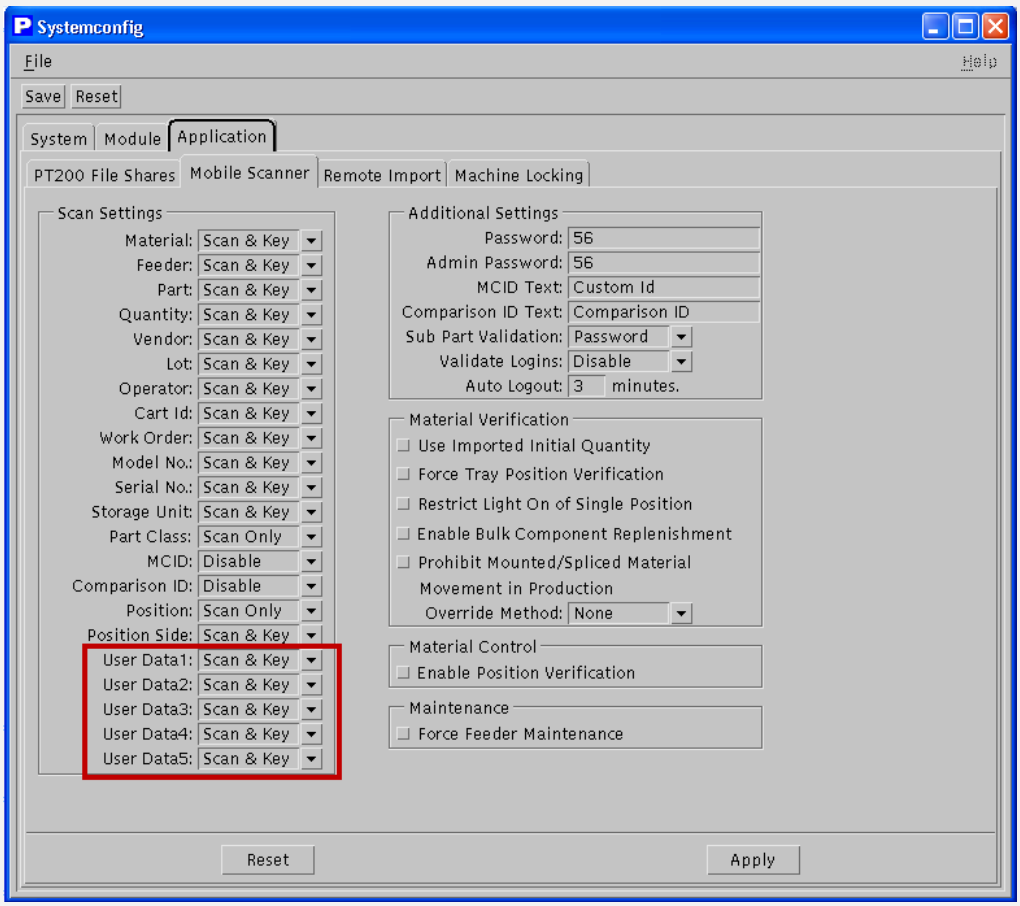
The user data is the information to be added to the major part attributes including vendor, lot and quantity.
For example, if you want to keep track of the scanned date of a particular reel, the user data fields are useful.

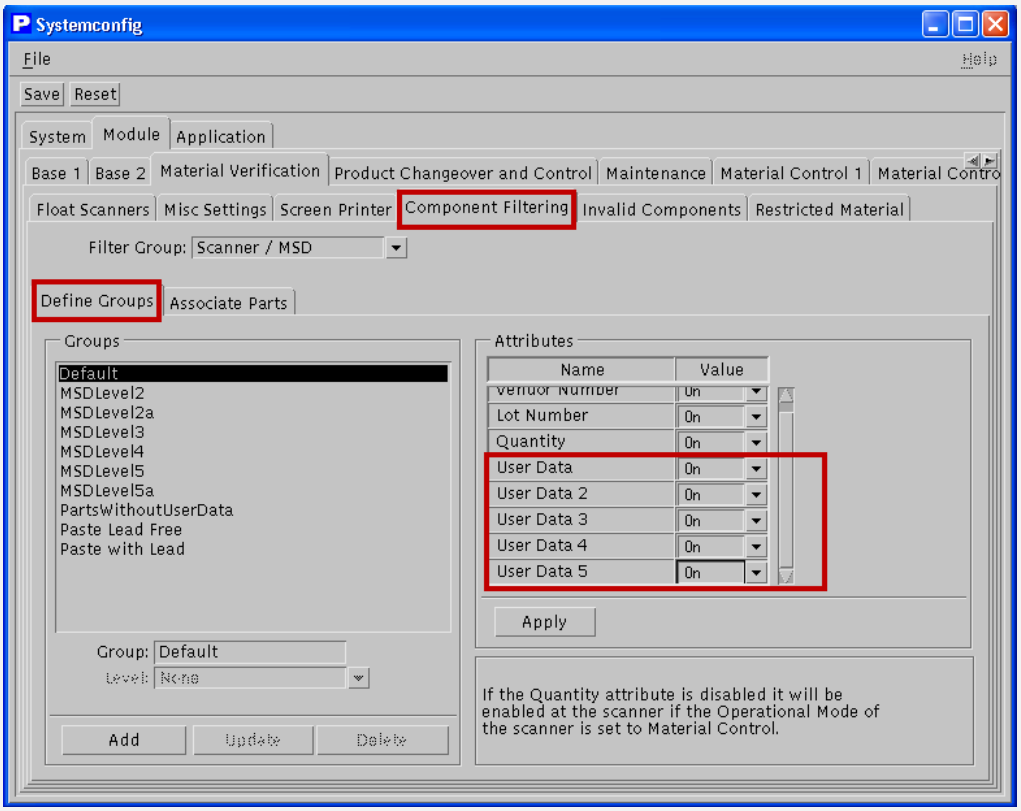
In PanaCIM , five additional data fields are supported for a particular part / reel barcode. This gives leverage to the user to specify additional customized data for a part / reel barcode. User can enter any type of data in the user data fields at the time of scanning a part / reel barcode.

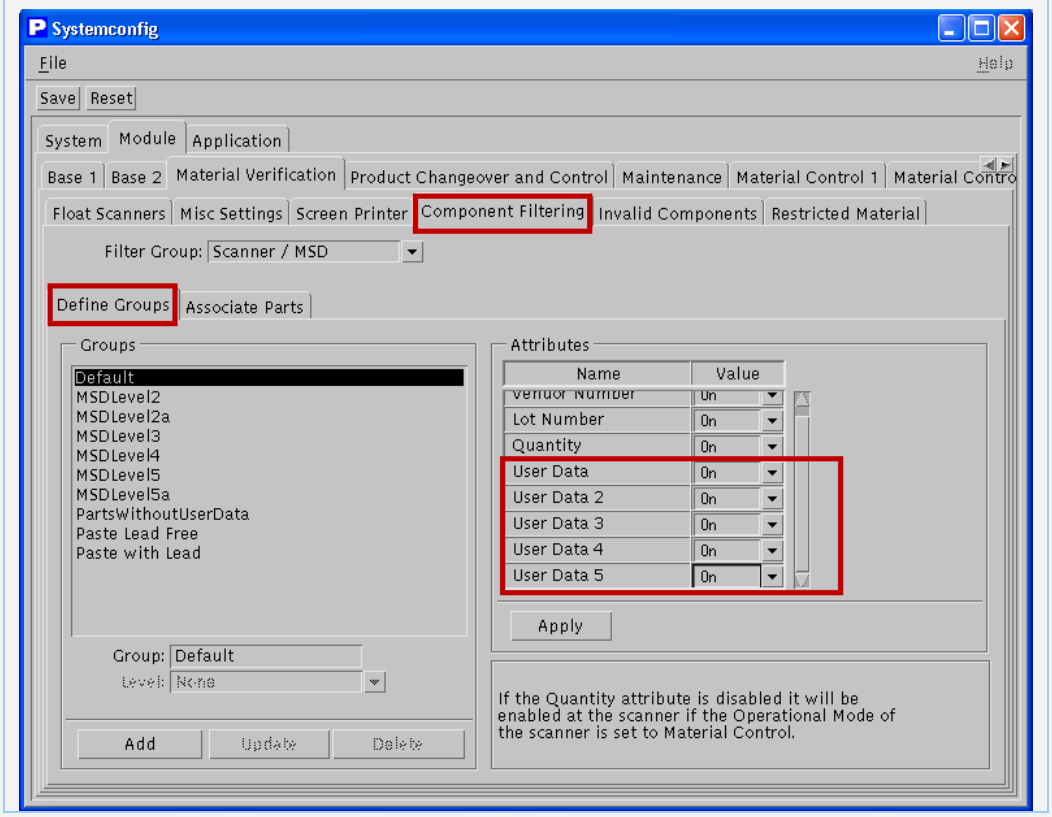
9.2 Settings

9.2.1 Configuration of Multi User Data

Refer to the action table below for enabling additional user data fields at the time of scanning a part / reel barcode using the mobile scanner:
* In the User Data 1 to 5, the same parser should not be defined.
  For details of the parser definition, refer to 7.2 Barcode Setting in the Base documentation.

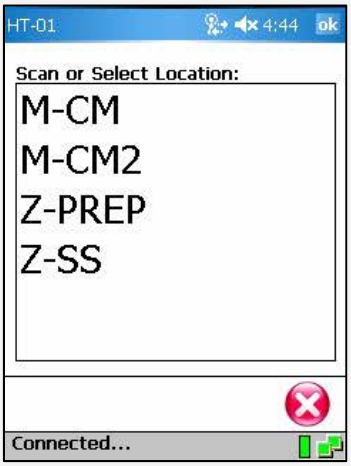
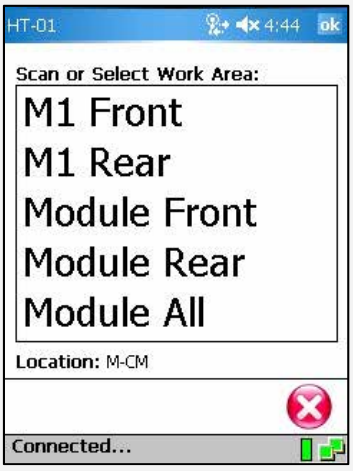
Step	Procedure
1	Launch **System Config** dialog and select **Application -> Mobile Scanner** tab. Following screen appears: 
2	Enable User Data1 to User Data5 fields by selecting one of the three options - **Scan Only**, **Scan & Key**, **Key Only** from underneath the **Scan Settings** area.
3	Click the **Save** button in the upper left of the screen to save the settings made.

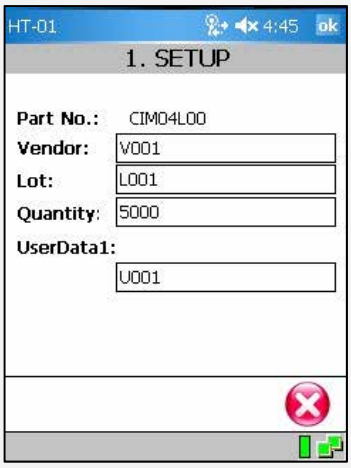
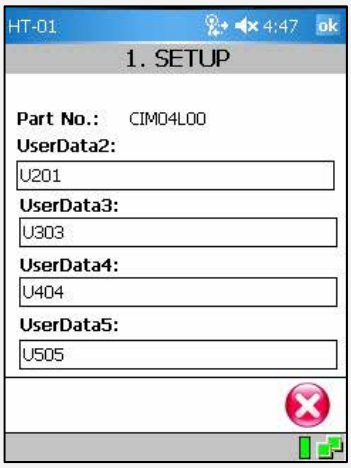
Step	Procedure
4	Activate the **Module** tab and then the **Component Filtering** tab. This is shown below: 
5	Enable User Data1 to User Data5 fields by selecting one of the three options - **Scan only**, **Scan and Key**, **Key Only** from underneath the **Scan Settings** area.
6	Click the **Apply** button, followed by the **Save** button at the top left of the screen to save the input method of the **User Data1** to **User Data5** fields.

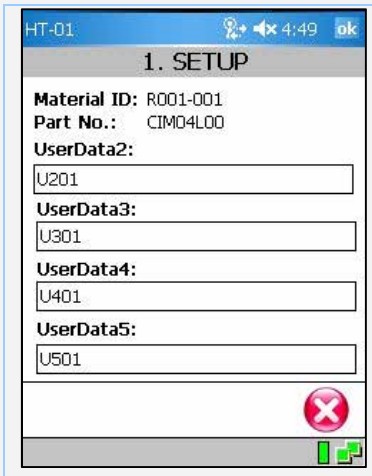
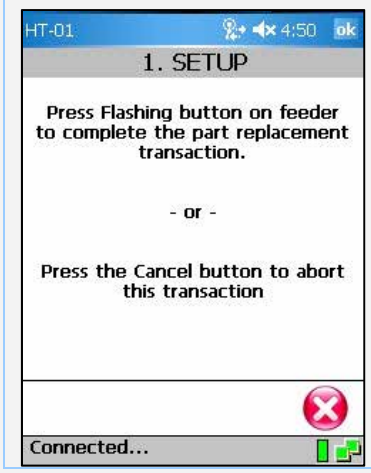
Step	Procedure
7	Click the **Module** - **Material Verification** - **Component Filtering** tab and select **Scanner/MSD** from the **Filter Group** drop-down list. Following screen appears: 
8	Select the **Default** group and then enable the **User Data1** to **User Data5** attributes by selecting **On** from the drop-down list.
9	Click the **Apply** button, followed by the **Save** button at the top left of the screen to save the settings made in the **Attributes** field.

9.3 Operation

9.3.1 Mobile Scanner Operation

Step	Procedure
1	Select **Part Scan** from the Production main menu screen. 
2	When the screen shown below appears, select the desired location from the list. 
3	Scan or select a work area on the screen below. 

Step	Procedure
4	The display changes to the **SETUP** screen as follows. 
5	Scan or enter a part number. When the following screen appears, enter the part attributes. 
6	Enter additional user data.  **Note:** In the MC mode, the material ID also appears as shown below.

Step	Procedure
	
7	When the following message appears, press the blinking feeder button to set the entered information to the feeder memory. 

* When the User Data1 to User Data5 have been set to On on the System Configuration dialog, the user data can be specified also when doing presetup.

9.3.2 Reports

Apart from **User Data** column, additional user data columns (**User Data 2** to **User Data 5**) appear in the following production analysis reports:

■ Material - Base

- Current Status Report
- Component Traceability Report
- History Report
- Transaction Trace Report
- Work Order Trace Report
- Loaded feeder report

■ Material Verification - Presetup

- Presetup Detail Report
- Modular Presetup Cart Current Contents
- Modular Presetup Cart Feeder History

■ Material Control

- Material Quantity Report
- Material Adjustment Report
- Material Usage Report
- Material Position History Report
- Material Movement Report
- Material Registration Report
- MSD Transaction Report
- Latest MSD Report
- Expired MSD Report

■ Traceability

- Panel to Component Traceability Report
- Barcode Trace Report
- Component to Panel Traceability Report

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10. APPENDIX - SCANNER OPERATION

This chapter explains operation of a mobile scanner.

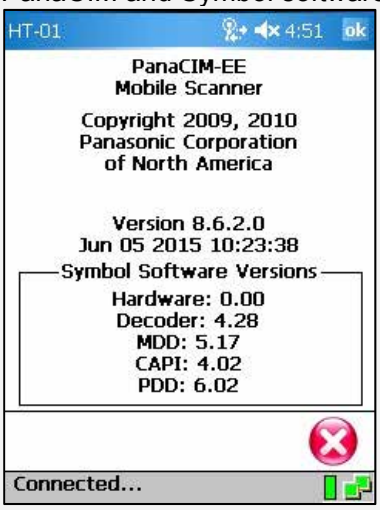
10.1 Overview

Mobile scanners used in material verification process are Motorola MC70/MC75/MC55.

10.2 Settings

10.2.1 Setting Menu

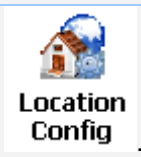
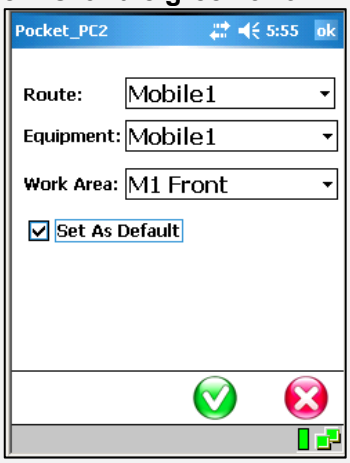
■ Check software version

Step	Procedure
1	Select  , then  .
2	PanaCIM and Symbol software version appear. 

■ Location Configuration

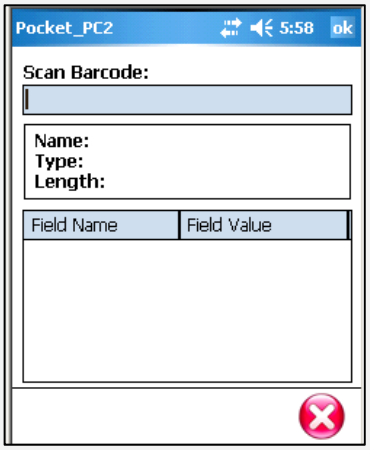
Although the Mobile Scanner is designed to move between work areas, it can be set to a default work area if it will generally remain in one work area.

Refer to the action table below to configure your default work area using the Mobile Scanner.

Step	Procedure
1	Select  and then select  .
2	a. Select **Route** from the drop-down list. b. Select **Equipment** from the drop-down list. c. Select **Work Area** from the drop-down list. d. Optionally, check **Set As Default** to set these as your default settings (Default Location Updated… message will display after clicking the green arrow). e. Click the **green** arrow. 
3	After finished setting, if switching to the **Scan or Select Location** screen, the screen for part scanning of the selected work area appears instead of the one for selecting the location and work area.

■ Barcode Test

The Barcode Test determines information the scanner reads when scanning a barcode. It also indicates how PanaCIM interprets the barcode. It is useful when PanaCIM is not able to process a barcode during production operation.

Step	Procedure
1	Select  and then select  .
2	Scan the Barcode.
3	The scanned barcode information and analyzed information appear on the screen.  ← Scanned barcode information ←Analyzed information
4	If the Field Name does not appear as expected, use System Configuration to verify that your barcode expression is correct or it has priority over other barcode expressions. If your barcode expression is not considered the priority, increase the value of your expression

■ Part Sorting Process

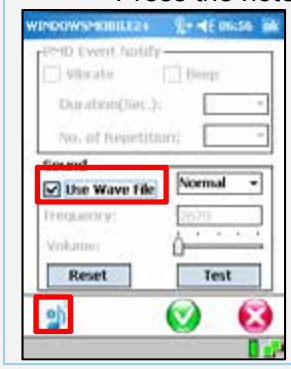
Use the Part Sorting option to indicate slot locations for a particular part. If a part is scanned during the PreSetup operation but only exists on one or more different tables, the scanner automatically displays the Part Sort screen to indicate the proper slot location of the part.

Refer to the action table below to select slot locations for a part using the Mobile Scanner.

Step	Procedure
1	Select  and then select  .
2	a. Select the **Route**. b. Select the **Module**. c. After the product setups appear, select the appropriate **Product**. d. Click the **green** checkmark. 
3	Scan or Enter a Part Number that should be included in the Product. All Machine and Slot locations should appear that contain this Part Number. 

■ Change of Scanner Sound

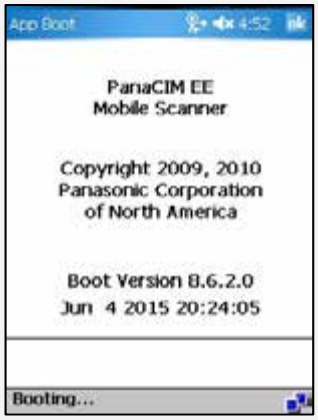
The scanner makes beeping sounds for notification of the operation results such as the time when scan is succeeded, scan is failed and operation is completed. Using a WAVE file, you can set such notification sound of the operation results from the WAVE file.

Step	Procedure
1	After selecting  , select  and  .
2	Perform the following operation. *The scanner must be connected with CIMC. • Mark the checkbox of 'Use Wave File' to enable. • Press the note icon. 
3	After the Wave file is downloaded, press the check button.  - Note - • You can hear the sound by pressing the Test button. -Normal: Sound when scan is succeeded -Error: Sound when scan is failed -Compleate: Sound when all setups or regular check is completed. • Pressing the Reset button enables you to reset the setting to default (WAVE file is not used).

10.3 Operation

10.3.1 Basic Operation

■ Start mobile scanner

Step	Procedure
1	Keep pressing the red button (power ON) on mobile scanner. Scanner start screen and starting message appears.
2	<Start PanaCIM> screen appears. 
3	<Start PanaCIM> screen appears for a few seconds and <Production> screen appears. 
4	After the scanner starts connecting with PanaCIM controller server, its progress appears.  **Note:** If the status does not change from connecting, contact the administrator or us.

■ Scan/input data

Use mobile scanner keyboard or scanner and input data. Set input method on 'Route Editor'.

<Material Verification main menu>



- **Setup**
  Verification work for feeder change at changeover, part exhaust during production, splicing and feeder jam.
  See "3. FUNCTION - PREVENTION FROM AN INCORRECT SET - PARTS SETUP" for more details.

- **Presetup**
  Perform and edit presetup, and set/remove carts.
  See "5. FUNCTION - PREVENTION FROM AN INCCORECT SET - PRESETUP" for presetup operation.

- **Setting/Tool**
  Check mobile scanner software version, barcode test and part sort.

Note: If offline unit/setup audit are not set, Presetup/Setup audit icon do not appear.

■ Handy scanner MC70 basic operation

● How to start

Press the power button to start the scanner.
If the power is not turned on, check the battery charge and if it is well set.



* How to start MC55 is the same as for MC70.

● How to reset

Use hard reset (Cold Boot) or soft reset (Warm Boot) to reset the scanner.
For hard reset, hold the power button and press [1] and [9] key together. Hold them pressed until the reboot screen appears on the terminal.
For soft reset, press the power button. Hold it on until the reboot screen appears on the terminal.



■ Reset scanner material verification software

If a scanner became unstable (screen does not change after scan or input or no reaction after selecting an icon, for example) while using it, it often recovers by resetting the software. Software can be reset in the following steps. If the scanner does not recover by resetting software, reset the scanner according to “How to reset” on the previous page.

If you wish to operate scanner’s OS (to check or edit IP setting, for example), follow the steps “How to go back to Windows mobile desktop”.
How to reset MC55 scanner material verification software is the same as resetting MC70.

● How to reset

Press [Phone Off] key a few times on scanner’s keyboard.
PanaCIM start screen appears. Stop pressing the key.



● How to go back to Windows mobile desktop

Press [Phone Off] key a few times on scanner’s keyboard.
After pressing it a few times, PanaCIM start screen appears, but hold the key. Desktop screen appears. Stop pressing the key.





11. APPENDIX - CREATE BARCODE

This chapter explains how to create barcodes that are to be read to switch line and work area from mobile scanner.

11.1 Overview

Create barcodes in the pre-defined format and read them by a scanner to switch lines or work areas from a mobile scanner easily.

Create barcodes according to the following sample and paste them on machines. Read them by a scanner to reduce operators' work load.

11.2 Settings

11.2.1 Configure Barcode Expressions

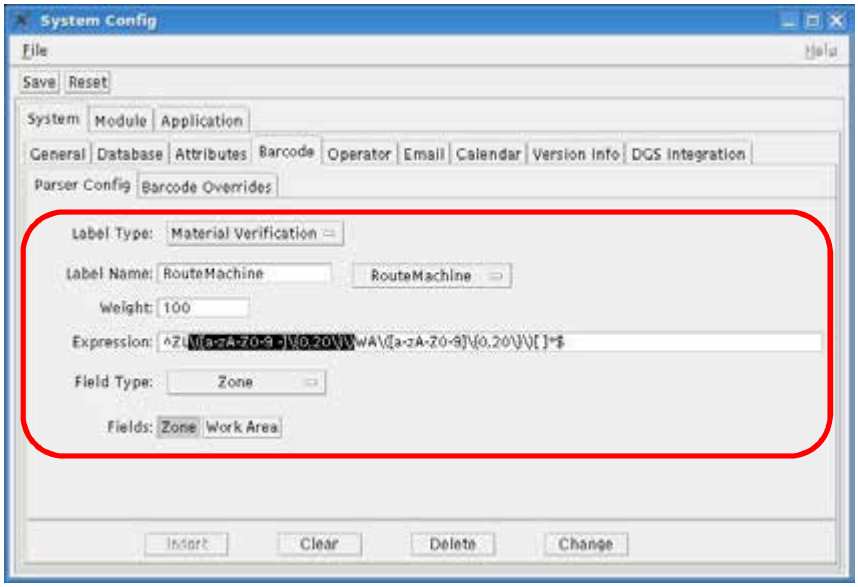
Add the following barcode definition (parser setting) at CIMC system setting→System→Barcode →Parser setting screen.

■ Barcode definition (Parser setting)

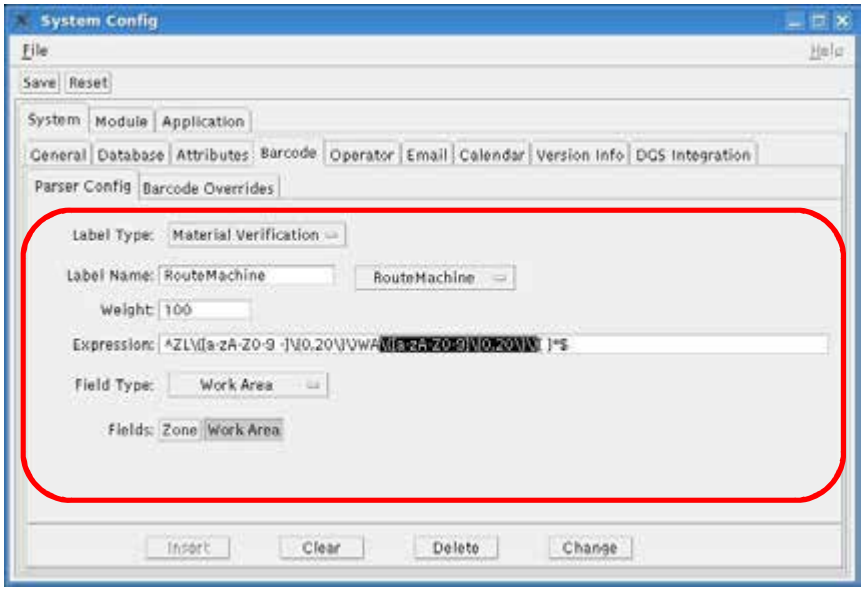
Add barcode definition (RouteMachine) as parser setting.
Define barcode in a format in which one barcode has two kinds of information (line name/work area name).

- Label type: Material Verification
- Label name: RouteMachine
- Priority: 100
- Expression: ^ZL¥([a-zA-Z0-9 -]¥{0,20¥}¥)WA¥([a-zA-Z0-9 -]¥{0,20¥}¥)[ ]*$

<Line name definition>



<Work area definition>



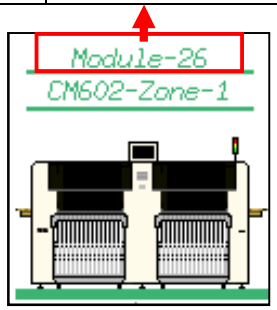
11.2.2 Form of Barcode (Sample)

Format of barcodes is shown below.

■ Barcode format (sample)

ZLModule-26LINEWAM1 Front

Definition	Line name definition		Work area definition	
Sample string	ZL	Module-26LINE	WA	M1 Front
Explanation	Line name definition prefix	Registered module (line) name	Work area definition prefix	Work area name
Number of characters	2 characters fixed	20 characters variable	2 characters fixed	20 characters variable



11.3 Operation

Example of barcodes (sample) is shown below.

11.3.1 Create Barcode (Sample)

Line		**26Line**
Work area	**Barcode character string**	**QR Code/Barcode**
M1 Front (Machine1 front)	ZLModule-26 LINEWAM1 Front	 
M1 Rear (Machine1 rear)	ZLModule-26 LINEWAM1 Rear	 
M2 Front (Machin2 front)	ZLModule-26 LINEWAM2 Front	 
M2 Rear (Machine2 rear)	ZLModule-26 LINEWAM2 Rear	 

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